

Vivante ISP8000 Series
ISP IP
Accessible Registers
for V6.2.x

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This document is compatible with
Vivante ISP8000 Series hardware version 6.2.x

VERISILICON

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Preface

This preface introduces this document, including the intended audience, conventions used, and reference documents for additional reading.

Intended Audience

This document is prepared for both hardware and software integrators who are familiar with system on chip (SoC) digital design and related support devices. Those who would benefit from this technical manual are:

- Engineers and managers who are evaluating this Vivante DC IP for use in a system
- Engineers who are designing this Vivante DC IP into a system

Using This Document

Vivante maintains a consistent register definition scheme across all its cores. For this reason, some field bits may have a field name defined in this document, but the field may not be in use for the core or specific core revision described in this document

You must not access register bits that are not defined in this document. All undefined bits within the valid register address range are designated as reserved register bits. These reserved bits are used for internal functions and any illegal operations may cause unpredictable behavior including, but not limited to, an IP/data bus hang.

Other conventions used in this document include:

- Hexadecimal numbers are indicated by the prefix "0x".
  For example, 0x32CF.
- Binary numbers are indicated by the prefix "0b".
  For example, 0b0011.0010.1100.1111.
- Rows with fields which are obsolete or not applicable for this core are not present or are grayed out.

References

See the following documents for other relevant information:

- Vivante ISP8000 Series Hardware Features: Feature set available with this IP core
- Vivante ISP8000 Series Hardware Integration Guide: Integration reference manual for this IP core

- Vivante ISP8000 Series Accessible Registers: This document
- Vivante ISP8000 Series Hardware Implementation Manual: Implementation reference material for the physical design implementation of this IP core

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1 Introduction

This document is a register reference of exposed AHB/APB accessible registers for this Vivante IP core and includes only registers which Vivante customers may expose to their customers.

The register description chapter provides details about these registers, including their addresses, meanings, read/write attributes, field bits, reset values, and field meanings.

Register and field descriptions are not verbose, as their functions may be discerned from the module and register and/or field name. Registers are read/write unless specified otherwise.

Reset values for some registers will change for each release. These include those registers which uniquely identify a given release for a Vivante core. The reset value for these fields can be ignored during test.

A field or register may be noted as double-buffered. Double buffered means there are two sets of registers/register bits. One set is the working set used for the current frame. The other set is for configuration, so when you configure a register, the configuration does not take effect until the next frame begins.

2 Register Description

VI_CCL

0x00000000 = Byte Address

Description: Clock Control Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
vi_ccl_dis	2:2	rw	'h0	Clock Control Logic disable 0: processing/cfg-clocks for all sub modules enabled 1: processing/cfg-clocks for all sub modules disabled without access to ID and VI_CCL register
vi_ccl_dis_status	1:1	r	'h0	Status of vi_ccl[2] bit (copy of vi_ccl[2])

ISP_ID_CUSTOM_ID

0x00000004 = Byte Address

Description: Customer ID

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_id_custom_id	31:0	r	'h500000A3	Shows Project ID, Each Project maps to a Unique Project ID 0x80: Project-0 0x81: Project-1 0x82: Project-2 0x83: Project-3

ISP_ID_PRODUCT_ID

0x00000008 = Byte Address

Description: Global control internal shadow register

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Sub-Field Name	Bits	rw	Reset	Description
productidaux	31:28	r	'h0	Product Aux
productidtype	27:24	r	'h0	Following is a Example for this 4bit field (GPU/VIP Using) , You can plan the values accordling to ISP 's Product RoadMap Product Type 0: ISP Only 1: DW Only 2: VSE Only 3: ISP + DW100 4: ISP + DW100 + VSE 5: ISP + DW200
productidnum	23:4	r	'h08000	Shows Product Number, Example: ISP8000L,ProductIDNum=0x08000 DW100,ProductIDNum=0x00100 DW200,ProductIDNum=0x00200
productidgradelevel	3:0	r	'h0	Product GradeLevel 0: None-no extra letter on the product name 1: N (Nano) 2: L (Lite) 3: D,Dual 4: N-D, Nano-Dual 5: L-D, Lite-Dual

ISP_ID_CHIP_ID

0x0000000c = Byte Address

Description: Chip ID

Sub-Field Name	Bits	rw	Reset	Description
isp_id_chip_id	31:0	r	'8000	ChipID ISPNano Series, ChipID=0x8000 ISP8000L Series, ChipID=0x8000 ISP8000 Series, ChipID=0x8000 DW100, ChipID=0x0100 DW200, ChipID=0x0200

VI_ICCL

0x00000010 = Byte Address

Description: Internal Clock Control Register

Sub-Field Name	Bits	rw	Reset	Description
cnrv2_clk_en	20:20	rw	'h1	cnrv2 clock enable 0: power safe ; 1: processing mode
stitch_3_clk_en	19:19	rw	'h1	unused
stitch_2_clk_en	18:18	rw	'h1	unused
reserved0	17:17	rw	'h1	reserved0
mcm_rd_clk_en	16:16	rw	'h1	MCM read clock enable 0: power safe ; 1: processing mode
mcm_wr_clk_en	15:15	rw	'h1	0: power safe ; 1: processing mode
stitch_1_clk_en	14:14	rw	'h1	unused

stitch_0_clk_en	13:13	rw	'h1	stitch0 clock enable 0: power safe ; 1: processing mode
vi_mipi2_clk_enable	12:12	rw	'h1	unused
vi_mipi_clk_enable	11:11	rw	'h1	unused
vi_smia_clk_enable	10:10	rw	'h1	unused
vi_simp_clk_enable	9:9	rw	'h1	unused
vi_ie_clk_enable	8:8	rw	'h1	unused
vi_mi_clk_enable	6:6	rw	'h1	Memory interface clock enable 0: power safe ; 1: processing mode
vi_jpeg_clk_enable	5:5	rw	'h1	unused
vi_srsz_clk_enable	4:4	rw	'h1	Self-picture resize clock enable 0: power safe ; 1: processing mode
vi_mrsz_clk_enable	3:3	rw	'h1	Main resize clock enable 0: power safe ; 1: processing mode
vi_cp_clk_enable	1:1	rw	'h1	CPROC processing clock enable 0: power safe ; 1: processing mode
vi_isp_clk_enable	0:0	rw	'h1	ISP processing clock enable 0: power safe ; 1: processing mode

VI_IRCL

0x00000014 = Byte Address

Description: Internal Reset Control Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
soft_rst_hw_mcm	23:23	rw	'h0	hw mcm software reset 0: Processing mode; 1: reset state
soft_rst_mp_ctrl	22:22	rw	'h0	mp_ctrl software reset 0: Processing mode; 1: reset state
soft_rst_cnrv2	21:21	rw	'h0	cnrv2 software reset 0: Processing mode; 1: reset state
soft_rst_mi_wrapper	20:20	rw	'h0	mi wrapper software reset 0: Processing mode; 1: reset state
soft_rst_stitch_3	19:19	rw	'h0	unused
soft_rst_stitch_2	18:18	rw	'h0	unused
soft_rst_fe	17:17	rw	'h0	0: Processing mode; 1: reset state
soft_rst_mcm_rd	16:16	rw	'h0	0: Processing mode; 1: reset state
soft_rst_mcm_wr	15:15	rw	'h0	MCM write software reset 0: Processing mode; 1: reset state
soft_rst_stitch_1	14:14	rw	'h0	unused

soft_rst_stitch_0	13:13	rw	'h0	stitch0 software reset 0: Processing mode; 1: reset state
vi_mipi2_soft_rst	12:12	rw	'h0	unused
vi_mipi_soft_rst	11:11	rw	'h0	unused
vi_smia_soft_rst	10:10	rw	'h0	unused
vi_simp_soft_rst	9:9	rw	'h0	unused
vi_ie_soft_rst	8:8	rw	'h0	unused
vi_marvin_rst	7:7	rw	'h0	Hardware reset of entire image signal processor 0: Processing mode; 1: reset state
vi_mi_soft_rst	6:6	rw	'h0	Memory interface software reset 0: Processing mode; 1: reset state
vi_jpeg_soft_rst	5:5	rw	'h0	unused
vi_srsz_soft_rst	4:4	rw	'h0	Self-picture resize software reset 0: Processing mode; 1: reset state
vi_mrsz_soft_rst	3:3	rw	'h0	Main-picture resize software reset 0: Processing mode; 1: reset state
vi_ycs_soft_rst	2:2	rw	'h0	Y/C splitter software reset 0: Processing mode; 1: reset state
vi_cp_soft_rst	1:1	rw	'h0	Color processing software reset 0: Processing mode; 1: reset state
vi_isp_soft_rst	0:0	rw	'h0	ISP software reset 0: Processing mode; 1: reset state

VI_DPCL

0x00000018 = Byte Address

Description: Data Path Control Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_only	31:31	rw	'h0	Selects output interface for fifo1 stream interface 0: YUV 1: only Y
if_select3	27:26	rw	'h0	unused
if_select2	25:24	rw	'h0	unused
next_sensor_id	23:22	rw	'h0	next frame id
sensor_id	21:20	rw	'h0	Present frame id [1:0]
reserved2	19:19	rw	'h0	reserved2
strm_mux	18:18	rw	'h0	1:raw out 0:yuv out
if_select1	17:16	rw	'h0	unused
reserved1	15:15	rw	'h0	unused

vi_chan_mode	14:12	rw	'h0	Enable Main Path, Self Path 1, Self Path 2. Note: More than one bit may be set to enable more than one path. Bit 12: Enable Main Path Bit 13: Enable Self Path 1 Bit 14: Enable Self Path 2
vi_dma_spmux	11:11	rw	'h0	unused
vi_dma_iemux	10:10	rw	'h0	unused
if_select0	9:8	rw	'h0	Input interface for sensor0 0: parallel interface 1: SMIA-interface (Reserved) 2: Unused 3: HDR Stitch output
vi_mp_ctrl_sel	7:7	rw	'h0	1:select mp ctrl data 0: not select mp ctrl
vi_dma_switch	6:4	rw	'h0	unused
sensor_id_h	3:2	rw	'h0	unused
vi_mp_mux	1:0	rw	'h0	unused

ISP_ID_ECO_ID

0x00000020 = Byte Address

Description: ECO version

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_id_eco_id	31:0	r	'h0	0x0: Formal release 0x1: 1st ECO release 0x2: 2nd ECO release ....

ISP_ID_CHIP_REVISION

0x00000024 = Byte Address

Description: Internal chip version, this value may vary between releases and can be ignored during test.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_id_chip_revision	31:0	r	'h00006200	Shows Tag Number Example: If Tag=isp8000_1_2_1_rc3z then, ChipRevision=0x0000_1213

ISP_ID_PATCH_REVISION

0x00000028 = Byte Address

Description: Internal tag version, this value may vary between releases and can be ignored during test.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_id_patch_revision	31:0	r	'd19	Shows the last letter of a Tag Example: If Tag=isp8000_1_2_1_rc3z then, PatchRevision=8'd26 "a" map to 8'd1 "b" map to 8'd2 ... "z" map to 8'd26

ISP_ID_CHIP_DATE

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0x0000002c = Byte Address

Description: Chip date value, this value may vary between releases and can be ignored during test.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_id_chip_date	31:0	r	'h20230717	Shows Date Value, such as 32'h20191225

ISP_ID_CHIP_TIME

0x00000030 = Byte Address

Description: Chip time value, this value may vary between releases and can be ignored during test.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_id_chip_time	31:0	r	'h1800	Shows Time Value, such as 32'h18365500

VI_ID_RSV0

0x00000034 = Byte Address

Description: marvin_id_rsv0

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
marvin_id_rsv0	31:0	r	'ha100	reserved id

VI_ID_RSV1

0x00000038 = Byte Address

Description: marvin_id_rsv1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
marvin_id_rsv1	31:0	r	'ha1	reserved id

VI_ID_RSV2

0x0000003c = Byte Address

Description: marvin_id_rsv2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
marvin_id_rsv2	31:0	r	'ha2	reserved id

VI_ID_RSV3

0x00000040 = Byte Address

Description: marvin_id_rsv3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
marvin_id_rsv3	31:0	r	'ha3	reserved id

VI_ID_RSV4

0x00000044 = Byte Address

Description: marvin_id_rsv4

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
marvin_id_rsv4	31:0	r	'ha4	reserved id

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VI_ID_RSV5

0x00000048 = Byte Address

Description: marvin_id_rsv5

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
marvin_id_rsv5	31:0	r	'ha5	reserved id

VI_ID_RSV6

0x0000004c = Byte Address

Description: marvin_id_rsv6

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
marvin_id_rsv6	31:0	r	'ha6	reserved id

VI_ID_RSV7

0x00000050 = Byte Address

Description: marvin_id_rsv7

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
marvin_id_rsv7	31:0	r	'ha7	reserved id

ISP_FLASH_CMD

0x00000260 = Byte Address

Description: Flash command

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
preflash_on	2:2	w	'h0	Preflash on 0: no effect 1: flash delay counter is started at next trigger event, No capture event is signaled to the sensor interface block.
flash_on	1:1	w	'h0	Flash on 0: no effect 1: flash delay counter is started at next trigger event A capture event is signaled to the sensor interface block.
prelight_on	0:0	w	'h0	Prelight on 0: prelight is switched off at next trigger event 1: prelight is switched on at next trigger event

ISP_FLASH_CONFIG

0x00000264 = Byte Address

Description: Flash config

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
fl_cap_delay	7:4	rw	'h0	Capture delayframe number (0 to 15) to be captured after trigger event

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fl_trig_src	3:3	rw	'h0	Trigger source for flash and prelight 0: use “vds_vsync” for trigger event (with evaluation of vs_in_edge) 1: use “fl_trig” for trigger event (positive edge)
fl_pol	2:2	rw	'h0	Polarity of flash related signals 0: flash_trig, prelight_trig are high active 1: flash_trig, prelight_trig are low active
fl_trig_pol	1:1	rw	'h0	VSYNC edge 0: use negative edge of “vds_vsync” if generating a trigger event 1: use positive edge of “vds_vsync” if generating a trigger event
prelight_mode	0:0	rw	'h0	Prelight mode 0: prelight is switched off at begin of flash 1: prelight is switched off at end of flash

ISP_FLASH_PREDIV

0x00000268 = Byte Address

Description: Flash Counter Pre-Divider

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
fl_pre_div	9:0	rw	'h0	Pre-divider for flush/preflash counter

ISP_FLASH_DELAY

0x0000026c = Byte Address

Description: Flash Delay

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
fl_delay	17:0	rw	'h0	Counter value for flash/preflash delay open_delay = (fl_delay + 1) * (fl_pre_div+1) / clk_isp Fl_delay = (open_delay * clk_isp) / (fl_pre_div+1) - 1

ISP_FLASH_TIME

0x00000270 = Byte Address

Description: Flash time

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
fl_time	17:0	rw	'h0	Counter value for flash/preflash time open_time = (fl_time + 1) * (fl_pre_div+1) / clk_isp Fl_time = (open_time * clk_isp) / (fl_pre_div+1) - 1

ISP_FLASH_MAXP

0x00000274 = Byte Address

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Description: Maximum value for flash or preflash

Sub-Field Name	Bits	rw	Reset	Description
fl_maxp	15:0	rw	'h0	Maximum period value for flash or preflashmax. flash/preflash period = 214 * (fl_maxp + 1) / clk_isp Fl_maxp = (max_period * clk_isp) / 214 - 1

ISP_TPG_CTRL

0x00000700 = Byte Address

Description: Test Pattern Generator Module

Sub-Field Name	Bits	rw	Reset	Description
tpg_frame_num	31:16	rw	'h0	0: all of the frame output is valid Other: when the frame_cnt <= frame_num, the output is valid
tpg_solution	11:10	rw	'h0	00: 1080P (1920x1080) 01: 720P (1280x720) 10: 4K (3840x2160) 11: user defined resolution and timing
tpg_max_sync	9:9	rw	'h0	Set to 1 to use VS/VDE and HS/HDE same behavior
tpg_def_sync	8:8	rw	'h0	Set to 1 to select the default sync output
tpg_color_depth	7:6	rw	'h0	Test Pattern Color Depth 00: 8-bit 01: 10-bit 10: 12-bit (pix_width)
tpg_cfa_pat	5:4	rw	'h0	Bayer pattern: 00: RGGB 01: GRBG 10: GBRB 11: BGGR
tpg_img_num	3:1	rw	'h0	Test Pattern Generator Image index 000: 3x3 color block 001: Color bar 010: Gray Bar 011: Highlighted grid 100: Random Generator Others: Reserved
tpg_enable	0:0	rw	'h0	Enable/Disable Test Pattern Generator 0: Disable the TPG 1: Enable the TPG

ISP_TPG_TOTAL_IN

0x00000704 = Byte Address

Description: The total clock

Sub-Field Name	Bits	rw	Reset	Description
tpg_htotal_in	27:14	rw	'h0	The total clock of horizontal
tpg_vtotal_in	13:0	rw	'h0	The total clock of vertical

ISP_TPG_ACT_IN

0x00000708 = Byte Address

Description: The available clock

Sub-Field Name	Bits	rw	Reset	Description
tpg_hact_in	27:14	rw	'h0	The available clock of horizontal
tpg_vact_in	13:0	rw	'h0	The available clock of vertical

ISP_TPG_FP_IN

0x0000070c = Byte Address

Description: The first valid

Sub-Field Name	Bits	rw	Reset	Description
tpg_fp_h_in	27:14	rw	'h0	The first valid of hs
tpg_fp_v_in	13:0	rw	'h0	The first valid of vs

ISP_TPG_BP_IN

0x00000710 = Byte Address

Description: The distance between positive edge of vs with positive edge of hde

Sub-Field Name	Bits	rw	Reset	Description
tpg_bp_h_in	27:14	rw	'h0	The distance between positive edge of vs with positive edge of hde
tpg_bp_v_in	13:0	rw	'h0	The distance between positive edge of vs with positive edge of vde

ISP_TPG_W_IN

0x00000714 = Byte Address

Description: Available clk

Sub-Field Name	Bits	rw	Reset	Description
tpg_hs_w_in	27:14	rw	'h0	Available clk of hs
tpg_vs_w_in	13:0	rw	'h0	Available clk of vs

ISP_TPG_GAP_IN

0x00000718 = Byte Address

Description: The gap of sub_picture

Sub-Field Name	Bits	rw	Reset	Description
tpg_pix_gap_in	27:14	rw	'h0	The width of sub_picture
tpg_line_gap_in	13:0	rw	'h0	The height of sub_picture

ISP_TPG_GAP_STD_IN

0x0000071c = Byte Address

Description: The gap stdio of sub_picture

Sub-Field Name	Bits	rw	Reset	Description

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tpg_pix_gap_std_in	13:0	rw	'h0	The gap of sub_picture

ISP_TPG_RANDOM_SEED

0x00000720 = Byte Address

Description: Random seed

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
tpg_random_seed	31:0	rw	'h0	Random seed

ISP_VSYNC_DELAY

0x00000730 = Byte Address

Description: VSYNC adjust signal

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
regs_vsync_sel	31:31	rw	'h0	base vsync select 0:the first pixel 1:the input vsync
vsync_delay	24:0	rw	'h0	VSYNC adjust number

GREEN_EQUILIBRATE_CTRL

0x00000750 = Byte Address

Description: green_equilibrate_control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
green_equilibrate_th	16:1	rw	'h0	Green equilibrate level threshold, 7 bit floating point precision, default: 0 = disable green equilibrate.
green_equilibrate_en	0:0	rw	'h0	Enable signal of green_equilibrate 0: disable green equilibrate (Default) 1: enable green equilibrate

GREEN_EQUILIBRATE_HCNT_DUMMY

0x00000754 = Byte Address

Description: Blank of last 3 lines

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
green_equilibrate_hcnt_dummy	15:0	rw	'h0	Blank of last 3 lines

GREEN_EQUILIBRATE_CTRL_SHD

0x00000758 = Byte Address

Description: green_equilibrate_control shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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green_equilibrate_th_shd	16:1	r	'h0	Green equilibrate level threshold, 7 bit floating point precision, default: 0 = disable green equilibrate.
green_equilibrate_en_shd	0:0	r	'h0	Enable signal of green_equilibrate 0: disable green equilibrate (Default) 1: enable green equilibrate.

CROP_EN_RAW

0x00000790 = Byte Address

Description: crop enable for raw

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop_en_raw	0:0	rw	'h0	crop enable for raw

CROP_X_DIR_RAW

0x00000794 = Byte Address

Description: Crop range in x direction for raw

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop_h_end_raw	28:16	rw	'h0	Selected picture end pixel position in the x-direction for raw.
crop_h_start_raw	12:0	rw	'h0	Selected picture start pixel position in the x-direction for raw.

CROP_Y_DIR_RAW

0x00000798 = Byte Address

Description: Crop range in y direction for raw

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop_v_end_raw	28:16	rw	'h0	Selected picture end pixel position in the y-direction for raw.
crop_v_start_raw	12:0	rw	'h0	Selected picture start pixel position in the y-direction for raw.

CROP_EN_RAW_SHD

0x0000079c = Byte Address

Description: crop enable for raw for raw_shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop_en_raw_shd	0:0	r	'h0	crop enable for raw

CROP_X_DIR_RAW_SHD

0x000007a0 = Byte Address

Description: Crop range in x direction for raw -shadow register

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Sub-Field Name	Bits	rw	Reset	Description
crop_h_end_raw_shd	28:16	r	'h0	Selected picture end pixel position in the x-direction for raw.
crop_h_start_raw_shd	12:0	r	'h0	Selected picture start pixel position in the x-direction for raw.

CROP_Y_DIR_RAW_SHD

0x000007a4 = Byte Address

Description: Crop range in y direction for raw -shadow register

Sub-Field Name	Bits	rw	Reset	Description
crop_v_end_raw_shd	28:16	r	'h0	Selected picture end pixel position in the y-direction for raw.
crop_v_start_raw_shd	12:0	r	'h0	Selected picture start pixel position in the y-direction for raw.

CPROC_CTRL

0x00000800 = Byte Address

Description: Global control register

Sub-Field Name	Bits	rw	Reset	Description
cproc_c_out_range	3:3	rw	'h0	Color processing chrominance pixel clipping range at output 0: CbCr_out clipping range [16..240] according to ITU-R BT.601 standard 1: Full UV_out clipping range [0..255]
cproc_y_in_range	2:2	rw	'h0	Color processing luminance input range (offset processing) 0: Y_in range [64..940] according to ITU-R BT.601 standard; offset of 64 will be subtracted from Y_in 1: Y_in full range [0..1023]; no offset will be subtracted from Y_in
cproc_y_out_range	1:1	rw	'h0	Color processing luminance output clipping range 0: Y_out clipping range [16..235]; offset of 16 is added to Y_out according to ITU-R BT.601 standard 1: Y_out clipping range [0..255]; no offset is added to Y_out
cproc_enable	0:0	rw	'h0	Color processing enable 0: Color processing is bypassed 1: Color processing is active 2 * 10 bit input data are truncated to 2 * 8 bit output data output data are rounded to 2 * 8 bit and clipping is active

CPROC_CONTRAST

0x00000804 = Byte Address

Description: Color Processing contrast register

Sub-Field Name	Bits	rw	Reset	Description
cproc_contrast	7:0	rw	'h80	Contrast adjustment value 00H equals x 0.0 ... 80H equals x 1.0 ... FFH equals x 1.992

CPROC_BRIGHTNESS

0x00000808 = Byte Address

Description: Color Processing brightness register

Sub-Field Name	Bits	rw	Reset	Description
cproc_brightness_1	9:8	rw	'h0	10bit version Brightness adjustment value 200H equals -512 ... 00H equals +0 ... 1FFH equals +511
cproc_brightness	7:0	rw	'h0	8bit version Brightness adjustment value 80H equals -128 ... 00H equals +0 ... 7FH equals +127

CPROC_SATURATION

0x0000080c = Byte Address

Description: Color Processing saturation register

Sub-Field Name	Bits	rw	Reset	Description
cproc_saturation	7:0	rw	'h80	80H equals -128 (200H equals -512)

CPROC_HUE

0x00000810 = Byte Address

Description: Color Processing hue register

Sub-Field Name	Bits	rw	Reset	Description
cproc_hue	7:0	rw	'h0	...

CPROC_CTRL_SHD

0x00000814 = Byte Address

Description: Global control shadow register

Sub-Field Name	Bits	rw	Reset	Description
cproc_c_out_range_shd	3:3	r	'h0	00H equals +0
cproc_y_in_range_shd	2:2	r	'h0	...
cproc_y_out_range_shd	1:1	r	'h0	7FH equals +127 (1FFH equals +511)

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cproc_enable_shd	0:0	r	'h0	Color processing enable 0: Color processing is bypassed 1: Color processing is active 2 * 10 bit input data are truncated to 2 * 8 bit output data output data are rounded to 2 * 8 bit and clipping is activeColor processing enable 0: Color processing is bypassed 1: Color processing is active 2 * 10 bit input data are truncated to 2 * 8 bit output data output data are rounded to 2 * 8 bit and clipping is active

CPROC_CONTRAST_SHD

0x00000818 = Byte Address

Description: Color Processing contrast Shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cproc_contrast_shd	7:0	r	'h80	Contrast adjustment value 00H equals x 0.0 ... 80H equals x 1.0 ... FFH equals x 1.992

CPROC_BRIGHTNESS_SHD

0x0000081c = Byte Address

Description: Color Processing brightness Shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cproc_brightness_2_shd	11:10	r	'h0	
cproc_brightness_1_shd	9:8	r	'h0	10bit version Brightness adjustment value 80H equals -512 ? 00H equals +0 ? 7FH equals +511
cproc_brightness_shd	7:0	r	'h0	8bit version Brightness adjustment value 80H equals -128 ... 00H equals +0 ... 7FH equals +127

CPROC_SATURATION_SHD

0x00000820 = Byte Address

Description: Color Processing saturation Shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cproc_saturation_shd	7:0	r	'h80	Saturation adjustment value 00H equals x 0.0 ... 80H equals x 1.0 ... FFH equals x 1.992

CPROC_HUE_SHD

0x00000824 = Byte Address

Description: Color Processing hue Shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

cproc_hue_shd	7:0	r	'h0	Hue adjustment value 80H equals -90 degrees ... 00H equals 0 degrees ... 7FH equals +87.188 degrees

ISP_STITCHING0_BLS_EXP_0_A_SHD

0x00000880 = Byte Address

Description: Black level for exposure 0 channel A shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_0_r_shd	31:16	r	'h100	Digital gain value for exposure 0 red pixel
stitching_bls_exp_0_a_shd	11:0	r	'h0	Black level value for exposure 0 channel A

ISP_STITCHING0_BLS_EXP_0_B_SHD

0x00000884 = Byte Address

Description: Black level for exposure 0 channel B shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_0_g_shd	31:16	r	'h100	Digital gain value for exposure 0 blue pixel
stitching_bls_exp_0_b_shd	11:0	r	'h0	Black level value for exposure 0 channel B

ISP_STITCHING0_BLS_EXP_0_C_SHD

0x00000888 = Byte Address

Description: Black level for exposure 0 channel C shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_0_gr_shd	31:16	r	'h100	Digital gain value for exposure 0 green red pixel
stitching_bls_exp_0_c_shd	11:0	r	'h0	Black level value for exposure 0 channel C

ISP_STITCHING0_BLS_EXP_0_D_SHD

0x0000088c = Byte Address

Description: Black level for exposure 0 channel D shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_0_gb_shd	31:16	r	'h100	Digital gain value for exposure 0 green blue pixel
stitching_bls_exp_0_d_shd	11:0	r	'h0	Black level value for exposure 0 channel D

ISP_STITCHING0_BLS_EXP_1_A_SHD

0x00000890 = Byte Address

Description: Black level for exposure 1 channel A shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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stitching_digital_gain_exp_1_r_shd	31:16	r	'h100	Digital gain value for exposure 1 red pixel
stitching_bls_exp_1_a_shd	11:0	r	'h0	Black level value for exposure 1 channel A

ISP_STITCHING0_BLS_EXP_1_B_SHD

0x00000894 = Byte Address

Description: Black level for exposure 1 channel B shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_1_g_shd	31:16	r	'h100	Digital gain value for exposure 1 blue pixel
stitching_bls_exp_1_b_shd	11:0	r	'h0	Black level value for exposure 1 channel B

ISP_STITCHING0_BLS_EXP_1_C_SHD

0x00000898 = Byte Address

Description: Black level for exposure 1 channel C shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_1_gr_shd	31:16	r	'h100	Digital gain value for exposure 1 green red pixel
stitching_bls_exp_1_c_shd	11:0	r	'h0	Black level value for exposure 1 channel C

ISP_STITCHING0_BLS_EXP_1_D_SHD

0x0000089c = Byte Address

Description: Black level for exposure 1 channel D shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_1_gb_shd	31:16	r	'h100	Digital gain value for exposure 1 green blue pixel
stitching_bls_exp_1_d_shd	11:0	r	'h0	Black level value for exposure 1 channel D

ISP_STITCHING0_BLS_EXP_2_A_SHD

0x000008a0 = Byte Address

Description: Black level for exposure 2 channel A shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_2_r_shd	31:16	r	'h100	Digital gain value for exposure 2 red pixel
stitching_bls_exp_2_a_shd	11:0	r	'h0	Black level value for exposure 2 channel A

ISP_STITCHING0_BLS_EXP_2_B_SHD

0x000008a4 = Byte Address

Description: Black level for exposure 2 channel B shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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stitching_digital_gain_exp_2_b_shd	31:16	r	'h100	Digital gain value for exposure 2 blue pixel
stitching_bls_exp_2_b_shd	11:0	r	'h0	Black level value for exposure 2 channel B

ISP_STITCHING0_BLS_EXP_2_C_SHD

0x000008a8 = Byte Address

Description: Black level for exposure 2 channel C shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_2_gr_shd	31:16	r	'h100	Digital gain value for exposure 2 green red pixel
stitching_bls_exp_2_c_shd	11:0	r	'h0	Black level value for exposure 2 channel C

ISP_STITCHING0_BLS_EXP_2_D_SHD

0x000008ac = Byte Address

Description: Black level for exposure 2 channel D shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_2_gb_shd	31:16	r	'h100	Digital gain value for exposure 2 green blue pixel
stitching_bls_exp_2_d_shd	11:0	r	'h0	Black level value for exposure 2 channel D

ISP_STITCHING0_TRANS_RANGE_LINEAR_SHD

0x000008b0 = Byte Address

Description: Linear transformation range start and end shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_trans_range_start_linear_shd	27:16	r	'h19a	Transmission range norm factor for linear mode
stitching_trans_range_norm_factor_mul_linear_shd	15:0	r	'he66	Transmission range start for linear mode

ISP_STITCHING0_TRANS_RANGE_NONLINEAR_SHD

0x000008b4 = Byte Address

Description: Nonlinear transformation range start and end shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_trans_range_start_nonlinear_shd	27:16	r	'h19a	Transmission range norm factor for nonlinear mode
stitching_trans_range_norm_factor_mul_nonlinear_shd	15:0	r	'he66	Transmission range start for nonlinear mode

ISP_STITCHING0_SAT_LEVEL_SHD

0x000008b8 = Byte Address

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Description: Stitching saturation level shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_short_ls_extend_bit_shd	31:28	r	'h0	Short extend bit
stitching_short_vs_extend_bit_shd	27:24	r	'h0	Very Short extend bit
stitching_veryshort_valid_thresh_shd	23:12	r	'h7ff	Very short valid threshold
stitching_veryshort_offset_val_shd	11:0	r	'h0	Short extend bit

ISP_STITCHING0_EXP0_AWB_GAIN_G_SHD

0x000008bc = Byte Address

Description: AWB gain for exposure 0 channel G shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_exp0_awb_gain_gr_shd	25:16	r	'h100	Stitching AWB GR gain for exposure 0
stitching_exp0_awb_gain_gb_shd	9:0	r	'h100	Stitching AWB GB gain for exposure 0

ISP_STITCHING0_EXP0_AWB_GAIN_RB_SHD

0x000008c0 = Byte Address

Description: AWB gain for exposure 0 channel RB shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_exp0_awb_gain_r_shd	25:16	r	'h100	Stitching AWB R gain for exposure 0
stitching_exp0_awb_gain_b_shd	9:0	r	'h100	Stitching AWB B gain for exposure 0

ISP_STITCHING0_EXP1_AWB_GAIN_G_SHD

0x000008c4 = Byte Address

Description: AWB gain for exposure 1 channel G shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_exp1_awb_gain_gr_shd	25:16	r	'h100	Stitching AWB GR gain for exposure 1
stitching_exp1_awb_gain_gb_shd	9:0	r	'h100	Stitching AWB GB gain for exposure 1

ISP_STITCHING0_EXP1_AWB_GAIN_RB_SHD

0x000008c8 = Byte Address

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Description: AWB gain for exposure 1 channel RB shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_exp1_awb_gain_r_shd	25:16	r	'h100	Stitching AWB R gain for exposure 1
stitching_exp1_awb_gain_b_shd	9:0	r	'h100	Stitching AWB B gain for exposure 1

ISP_STITCHING0_EXP2_AWB_GAIN_G_SHD

0x000008cc = Byte Address

Description: AWB gain for exposure 2 channel G shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_exp2_awb_gain_gr_shd	25:16	r	'h100	Stitching AWB GR gain for exposure 2
stitching_exp2_awb_gain_gb_shd	9:0	r	'h100	Stitching AWB GB gain for exposure 2

ISP_STITCHING0_EXP2_AWB_GAIN_RB_SHD

0x000008d0 = Byte Address

Description: AWB gain for exposure 2 channel RB shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_exp2_awb_gain_r_shd	25:16	r	'h100	Stitching AWB R gain for exposure 2
stitching_exp2_awb_gain_b_shd	9:0	r	'h100	Stitching AWB B gain for exposure 2

ISP_STITCHING0_LONG_SAT_PARAMS_SHD

0x000008d4 = Byte Address

Description: Stitching saturation params for long exposure shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_long_sat_thresh_shd	23:12	r	'h7ff	Stitching long saturation threshold
stitching_long_sat_combine_weight_shd	8:0	r	'h0	Stitching long combine weight

ISP_STITCHING0_BLS_EXP_OUT_0_SHD

0x000008d8 = Byte Address

Description: Stitching black level for output shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_bls_exp_out_a_shd	27:16	r	'h0	Black level value for Stitching output channel A

stitching_bls_exp_out_b_shd	11:0	r	'h0	Black level value for Stitching output channel B

ISP_STITCHING0_BLS_EXP_OUT_1_SHD

0x000008dc = Byte Address

Description: Stitching black level for output shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_bls_exp_out_c_shd	27:16	r	'h0	Black level value for Stitching output channel C
stitching_bls_exp_out_d_shd	11:0	r	'h0	Black level value for Stitching output channel D

ISP_PRE_FILT_CTRL

0x00004000 = Byte Address

Description: Pre-filter Mode control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
part_two_enable	23:23	rw	'h1	pre part enable
part_one_enable	22:22	rw	'h1	post part enable
soft_rst_pre_filt	21:21	rw	'h0	Soft reset 0: release 1: reset
demosaic_threshold	20:13	rw	'h4	Threshold for Bayer demosaicing texture detection. This value shifted left 4 bits is compared with the difference of the vertical and horizontal 12 bit wide texture indicators, to determine if the vertical or horizontal texture flag must be set. 0x00: maximum edge sensitivity 0xFF: no texture detection
stage1_select	12:9	rw	'h4	Green filter stage 1 select (range 0x0...0x8) 0x0: maximum blurring 0x4: Default 0x7: minimum blurring 0x8: filter stage1 bypass For a detailed description refer to chapter "Filter (Sharpen/Blur) Programming" in the Programming Guide, Part 2.
out_rgb_bayer_pattern	8:7	rw	'h0	0: RGGB 1: GRBG 2: GBRG 3: BGGR
rgbir_bayer_pattern	6:3	rw	'h0	0: BGGIR/BAYER PATTERN (Default) 1: GRIRG 2: RGGIR 3: GBIRG 4: GIRRG 5: IRGGB 6: GIRBG 7: IRGGR 8: RGIRB 9: GRBIR 10: IRBRG 11: BIRGR 12: BGIRR 13: GBRIR 14: IRRBG 15: RIRGB
green_filt_enable	2:2	rw	'h0	Enable/disable green channel filter 0: disable green filter (Default) 1: enable green filter
green_filt_mode	1:1	rw	'h1	0: green filter static mode (active filter factor = FILT_FAC_MID) 1: dynamic noise reduction/sharpen (Default)

isp_pre_filt_enable	0:0	rw	'h0	0: bypass pre_filt (Default) 1: enable pre_filt

ISP_PRE_FILT_BLS_A

0x00004004 = Byte Address

Description: Black level value of red channel

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_bls_a	15:0	rw	'h0	Black level value R

ISP_PRE_FILT_BLS_B

0x00004008 = Byte Address

Description: Black level value of green channel

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_bls_b	15:0	rw	'h0	Black level value G

ISP_PRE_FILT_BLS_C

0x0000400c = Byte Address

Description: Black level value of blue channel

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_bls_c	15:0	rw	'h0	Black level value B

ISP_PRE_FILT_BLS_D

0x00004010 = Byte Address

Description: Black level value of infrared channel

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_bls_d	15:0	rw	'h0	Black level value IR

ISP_PRE_FILT_GAIN_R

0x00004014 = Byte Address

Description: AWB gain of red channel

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_gain_r	9:0	rw	'h100	Auto white balance gain R channel of pre-filter

ISP_PRE_FILT_GAIN_G

0x00004018 = Byte Address

Description: AWB gain of green channel

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_gain_g	9:0	rw	'h100	Auto white balance gain B channel of pre-filter

ISP_PRE_FILT_GAIN_B

0x0000401c = Byte Address

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Description: AWB gain of blue channel

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_gain_b	9:0	rw	'h100	Auto white balance gain G channel of pre-filter

ISP_PRE_FILT_DPC_TH_R

0x00004020 = Byte Address

Description: Defect pixel correction threshold of red channel

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_dpc_th_med_r	31:16	rw	'hffff	Defect pixel correct median threshold of R channel
isp_pre_filt_dpc_th_avg_r	15:0	rw	'hffff	Defect pixel correct average threshold of R channel

ISP_PRE_FILT_DPC_TH_G

0x00004024 = Byte Address

Description: Defect pixel correction threshold of green channel

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_dpc_th_med_g	31:16	rw	'hffff	Defect pixel correct median threshold of G channel
isp_pre_filt_dpc_th_avg_g	15:0	rw	'hffff	Defect pixel correct average threshold of G channel

ISP_PRE_FILT_DPC_TH_B

0x00004028 = Byte Address

Description: Defect pixel correction threshold of blue channel

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_dpc_th_med_b	31:16	rw	'hffff	Defect pixel correct median threshold of B channel
isp_pre_filt_dpc_th_avg_b	15:0	rw	'hffff	Defect pixel correct average threshold of B channel

ISP_PRE_FILT_DPC_TH_IR

0x0000402c = Byte Address

Description: Defect pixel correction threshold of infrared channel

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_dpc_th_med_ir	31:16	rw	'hffff	Defect pixel correct median threshold of IR channel
isp_pre_filt_dpc_th_avg_ir	15:0	rw	'hffff	Defect pixel correct average threshold of IR channel

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ISP_PRE_FILT_CC_00_01

0x00004030 = Byte Address

Description: Remove IR and color correct matrix coefficient

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_00	26:16	rw	'h80	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_01	10:0	rw	'h0	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision

ISP_PRE_FILT_CC_02_03

0x00004034 = Byte Address

Description: Remove IR and color correct matrix coefficient

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_02	26:16	rw	'h0	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_03	10:0	rw	'h0	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision

ISP_PRE_FILT_CC_10_11

0x00004038 = Byte Address

Description: Remove IR and color correct matrix coefficient

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_10	26:16	rw	'h0	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_11	10:0	rw	'h80	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision

ISP_PRE_FILT_CC_12_13

0x0000403c = Byte Address

Description: Remove IR and color correct matrix coefficient

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_12	26:16	rw	'h0	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_13	10:0	rw	'h0	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision

ISP_PRE_FILT_CC_20_21

0x00004040 = Byte Address

Description: Remove IR and color correct matrix coefficient

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Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_20	26:16	rw	'h0	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_21	10:0	rw	'h0	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision

ISP_PRE_FILT_CC_22_23

0x00004044 = Byte Address

Description: Remove IR and color correct matrix coefficient

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_22	26:16	rw	'h80	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_23	10:0	rw	'h0	Infrared remove and color correction matrix coefficients, -4~3.998, 8 bit float point precision

ISP_PRE_FILT_CC_00_01_SHD

0x00004048 = Byte Address

Description: Remove IR and color correct matrix coefficient shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_00_shd	26:16	r	'h80	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_01_shd	10:0	r	'h0	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision

ISP_PRE_FILT_CC_02_03_SHD

0x0000404c = Byte Address

Description: Remove IR and color correct matrix coefficient shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_02_shd	26:16	r	'h0	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_03_shd	10:0	r	'h0	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision

ISP_PRE_FILT_CC_10_11_SHD

0x00004050 = Byte Address

Description: Remove IR and color correct matrix coefficient shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_10_shd	26:16	r	'h0	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_11_shd	10:0	r	'h80	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision

ISP_PRE_FILT_CC_12_13_SHD

0x00004054 = Byte Address

Description: Remove IR and color correct matrix coefficient shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_12_shd	26:16	r	'h0	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_13_shd	10:0	r	'h0	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision

ISP_PRE_FILT_CC_20_21_SHD

0x00004058 = Byte Address

Description: Remove IR and color correct matrix coefficient shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_20_shd	26:16	r	'h0	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_21_shd	10:0	r	'h0	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision

ISP_PRE_FILT_CC_22_23_SHD

0x0000405c = Byte Address

Description: Remove IR and color correct matrix coefficient shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_cc_22_shd	26:16	r	'h80	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision
isp_pre_filt_cc_23_shd	10:0	r	'h0	Infrared remove and color correction matrix coefficients shadow, -4~3.998, 8 bit float point precision

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ISP_PRE_FILT_IR_DES_PX1

0x00004060 = Byte Address

Description: Infrared de-saturation piecewise linear interpolation x-coordinate 1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_ir_des_px1	15:0	rw	'hffff	Infrared de-saturation piecewise linear interpolation x-coordinate start piont

ISP_PRE_FILT_IR_DES_PY1

0x00004080 = Byte Address

Description: Infrared de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_ir_des_py2	31:16	rw	'h0	infrared de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_ir_des_py1	15:0	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_IR_DES_PY2

0x00004084 = Byte Address

Description: Infrared de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_ir_des_py4	31:16	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_ir_des_py3	15:0	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_IR_DES_PY3

0x00004088 = Byte Address

Description: Infrared de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_ir_des_py6	31:16	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_ir_des_py5	15:0	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_IR_DES_PY4

0x0000408c = Byte Address

Description: Infrared de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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isp_pre_filt_ir_des_py8	31:16	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_ir_des_py7	15:0	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_IR_DES_PY5

0x00004090 = Byte Address

Description: Infrared de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_ir_des_py10	31:16	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_ir_des_py9	15:0	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_IR_DES_PY6

0x00004094 = Byte Address

Description: Infrared de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_ir_des_py12	31:16	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_ir_des_py11	15:0	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_IR_DES_PY7

0x00004098 = Byte Address

Description: Infrared de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_ir_des_py14	31:16	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_ir_des_py13	15:0	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_IR_DES_PY8

0x0000409c = Byte Address

Description: Infrared de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_ir_des_py16	31:16	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_ir_des_py15	15:0	rw	'h0	Infrared de-saturation piecewise linear interpolation y-coordinate point

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ISP_PRE_FILT_IR_DES_PD1

0x000040a0 = Byte Address

Description: Infrared de-saturation piecewise linear interpolation x-distance

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_ir_des_pd4	31:24	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd3	23:16	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd2	15:8	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd1	7:0	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points

ISP_PRE_FILT_IR_DES_PD2

0x000040a4 = Byte Address

Description: Infrared de-saturation piecewise linear interpolation x-distance

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_ir_des_pd8	31:24	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd7	23:16	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd6	15:8	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd5	7:0	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points

ISP_PRE_FILT_IR_DES_PD3

0x000040a8 = Byte Address

Description: Infrared de-saturation piecewise linear interpolation x-distance

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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isp_pre_filt_ir_des_pd12	31:24	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd11	23:16	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd10	15:8	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd9	7:0	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points

ISP_PRE_FILT_IR_DES_PD4

0x000040ac = Byte Address

Description: Infrared de-saturation piecewise linear interpolation x-distance

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_ir_des_pd16	31:24	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd15	23:16	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd14	15:8	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_ir_des_pd13	7:0	rw	'h0	Infrared de-saturation piecewise linear interpolation x-coordinate the distance between 2 points

ISP_PRE_FILT_L_DES_PX1

0x000040b0 = Byte Address

Description: IR remove de-saturation piecewise linear interpolation x-coordinate 1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_l_des_px1	15:0	rw	'hffff	IR remove de-saturation piecewise linear interpolation x-coordinate start piont

ISP_PRE_FILT_L_DES_PY1

0x000040d0 = Byte Address

Description: IR remove de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_l_des_py2	31:16	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_l_des_py1	15:0	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_L_DES_PY2

0x000040d4 = Byte Address

Description: IR remove de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_l_des_py4	31:16	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_l_des_py3	15:0	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_L_DES_PY3

0x000040d8 = Byte Address

Description: IR remove de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_l_des_py6	31:16	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_l_des_py5	15:0	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_L_DES_PY4

0x000040dc = Byte Address

Description: IR remove de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_l_des_py8	31:16	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_l_des_py7	15:0	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_L_DES_PY5

0x000040e0 = Byte Address

Description: IR remove de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_l_des_py10	31:16	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point

isp_pre_filt_l_des_py9	15:0	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_L_DES_PY6

0x000040e4 = Byte Address

Description: IR remove de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_l_des_py12	31:16	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_l_des_py11	15:0	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_L_DES_PY7

0x000040e8 = Byte Address

Description: IR remove de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_l_des_py14	31:16	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_l_des_py13	15:0	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_L_DES_PY8

0x000040ec = Byte Address

Description: IR remove de-saturation piecewise linear interpolation y-coordinate

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_l_des_py16	31:16	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point
isp_pre_filt_l_des_py15	15:0	rw	'h0	IR remove de-saturation piecewise linear interpolation y-coordinate point

ISP_PRE_FILT_L_DES_PD1

0x000040f0 = Byte Address

Description: IR remove de-saturation piecewise linear interpolation x-distance

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_l_des_pd4	31:24	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_l_des_pd3	23:16	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points

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isp_pre_filt_l_des_pd2	15:8	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_l_des_pd1	7:0	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points

ISP_PRE_FILT_L_DES_PD2

0x000040f4 = Byte Address

Description: IR remove de-saturation piecewise linear interpolation x-distance

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_l_des_pd8	31:24	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_l_des_pd7	23:16	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_l_des_pd6	15:8	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_l_des_pd5	7:0	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points

ISP_PRE_FILT_L_DES_PD3

0x000040f8 = Byte Address

Description: IR remove de-saturation piecewise linear interpolation x-distance

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_pre_filt_l_des_pd12	31:24	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_l_des_pd11	23:16	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_l_des_pd10	15:8	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_l_des_pd9	7:0	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points

ISP_PRE_FILT_L_DES_PD4

0x000040fc = Byte Address

Description: IR remove de-saturation piecewise linear interpolation x-distance

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_l_des_pd16	31:24	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_l_des_pd15	23:16	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_l_des_pd14	15:8	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points
isp_pre_filt_l_des_pd13	7:0	rw	'h0	IR remove de-saturation piecewise linear interpolation x-coordinate the distance between 2 points

ISP_PRE_FILT_THRESH_BL0

0x00004100 = Byte Address

Description: Blurring threshold 0

Sub-Field Name	Bits	rw	Reset	Description
prefilt_thresh_bl0	9:0	rw	'hd	unused

ISP_PRE_FILT_THRESH_BL1

0x00004104 = Byte Address

Description: Blurring threshold 1

Sub-Field Name	Bits	rw	Reset	Description
prefilt_thresh_bl1	9:0	rw	'h5	unused

ISP_PRE_FILT_THRESH_SH0

0x00004108 = Byte Address

Description: Sharpening threshold 0

Sub-Field Name	Bits	rw	Reset	Description
prefilt_thresh_sh0	9:0	rw	'h1a	unused

ISP_PRE_FILT_THRESH_SH1

0x0000410c = Byte Address

Description: Sharpening threshold 1

Sub-Field Name	Bits	rw	Reset	Description

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prefilt_thresh_sh1	9:0	rw	'h2c	unused

ISP_PRE_FILT_LUM_WEIGHT

0x00004110 = Byte Address

Description: Luminance weight function

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
lum_weight_gain	18:16	rw	'h2	unused
lum_weight_kink	15:8	rw	'h20	unused
lum_weight_min	7:0	rw	'h40	unused

ISP_PRE_FILT_FAC_SH1

0x00004114 = Byte Address

Description: Filter factor sharp1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
prefilt_fac_sh1	5:0	rw	'h10	unused

ISP_PRE_FILT_FAC_SH0

0x00004118 = Byte Address

Description: Filter factor sharp0

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
prefilt_fac_sh0	5:0	rw	'hc	unused

ISP_PRE_FILT_FAC_MID

0x0000411c = Byte Address

Description: Filter factor middle

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
prefilt_fac_mid	5:0	rw	'ha	unused

ISP_PRE_FILT_FAC_BL0

0x00004120 = Byte Address

Description: Parameter for blur 0 filter

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
prefilt_fac_bl0	5:0	rw	'h6	unused

ISP_PRE_FILT_FAC_BL1

0x00004124 = Byte Address

Description: Parameter for blur 1 filter

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
prefilt_fac_bl1	5:0	rw	'h2	unused

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ISP_PRE_FILT_IR_DENOISE_SW_03

0x00004128 = Byte Address

Description: IR denoise filter weight value 0~3

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_ir_denoise_sw3	31:24	rw	'h10	IR denoise filter coefficient3
isp_pre_filt_ir_denoise_sw2	23:16	rw	'h10	IR denoise filter coefficient2
isp_pre_filt_ir_denoise_sw1	15:8	rw	'h10	IR denoise filter coefficient1
isp_pre_filt_ir_denoise_sw0	7:0	rw	'h10	IR denoise filter coefficient0

ISP_PRE_FILT_IR_DENOISE_SW_45

0x0000412c = Byte Address

Description: IR denoise filter weight value 4~5

Sub-Field Name	Bits	rw	Reset	Description
isp_pre_filt_ir_denoise_sw5	15:8	rw	'h10	IR denoise filter coefficient5
isp_pre_filt_ir_denoise_sw4	7:0	rw	'h10	IR denoise filter coefficient4

ISP_PRE_FILT_H_SIZE

0x00004130 = Byte Address

Description: Pre filter image width

Sub-Field Name	Bits	rw	Reset	Description
pre_filt_h_size	14:0	rw	'h1000	Image width

ISP_PRE_FILT_V_SIZE

0x00004134 = Byte Address

Description: Pre filter image height

Sub-Field Name	Bits	rw	Reset	Description
pre_filt_v_size	13:0	rw	'hc00	Image height

ISP_PRE_FILT_DMY_HBLANK

0x00004138 = Byte Address

Description: Pre filter dummy hblank cycles

Sub-Field Name	Bits	rw	Reset	Description
pre_filt_dmy_hb	15:0	rw	'h20	Dummy lines hblank cycles
post_filt_dmy_hb	31:16	rw	'h20	Dummy lines hblank cycles of post

ISP_PRE_FILT_CTRL_SHD

0x0000413c = Byte Address

Description: Pre-filter Mode control shadow register

Sub-Field Name	Bits	rw	Reset	Description

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part_two_enable_shd	23:23	r	'h1	Pre-filter mode control Part 2 enable shadow
part_one_enable_shd	22:22	r	'h1	Pre-filter mode control Part 1 enable shadow
soft_rst_pre_filt_shd	21:21	r	'h0	Soft reset shadow 0: release 1: reset
demosaic_threshold_shd	20:13	r	'h4	Threshold for Bayer demosaicing texture detection. This value shifted left 4 bits is compared with the difference of the vertical and horizontal 12 bit wide texture indicators, to determine if the vertical or horizontal texture flag must be set. 0x00: maximum edge sensitivity 0xFF: no texture detection
stage1_select_shd	12:9	r	'h4	Green filter stage 1 select (range: 0x0...0x8) 0x0: maximum blurring 0x4: Default 0x7: minimum blurring 0x8: filter stage1 bypass For a detailed description refer to chapter "Filter (Sharpen/Blur) Programming" in the Programming Guide, Part 2.
out_rgb_bayer_pattern_shd	8:7	r	'h0	0: RGGB 1: GRBG 2: GBRG 3: BGGR
rgbir_bayer_pattern_shd	6:3	r	'h0	0: BGGIR 0 0: BAYER PATTERN (Default) 1: GRIRG 2: RGGIR 3: GBIRG 4: GIRRG 5: IRGGB 6: GIRBG 7: IRGGR 8: RGIRB 9: GRBIR 10: IRBRG 11: BIRGR 12: BGIRR 13: GBRIR 14: IRRBG 15: RIRGB
green_filt_enable_shd	2:2	r	'h0	Enable/disable green channel filter 0: disable green filter (Default) 1: enable green filter
green_filt_mode_shd	1:1	r	'h1	(Default)
isp_pre_filt_enable_shd	0:0	r	'h0	(Default)

MRSZ_CTRL

0x00000c00 = Byte Address

Description: Global control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop2_enable	12:12	rw	'h0	keep
auto_upd	10:10	rw	'h0	0: no update 1: update shadow registers at frame end
cfg_upd	9:9	w	'h0	Write 0: nothing happens Write 1: update shadow registers Read: Always 0
crop_enable	8:8	rw	'h0	0: Bypass crop 1: Enable picture crop

MRSZ_CTRL_SHD

0x00000c30 = Byte Address

Description: Global control shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop2_enable_shd	12:12	r	'h0	keep
crop_enable_shd	8:8	r	'h0	0: Bypass crop 1: Enable picture crop
scale_vc_up_shd	7:7	r	'h0	0: Vertical chrominance downscaling selected 1: Vertical chrominance upscaling selected
scale_vy_up_shd	6:6	r	'h0	0: Vertical luminance downscaling selected 1: Vertical luminance upscaling selected

MRSZ_CROP_X_DIR

0x00000c58 = Byte Address

Description: Crop range in x direction

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop_h_end	28:16	rw	'h0	Selected picture end pixel position in the x-direction.
crop_h_start	12:0	rw	'h0	Selected picture start pixel position in the x-direction.

MRSZ_CROP_Y_DIR

0x00000c5c = Byte Address

Description: Crop range in y direction

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop_v_end	28:16	rw	'h0	Selected picture end pixel position in the y-direction.
crop_v_start	12:0	rw	'h0	Selected picture start pixel position in the y-direction.

MRSZ_CROP_X_DIR_SHD

0x00000c60 = Byte Address

Description: Crop range in x direction -shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop_h_end_shd	28:16	r	'h0	Selected picture end pixel position in the x-direction.
crop_h_start_shd	12:0	r	'h0	Selected picture start pixel position in the x-direction.

MRSZ_CROP_Y_DIR_SHD

0x00000c64 = Byte Address

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Description: Crop range in y direction -shadow register

Sub-Field Name	Bits	rw	Reset	Description
crop_v_end_shd	28:16	r	'h0	Selected picture end pixel position in the y-direction.
crop_v_start_shd	12:0	r	'h0	Selected picture start pixel position in the y-direction.

MRSZ_FORMAT_CONV_CTRL

0x00000c6c = Byte Address

Description: Format conversion control

Sub-Field Name	Bits	rw	Reset	Description
rsz_pack_format	10:10	rw	'h0	0: YUV422/YUV420 planar 1: YUV422/YUV420 semi_planar
expand_8to10_method	9:9	rw	'h0	0: add two MSBs in two LSBs 1: add zeros in two LSBs
expand_8to10_enable	8:8	rw	'h0	0: disable 10 bit (add two zeros in two LSBs) 1: enable 10 bit
cfg_422nocosited	7:7	rw	'h0	0: YCbCr422 data are co-sited (Y0 Cb0 and Cr0 are sampled at the same position) 1: YCbCr422 data are non_co-sited (Cb and Cr samples are centered between Y samples) so modified interpolation is activated Note: the programmed value becomes effective immediately after this register is set. Therefore, only write to this register if no picture data is sent to the Self Path.
cfg_cbcr_full	6:6	rw	'h0	0: CbCr has a compressed range of [16..240] 1: CbCr has full range [0..255] Note: the programmed value becomes effective immediately after this register is set. Therefore, only write to this register if no picture data is sent to the Self Path.
cfg_y_full	5:5	rw	'h0	0: Y has a compressed range of [16..235] 1: Y has full range [0..255] Note: the programmed value becomes effective immediately after this register is set. Therefore, only write to this register if no picture data is sent to the Self Path.

rsz_output_format	4:2	rw	'h0	Main Resize output format 000: YCbCr 4:0:0 001: YCbCr 4:2:0 010: YCbCr 4:2:2 011: YCbCr 4:4:4 100: RGB565 101: RGB666 110: RGB888 Reserved
rsz_input_format	1:0	rw	'h0	Main resize input format 00: YCbCr 4:0:0 01: YCbCr 4:2:0 10: YCbCr 4:2:2 11: YCbCr 4:4:4

MRSZ_CROP2_X_DIR_Y_CHANNEL

0x00000ca8 = Byte Address

Description: mrsz_crop2_x_dir_y_channel

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop2_h_end_y_channel	28:16	rw	'h0	Crop horizontal end position,0-4095, crop_h_end >crop_h_start,after conv
crop2_h_start_y_channel	12:0	rw	'h0	Crop horizontal start position,0-4095,crop_h_end >crop_h_start,after conv

MRSZ_CROP2_Y_DIR_Y_CHANNEL

0x00000cac = Byte Address

Description: mrsz_crop2_y_dir_y_channel

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop2_v_end_y_channel	28:16	rw	'h0	Crop vertical end position, 0-4095,crop_v_end >crop_v_start,after conv
crop2_v_start_y_channel	12:0	rw	'h0	Crop vertical start position, 0-4095,crop_v_end >crop_v_start,after conv

MRSZ_CROP2_X_DIR_Y_CHANNEL_SHD

0x00000cb0 = Byte Address

Description: mrsz_crop2_x_dir_y_channel_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop2_h_end_y_channel_shd	28:16	r	'h0	Shadow register, Same as crop1_h_end_y_channel
crop2_h_start_y_channel_shd	12:0	r	'h0	Shadow register, Same as crop1_h_start_y_channel

MRSZ_CROP2_Y_DIR_Y_CHANNEL_SHD

0x00000cb4 = Byte Address

Description: mrsz_crop2_y_dir_y_channel_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop2_v_end_y_channel_shd	28:16	r	'h0	Shadow register, Same as crop1_v_end_y_channel

crop2_v_start_y_channel_shd	12:0	r	'h0	Shadow register, Same as crop1_v_start_y_channel

MRSZ_CROP2_X_DIR_C_CHANNEL

0x00000cb8 = Byte Address

Description: mrsz_crop2_x_dir_c_channel

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop2_h_end_c_channel	28:16	rw	'h0	Crop horizontal end position,0-4095, crop_h_end >crop_h_start,after conv
crop2_h_start_c_channel	12:0	rw	'h0	Crop horizontal start position,0-4095,crop_h_end >crop_h_start,after conv

MRSZ_CROP2_Y_DIR_C_CHANNEL

0x00000cbc = Byte Address

Description: mrsz_crop2_y_dir_c_channel

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop2_v_end_c_channel	28:16	rw	'h0	Crop vertical end position, 0-4095,crop_v_end >crop_v_start,after conv
crop2_v_start_c_channel	12:0	rw	'h0	Crop vertical start position, 0-4095,crop_v_end >crop_v_start,after conv

MRSZ_CROP2_X_DIR_C_CHANNEL_SHD

0x00000cc0 = Byte Address

Description: mrsz_crop2_x_dir_c_channel_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop2_h_end_c_channel_shd	28:16	r	'h0	Shadow register, Same as crop1_h_end_c_channel
crop2_h_start_c_channel_shd	12:0	r	'h0	Shadow register, Same as crop1_h_start_c_channel

MRSZ_CROP2_Y_DIR_C_CHANNEL_SHD

0x00000cc4 = Byte Address

Description: mrsz_crop2_y_dir_c_channel_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
crop2_v_end_c_channel_shd	28:16	r	'h0	Shadow register, Same as crop1_v_end_c_channel
crop2_v_start_c_channel_shd	12:0	r	'h0	Shadow register, Same as crop1_v_start_c_channel

MCM_HW_FRAME_CTRL

VERISILICON LEVEL B: CONFIDENTIAL- DISTRIBUTION RESTRICTED

0x00000d00 = Byte Address

Description: frame ctrl

Sub-Field Name	Bits	rw	Reset	Description
soft_rst	9:9	rw	'h0	the soft reset for mcm_hw
ready_s_err_info_clr	8:8	w	'h0	clr the error info
first_err_clr	7:7	rw	'h0	clear the error info
batch_cnt_clr	6:6	rw	'h0	clear the batch cnt
auto_upd	5:5	rw	'h0	auto update shadow register, when batch end
cfg_upd	4:4	w	'h0	immediately refresh the shadow register
frame_ratio_mode	2:1	rw	'h0	2'b00: process next by round-robin when previous frame is finished 2'b01:process by the frame ration spec. 2'b10: first input first process
mcm_hw_en	0:0	rw	'h0	enable

MCM_HW_FRAME_CTRL_SHD

0x00000d04 = Byte Address

Description: frame ctrl shd

Sub-Field Name	Bits	rw	Reset	Description
frame_ratio_mode	2:1	r	'h0	2'b00:first input first process 2'b01:process by the frame ration spec.
mcm_hw_en	0:0	r	'h0	enable

MCM_HW_FRAME_RATIO0

0x00000d10 = Byte Address

Description: frame ratio and id for seq 0-3

Sub-Field Name	Bits	rw	Reset	Description
sensor_id_seq03	31:28	rw	'h3	same with that of seq00
sensor_ratio_seq03	27:24	rw	'h1	same with that of seq00
sensor_id_seq02	23:20	rw	'h2	same with that of seq00
sensor_ratio_seq02	19:16	rw	'h1	same with that of seq00
sensor_id_seq01	15:12	rw	'h1	same with that of seq00
sensor_ratio_seq01	11:8	rw	'h1	same with that of seq00
sensor_id_seq00	7:4	rw	'h0	the sensor id for this seq
sensor_ratio_seq00	3:0	rw	'h1	the sensor ratio for this seq

MCM_HW_FRAME_RATIO1

0x00000d14 = Byte Address

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Description: frame ratio and id for seq 4-7

Sub-Field Name	Bits	rw	Reset	Description
sensor_id_seq03	31:28	rw	'h7	same with that of seq00
sensor_ratio_seq03	27:24	rw	'h1	same with that of seq00
sensor_id_seq02	23:20	rw	'h6	same with that of seq00
sensor_ratio_seq02	19:16	rw	'h1	same with that of seq00
sensor_id_seq01	15:12	rw	'h5	same with that of seq00
sensor_ratio_seq01	11:8	rw	'h1	same with that of seq00
sensor_id_seq00	7:4	rw	'h4	the sensor id for this seq
sensor_ratio_seq00	3:0	rw	'h1	the sensor ratio for this seq

MCM_HW_FRAME_RATIO2

0x00000d18 = Byte Address

Description: frame ratio and id for seq 8-11

Sub-Field Name	Bits	rw	Reset	Description
sensor_id_seq03	31:28	rw	'hB	same with that of seq00
sensor_ratio_seq03	27:24	rw	'h1	same with that of seq00
sensor_id_seq02	23:20	rw	'hA	same with that of seq00
sensor_ratio_seq02	19:16	rw	'h1	same with that of seq00
sensor_id_seq01	15:12	rw	'h9	same with that of seq00
sensor_ratio_seq01	11:8	rw	'h1	same with that of seq00
sensor_id_seq00	7:4	rw	'h8	the sensor id for this seq
sensor_ratio_seq00	3:0	rw	'h1	the sensor ratio for this seq

MCM_HW_FRAME_RATIO3

0x00000d1c = Byte Address

Description: frame ratio and id for seq 12-15

Sub-Field Name	Bits	rw	Reset	Description
sensor_id_seq03	31:28	rw	'hF	same with that of seq00
sensor_ratio_seq03	27:24	rw	'h1	same with that of seq00
sensor_id_seq02	23:20	rw	'hE	same with that of seq00
sensor_ratio_seq02	19:16	rw	'h1	same with that of seq00
sensor_id_seq01	15:12	rw	'hD	same with that of seq00
sensor_ratio_seq01	11:8	rw	'h1	same with that of seq00
sensor_id_seq00	7:4	rw	'hC	the sensor id for this seq
sensor_ratio_seq00	3:0	rw	'h1	the sensor ratio for this seq

VERISILICON LEVEL B: CONFIDENTIAL- DISTRIBUTION RESTRICTED

MCM_HW_S0_ADDR0

0x00000d20 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S0_ADDR1

0x00000d24 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S0_READY0

0x00000d28 = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S0_READY1

0x00000d2c = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S1_ADDR0

0x00000d30 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S1_ADDR1

VERISILICON LEVEL B: CONFIDENTIAL- DISTRIBUTION RESTRICTED

0x00000d34 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S1_READY0

0x00000d38 = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S1_READY1

0x00000d3c = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S2_ADDR0

0x00000d40 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S2_ADDR1

0x00000d44 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S2_READY0

VERISILICON LEVEL B: CONFIDENTIAL- DISTRIBUTION RESTRICTED

0x00000d48 = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S2_READY1

0x00000d4c = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S3_ADDR0

0x00000d50 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S3_ADDR1

0x00000d54 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S3_READY0

0x00000d58 = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

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MCM_HW_S3_READY1

0x00000d5c = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S4_ADDR0

0x00000d60 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S4_ADDR1

0x00000d64 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S4_READY0

0x00000d68 = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S4_READY1

0x00000d6c = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns

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reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S5_ADDR0

0x00000d70 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S5_ADDR1

0x00000d74 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S5_READY0

0x00000d78 = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S5_READY1

0x00000d7c = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S6_ADDR0

0x00000d80 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S6_ADDR1

0x00000d84 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S6_READY0

0x00000d88 = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S6_READY1

0x00000d8c = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S7_ADDR0

0x00000d90 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S7_ADDR1

0x00000d94 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S7_READY0

0x00000d98 = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S7_READY1

0x00000d9c = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S8_ADDR0

0x00000da0 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S8_ADDR1

0x00000da4 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S8_READY0

0x00000da8 = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S8_READY1

0x00000dac = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S9_ADDR0

0x00000db0 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S9_ADDR1

0x00000db4 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S9_READY0

0x00000db8 = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S9_READY1

0x00000dbc = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S10_ADDR0

0x00000dc0 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S10_ADDR1

0x00000dc4 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S10_READY0

0x00000dc8 = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S10_READY1

0x00000dcc = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S11_ADDR0

0x00000dd0 = Byte Address

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Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S11_ADDR1

0x00000dd4 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S11_READY0

0x00000dd8 = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S11_READY1

0x00000ddc = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S12_ADDR0

0x00000de0 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S12_ADDR1

0x00000de4 = Byte Address

Description: fe dma address

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Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S12_READY0

0x00000de8 = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S12_READY1

0x00000dec = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S13_ADDR0

0x00000df0 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S13_ADDR1

0x00000df4 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S13_READY0

0x00000df8 = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description

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reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S13_READY1

0x00000dfc = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S14_ADDR0

0x00000e00 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S14_ADDR1

0x00000e04 = Byte Address

Description: fe dma address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S14_READY0

0x00000e08 = Byte Address

Description: fe dma ready

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S14_READY1

0x00000e0c = Byte Address

Description: fe dma ready

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Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S15_ADDR0

0x00000e10 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S15_ADDR1

0x00000e14 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	31:4	rw	'h0	command buffer address base in memory range

MCM_HW_S15_READY0

0x00000e18 = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_S15_READY1

0x00000e1c = Byte Address

Description: fe dma ready

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ready	16:16	rw	'h0	Set to 1' b1 to trigger fe(command buffer) to work when in turns
reg_fe_dma_count_qw	15:0	rw	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

MCM_HW_FRAME_RATIO0_SHD

0x00000e20 = Byte Address

Description: frame ratio and id for seq

Sub-Field Name	Bits	rw	Reset	Description
frame_ratio_shd	31:0	r	'h31211101	shadow of seq ratio, refer to MCM_HW_FRAME_RATIO0

MCM_HW_FRAME_RATIO1_SHD

0x00000e24 = Byte Address

Description: frame ratio and id for seq

Sub-Field Name	Bits	rw	Reset	Description
frame_ratio_shd	31:0	r	'h71615141	shadow of seq ratio, refer to MCM_HW_FRAME_RATIO1

MCM_HW_FRAME_RATIO2_SHD

0x00000e28 = Byte Address

Description: frame ratio and id for seq

Sub-Field Name	Bits	rw	Reset	Description
frame_ratio_shd	31:0	r	'hB1A19181	shadow of seq ratio, refer to MCM_HW_FRAME_RATIO2

MCM_HW_FRAME_RATIO3_SHD

0x00000e2c = Byte Address

Description: frame ratio and id for seq

Sub-Field Name	Bits	rw	Reset	Description
frame_ratio_shd	31:0	r	'hF1E1D1C1	shadow of seq ratio, refer to MCM_HW_FRAME_RATIO3

MCM_HW_BATCH_CNT

0x00000e30 = Byte Address

Description: batch counter

Sub-Field Name	Bits	rw	Reset	Description
batch_cnt	31:0	r	'h0	total run batch number after power on or last clear

MCM_HW_BATCH_FRAME_CNT

0x00000e34 = Byte Address

Description: frame count inside a batch

Sub-Field Name	Bits	rw	Reset	Description

batch_frame_cnt	23:12	r	'h0	indicate the frame info in the current batch. It is incremented when every frame-done and reset when batch start Bit[23:20]: sensor_id Bit[19:16]:remaining frames for the sensor_id Bit[15:12]: the sequence num of the sensoid
first_err_batch_frm_cnt	11:0	r	'h0	the frame info when first err irq happens, valid if first_err_flag == 1 Bit[11:8]: sensor_id Bit[7:4]:remaining frames for the sensor_id Bit[3:0]: the sequence num of the sensoid

MCM_HW_BATCH_FIRST_ERR_BATCH_CNT

0x00000e38 = Byte Address

Description: the batch idx when first err irq happens, valid if first_err_flag == 1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
first_err_batch_cnt	31:0	r	'h0	the batch idx when first err irq happens, valid if first_err_flag == 1

MCM_HW_BATCH_ERR_EN

0x00000e3c = Byte Address

Description: batch error enable

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
batch_err_enable	17:0	rw	'h0	err flag generated enable

MCM_HW_BATCH_FIRST_ERR_FLAG

0x00000e40 = Byte Address

Description: first_err_flag

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
first_err_flag	0:0	r	'h0	first err flag

MCM_HW_BATCH_FIRST_ERR_SATUS

0x00000e44 = Byte Address

Description: first_err_satus

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
first_err_status	17:0	r	'h0	when first error occur, indicate which err

MCM_HW_READY_S_ERR_STATUS

0x00000e48 = Byte Address

Description: ready_s_err_status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ready_id_fifo_full	25:25	r	'h0	the full signal of the fifo

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ready_id_fifo_empty	24:24	r	'h1	the empty signal of the fifo
ready_id_cnt	23:16	r	'h0	the count of the ready in internal ready fifo
ready_1_st	9:9	r	'h0	the status of the corresponding ready1 register
ready_0_st	8:8	r	'h0	the status of the corresponding ready0 register
ready_1_wr	7:7	r	'h0	1'b1 indicates ready1 is asserted when err happens
ready_0_wr	6:6	r	'h0	1'b1 indicates ready0 is asserted when err happens
ready_cnt	5:5	r	'h0	the ready write pointer of the sensor when err happen
sensor_id	4:1	r	'h0	the recorded sensor id
ready_s_err_status_en	0:0	r	'h0	the err flag 0: no err 1: err active

MCM_CTRL

0x00001200 = Byte Address

Description: MCM control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_sensor_mem_bypass	17:17	rw	'h1	mcm_sensor_mem_bypass: 0: use sensor latency memory in MCM 1: not use sensor latency memory in MCM
mcm_g2_wr1_fmt	16:14	rw	'h0	Controls the input format for MCM_G2 channel 1 with same definition as mcm_wr0_fmt
mcm_g2_wr0_fmt	13:11	rw	'h0	Controls the input format for MCM_G2 channel 0 with same definition as mcm_wr0_fmt
mcm_wr1_fmt	10:8	rw	'h0	Controls the input format for MCM channel 1 with same definition as mcm_wr0_fmt
mcm_wr0_fmt	7:5	rw	'h0	'b0xx: same with 'b100 'b100: not compress, truncating from MSB, by most bit width supported by post MI_MCM data interface 'b101: RAW20 compress 'b110: RAW24 compress, only for MCM V3 24bit version
mcm_bypass_switch	4:1	rw	'h0	Bypass switch 0000: sensor0 bypass 0001: sensor1 bypass ... 1111: sensor15 bypass if any
mcm_bypass_en	0:0	rw	'h0	Bypass enable 0: bypass disabled 1: bypass enabled

MCM_SIZE0

0x00001204 = Byte Address

Description: MCM dimension register 0

Sub-Field Name	Bits	rw	Reset	Description
mcm_height0	29:16	rw	'h0	Image height for write channel 0
mcm_width0	13:0	rw	'h0	Image width for write channel 0

MCM_SIZE1

0x00001208 = Byte Address

Description: MCM dimension register 1

Sub-Field Name	Bits	rw	Reset	Description
mcm_height1	29:16	rw	'h0	Image height for write channel 1
mcm_width1	13:0	rw	'h0	Image width for write channel 1

MCM_G2_SIZE0

0x0000120c = Byte Address

Description: MCM_G2 dimension register 0

Sub-Field Name	Bits	rw	Reset	Description
mcm_height0	29:16	rw	'h0	Image height for MCM_G2 write channel 0
mcm_width0	13:0	rw	'h0	Image width for MCM_G2 write channel 0

MCM_G2_SIZE1

0x00001210 = Byte Address

Description: MCM_G2 dimension register 1

Sub-Field Name	Bits	rw	Reset	Description
mcm_height1	29:16	rw	'h0	Image height for MCM_G2 write channel 1
mcm_width1	13:0	rw	'h0	Image width for MCM_G2 write channel 1

MCM_RD_CFG

0x00001280 = Byte Address

Description: MCM read format register

Sub-Field Name	Bits	rw	Reset	Description
mcm_rd_fmt	2:0	rw	'h0	'b0xx: same with 'b100 'b100: not decompress, truncating from MSB, by most bit width supported by post ISP data interface 'b101: RAW20 decompress for data from DMA 'b110: RAW24 decompress for data from DMA, only for MCM V3 24bit version

MCM_RETIMING0

0x00001284 = Byte Address

Description: MCM retiming control register 0

Sub-Field Name	Bits	rw	Reset	Description
mcm_vsync_blank	31:8	rw	'h0	Minimum count of clock from previous frame end to the VSYNC of the next frame
mcm_vsync_duration	7:0	rw	'h0	VSYNC active duration for retiming unit which indicate counts of clock

MCM_RETIMING1

0x00001288 = Byte Address

Description: MCM retiming control register 1

Sub-Field Name	Bits	rw	Reset	Description
mcm_hsync_blank	31:8	rw	'h0	Minimum count of clock from the previous line end to the HSYNC of the next line
mcm_hsync_preample	7:0	rw	'h0	Interval count of clock between VSYNC and the subsequent first HSYNC

MCM_WR_RETIMING0

0x00001290 = Byte Address

Description: MCM_WR retiming control register 0 for bypass

Sub-Field Name	Bits	rw	Reset	Description
mcm_vsync_blank	31:8	rw	'h0	Minimum count of clock from previous frame end to the VSYNC of the next frame
mcm_vsync_duration	7:0	rw	'h0	VSYNC active duration for retiming unit which indicate counts of clock

MCM_WR_RETIMING1

0x00001294 = Byte Address

Description: MCM_WR retiming control register 1 for bypass

Sub-Field Name	Bits	rw	Reset	Description
mcm_hsync_blank	31:8	rw	'h0	Minimum count of clock from the previous line end to the HSYNC of the next line
mcm_hsync_preample	7:0	rw	'h0	Interval count of clock between VSYNC and the subsequent first HSYNC

MI_CTRL

0x00001300 = Byte Address

Description: Global control register

Sub-Field Name	Bits	rw	Reset	Description

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regs_frame_end_no_rsp	25:25	rw	'h0	Disable frame_end with write response. 0: the frame end need write response back.(default) 1: the frame end need not write response back(for debug)
sp2_raw2_continuous	24:24	rw	'h0	Enables continuous mode. If set to 1, the same frame is read back over and over. 0: Disabled; 1: enabled
sp2_raw2_start	23:23	w	'h0	Enables DMA access 0: Disabled; 1: enabled
pp_dma_continuous	22:22	rw	'h0	Enables continuous mode. If set to 1, the same frame is read back over and over. 0: Disabled; 1: enabled
pp_dma_start	21:21	w	'h0	Enables DMA access 0: Disabled; 1: enabled
sp2_raw2_write_path_enable	20:20	rw	'h0	Enables sp2_raw2_write path 0: Disabled; 1: enabled
pp_write_path_enable	19:19	rw	'h0	Enables pp write path 0: Disabled; 1: enabled
mcm_g2_raw1_path_enable	18:18	rw	'h0	Enables mcm_g2_raw1 path 0: Disabled; 1: enabled
mcm_g2_raw0_path_enable	17:17	rw	'h0	Enables mcm_g2_raw0 path 0: Disabled; 1: enabled
mcm_raw_rdma_start_con	16:16	rw	'h0	Enables continuous mode. If set to 1, the same frame is read back over and over. A start pulse on dma_start is needed only for the first time. To stop continuous mode reset this bit (takes effect after the next frame end) or execute a soft reset. 0: Disabled; 1: enabled
mcm_raw_rdma_start	15:15	w	'h0	Enables DMA access 0: Disabled; 1: enabled
mcm_raw_rdma_path_enable	14:14	rw	'h0	Enables mcm_raw_rdma path 0: Disabled; 1: enabled
sp2_raw_rdma_start_con	13:13	rw	'h0	Enables continuous mode. If set to 1, the same frame is read back over and over. 0: Disabled; 1: enabled
sp2_raw_rdma_start	12:12	w	'h0	Enables DMA access 0: Disabled; 1: enabled
sp2_raw_rdma_path_enable	11:11	rw	'h0	Enables sp2_raw_rdma path 0: Disabled; 1: enabled
sp2_ycbcr_rdma_start_con	10:10	rw	'h0	Enables continuous mode. If set to 1, the same frame is read back over and over. 0: Disabled; 1: enabled
sp2_ycbcr_rdma_start	9:9	w	'h0	Enables DMA access 0: Disabled; 1: enabled

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sp2_ycbcr_rdma_path_enable	8:8	rw	'h0	Enables sp2_ycbcr_rdma path 0: Disabled; 1: enabled
mcm_raw1_path_enable	7:7	rw	'h0	Enables mcm_raw1 path 0: Disabled; 1: enabled
mcm_raw0_path_enable	6:6	rw	'h0	Enables mcm_raw0 path 0: Disabled; 1: enabled
sp2_raw_path_enable	5:5	rw	'h0	Enables sp2_raw path 0: Disabled; 1: enabled
sp2_ycbcr_path_enable	4:4	rw	'h0	Enables sp2_ycbcr path 0: Disabled; 1: enabled
sp1_ycbcr_path_enable	3:3	rw	'h0	Enables sp1_ycbcr path 0: Disabled; 1: enabled
mp_jdp_path_enable	2:2	rw	'h0	Enables mp_jdp path 0: Disabled; 1: enabled
mp_raw_path_enable	1:1	rw	'h0	Enables mp_raw path 0: Disabled; 1: enabled
mp_ycbcr_path_enable	0:0	rw	'h0	Enables mp_ycbcr path 0: Disabled; 1: enabled

MI_CTRL_SHD

0x00001304 = Byte Address

Description: Global control internal shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_raw2_write_path_enable	20:20	r	'h0	Enables sp2_raw2_write path 0: Disabled; 1: enabled
pp_write_path_enable	19:19	r	'h0	Enables pp write path 0: Disabled; 1: enabled
mcm_g2_raw1_path_enable	18:18	r	'h0	Enables mcm_raw1 path 0: Disabled; 1: enabled
mcm_g2_raw0_path_enable	17:17	r	'h0	Enables mcm_raw0 path 0: Disabled; 1: enabled
mcm_raw_rdma_start_con	16:16	r	'h0	Unused
mcm_raw_rdma_start	15:15	r	'h0	Unused
mcm_raw_rdma_path_enable	14:14	r	'h0	Unused
sp2_raw_rdma_start_con	13:13	r	'h0	Unused
sp2_raw_rdma_start	12:12	r	'h0	Unused
sp2_raw_rdma_path_enable	11:11	r	'h0	Unused
sp2_ycbcr_rdma_start_con	10:10	r	'h0	Unused
mcm_raw1_path_enable	9:9	r	'h0	Enables mcm_raw1 path 0: Disabled; 1: enabled
mcm_raw0_path_enable	8:8	r	'h0	Enables mcm_raw0 path 0: Disabled; 1: enabled

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sp2_raw_path_enable	5:5	r	'h0	Enables sp2_raw path 0: Disabled; 1: enabled
sp2_ycbcr_path_enable	4:4	r	'h0	Enables sp2_ycbcr path 0: Disabled; 1: enabled
sp1_ycbcr_path_enable	3:3	r	'h0	Enables sp1_ycbcr path 0: Disabled; 1: enabled
mp_jdp_path_enable	2:2	r	'h0	Enables mp_jdp path 0: Disabled; 1: enabled
mp_raw_path_enable	1:1	r	'h0	Enables mp_raw path 0: Disabled; 1: enabled
mp_ycbcr_path_enable	0:0	r	'h0	Enables mp_ycbcr path 0: Disabled; 1: enabled

MI_QOS

0x00001308 = Byte Address

Description: AXI Bus QOS value configuration

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_qos_rd	31:28	rw	'h0	QoS value of AXI bus for MCM read
mcm_qos_wr	27:24	rw	'h0	QoS value of AXI bus for MCM write
sp2_qos_rd	23:20	rw	'h0	QoS value of AXI bus bus for sp2 read
sp2_qos_wr	19:16	rw	'h0	QoS value of AXI bus bus for sp2 write
sp1_qos_wr	11:8	rw	'h0	QoS value of AXI bus bus for sp1 write
mp_qos_wr	3:0	rw	'h0	QoS value of AXI bus bus for mp write

MI_QOS2

0x0000130c = Byte Address

Description: AXI Bus QOS value configuration

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rt_qos_rd	7:4	rw	'h0	QoS value of AXI bus for RT read
rt_qos_wr	3:0	rw	'h0	QoS value of AXI bus for RT write

MI_MP_CTRL

0x00001310 = Byte Address

Description: MP control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

mp_init_offset_en	5:5	rw	'h0	Enables updating of the offset counters shadow registers for main path to the programmed register init values. MI_MP_Y/CB/CR_OFFS_CNT_INIT-> MI_MP_Y/CB/CR_OFFS_CNT_SHD The update will be executed either when a forced software update occurs (in register MI_MP_CTRL bit cfg_upd = 1) or when an automatic configuration update signal arrives at the MI input port. The latter is split into main and self picture, so only the corresponding main/self shadow registers are affected.
mp_init_base_en	4:4	rw	'h0	Enables updating of the base address and buffer size shadow registers for main picture to the programmed register init values. MI_MP_Y/CB/CR_BASE_AD_INIT-> MI_MP_Y/CB/CR_OFFS_CNT_SHD MI_MP_Y/CB/CR_SIZE_INIT-> MI_MP_Y/CB/CR_SIZE_SHD The update will be executed either when a forced software update occurs (in register MI_MP_CTRL bit cfg_upd = 1) or when an automatic config update signal arrives at the MI input port. The latter is split into main and self picture. So only the corresponding main/self shadow registers are affected. After a picture skip has been performed init_offset_en selects between skip restart and skip init mode (see bit skip in register MI_MP_CTRL).
mp_mi_cfg_upd	3:3	w	'h0	Forced configuration update. Leads to an immediate update of the shadow registers. Depending on the two init enable bits in the MI_MP_CTRL register (init_offset_en and init_base_en) the offset counter, base address and buffer size shadow registers are also updated.

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mp_mi_skip	2:2	w	'h0	Skip of current or next starting main picture: Aborts writing of main picture image data of the current frame to RAM (after the current burst transmission has been completed). Further main picture data up to the end of the current frame are discarded. No further macroblock line interrupt (mblk_line), no wrap around interrupt for main picture (wrap_mp_y/cb/cr) and no fill level interrupt (fill_mp_y) are generated. Skip does not affect the generation of the main path frame end interrupt (mp_frame_end). Skip does not affect the processing of self picture and its corresponding interrupts namely the self path frame end interrupt (sp_frame_end). The byte counter (register MI_BYTE_CNT) is not affected. It produces the correct number of JPEG or RAW data bytes at the end of the current (skipped) frame. After a skip has been performed, the offset counter for the main picture at the start of the following frame are set depending on the bit init_offset_en in register MI_MP_CTRL: - Skip restart mode (init_offset_en = 0) The offset counters of the main picture are restarted at the old start values of the previous skipped frame. - Skip init mode (init_offset_en = 1) The offset counters of the main picture are initialized with the register contents of the offset counter init registers without any additional forced software update or automatic config update.
mp_auto_update	1:1	rw	'h0	Automatic update of configuration registers for main path at frame end. 0: Disabled; 1: enabled
mp_pingpong_enable	0:0	rw	'h0	Pingpong mode of configuration registers for main path at frame end. 0: Disabled; 1: enabled

MI_MP_FMT

0x00001314 = Byte Address

Description: MI main path format

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_wr_jdp_dp_bit	17:17	rw	'h0	0: 32bit; 1: 64bit;

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mp_wr_yuv_word_aligned_msb_en	16:16	rw	'h0	0: 10bit yuv in LSB of 16bit; 1: 10bit yuv in MSB of 16bit;
mp_wr_yuv_word_aligned	15:15	rw	'h0	0: disable; 1:enable
mp_wr_yuv_nvy	14:13	rw	'h0	00: Put Y first => YC1C2 01: Put Y second => C1YC2 10: Put Y Third => C1C2Y
mp_wr_yuv_nv21	12:12	rw	'h0	0: Put U before V; 1: Put V before U
mp_wr_raw_aligned	11:10	rw	'h0	00: unaligned 01: aligned mode0 10: aligned mode1 11: Reserved
mp_wr_yuv_aligned	9:9	rw	'h0	0: unaligned; 1:aligned
mp_wr_raw_bit	8:6	rw	'h0	The bits count to be packed from MSB: 000: RAW8 001: RAW10 010: RAW12 011: RAW14 100: RAW16 110: RAW24(only for 24bit version) Other: Reserved.
mp_wr_yuv_str	5:4	rw	'h0	00: YUV/RGB SemiPlanar 01: YUV Interleave, RGB Raster 10: YUV/RGB Planar 11: Y Only
mp_wr_yuv_fmt	3:2	rw	'h0	00: YUV420 01: YUV422 10: YUV444 11: Reserved
mp_wr_yuv_bit	1:1	rw	'h0	0: 8 bit YUV/RGB; 1: 10 bit YUV/RGB
mp_wr_jdp_fmt	0:0	rw	'h0	Output format 0: DPCC; 1: JPEG

MI_MP_BUS_CFG

0x00001318 = Byte Address

Description: MP AXI Bus configuration

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_wr_swap_jdp	27:24	rw	'h0	Same as mp_wr_swap_y
mp_wr_swap_raw	23:20	rw	'h0	Same as mp_wr_swap_y
mp_wr_swap_v	19:16	rw	'h0	Same as mp_wr_swap_y
mp_wr_swap_u	15:12	rw	'h0	Same as mp_wr_swap_y
mp_wr_swap_y	11:8	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave or RGB raster mode share the same bits. Bit 0: to swap bytes Bit 1: to swap words Bit 2 to swap double words Bit 3 to swap four words
mp_rd_issue_cap	7:6	rw	'h0	Unused
mp_wr_issue_cap	5:4	rw	'h0	Maximum outstanding issue capability 00: 4 01: 8 10: 16 11: 32
mp_rd_burst_len	3:2	rw	'h0	Unused
mp_wr_burst_len	1:0	rw	'h0	Maximum burst length 00: 4 01: 8 10: 16 11: 32

MI_MP_BUS_ID

0x0000131c = Byte Address

Description: MP AXI Bus ID configuration

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_bus_sw_en	25:25	rw	'h0	Control whether BUS works, and set to 1'b1 before ISP works
mp_rd_id_en	24:24	rw	'h0	Unused
mp_rd_id_cfg	23:16	rw	'h0	Unused
mp_wr_id_en	8:8	rw	'h0	MP AXI ports write ID configuration enable
mp_wr_id_cfg	7:0	rw	'h0	MP AXI ports write ID configuration

MI_MP_BUS_TIMEO

0x00001320 = Byte Address

Description: MP AXI Bus control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_bus_timeo_en	31:31	rw	'h0	MP AXI bus time out waiting enable
mp_bus_timeo	30:1	rw	'h0	MP AXI bus time out waiting clock cycle numbers
mp_bus_timeo_interrupt_disable	0:0	rw	'h0	This interrupt function switch: 1'b0: enable 1'b1: disable

MI_MP_Y_BASE_AD_INIT

0x00001324 = Byte Address

Description: Base address for main picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_base_ad_init	31:4	rw	'h0	Base address of main picture Y component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_Y_SIZE_INIT

0x00001328 = Byte Address

Description: Size of main picture Y component in frame buffer(ring buffer)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

mp_y_size_init	28:4	rw	'h0	Size of main picture Y component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_Y_OFFS_CNT_INIT

0x0000132c = Byte Address

Description: Offset counter init value for main picture Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_offs_cnt_init	28:4	rw	'h0	Offset counter init value of main picture Y component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_Y_LLENGTH

0x00001330 = Byte Address

Description: Line length of main picture Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_llengh	31:0	rw	'h0	Line length of Main Path picture Y component or RGB picture in bytes, also known as line stride. This value should be aligned to bus width (16-byte), and greater than or equal to the valid data length (lval), except when mode is 4:2:x planar, where the value should be aligned to 2 * (bus width) (32 bytes). In planar mode the line length of the Cb and Cr component is assumed according to the YCbCr format, i.e. half for 4:2:x and the same size for 4:4:4. In semi planar 4:2:x mode, the line length of the Cb and Cr component is assumed to be the same size. Note: Line length always refers to the line length of the output image. This is particularly important when rotating. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path. Note: Refer to the Hardware Integration document for a detailed calculation of line length.

MI_MP_Y_PIC_WIDTH

0x00001334 = Byte Address

Description: Image width of the Y component in pixels for main path

Sub-Field Name	Bits	rw	Reset	Description
mp_y_pic_width	31:0	rw	'h0	Image width of the main picture Y component or RGB picture in pixels. For YCbCr 4:2:x and RGB 565 the image width must be a multiple of 2. If no line stride is used but flipping required, the image width must be a multiple of 8 for 4:2:x planar or 4 for 4:4:4 planar/4:2:x semi planar. There are no restrictions for RGB 888/666. In planar mode the image width of the Cb and Cr component is assumed according to the YCbCr format, i.e. half for 4:2:x and the same size for 4:4:4. In semi planar 4:2:x mode the image width of the Cb component (which includes Cr) is assumed the same size. In interleave mode no Cb/Cr image width is used. Note: Image width always refers to the picture width of the output image. This is particularly important when rotating. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path.

MI_MP_Y_PIC_HEIGHT

0x00001338 = Byte Address

Description: Image height of the Y component in pixels for main path

Sub-Field Name	Bits	rw	Reset	Description
mp_y_pic_height	31:0	rw	'h0	Image height of the y component or RGB picture in pixels. In planar and semi planar modes, the image width of the Cb and Cr component is assumed according to the YCbCr format, i.e. half for 4:2:0 and the same for 4:2:2 and 4:4:4. Note: Image height always refers to the picture height of the output image. This is particularly important when rotating. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path.

MI_MP_Y_PIC_SIZE

0x0000133c = Byte Address

Description: Image memory size of the Y component in bytes for main path

Sub-Field Name	Bits	rw	Reset	Description
mp_y_pic_size	31:0	rw	'h0	Image size of the Y component or RGB picture in bytes which has to be the Y line length multiplied by the Y image height (mp_y_llength * mp_y_pic_height). In planar mode, the image size of the Cb and Cr component is assumed according to the YCbCr format, i.e., a quarter for 4:2:0, half for 4:2:2 and the same for 4:4:4. In semi planar mode, the image size of the Cb and Cr component is assumed half for 4:2:0 and the same size for 4:2:2. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path.

MI_MP_CB_BASE_AD_INIT

0x00001340 = Byte Address

Description: Base address for main picture Cb component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
mp_cb_base_ad_init	31:4	rw	'h0	Base address of main picture Cb component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_MP_CB_SIZE_INIT

0x00001344 = Byte Address

Description: Size of main picture Cb component ring buffer

Sub-Field Name	Bits	rw	Reset	Description

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mp_cb_size_init	27:4	rw	'h0	Size of main picture Cb component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_MP_CB_OFFS_CNT_INIT

0x00001348 = Byte Address

Description: Offset counter init value for main picture Cb component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_cb_offs_cnt_init	27:4	rw	'h0	Offset counter init value of main picture Cb component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect. Check exception handling in skip modes.

MI_MP_CR_BASE_AD_INIT

0x0000134c = Byte Address

Description: Base address for main picture Cr component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_cr_base_ad_init	31:4	rw	'h0	Base address of main picture Cr component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_MP_CR_SIZE_INIT

0x00001350 = Byte Address

Description: Size of main picture Cr component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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mp_cr_size_init	27:4	rw	'h0	Size of main picture Cr component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_MP_CR_OFFS_CNT_INIT

0x00001354 = Byte Address

Description: Offset counter init value for main picture Cr component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_cr_offs_cnt_init	27:4	rw	'h0	Offset counter init value of main picture Cr component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect. Check exception handling in skip modes.

MI_MP_Y_BASE_AD_INIT2

0x00001358 = Byte Address

Description: Base address 2 (ping pong) for main picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_base_ad_init2	31:4	rw	'h0	Second ping pong base address of main picture Y component buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_MP_CB_BASE_AD_INIT2

0x0000135c = Byte Address

Description: Base address 2 (pingpong) for main picture Cb component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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mp_cb_base_ad_init2	31:4	rw	'h0	Second ping pong base address of main picture Cb component buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_MP_CR_BASE_AD_INIT2

0x00001360 = Byte Address

Description: Base address 2 (pingpong) for main picture Cr component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_cr_base_ad_init2	31:4	rw	'h0	Second ping pong base address of main picture Cr component buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_MP_Y_OFFS_CNT_START

0x00001364 = Byte Address

Description: Offset counter start value for main picture Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_offs_cnt_start	28:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (main picture Y component). Updated at frame end.

MI_MP_CB_OFFS_CNT_START

0x00001368 = Byte Address

Description: Offset counter start value for main picture Cb component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_cb_offs_cnt_start	27:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (main picture Cb component). Updated at frame end.

MI_MP_CR_OFFS_CNT_START

0x0000136c = Byte Address

Description: Offset counter start value for main picture Cr component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_cr_offs_cnt_start	27:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (main picture Cr component). Updated at frame end.

MI_MP_Y_BASE_AD_SHD

0x00001370 = Byte Address

Description: Base address shadow register for main picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_base_ad	31:4	r	'h0	Base address of main picture Y component ring buffer.

MI_MP_Y_SIZE_SHD

0x00001374 = Byte Address

Description: Size shadow register of main picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_size	28:4	r	'h0	Size of main picture Y component ring buffer.

MI_MP_Y_OFFS_CNT_SHD

0x00001378 = Byte Address

Description: Current offset counter of main picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_offs_cnt	28:4	r	'h0	Current offset counter of main picture Y component for address for address generation. Note: Soft reset will reset the contents to the reset value.

MI_MP_CB_BASE_AD_SHD

0x0000137c = Byte Address

Description: Base address shadow register for main picture Cb component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_cb_base_ad	31:4	r	'h0	Base address of main picture Cb component ring buffer.

MI_MP_CB_SIZE_SHD

0x00001380 = Byte Address

Description: Size shadow register of main picture Cb component ring buffer

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Sub-Field Name	Bits	rw	Reset	Description
mp_cb_size	27:4	r	'h0	Size of main picture Cb component ring buffer.

MI_MP_CB_OFFS_CNT_SHD

0x00001384 = Byte Address

Description: Current offset counter of main picture Cb component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
mp_cb_offs_cnt	27:4	r	'h0	Current offset counter of main picture Cb component ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_MP_CR_BASE_AD_SHD

0x00001388 = Byte Address

Description: Base address shadow register for main picture Cr component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
mp_cr_base_ad	31:4	r	'h0	Base address of main picture Cr component ring buffer.

MI_MP_CR_SIZE_SHD

0x0000138c = Byte Address

Description: Size shadow register of main picture Cr component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
mp_cr_size	27:4	r	'h0	Size of main picture Cr component ring buffer.

MI_MP_CR_OFFS_CNT_SHD

0x00001390 = Byte Address

Description: Current offset counter of main picture Cr component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
mp_cr_offs_cnt	27:4	r	'h0	Current offset counter of main picture Cr component ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_MP_RAW_BASE_AD_INIT

0x00001394 = Byte Address

Description: Follow Y

Sub-Field Name	Bits	rw	Reset	Description

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mp_raw_base_ad_init	31:4	rw	'h0	Base address of main picture RAW component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_RAW_SIZE_INIT

0x00001398 = Byte Address

Description: Follow Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_raw_size_init	28:4	rw	'h0	Size of main picture RAW data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_RAW_OFFS_CNT_INIT

0x0000139c = Byte Address

Description: Follow Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_raw_offs_cnt_init	28:4	rw	'h0	Offset counter initial value of main picture RAW data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_RAW_LLENGTH

0x000013a0 = Byte Address

Description: Follow Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_raw_llengh	31:0	rw	'h0	Line length of Main Path picture RAW data in bytes, also known as line stride. If no line stride is used, line length must match image width. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path.

MI_MP_RAW_PIC_WIDTH

0x000013a4 = Byte Address

Description: Follow Y

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Sub-Field Name	Bits	rw	Reset	Description
mp_raw_pic_width	31:0	rw	'h0	Image width of the Main Path picture RAW component or RGB picture in pixels. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path.

MI_MP_RAW_PIC_HEIGHT

0x000013a8 = Byte Address

Description: Follow Y

Sub-Field Name	Bits	rw	Reset	Description
mp_raw_pic_height	31:0	rw	'h0	Image height of Main Path picture RAW data in pixels, also known as line stride. If no line stride is used, line length must match image width. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path.

MI_MP_RAW_PIC_SIZE

0x000013ac = Byte Address

Description: Follow Y

Sub-Field Name	Bits	rw	Reset	Description
mp_raw_pic_size	31:0	rw	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (mp_raw_llength * mp_raw_pic_height). If no line stride is used, line length must match image width. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path.

MI_MP_RAW_BASE_AD_SHD

0x000013b4 = Byte Address

Description: Follow Y

Sub-Field Name	Bits	rw	Reset	Description
mp_raw_base_ad	31:4	r	'h0	Base address of main picture raw data ring buffer.

MI_MP_RAW_SIZE_SHD

0x000013b8 = Byte Address

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Description: Follow Y

Sub-Field Name	Bits	rw	Reset	Description
mp_raw_size	28:4	r	'h0	Size of main picture raw data ring buffer.

MI_MP_RAW_OFFS_CNT_SHD

0x000013bc = Byte Address

Description: Follow Y

Sub-Field Name	Bits	rw	Reset	Description
mp_raw_offs_cnt	28:4	r	'h0	Current offset counter of main picture raw data ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_MP_JDP_BASE_AD_INIT

0x000013c0 = Byte Address

Description: Follow Y

Sub-Field Name	Bits	rw	Reset	Description
mp_jdp_base_ad_init	31:4	rw	'h0	Base address of Main Path picture JPEG or dpcc ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_JDP_SIZE_INIT

0x000013c4 = Byte Address

Description: Follow Y

Sub-Field Name	Bits	rw	Reset	Description
mp_jdp_size_init	28:4	rw	'h0	Size of Main Path picture JPEG or DPCC ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_JDP_OFFS_CNT_INIT

0x000013c8 = Byte Address

Description: Follow Y

Sub-Field Name	Bits	rw	Reset	Description

mp_jdp_offs_cnt_init	28:4	rw	'h0	Offset counter init value of Main Path picture JPEG or dpcc data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_JDP_LLENGTH

0x000013cc = Byte Address

Description: Follow Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_jdp_llengh	31:0	rw	'h0	Line length of Main Path picture JPEG data in pixels, also known as line stride. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path.

MI_MP_JDP_PIC_WIDTH

0x000013d0 = Byte Address

Description: Follow Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_jdp_pic_width	31:0	rw	'h0	Image width of the Main Path picture JPEG component picture in pixels. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path.

MI_MP_JDP_PIC_HEIGHT

0x000013d4 = Byte Address

Description: Follow Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_jdp_pic_height	31:0	rw	'h0	Image height of the Main Path picture JPEG component picture in pixels. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path.

MI_MP_JDP_PIC_SIZE

0x000013d8 = Byte Address

Description: Follow Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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mp_jdp_pic_size	31:0	rw	'h0	Image size of the Main Path picture JPEG component picture in bytes, which must be the JPEG line length multiplied by the JPEG image height. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the main path.

MI_MP_JDP_OFFS_CNT_START

0x000013dc = Byte Address

Description: Follow Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_jdp_offs_cnt_start	28:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (main picture JPEG or DPCC data). Updated at frame end.

MI_MP_JDP_BASE_AD_SHD

0x000013e0 = Byte Address

Description: Follow Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_jdp_base_ad	31:4	r	'h0	Base address of Main Path picture JPEG data ring buffer.

MI_MP_JDP_SIZE_SHD

0x000013e4 = Byte Address

Description: Follow Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_jdp_size	28:4	r	'h0	Size of Main Path picture JPEG data ring buffer.

MI_MP_JDP_OFFS_CNT_SHD

0x000013e8 = Byte Address

Description: Follow Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_jdp_offs_cnt	28:4	r	'h0	Current offset counter of Main Path picture JPEG or DPCC data ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_MP_STATUS_CLR

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0x000013ec = Byte Address

Description: FIFO full clear

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_jdp_fifo_full	4:4	rw	'h0	Clear status of JDP FIFO full flag in Main path
mp_raw_fifo_full	3:3	rw	'h0	Clear status of RAW FIFO full flag in Main path
mp_cr_fifo_full	2:2	rw	'h0	Clear status of Cr FIFO full flag in Main path
mp_cb_fifo_full	1:1	rw	'h0	Clear status of Cb FIFO full flag in Main path
mp_y_fifo_full	0:0	rw	'h0	Clear status of Y FIFO full flag in Main path

MI_MP_CTRL_STATUS

0x000013f0 = Byte Address

Description: FIFO Full STATUS, Y,U,V,RAW,JDP

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_jdp_fifo_full	4:4	r	'h0	FIFO full flag of JDP FIFO in Main path asserted since last clear
mp_raw_fifo_full	3:3	r	'h0	FIFO full flag of RAW FIFO in Main path asserted since last clear
mp_cr_fifo_full	2:2	r	'h0	FIFO full flag of Cr FIFO in Main path asserted since last clear
mp_cb_fifo_full	1:1	r	'h0	FIFO full flag of Cb FIFO in Main path asserted since last clear
mp_y_fifo_full	0:0	r	'h0	FIFO full flag of Y FIFO in Main path asserted since last clear

MI_MP_AXI_STATUS

0x000013f4 = Byte Address

Description: Bus status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
agsw_enc_pic_rdy	1:1	r	'h0	Memory Interface Bus Error
agsw_enc_buf_full	0:0	r	'h0	Memory Interface Bus Timeout

MI_MP_RAW_BYTE_CNT_STATUS

0x000013f8 = Byte Address

Description: Number of RAW data bytes of the last transmitted frame. Updated at frame end.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_raw_byte_cnt_status	27:0	r	'h0	Counter value which specifies the number of RAW data bytes of the last transmitted frame. Updated at frame end. Note: A soft reset will set the byte counter to zero.

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MI_MP_JPEG_BYTE_CNT_STATUS

0x000013fc = Byte Address

Description: Number of JPEG or DP data bytes of the last transmitted frame.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_jpeg_byte_cnt_status	27:0	r	'h0	Counter value which specifies the number of JPEG data bytes of the last transmitted frame. Updated at frame end. Note: A soft reset will set the byte counter to zero.

MI_MP_DPCC_BYTE_CNT_STATUS

0x00001400 = Byte Address

Description: Number of pixels of the defect pixel list generated by DPCC of the last transmitted frame.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_dpcc_byte_cnt_status	27:0	r	'h0	Counter value specifies the number of pixels of the defect pixel list generated by DPCC of the last transmitted frame. Updated at frame end. Note: A soft reset will set the byte counter to zero.

MI_MP_Y_LVAL_BYTES

0x00001408 = Byte Address

Description: MP Y Component line valid bytes number.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_width_bytes	15:0	rw	'h0	Sizes of bytes occupied by valid data of a line.

MI_MP_RAW_LVAL_BYTES

0x0000140c = Byte Address

Description: MP RAW Component line valid bytes number.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_raw_width_bytes	15:0	rw	'h0	Sizes of bytes occupied by valid data of a line.

MI_SP2_CTRL

0x000014e4 = Byte Address

Description: SP2 control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_rd_raw_cfg_update	9:9	w	'h0	Forced configuration update which leads to an immediate update of the shadow registers for this RAW read path.
sp2_rd_raw_auto_update	8:8	rw	'h0	unused

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sp2_rd_yuv_cfg_update	7:7	w	'h0	unused
sp2_rd_yuv_auto_update	6:6	rw	'h0	unused
sp2_init_offset_en	5:5	rw	'h0	Enables updating of the offset counters shadow registers for self path to the programmed register init values.
sp2_init_base_en	4:4	rw	'h0	Enables updating of the base address and buffer size shadow registers for self picture to the programmed register init values.
sp2_mi_cfg_upd	3:3	w	'h0	Forced configuration update which leads to an immediate update of the shadow registers. Depending on the init_offset_en and init_base_en enable bits in the MI_SP2_CTRL register, the offset counter, base address and buffer size shadow registers are also updated.
sp2_mi_skip	2:2	w	'h0	Skip of current or next starting main picture.
sp2_auto_update	1:1	rw	'h0	Automatic update of configuration registers for Self Path 2 at frame end. 0: disabled; 1: enabled
sp2_pingpong_enable	0:0	rw	'h0	Ping-pong mode of configuration registers for Self Path 2 at frame end. 0: disabled; 1: enabled

MI_SP2_FMT

0x000014e8 = Byte Address

Description: SP2 pixel format control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_wr_yuv_word_aligned_msb_en	29:29	rw	'h0	0: 10bit yuv in LSB of 16bit; 1: 10bit yuv in MSB of 16bit;
sp2_wr_yuv_word_aligned	28:28	rw	'h0	0: disable; 1:enable
sp2_rd_yuv_nvy	27:26	rw	'h0	00: Y channel in the first position 01: Y in the second 10: Y in the third 11: Reserved When in YUV444 or RGB888 interleave: 00: Y channel in the first and third position 01: Y channel in the second and fourth position
sp2_rd_yuv_nv21	25:25	rw	'h0	0: No swap 1: Swap Cr and Cb channel for semi-planar and YUV444 or RGB interleave and YUV422 interleave
sp2_rd_raw_aligned	24:23	rw	'h0	00: Unaligned 01: Aligned mode0 10: Aligned mode1 11: Reserved

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sp2_rd_yuv_aligned	22:22	rw	'h0	0: Unaligned; 1: Aligned
sp2_rd_raw_bit	21:19	rw	'h0	000: RAW8 001: RAW10 010: RAW12 011: RAW14 100: RAW16 101: RAW20, or Reserved
sp2_rd_yuv_str	18:17	rw	'h0	00: YUV/RGB Semi-Planar 01: YUV Interleave, RGB Raster 10: YUV/RGB Planar 11: Reserved
sp2_rd_yuv_fmt	16:15	rw	'h0	00: YUV420 01: YUV422 10: YUV444 11: Reserved
sp2_rd_yuv_bit	14:14	rw	'h0	0: 8 bit YUV/RGB 1: 10 bit YUV/RGB
sp2_wr_yuv_nvy	13:12	rw	'h0	00: Y channel in the first position 01: Y in the second 10: Y in the third 11: Reserved When in YUV444 or RGB888 interleave: 00: Y channel in the first and third position 01: Y channel in the second and forth position
sp2_wr_yuv_nv21	11:11	rw	'h0	0: No swap 1: Swap Cr and Cb channel for semi-planar and YUV444 or RGB interleave and YUV422 interleave
sp2_wr_raw_aligned	10:9	rw	'h0	00: Unaligned 01: Aligned mode0 10: Aligned mode1 11: Reserved
sp2_wr_yuv_aligned	8:8	rw	'h0	0: Unaligned; 1:Aligned
sp2_wr_raw_bit	7:5	rw	'h0	000: RAW8 001: RAW10 010: RAW12 011: RAW14 100: RAW16 101: RAW20, or Reserved
sp2_wr_yuv_str	4:3	rw	'h0	00: YUV/RGB SemiPlanar 01: YUV Interleave, RGB Raster 10: YUV/RGB Planar 11: Y Only
sp2_wr_yuv_fmt	2:1	rw	'h0	00: YUV420 01: YUV422 10: YUV444 11: Reserved
sp2_wr_yuv_bit	0:0	rw	'h0	0: 8 bit YUV/RGB 1: 10 bit YUV/RGB

MI_SP2_BUS_CFG

0x000014ec = Byte Address

Description: SP2 AXI Bus cfg

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_rd_swap_raw	31:28	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words

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sp2_rd_swap_v	27:24	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
sp2_rd_swap_u	23:20	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
sp2_rd_swap_y	19:16	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
sp2_wr_swap_raw	15:12	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
sp2_wr_swap_v	11:8	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
sp2_wr_swap_u	7:4	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
sp2_wr_swap_y	3:0	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words

MI_SP2_BUS_ID

0x000014f0 = Byte Address

Description: SP2 AXI Bus ID configuration

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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sp2_bus_sw_en	27:27	rw	'h0	Control whether BUS works, and set to 1'b1 before ISP works
sp2_rd_issue_cap	26:25	rw	'h0	Maximum outstanding issue capability. 00: 4 01: 8 10: 16 11: 32
sp2_wr_issue_cap	24:23	rw	'h0	Maximum outstanding issue capability. 00: 4 01: 8 10: 16 11: 32
sp2_rd_burst_len	22:21	rw	'h0	Maximum burst length. 00: 4 01: 8 10:16 11: Reserved
sp2_wr_burst_len	20:19	rw	'h0	Maximum burst length. 00: 4 01: 8 10:16 11: Reserved
sp2_rd_id_en	18:18	rw	'h0	Self Path 2 AXI ports read ID configuration enable
sp2_rd_id_cfg	17:10	rw	'h0	Self Path 2 AXI ports read ID configuration
sp2_wr_id_en	8:8	rw	'h0	Self Path 2 AXI ports write ID configuration enable
sp2_wr_id_cfg	7:0	rw	'h0	Self Path 2 AXI ports write ID configuration

MI_SP2_BUS_TIMEO

0x000014f4 = Byte Address

Description: SP2 AXI Bus ctrol register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_bus_timeo_en	31:31	rw	'h0	Self Path 2 AXI bus time out waiting enable
sp2_bus_timeo	30:1	rw	'h0	Self Path 2 AXI bus time out waiting clock cycle numbers
sp2_bus_timeo_interrupt_disable	0:0	rw	'h0	This interrupt function switch: 1'b0: enable 1'b1: disable

MI_SP2_Y_BASE_AD_INIT

0x000014f8 = Byte Address

Description: Base address for self path 2 picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_y_base_ad_init	31:4	rw	'h0	Base address of Self Path 2 picture Y component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_Y_SIZE_INIT

0x000014fc = Byte Address

Description: Size of self path 2 picture Y component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
sp2_y_size_init	28:4	rw	'h0	Size of Self Path 2 picture Y component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_Y_OFFS_CNT_INIT

0x00001500 = Byte Address

Description: Offset counter init value for self picture Y component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
sp2_y_offs_cnt_init	28:4	rw	'h0	Offset counter initial value of Self Path 2 picture Y component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_Y_LLENGTH

0x00001504 = Byte Address

Description: Line length of self picture Y

Sub-Field Name	Bits	rw	Reset	Description

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sp2_y_llengh	31:0	rw	'h0	Line length of Self Path 2 picture Y component or RGB picture in pixel, also known as line stride. If no line stride is used, line length must match image width. For Y component the line length in 4:2:x planar mode must be a multiple of 8, for all other component modes a multiple of 4 and for RGB 565 a multiple of 2. There are no restrictions for RGB 888/666. In planar mode, the line length of the Cb and Cr component is assumed according to the YCbCr format, i.e. half for 4:2:x and the same size for 4:4:4. In semi planar 4:2:x mode, the line length of the Cb and Cr component is assumed the same size. Note: Line length always refers to the line length of the output image. This is particularly important when rotating. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

MI_SP2_Y_PIC_WIDTH

0x00001508 = Byte Address

Description: Y component image width

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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sp2_y_pic_width	31:0	rw	'h0	Image width of the Self Path 2 picture Y component or RGB picture in pixels. For YCbCr 4:2:x and RGB 565, the image width must be a multiple of 2. If no line stride is used but flipping required, the image width must be a multiple of 8 for 4:2:x planar or 4 for 4:4:4 planar/4:2:x semi planar. There are no restrictions for RGB 888/666. In planar mode, the image width of the Cb and Cr component is assumed according to the YCbCr format, i.e. half for 4:2:x and the same size for 4:4:4. In semi planar 4:2:x mode, the image width of the Cb component (which includes Cr) is assumed the same size. In interleave mode, no Cb/Cr image width is used. Note: Image width always refers to the picture width of the output image. This is particularly important when rotating. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

MI_SP2_Y_PIC_HEIGHT

0x0000150c = Byte Address

Description: Y component image height

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_y_pic_height	31:0	rw	'h0	Image height of the Self Path 2 picture Y component or RGB picture in pixels. In planar and semi planar mode, the image height of the Cb and Cr component is assumed according to the YCbCr format, i.e. half for 4:2:0 and the same for 4:2:2 and 4:4:4. Note: Image height always refers to the picture height of the output image. This is particularly important when rotating. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

MI_SP2_Y_PIC_SIZE

0x00001510 = Byte Address

Description: Y component image memory size in bytes

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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sp2_y_pic_size	31:0	rw	'h0	Image size of the Self Path 2 picture Y component or RGB picture in bytes which has to be the Y line length multiplied by the Y image height (sp2_y_llength * sp2_y_pic_height). In planar mode the image size of the Cb and Cr component is assumed according to the YCbCr format, i.e. a quarter for 4:2:0, half for 4:2:2 and the same for 4:4:4. In semi planar mode the image size of the Cb and Cr component is assumed half for 4:2:0 and the same size for 4:2:2. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

MI_SP2_CB_BASE_AD_INIT

0x00001514 = Byte Address

Description: Base address for self path 2 picture Cb component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cb_base_ad_init	31:4	rw	'h0	Base address of Self Path 2 picture Cb component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_CB_SIZE_INIT

0x00001518 = Byte Address

Description: Size of self path 2 picture Cb component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cb_size_init	27:4	rw	'h0	Size of Self Path 2 picture Cb component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_CB_OFFS_CNT_INIT

0x0000151c = Byte Address

Description: Offset counter init value for self path 2 picture Cb component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cb_offs_cnt_init	27:4	rw	'h0	Offset counter initial value of Self Path 2 picture Cb component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_CR_BASE_AD_INIT

0x00001520 = Byte Address

Description: Base address for self path 2 picture Cr component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cr_base_ad_init	31:4	rw	'h0	Base address of Self Path 2 picture Cr component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_CR_SIZE_INIT

0x00001524 = Byte Address

Description: Size of self path 2 picture Cr component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cr_size_init	27:4	rw	'h0	Size of Self Path 2 picture Cr component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_CR_OFFS_CNT_INIT

0x00001528 = Byte Address

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Description: Offset counter init value for self path 2 picture Cr component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
sp2_cr_offs_cnt_init	27:4	rw	'h0	Offset counter initial value of Self Path 2 picture Cr component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_Y_BASE_AD_INIT2

0x0000152c = Byte Address

Description: Base address 2 (ping pong) for self path 2 picture Y component

Sub-Field Name	Bits	rw	Reset	Description
sp2_y_base_ad_init2	31:4	rw	'h0	Second ping pong base address of Self Path 2 picture Y component buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_CB_BASE_AD_INIT2

0x00001530 = Byte Address

Description: Base address 2 (pingpong) for self path 2 picture Cb component

Sub-Field Name	Bits	rw	Reset	Description
sp2_cb_base_ad_init2	31:4	rw	'h0	Second ping pong base address of Self Path 2 picture Cb component buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_CR_BASE_AD_INIT2

0x00001534 = Byte Address

Description: Base address 2 (pingpong) for self path 2 picture Cr component ring buffer

Sub-Field Name	Bits	rw	Reset	Description

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sp2_cr_base_ad_init2	31:4	rw	'h0	Second ping pong base address of Self Path 2 picture Cr component buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_Y_OFFS_CNT_START

0x00001538 = Byte Address

Description: Offset counter start value for self path 2 picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_y_offs_cnt_start	28:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (Self Path 2 picture Y component). Updated at frame end. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_CB_OFFS_CNT_START

0x0000153c = Byte Address

Description: Offset counter start value for self path 2 picture Cb component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cb_offs_cnt_start	27:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (Self Path 2 picture Cb component). Updated at frame end. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_CR_OFFS_CNT_START

0x00001540 = Byte Address

Description: Offset counter start value for self path 2 picture Cr component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cr_offs_cnt_start	27:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (Self Path 2 picture Cr component). Updated at frame end. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_SP2_Y_BASE_AD_SHD

0x00001544 = Byte Address

Description: Base address shadow register for self path 2 picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_y_base_ad	31:4	r	'h0	Base address of Self Path 2 picture Y component ring buffer.

MI_SP2_Y_SIZE_SHD

0x00001548 = Byte Address

Description: Size shadow register of self path 2 picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_y_size	28:4	r	'h0	Size of Self Path 2 picture Y component ring buffer.

MI_SP2_Y_OFFS_CNT_SHD

0x0000154c = Byte Address

Description: Current offset counter of self path 2 picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_y_offs_cnt	28:4	r	'h0	Current offset counter of Self Path 2 picture Y component, JPEG or RAW data ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_SP2_CB_BASE_AD_SHD

0x00001550 = Byte Address

Description: Base address shadow register for self path 2 picture Cb component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cb_base_ad	31:4	r	'h0	Base address of Self Path 2 picture Cb component ring buffer.

MI_SP2_CB_SIZE_SHD

0x00001554 = Byte Address

Description: Size shadow register of self path 2 picture Cb component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cb_size	27:4	r	'h0	Size of Self Path 2 picture Cb component ring buffer.

MI_SP2_CB_OFFS_CNT_SHD

0x00001558 = Byte Address

Description: Current offset counter of self path 2 picture Cb component ring buffer

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Sub-Field Name	Bits	rw	Reset	Description
sp2_cb_offs_cnt	27:4	r	'h0	Current offset counter of Self Path 2 picture Cb component ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_SP2_CR_BASE_AD_SHD

0x0000155c = Byte Address

Description: Base address shadow register for self path 2 picture Cr component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
sp2_cr_base_ad	31:4	r	'h0	Base address of Self Path 2picture Cr component ring buffer.

MI_SP2_CR_SIZE_SHD

0x00001560 = Byte Address

Description: Size shadow register of self path 2 picture Cr component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
sp2_cr_size	27:4	r	'h0	Size of Self Path 2 picture Cr component ring buffer.

MI_SP2_CR_OFFS_CNT_SHD

0x00001564 = Byte Address

Description: Current offset counter of self path 2 picture Cr component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
sp2_cr_offs_cnt	27:4	r	'h0	Current offset counter of Self Path 2 picture Cr component ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_SP2_RAW_BASE_AD_INIT

0x00001568 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_base_ad_init	31:4	rw	'h0	Base address of Self Path 2 picture RAW component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP2_RAW_SIZE_INIT

0x0000156c = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_size_init	28:4	rw	'h0	Size of Self Path 2 picture RAW data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP2_RAW_OFFS_CNT_INIT

0x00001570 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_offs_cnt_init	28:4	rw	'h0	Offset counter initial value of Self Path 2 picture RAW data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP2_RAW_LLENGTH

0x00001574 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_llengh	31:0	rw	'h0	Line length of Self Path 2 RAW data in bytes, also known as line stride. If no line stride is used, line length must match image width. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

MI_SP2_RAW_PIC_WIDTH

0x00001578 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_pic_width	31:0	rw	'h0	Image width of the RAW component picture in pixels. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

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MI_SP2_RAW_PIC_HEIGHT

0x0000157c = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_pic_height	31:0	rw	'h0	Image height of RAW component in pixels. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

MI_SP2_RAW_PIC_SIZE

0x00001580 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_pic_size	31:0	rw	'h0	Image size of the RAW component in pixels which has to be the Y line length multiplied by the RAW image height (sp2_raw_llength * sp2_raw_pic_height). Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

MI_SP2_RAW_OFFS_CNT_START

0x00001584 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_offs_cnt_start	28:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (Self Path 2 picture RAW data). Updated at frame end. Note: A soft reset resets the contents to the reset value.

MI_SP2_RAW_BASE_AD_SHD

0x0000158c = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_base_ad	31:4	r	'h0	Base address of Self Path 2 picture RAW data ring buffer.

MI_SP2_RAW_SIZE_SHD

0x00001590 = Byte Address

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Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_size	28:4	r	'h0	Size of Self Path 2 picture RAW data ring buffer.

MI_SP2_RAW_OFFS_CNT_SHD

0x00001594 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_offs_cnt	28:4	r	'h0	Current offset counter of Self Path 2 picture RAW data ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_SP2_DMA_Y_PIC_START_AD

0x00001598 = Byte Address

Description: Y component image start address

Sub-Field Name	Bits	rw	Reset	Description
sp2_dma_y_pic_start_ad	31:4	rw	'h0	Image start address of the Y component. Note: Must be a multiple of 4 in interleaved mode,

MI_SP2_DMA_Y_PIC_WIDTH

0x0000159c = Byte Address

Description: Y component image width

Sub-Field Name	Bits	rw	Reset	Description
sp2_dma_y_pic_width	14:0	rw	'h0	Image width of the Y component in pixels. For YCbCr 4:2:x, the image width must be a multiple of 2. In planar mode, the image width of the Cb and Cr component is assumed according to the YCbCr format, i.e. half for 4:2:x and the same size for 4:4:4. In semi planar 4:2:x mode, the image width of the Cb component (which includes Cr) is assumed the same size. In interleaved mode, no Cb/Cr image width is used.

MI_SP2_DMA_Y_PIC_LLENGTH

0x000015a0 = Byte Address

Description: Y component original line length

Sub-Field Name	Bits	rw	Reset	Description

sp2_dma_y_pic_llength	31:0	rw	'h0	Line length of the Y component of the original image in memory. For an uncropped image, where lines follow each other without offset (no line stride), line length must match image width. For the Y component, the line length in 4:2:x planar mode must be a multiple of 8; for all other component modes, the line length must be a multiple of 4. In planar mode, the line length of the Cb and Cr component is assumed according to the YCbCr format, i.e. half for 4:2:x and the same size for 4:4:4. In semi planar 4:2:x mode, the line length of the Cb component (which includes Cr) is assumed the same size. In interleaved mode, no Cb/Cr line length is used.

MI_SP2_DMA_Y_PIC_SIZE

0x000015a4 = Byte Address

Description: Y component image memory size in bytes

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_y_pic_size	27:0	rw	'h0	Image size of the Y component in bytes which has to be the Y line length multiplied by the Y image height (dma_y_llength * dma_y_pic_height). In planar mode, the image size of the Cb and Cr component is assumed according to the YCbCr format, i.e. a quarter for 4:2:0, half for 4:2:2 and the same for 4:4:4. In semi planar mode, the image size of the Cb component (which includes Cr) is assumed half for 4:2:0 and the same size for 4:2:2. In interleaved mode, no Cb/Cr image size is used.

MI_SP2_DMA_CB_PIC_START_AD

0x000015a8 = Byte Address

Description: Cb component image start address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_cb_pic_start_ad	31:4	rw	'h0	Image start address of the Cb component. Note: Must be a multiple of 2 in semi-planar mode.

MI_SP2_DMA_CR_PIC_START_AD

0x000015ac = Byte Address

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Description: Cr component image start address

Sub-Field Name	Bits	rw	Reset	Description
sp2_dma_cr_pic_start_ad	31:4	rw	'h0	Image start address of the Cr component.

MI_SP2_DMA_Y_PIC_START_AD_SHD

0x000015b0 = Byte Address

Description: Shadow register of Y component image start address

Sub-Field Name	Bits	rw	Reset	Description
sp2_dma_y_pic_start_ad	31:4	r	'h0	Image start address of the Y component. Note: Must be a multiple of 4 in interleaved mode.

MI_SP2_DMA_CB_PIC_START_AD_SHD

0x000015b4 = Byte Address

Description: Shadow register of Cb component image start address

Sub-Field Name	Bits	rw	Reset	Description
sp2_dma_cb_pic_start_ad	31:4	r	'h0	Image start address of the Cb component. Note: Must be a multiple of 2 in semi-planar mode.

MI_SP2_DMA_CR_PIC_START_AD_SHD

0x000015b8 = Byte Address

Description: Shadow register of Cr component image start address

Sub-Field Name	Bits	rw	Reset	Description
sp2_dma_cr_pic_start_ad	31:4	r	'h0	Image start address of the Cr component.

MI_SP2_DMA_RAW_PIC_START_AD

0x000015bc = Byte Address

Description: RAW component image start address

Sub-Field Name	Bits	rw	Reset	Description
sp2_dma_raw_pic_start_ad	31:4	rw	'h0	Image start address of the RAW component.

MI_SP2_DMA_RAW_PIC_WIDTH

0x000015c0 = Byte Address

Description: RAW component image width

Sub-Field Name	Bits	rw	Reset	Description
sp2_dma_raw_pic_width	14:0	rw	'h0	Image width of the RAW component in pixels.

MI_SP2_DMA_RAW_PIC_LLENGTH

0x000015c4 = Byte Address

Description: RAW component original line length

Sub-Field Name	Bits	rw	Reset	Description
sp2_dma_raw_pic_llength	31:0	rw	'h0	Line length of the RAW component in bytes of the original image in memory. For an uncropped image, where lines follow each other without offset (no line stride), line length must match image width.

MI_SP2_DMA_RAW_PIC_SIZE

0x000015c8 = Byte Address

Description: RAW component image memory size in bytes

Sub-Field Name	Bits	rw	Reset	Description
sp2_dma_raw_pic_size	27:0	rw	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (dma_raw_llength * dma_raw_pic_height).

MI_SP2_DMA_RAW_PIC_START_AD_SHD

0x000015cc = Byte Address

Description: Shadow register of RAW component image start address

Sub-Field Name	Bits	rw	Reset	Description
sp2_dma_raw_pic_start_ad	31:4	r	'h0	Image start address of the RAW component.

MI_SP2_STATUS_CLR

0x000015d0 = Byte Address

Description: FIFO full clear

Sub-Field Name	Bits	rw	Reset	Description
sp2_jdp_fifo_full	4:4	rw	'h0	Clear status of JDP FIFO full flag in Self Path 2
sp2_raw_fifo_full	3:3	rw	'h0	Clear status of RAW FIFO full flag in Self Path 2
sp2_cr_fifo_full	2:2	rw	'h0	Clear status of Cr FIFO full flag in Self Path 2
sp2_cb_fifo_full	1:1	rw	'h0	Clear status of Cb FIFO full flag in Self Path 2
sp2_y_fifo_full	0:0	rw	'h0	Clear status of Y FIFO full flag in Self Path 2

MI_SP2_CTRL_STATUS

0x000015d4 = Byte Address

Description: FIFO Full STATUS, Y,U,V,RAW,JDP

Sub-Field Name	Bits	rw	Reset	Description
sp2_jdp_fifo_full	4:4	r	'h0	FIFO full flag of JDP FIFO in Self Path 2 asserted since last clear

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sp2_raw_fifo_full	3:3	r	'h0	FIFO full flag of RAW FIFO in Self Path 2 asserted since last clear
sp2_cr_fifo_full	2:2	r	'h0	FIFO full flag of Cr FIFO in Self Path 2 asserted since last clear
sp2_cb_fifo_full	1:1	r	'h0	FIFO full flag of Cb FIFO in Self Path 2 asserted since last clear
sp2_y_fifo_full	0:0	r	'h0	FIFO full flag of Y FIFO in Self Path 2 asserted since last clear

MI_SP2_AXI_STATUS

0x000015d8 = Byte Address

Description: Bus status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
agsw_enc_pic_rdy	1:1	r	'h0	Encoder Picture Ready interrupt
agsw_enc_buf_full	0:0	r	'h0	Encoder Buffer Full interrupt

MI_SP2_DMA_YUV_STATUS

0x000015dc = Byte Address

Description: DMA yuv status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_yuv_active	0:0	r	'h0	If set, DMA YUV access is active.

MI_SP2_DMA_RAW_STATUS

0x000015e0 = Byte Address

Description: DMA raw status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_active	0:0	r	'h0	If set, DMA RAW access is active.

MI_SP2_DMA_Y_PIC_LVAL

0x000015e4 = Byte Address

Description: SP2_DMA Y Component line valid bytes number, 128 bit Alignment.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_y_width_bytes	15:0	w	'h0	Sizes occupied by valid data of a line; value should be a multiple of 16 (bus width = 16 bytes).

MI_SP2_DMA_RAW_PIC_LVAL

0x000015ec = Byte Address

Description: SP2_DMA RAW Component line valid bytes number, 128 bit Alignment.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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sp2_dma_raw_width_bytes	15:0	rw	'h0	Sizes occupied by valid data of a line; value should be a multiple of 16 (bus width = 16 bytes).

MI_SP2_Y_LVAL_BYTES

0x000015f0 = Byte Address

Description: SP2 Y Component line valid bytes number.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_y_width_bytes	15:0	rw	'h0	Sizes of bytes occupied by valid data of a line.

MI_MCM_CTRL

0x00001600 = Byte Address

Description: MCM Path MI control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_wr1_cfg_upd	7:7	w	'h0	Forced configuration update which leads to an immediate update of the shadow registers. Depending on the two init enable bits in the MI_MP_CTRL register (init_offset_en and init_base_en), the offset counter, base address and buffer size shadow registers are also updated.
mcm_rd_cfg_upd	6:6	w	'h0	Forced configuration update which leads to an immediate update of the shadow registers for the MCM read path.
mcm_rd_auto_update	5:5	rw	'h0	unused
mcm_init_offset_en	4:4	rw	'h0	Enables updating of the offset counters shadow registers for MCM path to the programmed register init values. The update will be executed either when a forced software update occurs (in register MI_MCM_CTRL bit cfg_upd = 1) or when an automatic config update signal arrives at the MI input port. The latter is split into main and self picture. So only the corresponding main/self shadow registers are affected. After a picture skip has been performed init_offset_en selects between skip restart and skip init mode (see bit skip in register MI_MCM_CTRL).

mcm_init_base_en	3:3	rw	'h0	Enables updating of the base address and buffer size shadow registers for main picture to the programmed register init values. The update will be executed either when a forced software update occurs (in register MI_MCM_CTRL bit cfg_upd = 1) or when an automatic configuration update signal arrives at the MI input port. The latter is split into main and self picture. So only the corresponding main/self shadow registers are affected.
mcm_wr_cfg_upd	2:2	w	'h0	Forced configuration update which leads to an immediate update of the shadow registers. Depending on the two init enable bits in the MI_MP_CTRL register (init_offset_en and init_base_en), the offset counter, base address and buffer size shadow registers are also updated.
mcm_wr_skip	1:1	w	'h0	Skip of current or next starting MCM picture.
mcm_wr_auto_update	0:0	rw	'h0	Automatic update of configuration registers for Self Path 1 at frame end. 0: Disabled; 1: Enabled

MI_MCM_FMT

0x00001604 = Byte Address

Description: MCM pixel format control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_wr1_fmt_aligned	17:16	rw	'h0	00: unaligned 01: aligned mode0 10: aligned mode1 11: Reserved
mcm_wr0_fmt_aligned	15:14	rw	'h0	00: unaligned 01: aligned mode0 10: aligned mode1 11: Reserved
mcm_rd_fmt_aligned	13:12	rw	'h0	00: unaligned 01: aligned mode0 10: aligned mode1 11: Reserved
mcm_wr1_raw_bit	11:8	rw	'h0	same with mcm_wr0_raw_bit
mcm_wr0_raw_bit	7:4	rw	'h0	The bits count to be packed from MSB: 000: RAW8 001: RAW10 010: RAW12 011: RAW14 100: RAW16 110: RAW24(only for 24bit version) Other: Reserved.

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mcm_rd_raw_bit	3:0	rw	'h0	The bits count to be unpacked from MSB: 000: RAW8 001: RAW10 010: RAW12 011: RAW14 100: RAW16 110: RAW24(only for 24bit version) Other: Reserved.

MI_MCM_BUS_CFG

0x00001608 = Byte Address

Description: MCM AXI Bus cfg

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_rd_swap_raw	19:16	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster mode share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
mcm_wr0_swap_raw	15:12	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster mode share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
mcm_wr1_swap_raw	11:8	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster mode share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
mcm_rd_issue_cap	7:6	rw	'h0	Maximum outstanding issue capability. 00: 4 01: 8 10: 16 11: 32
mcm_wr_issue_cap	5:4	rw	'h0	Maximum outstanding issue capability. 00: 4 01: 8 10: 16 11: 32
mcm_rd_burst_len	3:2	rw	'h0	Maximum burst length. 00: 4 01: 8 10: 16 11: Reserved
mcm_wr_burst_len	1:0	rw	'h0	Maximum burst length. 00: 4 01: 8 10: 16 11: Reserved

MI_MCM_BUS_ID

0x0000160c = Byte Address

Description: MCM AXI Bus ID configuration

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_bus_sw_en	27:27	rw	'h0	Control whether BUS works, and set to 1'b1 before ISP works
mcm_rd_id_en	26:26	rw	'h0	MCM AXI ports read ID configuration enable

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mcm_rd_id_cfg	25:18	rw	'h0	MCM AXI ports read ID configuration
mcm_wr1_id_en	8:8	rw	'h0	MCM AXI ports write1 ID configuration enable
mcm_wr1_id_cfg	7:0	rw	'h0	MCM AXI ports write1 ID configuration

MI_MCM_BUS_TIMEO

0x00001610 = Byte Address

Description: MCM AXI Bus control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_bus_timeo_en	31:31	rw	'h0	MCM AXI bus time out waiting enable
mcm_bus_timeo	30:1	rw	'h0	MCM AXI bus time out waiting clock cycle numbers
mcm_bus_timeo_interrupt_disable	0:0	rw	'h0	This interrupt function switch: 1'b0: enable 1'b1: disable

MI_MCM_RAW0_BASE_AD_INIT

0x00001614 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_base_ad_init	31:4	rw	'h0	Base address of MCM Path picture RAW0 component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MCM_RAW0_SIZE_INIT

0x00001618 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_size_init	28:4	rw	'h0	Size of MCM Path picture RAW0 data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MCM_RAW0_OFFS_CNT_INIT

0x0000161c = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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mcm_raw0_offs_cnt_init	28:4	rw	'h0	Offset counter initial value of MCM Path picture RAW data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MCM_RAW0_LLENGTH

0x00001620 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_llengh	31:0	rw	'h0	Line length of MCM Path picture RAW0 data in bytes, also known as line stride. If no line stride is used, line length must match image width. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the MCM path.

MI_MCM_RAW0_PIC_WIDTH

0x00001624 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_pic_width	31:0	rw	'h0	Image width of the MCM Path picture RAW component in pixels. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the MCM path.

MI_MCM_RAW0_PIC_HEIGHT

0x00001628 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_pic_height	31:0	rw	'h0	Image height of the RAW0 component in pixels. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the MCM path.

MI_MCM_RAW0_PIC_SIZE

0x0000162c = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw0_pic_size	31:0	rw	'h0	Image size of the RAW0 component in bytes which has to be the RAW0 line length multiplied by the RAW0 image height (mcm_raw0_llength * mcm_raw0_pic_height). If no line stride is used, line length must match image width. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the MCM path.

MI_MCM_RAW0_OFFS_CNT_START

0x00001630 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw0_offs_cnt_start	28:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (MCM Path picture RAW data). Updated at frame end. Note: A soft reset resets the contents to the reset value.

MI_MCM_RAW0_BASE_AD_SHD

0x00001634 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw0_base_ad	31:4	r	'h0	Base address of MCM Path picture RAW0 data ring buffer.

MI_MCM_RAW0_SIZE_SHD

0x00001638 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw0_size	28:4	r	'h0	Size of MCM Path picture RAW0 data ring buffer.

MI_MCM_RAW0_OFFS_CNT_SHD

0x0000163c = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description

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mcm_raw0_offs_cnt	28:4	r	'h0	Current offset counter of MCM Path picture RAW data ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_MCM_RAW1_BASE_AD_INIT

0x00001640 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_base_ad_init	31:4	rw	'h0	Base address of MCM Path picture RAW1 component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MCM_RAW1_SIZE_INIT

0x00001644 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_size_init	28:4	rw	'h0	Size of MCM Path picture RAW1 data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MCM_RAW1_OFFS_CNT_INIT

0x00001648 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_offs_cnt_init	28:4	rw	'h0	Offset counter initial value of MCM Path picture RAW1 data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MCM_RAW1_LLENGTH

0x0000164c = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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mcm_raw1_llengh	31:0	rw	'h0	Line length of MCM Path picture RAW1 data in bytes, also known as line stride. If no line stride is used, line length must match image width. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the MCM path.

MI_MCM_RAW1_PIC_WIDTH

0x00001650 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_pic_width	31:0	rw	'h0	Image width of the MCM picture RAW1 component in pixels. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the MCM path.

MI_MCM_RAW1_PIC_HEIGHT

0x00001654 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_pic_height	31:0	rw	'h0	Image height of the RAW1 component in pixels. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the MCM path.

MI_MCM_RAW1_PIC_SIZE

0x00001658 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_pic_size	31:0	rw	'h0	Image size of the RAW1 component in bytes which has to be the RAW1 line length multiplied by the RAW1 image height (mcm_raw1_llength * mcm_raw1_pic_height). If no line stride is used, line length must match image width. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the MCM path.

MI_MCM_RAW1_OFFS_CNT_START

0x0000165c = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_offs_cnt_start	28:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (MCM Path picture RAW1 data). Updated at frame end. Note: A soft reset resets the contents to the reset value.

MI_MCM_RAW1_BASE_AD_SHD

0x00001660 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_base_ad	31:4	r	'h0	Base address of MCM picture RAW1 data ring buffer.

MI_MCM_RAW1_SIZE_SHD

0x00001664 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_size	28:4	r	'h0	Size of MCM Path picture RAW1 data ring buffer.

MI_MCM_RAW1_OFFS_CNT_SHD

0x00001668 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_offs_cnt	28:4	r	'h0	Current offset counter of MCM Path picture RAW1 data ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_MCM_DMA_RAW_PIC_START_AD

0x0000166c = Byte Address

Description: RAW component image start address

Sub-Field Name	Bits	rw	Reset	Description
mcm_dma_raw_pic_start_ad	31:4	rw	'h0	Image start address of the RAW component.

MI_MCM_DMA_RAW_PIC_WIDTH

0x00001670 = Byte Address

Description: RAW component image width

Sub-Field Name	Bits	rw	Reset	Description
mcm_dma_raw_pic_width	14:0	rw	'h0	Image width of the RAW component in pixels.

MI_MCM_DMA_RAW_PIC_LLENGTH

0x00001674 = Byte Address

Description: RAW component original line length

Sub-Field Name	Bits	rw	Reset	Description
mcm_dma_raw_pic_llength	31:0	rw	'h0	Line length of the RAW component in bytes of the original image in memory. For an uncropped image, where lines follow each other without offset (no line stride), line length must match image width.

MI_MCM_DMA_RAW_PIC_SIZE

0x00001678 = Byte Address

Description: RAW component image memory size in bytes

Sub-Field Name	Bits	rw	Reset	Description
mcm_dma_raw_pic_size	27:0	rw	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (dma_raw_llength * dma_raw_pic_height).

MI_MCM_DMA_RAW_PIC_START_AD_SHD

0x0000167c = Byte Address

Description: Shadow register of RAW component image start address

Sub-Field Name	Bits	rw	Reset	Description
mcm_dma_raw_pic_start_ad	31:4	r	'h0	Image start address of the RAW component.

MI_MCM_STATUS_CLR

0x00001680 = Byte Address

Description: FIFO full clear

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_fifo_full	1:1	rw	'h0	Clear status of MCM RAW0 FIFO full flag in Main Path
mcm_raw0_fifo_full	0:0	rw	'h0	Clear status of MCM RAW0 FIFO full flag in Main Path

MI_MCM_CTRL_STATUS

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0x00001684 = Byte Address

Description: FIFO Full STATUS, Y,U,V,RAW,JDP

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_fifo_full	1:1	r	'h0	FIFO full flag of MCM RAW1 FIFO in Main Path asserted since last clear
mcm_raw0_fifo_full	0:0	r	'h0	FIFO full flag of MCM RAW0 FIFO in Main Path asserted since last clear

MI_MCM_AXI_STATUS

0x00001688 = Byte Address

Description: Bus status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
agsw_enc_pic_rdy	1:1	r	'h0	Encoder Picture Ready interrupt
agsw_enc_buf_full	0:0	r	'h0	Encoder Buffer Full interrupt

MI_MCM_DMA_STATUS

0x0000168c = Byte Address

Description: MCM_DMA status register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_dma_active	0:0	r	'h0	0: MCM DMA access is not active; 1: MCM DMA access is active

MI_MCM_DMA_RAW_PIC_LVAL

0x00001690 = Byte Address

Description: MCM_DMA RAW Component line valid bytes number, 128 bit Alignment.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_dma_raw_width_bytes	15:0	rw	'h0	Sizes occupied by valid data of a line; should be a multiple of 16 Note: bus width=16 bytes

MI_MCM_RAW0_LVAL_BYTES

0x00001694 = Byte Address

Description: MCM RAW0 Component line valid bytes number

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_width_bytes	15:0	rw	'h0	Sizes of bytes occupied by valid data of a line.

MI_MCM_RAW1_LVAL_BYTES

0x00001698 = Byte Address

Description: MCM RAW1 Component line valid bytes number

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_width_bytes	15:0	rw	'h0	Sizes of bytes occupied by valid data of a line.

MI_MCM_RAW2_LVAL_BYTES

0x0000169c = Byte Address

Description: MCM RAW2 Component line valid bytes number

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw2_width_bytes	15:0	rw	'h0	Sizes of bytes occupied by valid data of a line.

MI_MCM_RAW3_LVAL_BYTES

0x000016a0 = Byte Address

Description: MCM RAW3 Component line valid bytes number

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw3_width_bytes	15:0	rw	'h0	Sizes of bytes occupied by valid data of a line.

MI_IMSC

0x000016c0 = Byte Address

Description: Interrupt Mask

Sub-Field Name	Bits	rw	Reset	Description
mcm_dma_raw_ready	24:24	rw	'h0	Mask bit for mcm_dma RAW ready interrupt
sp2_dma_raw_ready	23:23	rw	'h0	Mask bit for sp2_dma RAW ready interrupt
sp2_dma_ycbcr_ready	22:22	rw	'h0	Mask bit for sp2_dma YCbCr ready interrupt
wrap_mcm_raw1	21:21	rw	'h0	Mask bit for MCM RAW1 address wrap interrupt
wrap_mcm_raw0	20:20	rw	'h0	Mask bit for MCM RAW0 address wrap interrupt
wrap_sp2_raw	19:19	rw	'h0	Mask bit for Self Path 2 RAW address wrap interrupt
wrap_sp2_cr	18:18	rw	'h0	Mask bit for Self Path 2 Cr address wrap interrupt
wrap_sp2_cb	17:17	rw	'h0	Mask bit for Self Path 2 Cb address wrap interrupt
wrap_sp2_y	16:16	rw	'h0	Mask bit for Self Path 2 Y address wrap interrupt
wrap_sp1_cr	15:15	rw	'h0	Mask bit for Self Path 1 Cr address wrap interrupt
wrap_sp1_cb	14:14	rw	'h0	Mask bit for Self Path 1 Cb address wrap interrupt
wrap_sp1_y	13:13	rw	'h0	Mask bit for Self Path 1 Y address wrap interrupt

wrap_mp_jdp	12:12	rw	'h0	Mask bit for Main Path JDP address wrap interrupt
wrap_mp_raw	11:11	rw	'h0	Mask bit for Main Path RAW address wrap interrupt
wrap_mp_cr	10:10	rw	'h0	Mask bit for Main Path Cr address wrap interrupt
wrap_mp_cb	9:9	rw	'h0	Mask bit for Main Path Cb address wrap interrupt
wrap_mp_y	8:8	rw	'h0	Mask bit for Main Path Y address wrap interrupt
mcm_raw1_frame_end	7:7	rw	'h0	Mask bit for MCM RAW1 end of frame interrupt
mcm_raw0_frame_end	6:6	rw	'h0	Mask bit for MCM RAW0 end of frame interrupt
sp2_raw_frame_end	5:5	rw	'h0	Mask bit for Self Path 2 RAW end of frame interrupt
sp2_ycbcr_frame_end	4:4	rw	'h0	Mask bit for Self Path 2 YCbCr end of frame interrupt
sp1_ycbcr_frame_end	3:3	rw	'h0	Mask bit for Self Path 1 YCbCr end of frame interrupt
mp_jdp_frame_end	2:2	rw	'h0	Mask bit for Main Path JDP end of frame interrupt
mp_raw_frame_end	1:1	rw	'h0	Mask bit for Main Path RAW end of frame interrupt
mp_ycbcr_frame_end	0:0	rw	'h0	Mask bit for Main Path YCbCr end of frame interrupt 0: Interrupt masked; 1: Interrupt active

MI_IMSC1

0x000016c4 = Byte Address

Description: Interrupt Mask

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_dma_raw_buf_full	26:26	rw	'h0	Mask bit for MCM DMA RAW buffer full interrupt
sp2_dma_raw_buf_full	25:25	rw	'h0	Mask bit for Self Path 2 DMA RAW buffer full interrupt
sp2_dma_v_buf_full	24:24	rw	'h0	Mask bit for Self Path 2 DMA V buffer full interrupt
sp2_dma_u_buf_full	23:23	rw	'h0	Mask bit for Self Path 2 DMA U buffer full interrupt
sp2_dma_y_buf_full	22:22	rw	'h0	Mask bit for Self Path 2 DMA Y buffer full interrupt

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mcm_raw1_buf_raw	21:21	rw	'h0	Mask bit for MCM RAW1 buffer full interrupt
mcm_raw0_buf_raw	20:20	rw	'h0	Mask bit for MCM RAW0 buffer full interrupt
sp2_raw_buf_full	19:19	rw	'h0	Mask bit for Self Path 2 RAW buffer full interrupt
sp2_v_buf_full	18:18	rw	'h0	Mask bit for Self Path 2 V buffer full interrupt
sp2_u_buf_full	17:17	rw	'h0	Mask bit for Self Path 2 U buffer full interrupt
sp2_y_buf_full	16:16	rw	'h0	Mask bit for Self Path 2 Y buffer full interrupt
sp1_v_buf_full	15:15	rw	'h0	Mask bit for Self Path 1 V buffer full interrupt
sp1_u_buf_full	14:14	rw	'h0	Mask bit for Self Path 1 U buffer full interrupt
sp1_y_buf_full	13:13	rw	'h0	Mask bit for Self Path 1 Y buffer full interrupt
mp_raw_buf_full	12:12	rw	'h0	Mask bit for Main Path RAW buffer full interrupt
mp_jdp_buf_full	11:11	rw	'h0	Mask bit for Main Path JDP buffer full interrupt
mp_v_buf_full	10:10	rw	'h0	Mask bit for Main Path V buffer full interrupt
mp_u_buf_full	9:9	rw	'h0	Mask bit for Main Path U buffer full interrupt
mp_y_buf_full	8:8	rw	'h0	Mask bit for Main Path Y buffer full interrupt
mi_mcm_bus_buserr	7:7	rw	'h0	Mask bit for Memory Interface MCM bus buserr interrupt
mi_mcm_bus_timeo	6:6	rw	'h0	Mask bit for Memory Interface MCM bus timeo interrupt
mi_sp2_bus_buserr	5:5	rw	'h0	Mask bit for Memory Interface Self Path 2 bus buserr interrupt
mi_sp2_bus_timeo	4:4	rw	'h0	Mask bit for Memory Interface Self Path 2 bus timeo interrupt
mi_sp1_bus_buserr	3:3	rw	'h0	Mask bit for Memory Interface Self Path 1 bus buserr interrupt
mi_sp1_bus_timeo	2:2	rw	'h0	Mask bit for Memory Interface Self Path 1 bus timeo interrupt
mi_mp_bus_buserr	1:1	rw	'h0	Mask bit for Memory Interface Main Path bus buserr interrupt
mi_mp_bus_timeo	0:0	rw	'h0	Mask bit for Memory Interface Main Path bus timeo interrupt

MI_ISR

0x000016c8 = Byte Address

Description: Raw Interrupt Status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_dma_raw_ready	24:24	w	'h0	Set mcm_dma RAW ready interrupt

sp2_dma_raw_ready	23:23	w	'h0	Set sp2_dma RAW ready interrupt
sp2_dma_ycbcr_ready	22:22	w	'h0	Set sp2_dma YCbCr ready interrupt
wrap_mcm_raw1	21:21	w	'h0	Set MCM RAW1 address wrap interrupt
wrap_mcm_raw0	20:20	w	'h0	Set MCM RAW0 address wrap interrupt
wrap_sp2_raw	19:19	w	'h0	Set Self Path 2 RAW address wrap interrupt
wrap_sp2_cr	18:18	w	'h0	Set Self Path 2 Cr address wrap interrupt
wrap_sp2_cb	17:17	w	'h0	Set Self Path 2 Cb address wrap interrupt
wrap_sp2_y	16:16	w	'h0	Set Self Path 2 Y address wrap interrupt
wrap_sp1_cr	15:15	w	'h0	Set Self Path 1 Cr address wrap interrupt
wrap_sp1_cb	14:14	w	'h0	Set Self Path 1 Cb address wrap interrupt
wrap_sp1_y	13:13	w	'h0	Set Self Path 1 Y address wrap interrupt
wrap_mp_jdp	12:12	w	'h0	Set Main Path JDP address wrap interrupt
wrap_mp_raw	11:11	w	'h0	Set Main Path RAW address wrap interrupt
wrap_mp_cr	10:10	w	'h0	Set Main Path Cr address wrap interrupt
wrap_mp_cb	9:9	w	'h0	Set Main Path Cb address wrap interrupt
wrap_mp_y	8:8	w	'h0	Set Main Path Y address wrap interrupt
mcm_raw1_frame_end	7:7	w	'h0	Set MCM RAW1 end of frame interrupt
mcm_raw0_frame_end	6:6	w	'h0	Set MCM RAW0 end of frame interrupt
sp2_raw_frame_end	5:5	w	'h0	Set Self Path 2 RAW end of frame interrupt
sp2_ycbcr_frame_end	4:4	w	'h0	Set Self Path 2 YCbCr end of frame interrupt
sp1_ycbcr_frame_end	3:3	w	'h0	Set Self Path 1 YCbCr end of frame interrupt
mp_jdp_frame_end	2:2	w	'h0	Set Main Path JDP end of frame interrupt
mp_raw_frame_end	1:1	w	'h0	Set Main Path RAW end of frame interrupt
mp_ycbcr_frame_end	0:0	w	'h0	Set Main Path YCbCr end of frame interrupt

MI_ISR1

0x000016cc = Byte Address

Description: Raw Interrupt Status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_dma_raw_buf_full	26:26	w	'h0	Set MCM DMA RAW buffer full interrupt
sp2_dma_raw_buf_full	25:25	w	'h0	Set Self Path2 DMA RAW buffer full interrupt
sp2_dma_v_buf_full	24:24	w	'h0	Set Self Path2 DMA V buffer full interrupt
sp2_dma_u_buf_full	23:23	w	'h0	Set Self Path2 DMA U buffer full interrupt
sp2_dma_y_buf_full	22:22	w	'h0	Set Self Path2 DMA Y buffer full interrupt
mcm_raw1_buf_raw	21:21	w	'h0	Set MCM RAW1 buffer full interrupt
mcm_raw0_buf_raw	20:20	w	'h0	Set MCM RAW0 buffer full interrupt

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sp2_raw_buf_full	19:19	w	'h0	Set Self Path2 RAW buffer full interrupt
sp2_v_buf_full	18:18	w	'h0	Set Self Path2 V buffer full interrupt
sp2_u_buf_full	17:17	w	'h0	Set Self Path2 U buffer full interrupt
sp2_y_buf_full	16:16	w	'h0	Set Self Path2 Y buffer full interrupt
sp1_v_buf_full	15:15	w	'h0	Set Self Path1 V buffer full interrupt
sp1_u_buf_full	14:14	w	'h0	Set Self Path1 U buffer full interrupt
sp1_y_buf_full	13:13	w	'h0	Set Self Path1 Y buffer full interrupt
mp_raw_buf_full	12:12	w	'h0	Set Main Path RAW buf full interrupt
mp_jdp_buf_full	11:11	w	'h0	Set Main Path JDP buf full interrupt
mp_v_buf_full	10:10	w	'h0	Set Main Path V buffer full interrupt
mp_u_buf_full	9:9	w	'h0	Set Main Path U buffer full interrupt
mp_y_buf_full	8:8	w	'h0	Set Main Path Y buffer full interrupt
mi_mcm_bus_buserr	7:7	w	'h0	Set Memory Interface MCM bus buserr interrupt
mi_mcm_bus_timeo	6:6	w	'h0	Set Memory Interface MCM bus timeout interrupt
mi_sp2_bus_buserr	5:5	w	'h0	Set Memory Interface Self Path2 bus buserr interrupt
mi_sp2_bus_timeo	4:4	w	'h0	Set Memory Interface Self Path2 bus timeout interrupt
mi_sp1_bus_buserr	3:3	w	'h0	Set Memory Interface Self Path1 bus buserr interrupt
mi_sp1_bus_timeo	2:2	w	'h0	Set Memory Interface Self Path1 bus timeout interrupt
mi_mp_bus_buserr	1:1	w	'h0	Set Memory Interface Main Path bus buserr interrupt
mi_mp_bus_timeo	0:0	w	'h0	Set Memory Interface Main Path bus timeout interrupt

MI_MIS

0x000016d0 = Byte Address

Description: Masked Interrupt Status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_dma_raw_ready	24:24	r	'h0	Masked status of mcm_dma RAW ready interrupt
sp2_dma_raw_ready	23:23	r	'h0	Masked status of sp2_dma RAW ready interrupt

sp2_dma_ycbcr_ready	22:22	r	'h0	Masked status of sp2_dma YCbCr ready interrupt
wrap_mcm_raw1	21:21	r	'h0	Masked status of MCM RAW1 address wrap interrupt
wrap_mcm_raw0	20:20	r	'h0	Masked status of MCM RAW0 address wrap interrupt
wrap_sp2_raw	19:19	r	'h0	Masked status of Self Path2 RAW address wrap interrupt
wrap_sp2_cr	18:18	r	'h0	Masked status of Self Path2 Cr address wrap interrupt
wrap_sp2_cb	17:17	r	'h0	Masked status of Self Path2 Cb address wrap interrupt
wrap_sp2_y	16:16	r	'h0	Masked status of Self Path2 y address wrap interrupt
wrap_sp1_cr	15:15	r	'h0	Masked status of Self Path1 Cr address wrap interrupt
wrap_sp1_cb	14:14	r	'h0	Masked status of Self Path1 Cb address wrap interrupt
wrap_sp1_y	13:13	r	'h0	Masked status of Self Path1 y address wrap interrupt
wrap_mp_jdp	12:12	r	'h0	Masked status of Main Path JDP address wrap interrupt
wrap_mp_raw	11:11	r	'h0	Masked status of Main Path RAW address wrap interrupt
wrap_mp_cr	10:10	r	'h0	Masked status of Main Path Cr address wrap interrupt
wrap_mp_cb	9:9	r	'h0	Masked status of Main Path Cb address wrap interrupt
wrap_mp_y	8:8	r	'h0	Masked status of Main Path Y address wrap interrupt
mcm_raw1_frame_end	7:7	r	'h0	Masked status of MCM RAW1 end of frame interrupt
mcm_raw0_frame_end	6:6	r	'h0	Masked status of MCM RAW0 end of frame interrupt
sp2_raw_frame_end	5:5	r	'h0	Masked status of Self Path2 RAW end of frame interrupt
sp2_ycbcr_frame_end	4:4	r	'h0	Masked status of Self Path2 YCbCr end of frame interrupt

sp1_ycbcr_frame_end	3:3	r	'h0	Masked status of Self Path1 YCbCr end of frame interrupt
mp_jdp_frame_end	2:2	r	'h0	Masked status of Main Path JDP end of frame interrupt
mp_raw_frame_end	1:1	r	'h0	Masked status of Main Path RAW end of frame interrupt
mp_ycbcr_frame_end	0:0	r	'h0	Masked status of Main Path YCbCr end of frame interrupt

MI_MIS1

0x000016d4 = Byte Address

Description: Masked Interrupt Status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_dma_raw_buf_full	26:26	r	'h0	Masked status of MCM DMA RAW buffer full interrupt
sp2_dma_raw_buf_full	25:25	r	'h0	Masked status of Self Path2 DMA RAW buffer full interrupt
sp2_dma_v_buf_full	24:24	r	'h0	Masked status of Self Path2 DMA V buffer full interrupt
sp2_dma_u_buf_full	23:23	r	'h0	Masked status of Self Path2 DMA U buffer full interrupt
sp2_dma_y_buf_full	22:22	r	'h0	Masked status of Self Path2 DMA Y buffer full interrupt
mcm_raw1_buf_raw	21:21	r	'h0	Masked status of MCM RAW1 buffer full interrupt
mcm_raw0_buf_raw	20:20	r	'h0	Masked status of MCM RAW0 buffer full interrupt
sp2_raw_buf_full	19:19	r	'h0	Masked status of Self Path2 RAW buffer full interrupt
sp2_v_buf_full	18:18	r	'h0	Masked status of Self Path2 V buffer full interrupt
sp2_u_buf_full	17:17	r	'h0	Masked status of Self Path2 U buffer full interrupt
sp2_y_buf_full	16:16	r	'h0	Masked status of Self Path2 Y buffer full interrupt
sp1_v_buf_full	15:15	r	'h0	Masked status of Self Path1 V buffer full interrupt
sp1_u_buf_full	14:14	r	'h0	Masked status of Self Path1 U buffer full interrupt

sp1_y_buf_full	13:13	r	'h0	Masked status of Self Path1 Y buffer full interrupt
mp_raw_buf_full	12:12	r	'h0	Masked status of Main Path RAW buffer full interrupt
mp_jdp_buf_full	11:11	r	'h0	Masked status of Main Path JDP buffer full interrupt
mp_v_buf_full	10:10	r	'h0	Masked status of Main Path V buffer full interrupt
mp_u_buf_full	9:9	r	'h0	Masked status of Main Path U buffer full interrupt
mp_y_buf_full	8:8	r	'h0	Masked status of Main Path Y buffer full interrupt
mi_mcm_bus_buserr	7:7	r	'h0	Masked status of Memory Interface MCM bus buserr interrupt
mi_mcm_bus_timeo	6:6	r	'h0	Masked status of Memory Interface MCM bus timeout interrupt
mi_sp2_bus_buserr	5:5	r	'h0	Masked status of Memory Interface Self Path2 bus buserr interrupt
mi_sp2_bus_timeo	4:4	r	'h0	Masked status of Memory Interface Self Path2 bus timeout interrupt
mi_sp1_bus_buserr	3:3	r	'h0	Masked status of Memory Interface Self Path1 bus buserr interrupt
mi_sp1_bus_timeo	2:2	r	'h0	Masked status of Memory Interface Self Path1 bus timeout interrupt
mi_mp_bus_buserr	1:1	r	'h0	Masked status of Memory Interface Main Path bus buserr interrupt
mi_mp_bus_timeo	0:0	r	'h0	Masked status of Memory Interface Main Path bus timeout interrupt

MI_ICR

0x000016d8 = Byte Address

Description: Interrupt Clear Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_dma_raw_ready	24:24	w	'h0	Clear mcm_dma RAW ready interrupt
sp2_dma_raw_ready	23:23	w	'h0	Clear sp2_dma RAW ready interrupt
sp2_dma_ycbcr_ready	22:22	w	'h0	Clear sp2_dma YCbCr ready interrupt
wrap_mcm_raw1	21:21	w	'h0	Clear MCM RAW1 address wrap interrupt
wrap_mcm_raw0	20:20	w	'h0	Clear MCM RAW0 address wrap interrupt

wrap_sp2_raw	19:19	w	'h0	Clear Self Path2 RAW address wrap interrupt
wrap_sp2_cr	18:18	w	'h0	Clear Self Path2 Cr address wrap interrupt
wrap_sp2_cb	17:17	w	'h0	Clear Self Path2 Cb address wrap interrupt
wrap_sp2_y	16:16	w	'h0	Clear Self Path2 y address wrap interrupt
wrap_sp1_cr	15:15	w	'h0	Clear Self Path1 Cr address wrap interrupt
wrap_sp1_cb	14:14	w	'h0	Clear Self Path1 Cb address wrap interrupt
wrap_sp1_y	13:13	w	'h0	Clear Self Path1 Y address wrap interrupt
wrap_mp_jdp	12:12	w	'h0	Clear Main Path JDP address wrap interrupt
wrap_mp_raw	11:11	w	'h0	Clear Main Path RAW address wrap interrupt
wrap_mp_cr	10:10	w	'h0	Clear Main Path Cr address wrap interrupt
wrap_mp_cb	9:9	w	'h0	Clear Main Path Cb address wrap interrupt
wrap_mp_y	8:8	w	'h0	Clear Main Path Y address wrap interrupt
mcm_raw1_frame_end	7:7	w	'h0	Clear MCM RAW1 end of frame interrupt
mcm_raw0_frame_end	6:6	w	'h0	Clear MCM RAW0 end of frame interrupt
sp2_raw_frame_end	5:5	w	'h0	Clear Self Path2 RAW end of frame interrupt
sp2_ycbcr_frame_end	4:4	w	'h0	Clear Self Path2 YCbCr end of frame interrupt
sp1_ycbcr_frame_end	3:3	w	'h0	Clear Self Path1 YCbCr end of frame interrupt
mp_jdp_frame_end	2:2	w	'h0	Clear Main Path JDP end of frame interrupt
mp_raw_frame_end	1:1	w	'h0	Clear Main Path RAW end of frame interrupt
mp_ycbcr_frame_end	0:0	w	'h0	Clear Main Path YCbCr end of frame interrupt

MI_ICR1

0x000016dc = Byte Address

Description: Interrupt Clear Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_dma_raw_buf_full	26:26	w	'h0	Clear MCM DMA RAW buffer full interrupt
sp2_dma_raw_buf_full	25:25	w	'h0	Clear Self Path2 DMA RAW buffer full interrupt
sp2_dma_v_buf_full	24:24	w	'h0	Clear Self Path2 DMA V buffer full interrupt
sp2_dma_u_buf_full	23:23	w	'h0	Clear Self Path2 DMA U buffer full interrupt
sp2_dma_y_buf_full	22:22	w	'h0	Clear Self Path2 DMA Y buffer full interrupt
mcm_raw1_buf_raw	21:21	w	'h0	Clear MCM RAW1 buffer full interrupt
mcm_raw0_buf_raw	20:20	w	'h0	Clear MCM RAW0 buffer full interrupt
sp2_raw_buf_full	19:19	w	'h0	Clear Self Path2 RAW buffer full interrupt
sp2_v_buf_full	18:18	w	'h0	Clear Self Path2 V buffer full interrupt
sp2_u_buf_full	17:17	w	'h0	Clear Self Path2 U buffer full interrupt
sp2_y_buf_full	16:16	w	'h0	Clear Self Path2 Y buffer full interrupt

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sp1_v_buf_full	15:15	w	'h0	Clear Self Path1 V buffer full interrupt
sp1_u_buf_full	14:14	w	'h0	Clear Self Path1 U buffer full interrupt
sp1_y_buf_full	13:13	w	'h0	Clear Self Path1 Y buffer full interrupt
mp_raw_buf_full	12:12	w	'h0	Clear Main Path RAW buffer full interrupt
mp_jdp_buf_full	11:11	w	'h0	Clear Main Path JDP buffer full interrupt
mp_v_buf_full	10:10	w	'h0	Clear Main Path V buffer full interrupt
mp_u_buf_full	9:9	w	'h0	Clear Main Path U buffer full interrupt
mp_y_buf_full	8:8	w	'h0	Clear Main Path Y buffer full interrupt
mi_mcm_bus_buserr	7:7	w	'h0	Clear Memory Interface MCM bus buserr interrupt
mi_mcm_bus_timeo	6:6	w	'h0	Clear Memory Interface MCM bus timeout interrupt
mi_sp2_bus_buserr	5:5	w	'h0	Clear Memory Interface Self Path2 bus buserr interrupt
mi_sp2_bus_timeo	4:4	w	'h0	Clear Memory Interface Self Path2 bus timeout interrupt
mi_sp1_bus_buserr	3:3	w	'h0	Clear Memory Interface Self Path1 bus buserr interrupt
mi_sp1_bus_timeo	2:2	w	'h0	Clear Memory Interface Self Path1 bus timeout interrupt
mi_mp_bus_buserr	1:1	w	'h0	Clear Memory Interface Main Path bus buserr interrupt
mi_mp_bus_timeo	0:0	w	'h0	Clear Memory Interface Main Path bus timeout interrupt

MI_RIS

0x000016e0 = Byte Address

Description: Interrupt Set Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_dma_raw_ready	24:24	r	'h0	RAW status of mcm_dma RAW ready interrupt
sp2_dma_raw_ready	23:23	r	'h0	RAW status of sp2_dma RAW ready interrupt
sp2_dma_ycbcr_ready	22:22	r	'h0	RAW status of sp2_dma YCbCr ready interrupt
wrap_mcm_raw1	21:21	r	'h0	RAW status of MCM RAW1 address wrap interrupt
wrap_mcm_raw0	20:20	r	'h0	RAW status of MCM RAW0 address wrap interrupt
wrap_sp2_raw	19:19	r	'h0	RAW status of Self Path2 RAW address wrap interrupt

wrap_sp2_cr	18:18	r	'h0	RAW status of Self Path2 Cr address wrap interrupt
wrap_sp2_cb	17:17	r	'h0	RAW status of Self Path2 Cb address wrap interrupt
wrap_sp2_y	16:16	r	'h0	RAW status of Self Path2 Y address wrap interrupt
wrap_sp1_cr	15:15	r	'h0	RAW status of Self Path1 Cr address wrap interrupt
wrap_sp1_cb	14:14	r	'h0	RAW status of Self Path1 Cb address wrap interrupt
wrap_sp1_y	13:13	r	'h0	RAW status of Self Path1 Y address wrap interrupt
wrap_mp_jdp	12:12	r	'h0	RAW status of Main Path JDP address wrap interrupt
wrap_mp_raw	11:11	r	'h0	RAW status of Main Path RAW address wrap interrupt
wrap_mp_cr	10:10	r	'h0	RAW status of Main Path Cr address wrap interrupt
wrap_mp_cb	9:9	r	'h0	RAW status of Main Path Cb address wrap interrupt
wrap_mp_y	8:8	r	'h0	RAW status of Main Path Y address wrap interrupt
mcm_raw1_frame_end	7:7	r	'h0	RAW status of MCM RAW1 end of frame interrupt
mcm_raw0_frame_end	6:6	r	'h0	RAW status of MCM RAW0 end of frame interrupt
sp2_raw_frame_end	5:5	r	'h0	RAW status of Self Path2 RAW end of frame interrupt
sp2_ycbcr_frame_end	4:4	r	'h0	RAW status of Self Path2 YCbCr end of frame interrupt
sp1_ycbcr_frame_end	3:3	r	'h0	RAW status of Self Path1 YCbCr end of frame interrupt
mp_jdp_frame_end	2:2	r	'h0	RAW status of Main Path JDP end of frame interrupt
mp_raw_frame_end	1:1	r	'h0	RAW status of Main Path RAW end of frame interrupt
mp_ycbcr_frame_end	0:0	r	'h0	RAW status of Main Path YCbCr end of frame interrupt

MI_RIS1

0x000016e4 = Byte Address

Description: Interrupt Set Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_dma_raw_buf_full	26:26	r	'h0	RAW status of MCM DMA RAW buffer full interrupt
sp2_dma_raw_buf_full	25:25	r	'h0	RAW status of Self Path2 DMA RAW buffer full interrupt
sp2_dma_v_buf_full	24:24	r	'h0	RAW status of Self Path2 DMA V buffer full interrupt
sp2_dma_u_buf_full	23:23	r	'h0	RAW status of Self Path2 DMA U buffer full interrupt
sp2_dma_y_buf_full	22:22	r	'h0	RAW status of Self Path2 DMA Y buffer full interrupt
mcm_raw1_buf_raw	21:21	r	'h0	RAW status of MCM RAW1 buffer full interrupt
mcm_raw0_buf_raw	20:20	r	'h0	RAW status of MCM RAW0 buffer full interrupt
sp2_raw_buf_full	19:19	r	'h0	RAW status of Self Path2 RAW buffer full interrupt
sp2_v_buf_full	18:18	r	'h0	RAW status of Self Path2 V buffer full interrupt
sp2_u_buf_full	17:17	r	'h0	RAW status of Self Path2 U buffer full interrupt
sp2_y_buf_full	16:16	r	'h0	RAW status of Self Path2 Y buffer full interrupt
sp1_v_buf_full	15:15	r	'h0	RAW status of Self Path1 V buffer full interrupt
sp1_u_buf_full	14:14	r	'h0	RAW status of Self Path1 U buffer full interrupt
sp1_y_buf_full	13:13	r	'h0	RAW status of Self Path1 Y buffer full interrupt
mp_raw_buf_full	12:12	r	'h0	RAW status of Main Path RAW buffer full interrupt
mp_jdp_buf_full	11:11	r	'h0	RAW status of Main Path JDP buffer full interrupt
mp_v_buf_full	10:10	r	'h0	RAW status of Main Path V buffer full interrupt
mp_u_buf_full	9:9	r	'h0	RAW status of Main Path U buffer full interrupt
mp_y_buf_full	8:8	r	'h0	RAW status of Main Path Y buffer full interrupt
mi_mcm_bus_buserr	7:7	r	'h0	RAW status of Memory Interface MCM bus buserr interrupt
mi_mcm_bus_timeo	6:6	r	'h0	RAW status of Memory Interface MCM bus timeout interrupt
mi_sp2_bus_buserr	5:5	r	'h0	RAW status of Memory Interface Self Path2 bus buserr interrupt

mi_sp2_bus_timeo	4:4	r	'h0	RAW status of Memory Interface Self Path2 bus timeout interrupt
mi_sp1_bus_buserr	3:3	r	'h0	RAW status of Memory Interface Self Path1 bus buserr interrupt
mi_sp1_bus_timeo	2:2	r	'h0	RAW status of Memory Interface Self Path1 bus timeout interrupt
mi_mp_bus_buserr	1:1	r	'h0	RAW status of Memory Interface Main Path bus buserr interrupt
mi_mp_bus_timeo	0:0	r	'h0	RAW status of Memory Interface Main Path bus timeout interrupt

MI_IMSC2

0x000016e8 = Byte Address

Description: Interrupt Set Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ppw_y_buf_full	26:26	rw	'h0	status of ppw y buf full ready interrupt
ppw_u_buf_full	25:25	rw	'h0	status of ppw u buf full ready interrupt
ppw_v_buf_full	24:24	rw	'h0	status of ppw v buf full ready interrupt
ppr_y_buf_full	23:23	rw	'h0	status of ppr y buf full ready interrupt
sp2_raw2_w_buf_full	22:22	rw	'h0	status of spw_raw2 wr buf full interrupt
sp2_raw2_r_buf_full	21:21	rw	'h0	status of spw_raw2 rd buf full interrupt
hdr_w_buf_full	20:20	rw	'h0	status of mi hdr wr buf full interrupt
hdr_r_buf_full	19:19	rw	'h0	status of mi hdr rd buf full interrupt
sp2_raw2_dma_ready	18:18	rw	'h0	status of spw_raw2 ready interrupt
ppr_dma_ready	17:17	rw	'h0	status of pp_dma yuv ready interrupt
wrap_sp2_raw	16:16	rw	'h0	status of sp2_raw2 address wrap interrupt
wrap_ppw_cr	15:15	rw	'h0	status of ppw_cr address wrap interrupt
wrap_ppw_cb	14:14	rw	'h0	status of ppw_cb address wrap interrupt
wrap_ppw_y	13:13	rw	'h0	status of ppw_y address wrap interrupt
sp2_raw2_frame_end	12:12	rw	'h0	status of spp2_raw2 end of frame interrupt
ppw_frame_end	11:11	rw	'h0	status of ppw end of frame interrupt
hdr_vs_dma_ready	10:10	rw	'h0	status of hdr_vs ready interrupt
hdr_s_dma_ready	9:9	rw	'h0	status of hdr_s ready interrupt
hdr_l_dma_ready	8:8	rw	'h0	status of hdr_l ready interrupt
wrap_hdr_vs	7:7	rw	'h0	status of hdr_vs address wrap interrupt
wrap_hdr_s	6:6	rw	'h0	status of hdr_s address wrap interrupt
wrap_hdr_l	5:5	rw	'h0	status of hdr_l address wrap interrupt

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hdr_vs_frame_end	4:4	rw	'h0	status of hdr_vs end of frame interrupt
hdr_s_frame_end	3:3	rw	'h0	status of hdr_s end of frame interrupt
hdr_l_frame_end	2:2	rw	'h0	status of hdr_l end of frame interrupt
mi_rt_bus_buserr	1:1	rw	'h0	status of Memory Interface RT Path bus buserr interrupt
mi_rt_bus_timeo	0:0	rw	'h0	status of Memory Interface RT Path bus timeout interrupt

MI_ISR2

0x000016ec = Byte Address

Description: Interrupt Set Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ppw_y_buf_full	26:26	w	'h0	status of ppw y buf full ready interrupt
ppw_u_buf_full	25:25	w	'h0	status of ppw u buf full ready interrupt
ppw_v_buf_full	24:24	w	'h0	status of ppw v buf full ready interrupt
ppr_y_buf_full	23:23	w	'h0	status of ppr y buf full ready interrupt
sp2_raw2_w_buf_full	22:22	w	'h0	status of spw_raw2 wr buf full interrupt
sp2_raw2_r_buf_full	21:21	w	'h0	status of spw_raw2 rd buf full interrupt
hdr_w_buf_full	20:20	w	'h0	status of mi hdr wr buf full interrupt
hdr_r_buf_full	19:19	w	'h0	status of mi hdr rd buf full interrupt
sp2_raw2_dma_ready	18:18	w	'h0	status of spw_raw2 ready interrupt
ppr_dma_ready	17:17	w	'h0	status of pp_dma yuv ready interrupt
wrap_sp2_raw	16:16	w	'h0	status of sp2_raw2 address wrap interrupt
wrap_ppw_cr	15:15	w	'h0	status of ppw_cr address wrap interrupt
wrap_ppw_cb	14:14	w	'h0	status of ppw_cb address wrap interrupt
wrap_ppw_y	13:13	w	'h0	status of ppw_y address wrap interrupt
sp2_raw2_frame_end	12:12	w	'h0	status of spp2_raw2 end of frame interrupt
ppw_frame_end	11:11	w	'h0	status of ppw end of frame interrupt
hdr_vs_dma_ready	10:10	w	'h0	status of hdr_vs ready interrupt
hdr_s_dma_ready	9:9	w	'h0	status of hdr_s ready interrupt
hdr_l_dma_ready	8:8	w	'h0	status of hdr_l ready interrupt
wrap_hdr_vs	7:7	w	'h0	status of hdr_vs address wrap interrupt
wrap_hdr_s	6:6	w	'h0	status of hdr_s address wrap interrupt
wrap_hdr_l	5:5	w	'h0	status of hdr_l address wrap interrupt
hdr_vs_frame_end	4:4	w	'h0	status of hdr_vs end of frame interrupt
hdr_s_frame_end	3:3	w	'h0	status of hdr_s end of frame interrupt

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hdr_l_frame_end	2:2	w	'h0	status of hdr_l end of frame interrupt
mi_rt_bus_buserr	1:1	w	'h0	status of Memory Interface RT Path bus buserr interrupt
mi_rt_bus_timeo	0:0	w	'h0	status of Memory Interface RT Path bus timeout interrupt

MI_MIS2

0x000016f0 = Byte Address

Description: Interrupt Set Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ppw_y_buf_full	26:26	r	'h0	status of ppw y buf full ready interrupt
ppw_u_buf_full	25:25	r	'h0	status of ppw u buf full ready interrupt
ppw_v_buf_full	24:24	r	'h0	status of ppw v buf full ready interrupt
ppr_y_buf_full	23:23	r	'h0	status of ppr y buf full ready interrupt
sp2_raw2_w_buf_full	22:22	r	'h0	status of spw_raw2 wr buf full interrupt
sp2_raw2_r_buf_full	21:21	r	'h0	status of spw_raw2 rd buf full interrupt
hdr_w_buf_full	20:20	r	'h0	status of mi hdr wr buf full interrupt
hdr_r_buf_full	19:19	r	'h0	status of mi hdr rd buf full interrupt
sp2_raw2_dma_ready	18:18	r	'h0	status of spw_raw2 ready interrupt
ppr_dma_ready	17:17	r	'h0	status of pp_dma yuv ready interrupt
wrap_sp2_raw	16:16	r	'h0	status of sp2_raw2 address wrap interrupt
wrap_ppw_cr	15:15	r	'h0	status of ppw_cr address wrap interrupt
wrap_ppw_cb	14:14	r	'h0	status of ppw_cb address wrap interrupt
wrap_ppw_y	13:13	r	'h0	status of ppw_y address wrap interrupt
sp2_raw2_frame_end	12:12	r	'h0	status of spp2_raw2 end of frame interrupt
ppw_frame_end	11:11	r	'h0	status of ppw end of frame interrupt
hdr_vs_dma_ready	10:10	r	'h0	status of hdr_vs ready interrupt
hdr_s_dma_ready	9:9	r	'h0	status of hdr_s ready interrupt
hdr_l_dma_ready	8:8	r	'h0	status of hdr_l ready interrupt
wrap_hdr_vs	7:7	r	'h0	status of hdr_vs address wrap interrupt
wrap_hdr_s	6:6	r	'h0	status of hdr_s address wrap interrupt
wrap_hdr_l	5:5	r	'h0	status of hdr_l address wrap interrupt
hdr_vs_frame_end	4:4	r	'h0	status of hdr_vs end of frame interrupt
hdr_s_frame_end	3:3	r	'h0	status of hdr_s end of frame interrupt
hdr_l_frame_end	2:2	r	'h0	status of hdr_l end of frame interrupt
mi_rt_bus_buserr	1:1	r	'h0	status of Memory Interface RT Path bus buserr interrupt

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mi_rt_bus_timeo	0:0	r	'h0	status of Memory Interface RT Path bus timeout interrupt

MI_ICR2

0x000016f4 = Byte Address

Description: Interrupt Set Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ppw_y_buf_full	26:26	w	'h0	status of ppw y buf full ready interrupt
ppw_u_buf_full	25:25	w	'h0	status of ppw u buf full ready interrupt
ppw_v_buf_full	24:24	w	'h0	status of ppw v buf full ready interrupt
ppr_y_buf_full	23:23	w	'h0	status of ppr y buf full ready interrupt
sp2_raw2_w_buf_full	22:22	w	'h0	status of spw_raw2 wr buf full interrupt
sp2_raw2_r_buf_full	21:21	w	'h0	status of spw_raw2 rd buf full interrupt
hdr_w_buf_full	20:20	w	'h0	status of mi hdr wr buf full interrupt
hdr_r_buf_full	19:19	w	'h0	status of mi hdr rd buf full interrupt
sp2_raw2_dma_ready	18:18	w	'h0	status of spw_raw2 ready interrupt
ppr_dma_ready	17:17	w	'h0	status of pp_dma yuv ready interrupt
wrap_sp2_raw	16:16	w	'h0	status of sp2_raw2 address wrap interrupt
wrap_ppw_cr	15:15	w	'h0	status of ppw_cr address wrap interrupt
wrap_ppw_cb	14:14	w	'h0	status of ppw_cb address wrap interrupt
wrap_ppw_y	13:13	w	'h0	status of ppw_y address wrap interrupt
sp2_raw2_frame_end	12:12	w	'h0	status of spp2_raw2 end of frame interrupt
ppw_frame_end	11:11	w	'h0	status of ppw end of frame interrupt
hdr_vs_dma_ready	10:10	w	'h0	status of hdr_vs ready interrupt
hdr_s_dma_ready	9:9	w	'h0	status of hdr_s ready interrupt
hdr_l_dma_ready	8:8	w	'h0	status of hdr_l ready interrupt
wrap_hdr_vs	7:7	w	'h0	status of hdr_vs address wrap interrupt
wrap_hdr_s	6:6	w	'h0	status of hdr_s address wrap interrupt
wrap_hdr_l	5:5	w	'h0	status of hdr_l address wrap interrupt
hdr_vs_frame_end	4:4	w	'h0	status of hdr_vs end of frame interrupt
hdr_s_frame_end	3:3	w	'h0	status of hdr_s end of frame interrupt
hdr_l_frame_end	2:2	w	'h0	status of hdr_l end of frame interrupt
mi_rt_bus_buserr	1:1	w	'h0	status of Memory Interface RT Path bus buserr interrupt
mi_rt_bus_timeo	0:0	w	'h0	status of Memory Interface RT Path bus timeout interrupt

MI_RIS2

0x000016f8 = Byte Address

Description: Interrupt Set Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ppw_y_buf_full	26:26	r	'h0	status of ppw y buf full ready interrupt
ppw_u_buf_full	25:25	r	'h0	status of ppw u buf full ready interrupt
ppw_v_buf_full	24:24	r	'h0	status of ppw v buf full ready interrupt
ppr_y_buf_full	23:23	r	'h0	status of ppr y buf full ready interrupt
sp2_raw2_w_buf_full	22:22	r	'h0	status of spw_raw2 wr buf full interrupt
sp2_raw2_r_buf_full	21:21	r	'h0	status of spw_raw2 rd buf full interrupt
hdr_w_buf_full	20:20	r	'h0	status of mi hdr wr buf full interrupt
hdr_r_buf_full	19:19	r	'h0	status of mi hdr rd buf full interrupt
sp2_raw2_dma_ready	18:18	r	'h0	status of spw_raw2 ready interrupt
ppr_dma_ready	17:17	r	'h0	status of pp_dma yuv ready interrupt
wrap_sp2_raw	16:16	r	'h0	status of sp2_raw2 address wrap interrupt
wrap_ppw_cr	15:15	r	'h0	status of ppw_cr address wrap interrupt
wrap_ppw_cb	14:14	r	'h0	status of ppw_cb address wrap interrupt
wrap_ppw_y	13:13	r	'h0	status of ppw_y address wrap interrupt
sp2_raw2_frame_end	12:12	r	'h0	status of spp2_raw2 end of frame interrupt
ppw_frame_end	11:11	r	'h0	status of ppw end of frame interrupt
hdr_vs_dma_ready	10:10	r	'h0	status of hdr_vs ready interrupt
hdr_s_dma_ready	9:9	r	'h0	status of hdr_s ready interrupt
hdr_l_dma_ready	8:8	r	'h0	status of hdr_l ready interrupt
wrap_hdr_vs	7:7	r	'h0	status of hdr_vs address wrap interrupt
wrap_hdr_s	6:6	r	'h0	status of hdr_s address wrap interrupt
wrap_hdr_l	5:5	r	'h0	status of hdr_l address wrap interrupt
hdr_vs_frame_end	4:4	r	'h0	status of hdr_vs end of frame interrupt
hdr_s_frame_end	3:3	r	'h0	status of hdr_s end of frame interrupt
hdr_l_frame_end	2:2	r	'h0	status of hdr_l end of frame interrupt
mi_rt_bus_buserr	1:1	r	'h0	status of Memory Interface RT Path bus buserr interrupt
mi_rt_bus_timeo	0:0	r	'h0	status of Memory Interface RT Path bus timeout interrupt

MI_MP_Y_IRQ_OFFS_INIT

0x00001700 = Byte Address

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Description: Fill level interrupt offset value for main picture Y

Sub-Field Name	Bits	rw	Reset	Description
mp_y_irq_offs_init	28:3	rw	'h0	Reaching this programmed value by the current offset counter for addressing Main Path picture Y component data leads to generation of fill level interrupt fill_mp_y. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_JDP_IRQ_OFFS_INIT

0x00001704 = Byte Address

Description: Fill level interrupt offset value for main picture JPEG

Sub-Field Name	Bits	rw	Reset	Description
mp_jdp_irq_offs_init	28:3	rw	'h0	Reaching this programmed value by the current offset counter for addressing Main Path picture JPEG component data leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_RAW_IRQ_OFFS_INIT

0x00001708 = Byte Address

Description: Fill level interrupt offset value for main picture RAW

Sub-Field Name	Bits	rw	Reset	Description
mp_raw_irq_offs_init	28:3	rw	'h0	Reaching this programmed value by the current offset counter for addressing Main Path picture RAW component data leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP1_Y_IRQ_OFFS_INIT

0x0000170c = Byte Address

Description: Follow Main Path

Sub-Field Name	Bits	rw	Reset	Description
sp1_y_irq_offs_init	28:3	rw	'h0	Reaching this programmed value by the current offset counter for addressing Self Path 1 picture Y component data leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP2_Y_IRQ_OFFS_INIT

0x00001710 = Byte Address

Description: Follow Main Path

Sub-Field Name	Bits	rw	Reset	Description
sp2_y_irq_offs_init	28:3	rw	'h0	Reaching this programmed value by the current offset counter for addressing Self Path 2 picture Y component data leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP2_RAW_IRQ_OFFS_INIT

0x00001714 = Byte Address

Description: Follow Main Path

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_irq_offs_init	28:3	rw	'h0	Reaching this programmed value by the current offset counter for addressing Self Path 2 picture RAW component data leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MCM_RAW0_IRQ_OFFS_INIT

0x00001718 = Byte Address

Description: Follow Main Path

Sub-Field Name	Bits	rw	Reset	Description

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mcm_raw0_irq_offs_init	28:3	rw	'h0	Reaching this programmed value by the current offset counter for addressing MCM picture RAW0 component data leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MCM_RAW1_IRQ_OFFS_INIT

0x0000171c = Byte Address

Description: Follow Main Path

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_irq_offs_init	28:3	rw	'h0	Reaching this programmed value by the current offset counter for addressing MCM picture RAW1 component data leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_Y_IRQ_OFFS_SHD

0x00001720 = Byte Address

Description: Shadow register of fill level interrupt offset value for main picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_irq_offs	28:3	r	'h0	Reaching this offset value by the current offset counter for addressing main picture Y component leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_JDP_IRQ_OFFS_SHD

0x00001724 = Byte Address

Description: Shadow register of fill level interrupt offset value for main picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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mp_jdp_irq_offs	28:3	r	'h0	Reaching this offset value by the current offset counter for addressing main picture JPEG data leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MP_RAW_IRQ_OFFS_SHD

0x00001728 = Byte Address

Description: Shadow register of fill level interrupt offset value for main picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_raw_irq_offs	28:3	r	'h0	Reaching this offset value by the current offset counter for addressing main picture RAW data leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP1_Y_IRQ_OFFS_SHD

0x0000172c = Byte Address

Description: Shadow register of fill level interrupt offset value for main picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp1_y_irq_offs	28:3	r	'h0	Reaching this offset value by the current offset counter for addressing Self Path 1 picture Y component leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP2_Y_IRQ_OFFS_SHD

0x00001730 = Byte Address

Description: Shadow register of fill level interrupt offset value for main picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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sp2_y_irq_offs	28:3	r	'h0	Reaching this offset value by the current offset counter for addressing Self Path 2 picture Y component leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP2_RAW_IRQ_OFFS_SHD

0x00001734 = Byte Address

Description: Shadow register of fill level interrupt offset value for main picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_raw_irq_offs	28:3	r	'h0	Reaching this offset value by the current offset counter for addressing Self Path 2 picture RAW component leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MCM_RAW0_IRQ_OFFS_SHD

0x00001738 = Byte Address

Description: Shadow register of fill level interrupt offset value for main picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_irq_offs	28:3	r	'h0	Reaching this offset value by the current offset counter for addressing MCM picture RAW0 data leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_MCM_RAW1_IRQ_OFFS_SHD

0x0000173c = Byte Address

Description: Shadow register of fill level interrupt offset value for main picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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mcm_raw1_irq_offs	28:3	r	'h0	Reaching this offset value by the current offset counter for addressing MCM picture RAW1 data leads to generation of fill level interrupt. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP2_DMA_RAW_PIC_WIDTH_SHD

0x00001740 = Byte Address

Description: RAW component image width

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_pic_width	14:0	r	'h0	Image width of the RAW component in pixels.

MI_SP2_DMA_RAW_PIC_LLENGTH_SHD

0x00001744 = Byte Address

Description: RAW component original line length

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_pic_llength	31:0	r	'h0	Line length of the RAW component in bytes of the original image in memory. For an uncropped image, where lines follow each other without offset (no line stride), line length must match image width.

MI_SP2_DMA_RAW_PIC_SIZE_SHD

0x00001748 = Byte Address

Description: RAW component image memory size in bytes

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_pic_size	27:0	r	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (dma_raw_llength * dma_raw_pic_height).

MI_SP2_DMA_RAW_PIC_LVAL_SHD

0x0000174c = Byte Address

Description: SP2_DMA RAW Component line valid bytes number, 128 bit Alignment.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_width_bytes	15:0	r	'h0	Sizes occupied by valid data of a line; value should be a multiple of 16 (bus width = 16 bytes).

MI_ARGB_PADDING

0x00001750 = Byte Address

Description: the padding data for ARGB 8-8-8-8 format, and use bit[1:0] when 10bit ARGB 10-10-10-2.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
argb_padding	7:0	rw	'h0	the padding data for ARGB 8-8-8-8 format, and use bit[1:0] when 10bit ARGB 10-10-10-2.

ISP_CTRL

0x00000400 = Byte Address

Description: ISP Global control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
disable_isp_clk	31:31	rw	'h1	1: disable ISP clock of SRAM.
disable_nr_reloc	28:28	rw	'h0	1: disable NR relocation.
pp_write_sel_pdaf	26:26	rw	'h0	1: select pdaf data as output 0: use bit23 pp_write_sel for output select
pp_write_sel	23:23	rw	'h0	1: select yuv output 0: select raw output
ir_raw_out	22:22	rw	'h0	0: disable 1: IR data output from main path
regs_wdrth_data_mux	20:20	rw	'h0	1'b1: use wdrth data from outside, 1'b0: use wdrth output data
digi_gain_en	19:19	rw	'h0	0:digital gain disable 1:digital gain enable
dpf_raw_out	18:18	rw	'h0	0: Disable 1: Enable DPF raw output (from March 2018)
cnr_en	17:17	rw	'h0	0: Disable 1: CNR is enabled. (from March 2018)
enable_12bit_unpack	16:16	rw	'h0	0:disable 1: enable 12-bit unpack for DMA read
enable_12bit_packed	15:15	rw	'h0	0:disable 1: enable 12-bit pack
csm_c_range	14:14	rw	'h0	Color Space Matrix chrominance clipping range for ISP output 0: CbCr range 64..960 [16..240] according to ITU-R BT.601 standard 1: full UV range 0..1023 [0..255] Numbers in brackets are for 8 bit resolution. This bit also configures the YCbCr sequence align block accordingly.

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csm_y_range	13:13	rw	'h0	Color Space Matrix luminance clipping range for ISP output 0: Y range 64..940 [16..235] according to ITU-R BT.601 standard 1: full Y range 0..1023 [0..255] Numbers in brackets are for 8 bit resolution. This bit also configures the YCbCr sequence align block accordingly.
flash_mode	12:12	rw	'h0	0: sensor interface works independently from flash control unit 1: one frame is captured when signaled by flash control unit
gamma_out_enable	11:11	rw	'h0	Gamma ON/OFF
gen_cfg_upd	10:10	w	'h0	1: generate frame synchronous configuration signal at the output of ISP for shadow registers of the following processing modules, write only
cfg_upd	9:9	w	'h0	1: immediately configure (update) shadow registers, write only
gen_cfg_upd_fix	8:8	rw	'h0	1: make gen_cfg_upd a level signal, effect since then. 0: gen_cfg_upd is a pulse signal, only effect for current frame.
awb_enable	7:7	rw	'h0	Auto white balance ON/OFF
gamma_in_enable	6:6	rw	'h0	Sensor De-gamma ON/OFF
inform_enable	4:4	rw	'h0	input formatter 0: disabled; 1: enabled The ISP input formatter is enabled or disabled by this bit immediately, But always starts or stops acquisition frame synchronously.
isp_mode	3:1	rw	'h0	000 - RAW picture with BT.601 sync (ISP bypass) 001 - ITU-R BT.656 (YUV with embedded sync) 010 - ITU-R BT.601 (YUV input with H/Vsync signals) 011 - Bayer RGB processing with H/Vsync signals 100 - data mode (ISP bypass, sync signals interpreted as data enable) 101 - Bayer RGB processing with BT.656 synchronization 110 - RAW picture with ITU-R BT.656 synchronization (ISP bypass) 111 - reserved Side effect: If RAW, BT.601, BT.656, or data mode is selected, the clock of the ISP SRAMs (ISP line buffer, Lens Shading, Bad Pixel) is switched off. Only in Bayer RGB mode the clock to the SRAMs is enabled. This further reduces power consumption.

isp_enable	0:0	rw	'h0	ISP data output 0: disabled ; 1: enabled Controls output formatter frame synchronously, if isp_gen_cfg_upd is Used to activate this bit. For immediate update isp_cfg_upd must be used.

ISP_ACQ_PROP

0x00000404 = Byte Address

Description: ISP acquisition properties

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_en	31:31	rw	'h0	in HDR mode, ISP input data bit width is 20
sample_edge_1	29:29	rw	'h0	Sample edge control for sensor 2 0: use input sclk 1: add half cycle delay of sclk
sample_edge_0	28:28	rw	'h0	Sample edge control for sensor 1 0: use input sclk 1: add half cycle delay of sclk
two_cycle_data_sel	27:26	rw	'h0	two_cycle_data_sel[2:0] 0:select the same bits of data0 and data1 1: select all bits of data0 and data1 low bits
change_two_cycle_data_sequence	25:25	rw	'h0	change_two_cycle_data_seq 0: data0 is high bits, data1 is low bits 1: data1 is high bits, data0 is low bits
two_cycle_merge_enable	24:24	rw	'h0	two_cycle_merge_enable 0: not merge 1: two sequence data merge to one pixel
input_bayer_format	23:20	rw	'h0	input_bayer_fomat 0: no more than bit16 1~5: reserved 6: bit18 7: bit20
pin_mapping	19:17	rw	'h0	Bit mapping from LSB to MSB: 000 - normal 12-bit external Interface 001 - mapping Low 10 bit to High 10 bits, append 2 zeroes as LSBs 010 - mapping Low 8 bit to High 8 bits, append 4 zeroes as LSBs 011 - mapping Middle 8 bit to High 8 bits, append 4 zeroes as LSBs 100...111 - Reserved (from March 2018)
yuv_dma_sel	16:16	rw	'h0	0: use align or conversion data for isp_is input. 1: use DMA yuv read data for isp_is input. (from March 2018)
rgb_dma_sel	15:15	rw	'h0	0: use input formatter data for latency fifo. 1: use DMA rgb read data for latency fifo.. (from March 2018)
input_selection	14:12	rw	'h0	reserved

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field_inv	11:11	rw	'h0	0: do not swap odd and even fields 1: swap odd and even fields
field_selection	10:9	rw	'h0	00- sample all fields (don't care about fields) 01- sample only even fields 10- sample only odd fields 11- reserved
ccir_seq	8:7	rw	'h0	00- YCbYCr 01- YCrYCb 10- CbYCrY 11- CrYCbY
conv_422	6:5	rw	'h0	00- cosited color subsampling Y0Cb0Cr0 – Y1 01- interleaved color subsampling Y0Cb0 – Y1Cr1 (not recommended) 10- non-cosited color subsampling Y0Cb(0+1)/2 – Y1Cr(0+1)/2 11- reserved
bayer_pat	4:3	rw	'h0	Color components from sensor, starting with top left position in sampled frame (reprogram with ISP_ACQ_H_OFFS, ISP_ACQ_V_OFFS) 00- first line: RGRG..., second line: GBGB..., etc. 01- first line: GRGR..., second line: BGBG..., etc. 10- first line: GBGB..., second line: RGRG..., etc. 11- first line: BGBG..., second line: GRGR..., etc. This configuration applies for the black level area after cropping by the input formatter.
vsync_pol	2:2	rw	'h0	Vertical sync polarity 0: high active; 1: low active
hsync_pol	1:1	rw	'h0	Horizontal sync polarity 0: high active; 1: low active
sample_edge	0:0	r	'h0	0- negative edge sampling; 1- positive edge sampling

ISP_ACQ_H_OFFS

0x00000408 = Byte Address

Description: Horizontal input offset

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
acq_h_offs	14:0	rw	'h0	Horizontal sample offset in 8-bit samples (YUV: 4 samples = 2 pixels)

ISP_ACQ_V_OFFS

0x0000040c = Byte Address

Description: Vertical input offset

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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acq_v_offs	13:0	rw	'h0	Vertical sample offset in lines

ISP_ACQ_H_SIZE

0x00000410 = Byte Address

Description: Horizontal input size

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
acq_h_size	14:0	rw	'h1000	Horizontal sample size in 12-bit samples YUV input: 2 samples = 1 pixel, else 1 sample = 1 pixel; So in YUV mode ACQ_H_SIZE must be twice as large as horizontal image size. Horizontal image size must always be even except in raw picture mode; if an odd size is programmed, the value will be truncated to an even size.

ISP_ACQ_V_SIZE

0x00000414 = Byte Address

Description: Vertical input size

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
acq_v_size	13:0	rw	'hc00	Vertical sample size in lines

ISP_ACQ_NR_FRAMES

0x00000418 = Byte Address

Description: Number of frames to be captured

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
acq_nr_frames	9:0	rw	'h0	Number of input frames to be sampled (0 = continuous)

ISP_DGAIN_RB

0x000004f0 = Byte Address

Description: digital gain of red blue

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dgain_r	31:16	rw	'h100	digital gain of red channel
dgain_b	15:0	rw	'h100	digital gain of blue channel

ISP_DGAIN_G

0x000004f4 = Byte Address

Description: digital gain of green

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dgain_gr	31:16	rw	'h100	digital gain of red line green channel

dgain_gb	15:0	rw	'h100	digital gain of blue line green channel

ISP_AWB_PROP

0x00000510 = Byte Address

Description: Auto white balance properties

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
awb_meas_mode	31:31	rw	'h0	0: near white discrimination mode using YCbCr color space 1: RGB based measurement mode
awb_max_en	2:2	rw	'h0	0: disable; 1: enable Not valid in RGB measurement mode.
awb_mode	1:0	rw	'h0	00: no measurement 01: reserved 10: measurement of YCbCr means (AWB_MEAS_MODE = 0) or RGB means (AWB_MEAS_MODE = 1) 11: reserved

ISP_AWB_H_OFFS

0x00000514 = Byte Address

Description: Auto white balance horizontal offset of measure window

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
h_offs	12:0	rw	'h0	Horizontal window offset in pixels

ISP_AWB_V_OFFS

0x00000518 = Byte Address

Description: Auto white balance vertical offset of measure window

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
v_offs	12:0	rw	'h0	Vertical window offset in lines

ISP_AWB_H_SIZE

0x0000051c = Byte Address

Description: Auto white balance horizontal window size

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
h_size	13:0	rw	'h0	Horizontal measurement of window size in pixels

ISP_AWB_V_SIZE

0x00000520 = Byte Address

Description: Auto white balance vertical window size

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
v_size	13:0	rw	'h0	Vertical measurement of window size in lines

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ISP_AWB_FRAMES

0x00000524 = Byte Address

Description: Auto white balance mean value over multiple frames

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
awb_frames	2:0	rw	'h0	((Number of frames) - 1) used for mean value calculation (a value of 0 means 1 frame; a value of 7 means 8 frames)

ISP_AWB_REF

0x00000528 = Byte Address

Description: Auto white balance reference Cb/Cr values

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
awb_ref_cr__max_r	15:8	rw	'h80	Reference Cr value for AWB regulation; target for AWB - maximum red value, if RGB measurement mode is selected
awb_ref_cb__max_b	7:0	rw	'h80	Reference Cb value for AWB regulation; target for AWB - maximum red value, if RGB measurement mode is selected

ISP_AWB_THRESH

0x0000052c = Byte Address

Description: Auto white balance threshold values

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
awb_max_y	31:24	rw	'he9	Luminance maximum value, only consider pixels with luminance smaller than threshold for the White Balance measurement (must be enabled by register AWB_MODE bit AWB_MAX_EN). Not valid for RGB measurement mode.
awb_min_y__max_g	23:16	rw	'hc0	Luminance minimum value, only consider pixels with luminance greater than threshold for the White Balance measurement - Maximum green value, if RGB measurement mode is selected
awb_max_csum	15:8	rw	'h10	Chrominance sum maximum value, only consider pixels with Cb+Cr smaller than threshold for White Balance measurements

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awb_min_c	7:0	rw	'h10	Chrominance minimum value, only consider pixels with Cb/Cr each greater than threshold value for White Balance measurements

ISP_AWB_GAIN_G

0x00000538 = Byte Address

Description: Auto white balance gain green

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
awb_gain_gr_high	29:26	rw	'h1	Gain value for green component for rccb, 4 bit int part
awb_gain_gr	25:16	rw	'h100	Gain value for green component in red line 100h = 1, Unsigned integer value, range 0 to 4 with 8 bit fractional part. For RCCB support, 10 bit fractional part
awb_gain_gb_high	13:10	rw	'h1	Gain value for green component for rccb, 4 bit int part
awb_gain_gb	9:0	rw	'h100	Gain value for green component in blue line 100h = 1, Unsigned integer value, range 0 to 4 with 8 bit fractional part. For RCCB support, 10 bit fractional part

ISP_AWB_GAIN_RB

0x0000053c = Byte Address

Description: Auto white balance gain red and blue

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
awb_gain_r_high	29:26	rw	'h1	
awb_gain_r	25:16	rw	'h100	Gain value for red component 100h = 1, Unsigned integer value, range 0 to 4 with 8 bit fractional part. For RCCB support, 10 bit fractional part
awb_gain_b_high	13:10	rw	'h1	
awb_gain_b	9:0	rw	'h100	Gain value for blue component 100h = 1, Unsigned integer value, range 0 to 4 with 8 bit fractional part. For RCCB support, 10 bit fractional part

ISP_AWB_WHITE_CNT

0x00000540 = Byte Address

Description: Auto white balance white pixel count

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

awb_white_cnt	27:0	r	'h0	White pixel count, number of "white pixels" found during last measurement, I.e. pixels included in mean value calculation

ISP_AWB_MEAN

0x00000544 = Byte Address

Description: Auto white balance measured mean value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
awb_mean_y__g	23:16	r	'h0	Mean value of Cb within window and frames- Mean value of blue, if RGB measurement mode is selected.
awb_mean_cb__b	15:8	r	'h0	Mean value of Cr within window and frames- Mean value of red, if RGB measurement mode is selected.
awb_mean_cr__r	7:0	r	'h0	Mean value of Y within window and frames- mean value of green, if RGB measurement mode is selected.

ISP_CC_COEFF_0

0x00000570 = Byte Address

Description: Coefficient 0 for color space conversion

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cc_coeff0	8:0	rw	'h21	Coefficient 0 for color space conversion

ISP_CC_COEFF_1

0x00000574 = Byte Address

Description: Coefficient 1 for color space conversion

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cc_coeff1	8:0	rw	'h40	Coefficient 1 for color space conversion

ISP_CC_COEFF_2

0x00000578 = Byte Address

Description: Coefficient 2 for color space conversion

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cc_coeff2	8:0	rw	'hd	Coefficient 2 for color space conversion

ISP_CC_COEFF_3

0x0000057c = Byte Address

Description: Coefficient 3 for color space conversion

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

cc_coeff3	8:0	rw	'h1ed	Coefficient 3 for color space conversion

ISP_CC_COEFF_4

0x00000580 = Byte Address

Description: Coefficient 4 for color space conversion

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cc_coeff4	8:0	rw	'h1db	Coefficient 4 for color space conversion

ISP_CC_COEFF_5

0x00000584 = Byte Address

Description: Coefficient 5 for color space conversion

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cc_coeff5	8:0	rw	'h38	Coefficient 5 for color space conversion

ISP_CC_COEFF_6

0x00000588 = Byte Address

Description: Coefficient 6 for color space conversion

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cc_coeff6	8:0	rw	'h38	Coefficient 6 for color space conversion

ISP_CC_COEFF_7

0x0000058c = Byte Address

Description: Coefficient 7 for color space conversion

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cc_coeff7	8:0	rw	'h1d1	Coefficient 7 for color space conversion

ISP_CC_COEFF_8

0x00000590 = Byte Address

Description: Coefficient 8 for color space conversion

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cc_coeff8	8:0	rw	'h1f7	Coefficient 8 for color space conversion

ISP_OUT_H_OFFS

0x00000594 = Byte Address

Description: Horizontal offset of output window

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
out_h_offs	13:0	rw	'h0	Horizontal pic offset in lines

ISP_OUT_V_OFFS

0x00000598 = Byte Address

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Description: Vertical offset of output window

Sub-Field Name	Bits	rw	Reset	Description
out_v_offs	13:0	rw	'h0	Vertical pic offset in lines

ISP_OUT_H_SIZE

0x0000059c = Byte Address

Description: Output horizontal picture size

Sub-Field Name	Bits	rw	Reset	Description
out_h_size	14:0	rw	'h1000	Horizontal picture size in pixel if ISP_MODE is set to: 001-(ITU-R BT.656 YUV), 010-(ITU-R BT.601 YUV), 011-(ITU-R BT.601 Bayer RGB), 101-(ITU-R BT.656 Bayer RGB) Only even numbers are accepted, because complete quadruples of YUYV(YCbYCr) are needed for the 422 output. If an odd size is programmed, the value will be truncated to an even size.

ISP_OUT_V_SIZE

0x000005a0 = Byte Address

Description: Output vertical picture size

Sub-Field Name	Bits	rw	Reset	Description
out_v_size	13:0	rw	'hc00	Vertical picture size in lines

ISP_DEMOSAIC

0x000005a4 = Byte Address

Description: Demosaic parameters

Sub-Field Name	Bits	rw	Reset	Description
demosaic_bypass	10:10	rw	'h0	0: normal operation for RGB Bayer Pattern input 1: demosaicing bypass for Black&White input data
demosaic_th	7:0	rw	'h4	Threshold for Bayer demosaicing texture detection. This value shifted left 4bit is compared with the difference of the vertical and horizontal 12-Bit wide texture indicators, to decide if the vertical or horizontal texture flag must be set. 0x00: maximum edge sensitivity 0xFF: no texture detection

ISP_FLAGS_SHD

0x000005a8 = Byte Address

Description: Flags (current status) of certain signals and Shadow regs for enable signals

Sub-Field Name	Bits	rw	Reset	Description
s_hsync	31:31	r	'h0	State of ISP input port s_hsync, for test purposes
s_vsync	30:30	r	'h0	State of ISP input port s_vsync, for test purposes
s_data	27:16	r	'h0	State of ISP input port s_data, for test purposes
inform_field	2:2	r	'h0	Current field information (0=odd, 1=even)
inform_en_shd	1:1	r	'h0	Input formatter enable shadow register
isp_on_shd	0:0	r	'h0	ISP enable shadow register Shows if ISP currently outputs data (1) or not (0)

ISP_OUT_H_OFFS_SHD

0x000005ac = Byte Address

Description: Current horizontal offset of output window (shadow register)

Sub-Field Name	Bits	rw	Reset	Description
out_h_offs_shd	13:0	r	'h0	Current horizontal picture offset in lines.

ISP_OUT_V_OFFS_SHD

0x000005b0 = Byte Address

Description: Current vertical offset of output window (shadow register)

Sub-Field Name	Bits	rw	Reset	Description
out_v_offs_shd	13:0	r	'h0	Current vertical picture offset in lines.

ISP_OUT_H_SIZE_SHD

0x000005b4 = Byte Address

Description: Current output horizontal picture size (shadow register)

Sub-Field Name	Bits	rw	Reset	Description
out_h_size_shd	14:0	r	'h1000	Current horizontal picture size in pixels.

ISP_OUT_V_SIZE_SHD

0x000005b8 = Byte Address

Description: Current output vertical picture size (shadow register)

Sub-Field Name	Bits	rw	Reset	Description
out_v_size_shd	13:0	r	'hc00	Current vertical picture size in lines.

ISP_IMSC

0x000005bc = Byte Address

Description: Interrupt mask

Sub-Field Name	Bits	rw	Reset	Description
imsc_sta_dma_error	30:30	rw	'h0	0: mask out; 1: enable interrupt
imsc_fifo1_frame_end	29:29	rw	'h0	0: mask out; 1: enable interrupt
imsc_sensor0_dataloss	26:26	rw	'h0	0: mask out; 1: enable interrupt
imsc_sensor1_dataloss	25:25	rw	'h0	0: mask out; 1: enable interrupt
imsc_sensor2_dataloss	24:24	rw	'h0	0: mask out; 1: enable interrupt
imsc_sensor3_dataloss	23:23	rw	'h0	0: mask out; 1: enable interrupt
imsc_ae2_dataloss	22:22	rw	'h0	0: mask out; 1: enable interrupt
imsc_awb2_dataloss	21:21	rw	'h0	0: mask out; 1: enable interrupt
imsc_tdnr	20:20	rw	'h0	0: mask out; 1: enable interrupt
imsc_vsm_done	19:19	rw	'h0	0: mask out; 1: enable interrupt
imsc_exp_end	18:18	rw	'h0	0: mask out; 1: enable interrupt
imsc_fl_cap	17:17	rw	'h0	0: mask out; 1: enable interrupt
imsc_tdnr1	16:16	rw	'h0	0: mask out; 1: enable interrupt
imsc_hist_end	15:15	rw	'h0	0: mask out; 1: enable interrupt
imsc_afm_fin	14:14	rw	'h0	0: mask out; 1: enable interrupt
imsc_afm_lum_of	13:13	rw	'h0	0: mask out; 1: enable interrupt
imsc_afm_sum_of	12:12	rw	'h0	0: mask out; 1: enable interrupt
imsc_sh_off	11:11	rw	'h0	0: mask out; 1: enable interrupt
imsc_sh_on	10:10	rw	'h0	0: mask out; 1: enable interrupt
imsc_fl_off	9:9	rw	'h0	0: mask out; 1: enable interrupt
imsc_fl_on	8:8	rw	'h0	0: mask out; 1: enable interrupt
imsc_h_start	7:7	rw	'h0	0: mask out; 1: enable interrupt
imsc_v_start	6:6	rw	'h0	0: mask out; 1: enable interrupt
imsc_frame_in	5:5	rw	'h0	0: mask out; 1: enable interrupt
imsc_awb_done	4:4	rw	'h0	0: mask out; 1: enable interrupt
imsc_size_err	3:3	rw	'h0	0: mask out; 1: enable interrupt
imsc_dataloss	2:2	rw	'h0	0: mask out; 1: enable interrupt
imsc_frame	1:1	rw	'h0	0: mask out; 1: enable interrupt
imsc_isp_off	0:0	rw	'h0	0: mask out; 1: enable interrupt

ISP_RIS

0x000005c0 = Byte Address

Description: Raw interrupt status

Sub-Field Name	Bits	rw	Reset	Description

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ris_sta_dma_error	30:30	r	'h0	statistic buffer write to DDR not completed before next frame start.
ris_fifo1_frame_end	29:29	r	'h0	fifo1 frame end interrupt
ris_sensor0_dataloss	26:26	r	'h0	sensor0 data loss
ris_sensor1_dataloss	25:25	r	'h0	sensor1 data loss
ris_sensor2_dataloss	24:24	r	'h0	sensor2 data loss
ris_sensor3_dataloss	23:23	r	'h0	sensor3 data loss
ris_ae2_dataloss	22:22	r	'h0	aev2 data loss
ris_awb2_dataloss	21:21	r	'h0	awb2 data loss
ris_tdnr	20:20	r	'h0	TDNR reference frame FIFO empty
ris_vsm_done	19:19	r	'h0	VSM module complete
ris_exp_end	18:18	r	'h0	Exposure measurement complete
ris_fl_cap	17:17	r	'h0	Signaling captured frame
ris_tdnr1	16:16	r	'h0	write cover read
ris_hist_end	15:15	r	'h0	Histogram measurement ready. (Old or new histogram measurement)
ris_afm_fin	14:14	r	'h0	AF measurement finished: this interrupt is set when the first complete frame is calculated after enabling the AF measurement
ris_afm_lum_of	13:13	r	'h0	Auto focus luminance overflow
ris_afm_sum_of	12:12	r	'h0	Auto focus sum overflow
ris_sh_off	11:11	r	'h0	Mechanical shutter is switched off
ris_sh_on	10:10	r	'h0	Mechanical shutter is switched on
ris_fl_off	9:9	r	'h0	Flash light is switched off
ris_fl_on	8:8	r	'h0	Flash light is switched on
ris_h_start	7:7	r	'h0	Start edge of h_sync
ris_v_start	6:6	r	'h0	Start edge of v_sync
ris_frame_in	5:5	r	'h0	Sampled input frame is complete
ris_awb_done	4:4	r	'h0	White balancing measurement cycle is complete; results can be read out
ris_size_err	3:3	r	'h0	Picture size violation occurred; incorrect programming
ris_dataloss	2:2	r	'h0	Loss of data occurred within a line, processing failure
ris_frame	1:1	r	'h0	Frame was completely put out
ris_isp_off	0:0	r	'h0	ISP output was disabled (vsynced) due to f_cnt reached or manual

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ISP_MIS

0x000005c4 = Byte Address

Description: Masked interrupt status

Sub-Field Name	Bits	rw	Reset	Description
mis_sta_dma_error	30:30	r	'h0	statistic buffer write to DDR not completed before next frame start.
mis_fifo1_frame_end	29:29	r	'h0	fifo1 frame end interrupt
mis_sensor0_dataloss	26:26	r	'h0	sensor0 data loss
mis_sensor1_dataloss	25:25	r	'h0	sensor1 data loss
mis_sensor2_dataloss	24:24	r	'h0	sensor2 data loss
mis_sensor3_dataloss	23:23	r	'h0	sensor3 data loss
mis_ae2_dataloss	22:22	r	'h0	aev2 data loss
mis_awb2_dataloss	21:21	r	'h0	awb2 data loss
mis_tdnr	20:20	r	'h0	TDNR reference frame FIFO empty
mis_vsm_done	19:19	r	'h0	VSM module complete
mis_exp_end	18:18	r	'h0	Exposure measurement complete
mis_fl_cap	17:17	r	'h0	Signaling captured frame
mis_tdnr1	16:16	r	'h0	write cover read
mis_hist_end	15:15	r	'h0	Histogram measurement ready. (Old or new histogram measurement)
mis_afm_fin	14:14	r	'h0	AF measurement finished: this interrupt is set when the first complete frame is calculated after enabling the AF measurement
mis_afm_lum_of	13:13	r	'h0	Auto focus luminance overflow
mis_afm_sum_of	12:12	r	'h0	Auto focus sum overflow
mis_sh_off	11:11	r	'h0	Mechanical shutter is switched off
mis_sh_on	10:10	r	'h0	Mechanical shutter is switched on
mis_fl_off	9:9	r	'h0	Flash light is switched off
mis_fl_on	8:8	r	'h0	Flash light is switched on
mis_h_start	7:7	r	'h0	Start edge of h_sync
mis_v_start	6:6	r	'h0	Start edge of v_sync
mis_frame_in	5:5	r	'h0	Sampled input frame is complete
mis_awb_done	4:4	r	'h0	White balancing measurement cycle is complete; results can be read out
mis_size_err	3:3	r	'h0	Picture size violation occurred; incorrect programming

mis_dataloss	2:2	r	'h0	Loss of data occurred within a line, processing failure
mis_frame	1:1	r	'h0	Frame was completely put out
mis_isp_off	0:0	r	'h0	ISP output was disabled (vsynced) due to f_cnt reached or manual

ISP_ICR

0x000005c8 = Byte Address

Description: Interrupt clear register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
icr_sta_dma_error	30:30	w	'h0	statistic buffer write to DDR not completed before next frame start.
icr_fifo1_frame_end	29:29	w	'h0	fifo1 frame end interrupt
icr_sensor0_dataloss	26:26	w	'h0	Clear interrupt
icr_sensor1_dataloss	25:25	w	'h0	Clear interrupt
icr_sensor2_dataloss	24:24	w	'h0	Clear interrupt
icr_sensor3_dataloss	23:23	w	'h0	Clear interrupt
icr_ae2_dataloss	22:22	w	'h0	Clear interrupt
icr_awb2_dataloss	21:21	w	'h0	Clear interrupt
icr_tdnr	20:20	w	'h0	Clear interrupt
icr_vsm_done	19:19	w	'h0	Clear interrupt
icr_exp_end	18:18	w	'h0	Clear interrupt
icr_fl_cap	17:17	w	'h0	Clear interrupt
icr_tdnr1	16:16	w	'h0	Clear interrupt
icr_hist_end	15:15	w	'h0	Clear interrupt
icr_afm_fin	14:14	w	'h0	Clear interrupt
icr_afm_lum_of	13:13	w	'h0	Clear interrupt
icr_afm_sum_of	12:12	w	'h0	Clear interrupt
icr_sh_off	11:11	w	'h0	Clear interrupt
icr_sh_on	10:10	w	'h0	Clear interrupt
icr_fl_off	9:9	w	'h0	Clear interrupt
icr_fl_on	8:8	w	'h0	Clear interrupt
icr_h_start	7:7	w	'h0	Clear interrupt
icr_v_start	6:6	w	'h0	Clear interrupt
icr_frame_in	5:5	w	'h0	Clear interrupt
icr_awb_done	4:4	w	'h0	Clear interrupt
icr_size_err	3:3	w	'h0	Clear interrupt

icr_dataloss	2:2	w	'h0	Clear interrupt
icr_frame	1:1	w	'h0	Clear interrupt
icr_isp_off	0:0	w	'h0	Clear interrupt

ISP_ISR

0x000005cc = Byte Address

Description: Interrupt set register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isr_sta_dma_error	30:30	w	'h0	statistic buffer write to DDR not completed before next frame start.
isr_fifo1_frame_end	29:29	w	'h0	fifo1 frame end interrupt
isr_sensor0_dataloss	26:26	w	'h0	Set interrupt
isr_sensor1_dataloss	25:25	w	'h0	Set interrupt
isr_sensor2_dataloss	24:24	w	'h0	Set interrupt
isr_sensor3_dataloss	23:23	w	'h0	Set interrupt
isr_ae2_dataloss	22:22	w	'h0	Set interrupt
isr_awb2_dataloss	21:21	w	'h0	Set interrupt
isr_tdnr	20:20	w	'h0	Set interrupt
isr_vsm_done	19:19	w	'h0	Set interrupt
isr_exp_end	18:18	w	'h0	Set interrupt
isr_fl_cap	17:17	w	'h0	Set interrupt
isr_tdnr1	16:16	w	'h0	Set interrupt
isr_hist_end	15:15	w	'h0	Set interrupt
isr_afm_fin	14:14	w	'h0	Set interrupt
isr_afm_lum_of	13:13	w	'h0	Set interrupt
isr_afm_sum_of	12:12	w	'h0	Set interrupt
isr_sh_off	11:11	w	'h0	Set interrupt
isr_sh_on	10:10	w	'h0	Set interrupt
isr_fl_off	9:9	w	'h0	Set interrupt
isr_fl_on	8:8	w	'h0	Set interrupt
isr_h_start	7:7	w	'h0	Set interrupt
isr_v_start	6:6	w	'h0	Set interrupt
isr_frame_in	5:5	w	'h0	Set interrupt
isr_awb_done	4:4	w	'h0	Set interrupt
isr_size_err	3:3	w	'h0	Set interrupt
isr_dataloss	2:2	w	'h0	Set interrupt

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isr_frame	1:1	w	'h0	Set interrupt
isr_isp_off	0:0	w	'h0	Set interrupt

ISP_CT_COEFF_0

0x000005d0 = Byte Address

Description: Cross-talk configuration register (color correction matrix)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_coeff0_high	11:11	rw	'h0	Coefficient for cross talk matrix.if use bit 11,Values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff0	10:0	rw	'h80	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_1

0x000005d4 = Byte Address

Description: Cross-talk configuration register (color correction matrix)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_coeff1_high	11:11	rw	'h0	Coefficient for cross talk matrix.if use bit 11,Values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff1	10:0	rw	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_2

0x000005d8 = Byte Address

Description: Cross-talk configuration register (color correction matrix)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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ct_coeff2_high	11:11	rw	'h0	Coefficient for cross talk matrix.if use bit 11,Values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff2	10:0	rw	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_3

0x000005dc = Byte Address

Description: Cross-talk configuration register (color correction matrix)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_coeff3_high	11:11	rw	'h0	Coefficient for cross talk matrix.if use bit 11,Values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff3	10:0	rw	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_4

0x000005e0 = Byte Address

Description: Cross-talk configuration register (color correction matrix)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_coeff4_high	11:11	rw	'h0	Coefficient for cross talk matrix.if use bit 11,Values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ct_coeff4	10:0	rw	'h80	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_5

0x000005e4 = Byte Address

Description: Cross-talk configuration register (color correction matrix)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_coeff5_high	11:11	rw	'h0	Coefficient for cross talk matrix.if use bit 11,Values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff5	10:0	rw	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_6

0x000005e8 = Byte Address

Description: Cross-talk configuration register (color correction matrix)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_coeff6_high	11:11	rw	'h0	Coefficient for cross talk matrix.if use bit 11,Values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff6	10:0	rw	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_7

0x000005ec = Byte Address

Description: Cross-talk configuration register (color correction matrix)

Sub-Field Name	Bits	rw	Reset	Description
ct_coeff7_high	11:11	rw	'h0	Coefficient for cross talk matrix.if use bit 11,Values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff7	10:0	rw	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_8

0x000005f0 = Byte Address

Description: Cross-talk configuration register (color correction matrix)

Sub-Field Name	Bits	rw	Reset	Description
ct_coeff8_high	11:11	rw	'h0	Coefficient for cross talk matrix.if use bit 11,Values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff8	10:0	rw	'h80	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_ERR

0x0000063c = Byte Address

Description: ISP error register

Sub-Field Name	Bits	rw	Reset	Description
outform_size_err	2:2	r	'h0	Size error is generated in outmux submodule
is_size_err	1:1	r	'h0	Size error is generated in image stabilization submodule
inform_size_err	0:0	r	'h0	Size error is generated in inform submodule

ISP_ERR_CLR

0x00000640 = Byte Address

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Description: ISP error clear register

Sub-Field Name	Bits	rw	Reset	Description
outform_size_err_clr	2:2	w	'h0	Size error is cleared
is_size_err_clr	1:1	w	'h0	Size error is cleared
inform_size_err_clr	0:0	w	'h0	Size error is cleared

ISP_FRAME_COUNT

0x00000644 = Byte Address

Description: Frame counter

Sub-Field Name	Bits	rw	Reset	Description
frame_count	9:0	r	'h0	Current frame count of processing

ISP_CT_OFFSET_R

0x00000648 = Byte Address

Description: Cross-talk offset red

Sub-Field Name	Bits	rw	Reset	Description
ct_offset_r	11:0	rw	'h0	Red offset for the cross talk matrix. Two's complement integer number ranging from [-2048 (0x800) to 2047 (0x7FF)]. 0 is represented as 0x000.

ISP_CT_OFFSET_G

0x0000064c = Byte Address

Description: Cross-talk offset green

Sub-Field Name	Bits	rw	Reset	Description
ct_offset_g	11:0	rw	'h0	Green offset for the cross talk matrix. Two's complement integer number ranging from [-2048 (0x800) to 2047 (0x7FF)]. 0 is represented as 0x000.

ISP_CT_OFFSET_B

0x00000650 = Byte Address

Description: Cross-talk offset blue

Sub-Field Name	Bits	rw	Reset	Description
ct_offset_b	11:0	rw	'h0	Blue offset for the cross talk matrix. Two's complement integer number ranging from [-2048 (0x800) to 2047 (0x7FF)]. 0 is represented as 0x000.

ISP_SH_CTRL

0x00000680 = Byte Address

Description: Mechanical shutter control

Sub-Field Name	Bits	rw	Reset	Description
sh_open_pol	4:4	rw	'h0	Shutter_open polarity 0: high active; 1: low active
sh_trig_pol	3:3	rw	'h0	Mechanical shutter trigger edge 0: use negative edge of trigger signal 1: use positive edge of trigger signal
sh_trig_src	2:2	rw	'h0	Mechanical shutter trigger source 0: use “vds_vsync” for trigger event 1: use “shutter_trig” for trigger event
sh_rep_en	1:1	rw	'h0	Mechanical shutter repetition enable 0: shutter is opened only once 1: shutter is opened with the repetition rate of the trigger signal
sh_en	0:0	rw	'h0	Mechanical shutter 0: disable; 1: enable

ISP_SH_PREDIV

0x00000684 = Byte Address

Description: Mechanical Shutter Counter Pre-Divider

Sub-Field Name	Bits	rw	Reset	Description
sh_pre_div	9:0	rw	'h0	Pre-divider for mechanical shutter open_delay and open_time counter.

ISP_SH_DELAY

0x00000688 = Byte Address

Description: Delay register

Sub-Field Name	Bits	rw	Reset	Description
sh_delay	19:0	rw	'h0	Counter value for delay open_delay = (sh_delay + 1) * (fl_pre_div+1) / clk_isp sh_delay = (open_delay * clk_isp) / (sh_pre_div+1) – 1

ISP_SH_TIME

0x0000068c = Byte Address

Description: Time register

Sub-Field Name	Bits	rw	Reset	Description
sh_time	19:0	rw	'h0	Counter value for time open_time = (sh_time + 1) * (fl_pre_div+1) / clk_isp sh_time = (open_time * clk_isp) / (sh_pre_div+1) - 1

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ISP_DGAIN_RB_SHD

0x00000690 = Byte Address

Description: digital gain of red blue

Sub-Field Name	Bits	rw	Reset	Description
dgain_r_shd	31:16	r	'h100	digital gain of red channel
dgain_b_shd	15:0	r	'h100	digital gain of blue channel

ISP_DGAIN_G_SHD

0x00000694 = Byte Address

Description: digital gain of green

Sub-Field Name	Bits	rw	Reset	Description
dgain_gr_shd	31:16	r	'h100	digital gain of red line green channel
dgain_gb_shd	15:0	r	'h100	digital gain of blue line green channel

ISP_CT_COEFF_0_SHD

0x00000698 = Byte Address

Description: Cross-talk configuration register(color correction matrix)

Sub-Field Name	Bits	rw	Reset	Description
ct_coeff0_high_shd	11:11	r	'h0	Coefficient for cross talk matrix.if use bit 11, values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff0_shd	10:0	r	'h80	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_1_SHD

0x0000069c = Byte Address

Description: Cross-talk configuration register(color correction matrix)

Sub-Field Name	Bits	rw	Reset	Description

ct_coeff1_high_shd	11:11	r	'h0	Coefficient for cross talk matrix.if use bit 11, values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff1_shd	10:0	r	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_2_SHD

0x000006a0 = Byte Address

Description: Cross-talk configuration register(color correction matrix)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_coeff2_high_shd	11:11	r	'h0	Coefficient for cross talk matrix.if use bit 11, values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff2_shd	10:0	r	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_3_SHD

0x000006a4 = Byte Address

Description: Cross-talk configuration register(color correction matrix)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_coeff3_high_shd	11:11	r	'h0	Coefficient for cross talk matrix.if use bit 11, values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

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ct_coeff3_shd	10:0	r	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_4_SHD

0x000006a8 = Byte Address

Description: Cross-talk configuration register(color correction matrix)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_coeff4_high_shd	11:11	r	'h0	Coefficient for cross talk matrix.if use bit 11, values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff4_shd	10:0	r	'h80	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_5_SHD

0x000006ac = Byte Address

Description: Cross-talk configuration register(color correction matrix)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_coeff5_high_shd	11:11	r	'h0	Coefficient for cross talk matrix.if use bit 11, values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff5_shd	10:0	r	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_6_SHD

0x000006b0 = Byte Address

Description: Cross-talk configuration register(color correction matrix)

Sub-Field Name	Bits	rw	Reset	Description
ct_coeff6_high_shd	11:11	r	'h0	Coefficient for cross talk matrix.if use bit 11, values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff6_shd	10:0	r	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_7_SHD

0x000006b4 = Byte Address

Description: Cross-talk configuration register(color correction matrix)

Sub-Field Name	Bits	rw	Reset	Description
ct_coeff7_high_shd	11:11	r	'h0	Coefficient for cross talk matrix.if use bit 11, values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.
ct_coeff7_shd	10:0	r	'h0	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_COEFF_8_SHD

0x000006b8 = Byte Address

Description: Cross-talk configuration register(color correction matrix)

Sub-Field Name	Bits	rw	Reset	Description
ct_coeff8_high_shd	11:11	r	'h0	Coefficient for cross talk matrix.if use bit 11, values are 12-bit signed fixed-point numbers with 5 bit integer and 7 bit fractional part, ranging from -16 (0x800) to +15.992 (0x7FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ct_coeff8_shd	10:0	r	'h80	Coefficient for cross talk matrix. Values are 11-bit signed fixed-point numbers with 4 bit integer and 7 bit fractional part, ranging from -8 (0x400) to +7.992 (0x3FF). 0 is represented by 0x000 and a coefficient value of 1 as 0x080.

ISP_CT_OFFSET_R_SHD

0x000006bc = Byte Address

Description: Cross-talk offset red shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_offset_r_shd	11:0	r	'h0	Red offset for the cross talk matrix. Two's complement integer number ranging from [-2048 (0x800) to 2047 (0x7FF)]. 0 is represented as 0x000.Red offset for the cross talk matrix.

ISP_CT_OFFSET_G_SHD

0x000006c0 = Byte Address

Description: Cross-talk offset green shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_offset_g_shd	11:0	r	'h0	Green offset for the cross talk matrix. Two's complement integer number ranging from [-2048 (0x800) to 2047 (0x7FF)]. 0 is represented as 0x000.Red offset for the cross talk matrix.

ISP_CT_OFFSET_B_SHD

0x000006c4 = Byte Address

Description: Cross-talk offset blue shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ct_offset_b_shd	11:0	r	'h0	Blue offset for the cross talk matrix. Two's complement integer number ranging from [-2048 (0x800) to 2047 (0x7FF)]. 0 is represented as 0x000.Red offset for the cross talk matrix.

ISP_AWB_GAIN_G_SHD

0x000006c8 = Byte Address

Description: Auto white balance gain green shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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awb_gain_gr_high_shd	29:26	r	'h1	Gain value for green component for rccb, 4 bit int part
awb_gain_gr_shd	25:16	r	'h100	Gain value for green component in red line 100h = 1, Unsigned integer value, range 0 to 4 with 8 bit fractional part. For RCCB support, 10 bit fractional part
awb_gain_gb_high_shd	13:10	r	'h1	Gain value for green component for rccb, 4 bit int part
awb_gain_gb_shd	9:0	r	'h100	Gain value for green component in blue line 100h = 1, Unsigned integer value, range 0 to 4 with 8 bit fractional part. For RCCB support, 10 bit fractional part

ISP_AWB_GAIN_RB_SHD

0x000006cc = Byte Address

Description: Auto white balance gain red and blue shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
awb_gain_r_high_shd	29:26	r	'h1	
awb_gain_r_shd	25:16	r	'h100	Gain value for red component 100h = 1, Unsigned integer value, range 0 to 4 with 8 bit fractional part. For RCCB support, 10 bit fractional part
awb_gain_b_high_shd	13:10	r	'h1	
awb_gain_b_shd	9:0	r	'h100	Gain value for blue component 100h = 1, Unsigned integer value, range 0 to 4 with 8 bit fractional part. For RCCB support, 10 bit fractional part

ISP_STA_DMA_CTRL

0x000006f0 = Byte Address

Description: control of writing statistic data to DDR

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitch_sta_dma_en	2:2	rw	'h0	1: start transferring a frame. The sta_dma module will reset this bit and halt after completing one frame, if the bit sta_dma_mode is set to 1
sta_dma_en	0:0	rw	'h0	1: start transferring a frame. The sta_dma module will reset this bit and halt after completing one frame, if the bit sta_dma_mode is set to 1

ISP_AFM_CTRL

0x00002000 = Byte Address

Description: Auto Focus Measurement Unit Control Register

Sub-Field Name	Bits	rw	Reset	Description
afm_en	0:0	rw	'h0	AF measurement enable 0: disable; 1: enable Writing a 1 to this register starts a new measurement and resets the afm_fin (measurement finished) interrupt to 0. As long as the afm_en is 1, the AFM unit calculates new sharpness values for each frame.

ISP_AFM_LT_A

0x00002004 = Byte Address

Description: Measure Window A Top Left Corner

Sub-Field Name	Bits	rw	Reset	Description
a_h_l	28:16	rw	'h0	First pixel of window A (horizontal left row)Value must be greater than or equal to 5
a_v_t	12:0	rw	'h0	First line of window A (vertical top line)Value must be greater than or equal to 2

ISP_AFM_RB_A

0x00002008 = Byte Address

Description: Measure Window A Bottom Right Corner

Sub-Field Name	Bits	rw	Reset	Description
a_h_r	28:16	rw	'h0	Last pixel of window A (horizontal right row)
a_v_b	12:0	rw	'h0	Last line of window A (vertical bottom line) Value must be lower than (number of lines - 2)

ISP_AFM_LT_B

0x0000200c = Byte Address

Description: Measure Window B Top Left Corner

Sub-Field Name	Bits	rw	Reset	Description
b_h_l	28:16	rw	'h0	First pixel of window B (horizontal left row) Value must be greater than or equal to 5
b_v_t	12:0	rw	'h0	First line of window B (vertical top line) Value must be greater than or equal to 2

ISP_AFM_RB_B

0x00002010 = Byte Address

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Description: Measure Window B Bottom Right Corner

Sub-Field Name	Bits	rw	Reset	Description
b_h_r	28:16	rw	'h0	Last pixel of window B (horizontal right row)
b_v_b	12:0	rw	'h0	Last line of window B (vertical bottom line) Value must be lower than (number of lines - 2)

ISP_AFM_LT_C

0x00002014 = Byte Address

Description: Measure Window C Top Left Corner

Sub-Field Name	Bits	rw	Reset	Description
isp_h_l	28:16	rw	'h0	First pixel of window C (horizontal left row) Value must be greater than or equal to 5
isp_v_t	12:0	rw	'h0	First line of window C (vertical top line) Value must be greater than or equal to 2

ISP_AFM_RB_C

0x00002018 = Byte Address

Description: Measure Window C Bottom Right Corner

Sub-Field Name	Bits	rw	Reset	Description
isp_h_r	28:16	rw	'h0	Last pixel of window C (horizontal right row)
isp_v_b	12:0	rw	'h0	Last line of window C (vertical bottom line) Value must be lower than (number of lines - 2)

ISP_AFM_THRESH

0x0000201c = Byte Address

Description: Auto Focus Measurement Threshold Register

Sub-Field Name	Bits	rw	Reset	Description
afm_thresh	15:0	rw	'h0	AF measurement threshold,This register defines a threshold which can be used for minimizing the influence of noise in the measurement result.

ISP_AFM_VAR_SHIFT

0x00002020 = Byte Address

Description: Auto Focus Measurement Variable Shift Register

Sub-Field Name	Bits	rw	Reset	Description

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lum_var_shift	18:16	rw	'h0	Variable shift for luminance summation. The lum_var_shift defines the number of bits for the shift operation of the value of the current pixel before summation. The shift operation is used to avoid a luminance sum overflow.
sum_var_shift	2:0	rw	'h0	Variable shift for AF measurement. The afm_var_shift defines the number of bits for the shift operation at the end of the calculation chain. The shift operation is used to avoid an AF measurement sum overflow.

ISP_AFM_SUM_A

0x00002024 = Byte Address

Description: Window A Sharpness Value Status Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_sum_a	31:0	r	'h0	Sharpness value of window A

ISP_AFM_SUM_B

0x00002028 = Byte Address

Description: Window B Sharpness Value Status Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_sum_b	31:0	r	'h0	Sharpness value of window B

ISP_AFM_SUM_C

0x0000202c = Byte Address

Description: Window C Sharpness Value Status Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_sum_c	31:0	r	'h0	Sharpness value of window C

ISP_AFM_LUM_A

0x00002030 = Byte Address

Description: Window A Luminance Value Status Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_lum_a	23:0	r	'h0	Luminance value of window A

ISP_AFM_LUM_B

0x00002034 = Byte Address

Description: Window B Luminance Value Status Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_lum_b	23:0	r	'h0	Luminance value of window B

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ISP_AFM_LUM_C

0x00002038 = Byte Address

Description: Window C Luminance Value Status Register

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_lum_c	23:0	r	'h0	Luminance value of window C

ISP_LSC_CTRL

0x00002200 = Byte Address

Description: Lens Shade Control Register

Sub-Field Name	Bits	rw	Reset	Description
lsc_en	0:0	rw	'h0	0: activation request for lens shading correction 1: deactivation request for lens shading correction Activation/Deactivation is an object of a shadowing mechanism. The current status is visible at ISP_LSC_STATUS::lsc_enable_status

ISP_LSC_R_TABLE_ADDR

0x00002204 = Byte Address

Description: Table RAM Address for Red Component

Sub-Field Name	Bits	rw	Reset	Description
rambuf_r_addr_1	10:9	rw	'h0	Table address in RAM for samples of the R color component. Valid addresses are in the range 0 to 152. Will be automatically incremented by each read or write access to the table.
rambuf_r_addr	8:0	rw	'h0	Table address in RAM for samples of the R color component. Valid addresses are in the range 0 to 152. Will be automatically incremented by each read or write access to the table.

ISP_LSC_GR_TABLE_ADDR

0x00002208 = Byte Address

Description: Table RAM Address for Green (Red) Component

Sub-Field Name	Bits	rw	Reset	Description

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rambuf_gr_addr_1	10:9	rw	'h0	Table address in RAM for samples of the GR color component. Valid addresses are in the range 0 to 152. Will be automatically incremented by each read or write access to the table.
rambuf_gr_addr	8:0	rw	'h0	Table address in RAM for samples of the GR color component. Valid addresses are in the range 0 to 152. Will be automatically incremented by each read or write access to the table.

ISP_LSC_B_TABLE_ADDR

0x0000220c = Byte Address

Description: Table RAM Address for Blue Component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rambuf_b_addr_1	10:9	rw	'h0	Table address in RAM for samples of the B color component. Valid addresses are in the range 0 to 152. Will be automatically incremented by each read or write access to the table.
rambuf_b_addr	8:0	rw	'h0	Table address in RAM for samples of the B color component. Valid addresses are in the range 0 to 152. Will be automatically incremented by each read or write access to the table.

ISP_LSC_GB_TABLE_ADDR

0x00002210 = Byte Address

Description: Table RAM Address for Green (Blue) Component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rambuf_gb_addr_1	10:9	rw	'h0	Table address in RAM for samples of the GB color component. Valid addresses are in the range 0 to 152.Will be automatically incremented by each read or write access to the table.
rambuf_gb_addr	8:0	rw	'h0	Table address in RAM for samples of the GB color component. Valid addresses are in the range 0 to 152.Will be automatically incremented by each read or write access to the table.

ISP_LSC_R_SAMPLE_TABLE_DATA

0x00002214 = Byte Address

Description: Sample Table Red

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
r_sample_0_low	11:0	rw	'h0	Correction factor for R color component at 0 sample point. Fixed point number: 2 bits integer with 10-bit fractional part Valid range: 1..3.999
r_sample_1_low	23:12	rw	'h0	Correction factor for R color component at 1 sample point. Fixed point number: 2 bits integer with 10-bit fractional part Valid range: 1..3.999
r_sample_0_high	25:24	rw	'h0	Correction factor for R color component at 0 sample point. Two MSB interger for U4.10
r_sample_1_high	29:28	rw	'h0	Correction factor for R color component at 1 sample point. Two MSB interger for U4.10

ISP_LSC_GR_SAMPLE_TABLE_DATA

0x00002218 = Byte Address

Description: Sample Table Green (Red)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gr_sample_0_low	11:0	rw	'h0	Correction factor for Gr color component at 0 sample point. Fixed point number: 2 bits integer with 10-bit fractional part Valid range: 1..3.999
gr_sample_1_low	23:12	rw	'h0	Correction factor for Gr color component at 1 sample point. Fixed point number: 2 bits integer with 10-bit fractional part Valid range: 1..3.999
gr_sample_0_high	25:24	rw	'h0	Correction factor for Gr color component at 0 sample point. Two MSB interger for U4.10
gr_sample_1_high	29:28	rw	'h0	Correction factor for Gr color component at 1 sample point. Two MSB interger for U4.10

ISP_LSC_B_SAMPLE_TABLE_DATA

0x0000221c = Byte Address

Description: Sample Table Blue

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
b_sample_0_low	11:0	rw	'h0	Correction factor for B color component at 0 sample point. Fixed point number: 2 bits integer with 10-bit fractional part Valid range: 1..3.999

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b_sample_1_low	23:12	rw	'h0	Correction factor for B color component at 1 sample point. Fixed point number: 2 bits integer with 10-bit fractional part Valid range: 1..3.999
b_sample_0_high	25:24	rw	'h0	Correction factor for B color component at 0 sample point. Two MSB interger for U4.10
b_sample_1_high	29:28	rw	'h0	Correction factor for B color component at 1 sample point. Two MSB interger for U4.10

ISP_LSC_GB_SAMPLE_TABLE_DATA

0x00002220 = Byte Address

Description: Sample Table Green (Blue)

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gb_sample_0_low	11:0	rw	'h0	Correction factor for Gb color component at 0 sample point. Fixed point number: 2 bits integer with 10-bit fractional part Valid range: 1..3.999
gb_sample_1_low	23:12	rw	'h0	Correction factor for Gb color component at 1 sample point. Fixed point number: 2 bits integer with 10-bit fractional part Valid range: 1..3.999
gb_sample_0_high	25:24	rw	'h0	Correction factor for Gb color component at 0 sample point. Two MSB interger for U4.10
gb_sample_1_high	29:28	rw	'h0	Correction factor for Gb color component at 1 sample point. Two MSB interger for U4.10

ISP_LSC_XGRAD_01

0x00002224 = Byte Address

Description: Gradient Table X - Sectors 0 and 1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
xgrad_1	27:16	rw	'h0	Factor for x-gradient calculation of sector 1
xgrad_0	11:0	rw	'h0	Factor for x-gradient calculation of sector 0

ISP_LSC_XGRAD_23

0x00002228 = Byte Address

Description: Gradient Table X - Sectors 2 and 3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
xgrad_3	27:16	rw	'h0	Factor for x-gradient calculation of sector 3
xgrad_2	11:0	rw	'h0	Factor for x-gradient calculation of sector 2

ISP_LSC_XGRAD_45

0x0000222c = Byte Address

Description: Gradient Table X - Sectors 4 and 5

Sub-Field Name	Bits	rw	Reset	Description
xgrad_5	27:16	rw	'h0	Factor for x-gradient calculation of sector 5
xgrad_4	11:0	rw	'h0	Factor for x-gradient calculation of sector 4

ISP_LSC_XGRAD_67

0x00002230 = Byte Address

Description: Gradient Table X - Sectors 6 and 7

Sub-Field Name	Bits	rw	Reset	Description
xgrad_7	27:16	rw	'h0	Factor for x-gradient calculation of sector 7
xgrad_6	11:0	rw	'h0	Factor for x-gradient calculation of sector 6

ISP_LSC_YGRAD_01

0x00002234 = Byte Address

Description: Gradient Table Y - Sectors 0 and 1

Sub-Field Name	Bits	rw	Reset	Description
ygrad_1	27:16	rw	'h0	Factor for y-gradient calculation of sector 1
ygrad_0	11:0	rw	'h0	Factor for y-gradient calculation of sector 0

ISP_LSC_YGRAD_23

0x00002238 = Byte Address

Description: Gradient Table Y - Sectors 2 and 3

Sub-Field Name	Bits	rw	Reset	Description
ygrad_3	27:16	rw	'h0	Factor for y-gradient calculation of sector 3
ygrad_2	11:0	rw	'h0	Factor for y-gradient calculation of sector 2

ISP_LSC_YGRAD_45

0x0000223c = Byte Address

Description: Gradient Table Y - Sectors 4 and 5

Sub-Field Name	Bits	rw	Reset	Description
ygrad_5	27:16	rw	'h0	Factor for y-gradient calculation of sector 5
ygrad_4	11:0	rw	'h0	Factor for y-gradient calculation of sector 4

ISP_LSC_YGRAD_67

0x00002240 = Byte Address

Description: radient Table Y - Sectors 6 and 7

Sub-Field Name	Bits	rw	Reset	Description
ygrad_7	27:16	rw	'h0	Factor for y-gradient calculation of sector 7
ygrad_6	11:0	rw	'h0	Factor for y-gradient calculation of sector 6

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ISP_LSC_XSIZE_01

0x00002244 = Byte Address

Description: Size Table - Sectors 0 and 1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_1	25:16	rw	'h0	Sector size 1 in x-direction
x_size_0	9:0	rw	'h0	Sector size 0 in x-direction

ISP_LSC_XSIZE_23

0x00002248 = Byte Address

Description: Size Table - Sectors 2 and 3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_3	25:16	rw	'h0	Sector size 3 in x-direction
x_size_2	9:0	rw	'h0	Sector size 2 in x-direction

ISP_LSC_XSIZE_45

0x0000224c = Byte Address

Description: Size Table - Sectors 4 and 5

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_5	25:16	rw	'h0	Sector size 5 in x-direction
x_size_4	9:0	rw	'h0	Sector size 4 in x-direction

ISP_LSC_XSIZE_67

0x00002250 = Byte Address

Description: Size Table - Sectors 6 and 7

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_7	25:16	rw	'h0	Sector size 7 in x-direction
x_size_6	9:0	rw	'h0	Sector size 6 in x-direction

ISP_LSC_YSIZE_01

0x00002254 = Byte Address

Description: Size Table Y Direction - Sectors 0 and 1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
y_size_1	25:16	rw	'h0	Sector size 1 in y-direction
y_size_0	9:0	rw	'h0	Sector size 0 in y-direction

ISP_LSC_YSIZE_23

0x00002258 = Byte Address

Description: Size Table Y Direction - Sectors 2 and 3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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y_size_3	25:16	rw	'h0	Sector size 3 in y-direction
y_size_2	9:0	rw	'h0	Sector size 2 in y-direction

ISP_LSC_YSIZE_45

0x0000225c = Byte Address

Description: Size Table Y Direction - Sectors 4 and 5

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
y_size_5	25:16	rw	'h0	Sector size 5 in y-direction
y_size_4	9:0	rw	'h0	Sector size 4 in y-direction

ISP_LSC_YSIZE_67

0x00002260 = Byte Address

Description: Size Table Y Direction - Sectors 6 and 7

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
y_size_7	25:16	rw	'h0	Sector size 7 in y-direction
y_size_6	9:0	rw	'h0	Sector size 6 in y-direction

ISP_LSC_TABLE_SEL

0x00002264 = Byte Address

Description: Lens Shade Table Set Selection

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
table_sel	0:0	rw	'h0	0: next active table set is table set 0. 1: next active table set is table set 1. Table selection is object of a shadowing mechanism. The current status is visible at ISP_LSC_STATUS::active_table.

ISP_LSC_STATUS_SHD

0x00002268 = Byte Address

Description: Lens Shade Status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
table_sel_shd	1:1	r	'h0	Current active table set number 0: table set 0 1: table set 1
lsc_en_shd	0:0	r	'h0	Current lens shading correction state 0: off; 1: on

ISP_LSC_XGRAD_89

0x00002270 = Byte Address

Description: Gradient Table X - Sectors 8 and 9

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Sub-Field Name	Bits	rw	Reset	Description
xgrad_9	27:16	rw	'h0	Factor for x-gradient calculation of sector 9
xgrad_8	11:0	rw	'h0	Factor for x-gradient calculation of sector 8

ISP_LSC_XGRAD_1011

0x00002274 = Byte Address

Description: Gradient Table X - Sectors 10 and 11

Sub-Field Name	Bits	rw	Reset	Description
xgrad_11	27:16	rw	'h0	Factor for x-gradient calculation of sector 11
xgrad_10	11:0	rw	'h0	Factor for x-gradient calculation of sector 10

ISP_LSC_XGRAD_1213

0x00002278 = Byte Address

Description: Gradient Table X - Sectors 12 and 13

Sub-Field Name	Bits	rw	Reset	Description
xgrad_13	27:16	rw	'h0	Factor for x-gradient calculation of sector 13
xgrad_12	11:0	rw	'h0	Factor for x-gradient calculation of sector 12

ISP_LSC_XGRAD_1415

0x0000227c = Byte Address

Description: Gradient Table X - Sectors 14 and 15

Sub-Field Name	Bits	rw	Reset	Description
xgrad_15	27:16	rw	'h0	Factor for x-gradient calculation of sector 15
xgrad_14	11:0	rw	'h0	Factor for x-gradient calculation of sector 14

ISP_LSC_XGRAD_1617

0x00002280 = Byte Address

Description: Gradient Table X - Sectors 16 and 17

Sub-Field Name	Bits	rw	Reset	Description
xgrad_17	27:16	rw	'h0	Factor for x-gradient calculation of sector 17
xgrad_16	11:0	rw	'h0	Factor for x-gradient calculation of sector 16

ISP_LSC_XGRAD_1819

0x00002284 = Byte Address

Description: Gradient Table X - Sectors 18 and 19

Sub-Field Name	Bits	rw	Reset	Description
xgrad_19	27:16	rw	'h0	Factor for x-gradient calculation of sector 19
xgrad_18	11:0	rw	'h0	Factor for x-gradient calculation of sector 18

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ISP_LSC_XGRAD_2021

0x00002288 = Byte Address

Description: Gradient Table X - Sectors 20 and 21

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
xgrad_21	27:16	rw	'h0	Factor for x-gradient calculation of sector 21
xgrad_20	11:0	rw	'h0	Factor for x-gradient calculation of sector 20

ISP_LSC_XGRAD_2223

0x0000228c = Byte Address

Description: Gradient Table X - Sectors 22 and 23

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
xgrad_23	27:16	rw	'h0	Factor for x-gradient calculation of sector 23
xgrad_22	11:0	rw	'h0	Factor for x-gradient calculation of sector 22

ISP_LSC_XGRAD_2425

0x00002290 = Byte Address

Description: Gradient Table X - Sectors 24 and 25

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
xgrad_25	27:16	rw	'h0	Factor for x-gradient calculation of sector 25
xgrad_24	11:0	rw	'h0	Factor for x-gradient calculation of sector 24

ISP_LSC_XGRAD_2627

0x00002294 = Byte Address

Description: Gradient Table X - Sectors 26 and 27

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
xgrad_27	27:16	rw	'h0	Factor for x-gradient calculation of sector 27
xgrad_26	11:0	rw	'h0	Factor for x-gradient calculation of sector 26

ISP_LSC_XGRAD_2829

0x00002298 = Byte Address

Description: Gradient Table X - Sectors 28 and 29

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
xgrad_29	27:16	rw	'h0	Factor for x-gradient calculation of sector 29
xgrad_28	11:0	rw	'h0	Factor for x-gradient calculation of sector 28

ISP_LSC_XGRAD_3031

0x0000229c = Byte Address

Description: Gradient Table X - Sectors 30 and 31

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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xgrad_31	27:16	rw	'h0	Factor for x-gradient calculation of sector 31
xgrad_30	11:0	rw	'h0	Factor for x-gradient calculation of sector 30

ISP_LSC_XSIZE_89

0x000022a0 = Byte Address

Description: Size Table - Sectors 8 and 9

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_9	25:16	rw	'h0	Sector size 9 in x-direction
x_size_8	9:0	rw	'h0	Sector size 8 in x-direction

ISP_LSC_XSIZE_1011

0x000022a4 = Byte Address

Description: Size Table - Sectors 10 and 11

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_11	25:16	rw	'h0	Sector size 11 in x-direction
x_size_10	9:0	rw	'h0	Sector size 10 in x-direction

ISP_LSC_XSIZE_1213

0x000022a8 = Byte Address

Description: Size Table - Sectors 12 and 13

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_13	25:16	rw	'h0	Sector size 13 in x-direction
x_size_12	9:0	rw	'h0	Sector size 12 in x-direction

ISP_LSC_XSIZE_1415

0x000022ac = Byte Address

Description: Size Table - Sectors 14 and 15

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_15	25:16	rw	'h0	Sector size 15 in x-direction
x_size_14	9:0	rw	'h0	Sector size 14 in x-direction

ISP_LSC_XSIZE_1617

0x000022b0 = Byte Address

Description: Size Table - Sectors 16 and 17

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_17	25:16	rw	'h0	Sector size 17 in x-direction
x_size_16	9:0	rw	'h0	Sector size 16 in x-direction

ISP_LSC_XSIZE_1819

0x000022b4 = Byte Address

Description: Size Table - Sectors 18 and 19

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_19	25:16	rw	'h0	Sector size 19 in x-direction
x_size_18	9:0	rw	'h0	Sector size 18 in x-direction

ISP_LSC_XSIZE_2021

0x000022b8 = Byte Address

Description: Size Table - Sectors 20 and 21

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_21	25:16	rw	'h0	Sector size 21 in x-direction
x_size_20	9:0	rw	'h0	Sector size 20 in x-direction

ISP_LSC_XSIZE_2223

0x000022bc = Byte Address

Description: Size Table - Sectors 22 and 23

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_23	25:16	rw	'h0	Sector size 23 in x-direction
x_size_22	9:0	rw	'h0	Sector size 22 in x-direction

ISP_LSC_XSIZE_2425

0x000022c0 = Byte Address

Description: Size Table - Sectors 24 and 25

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_25	25:16	rw	'h0	Sector size 25 in x-direction
x_size_24	9:0	rw	'h0	Sector size 24 in x-direction

ISP_LSC_XSIZE_2627

0x000022c4 = Byte Address

Description: Size Table - Sectors 26 and 27

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_27	25:16	rw	'h0	Sector size 27 in x-direction
x_size_26	9:0	rw	'h0	Sector size 26 in x-direction

ISP_LSC_XSIZE_2829

0x000022c8 = Byte Address

Description: Size Table - Sectors 28 and 29

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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x_size_29	25:16	rw	'h0	Sector size 29 in x-direction
x_size_28	9:0	rw	'h0	Sector size 28 in x-direction

ISP_LSC_XSIZE_3031

0x000022cc = Byte Address

Description: Size Table - Sectors 30 and 31

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
x_size_31	25:16	rw	'h0	Sector size 31 in x-direction
x_size_30	9:0	rw	'h0	Sector size 30 in x-direction

ISP_LSC_YGRAD_89

0x000022d0 = Byte Address

Description: Gradient Table Y - Sectors 8 and 9

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ygrad_9	27:16	rw	'h0	Factor for y-gradient calculation of sector 9
ygrad_8	11:0	rw	'h0	Factor for y-gradient calculation of sector 8

ISP_LSC_YGRAD_1011

0x000022d4 = Byte Address

Description: Gradient Table Y - Sectors 10 and 11

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ygrad_11	27:16	rw	'h0	Factor for y-gradient calculation of sector 11
ygrad_10	11:0	rw	'h0	Factor for y-gradient calculation of sector 10

ISP_LSC_YGRAD_1213

0x000022d8 = Byte Address

Description: Gradient Table Y - Sectors 12 and 13

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ygrad_13	27:16	rw	'h0	Factor for y-gradient calculation of sector 13
ygrad_12	11:0	rw	'h0	Factor for y-gradient calculation of sector 12

ISP_LSC_YGRAD_1415

0x000022dc = Byte Address

Description: Gradient Table Y - Sectors 14 and 15

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ygrad_15	27:16	rw	'h0	Factor for y-gradient calculation of sector 15
ygrad_14	11:0	rw	'h0	Factor for y-gradient calculation of sector 14

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ISP_LSC_YSIZE_89

0x000022e0 = Byte Address

Description: size table Y direction - Sectors 8 and 9

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
y_size_9	25:16	rw	'h0	Sector size 9 in y-direction
y_size_8	9:0	rw	'h0	Sector size 8 in y-direction

ISP_LSC_YSIZE_1011

0x000022e4 = Byte Address

Description: size table Y direction - Sectors 10 and 11

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
y_size_11	25:16	rw	'h0	Sector size 11 in y-direction
y_size_10	9:0	rw	'h0	Sector size 10 in y-direction

ISP_LSC_YSIZE_1213

0x000022e8 = Byte Address

Description: size table Y direction - Sectors 12 and 13

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
y_size_13	25:16	rw	'h0	Sector size 13 in y-direction
y_size_12	9:0	rw	'h0	Sector size 12 in y-direction

ISP_LSC_YSIZE_1415

0x000022ec = Byte Address

Description: size table Y direction - Sectors 14 and 15

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
y_size_15	25:16	rw	'h0	Sector size 15 in y-direction
y_size_14	9:0	rw	'h0	Sector size 14 in y-direction

ISP_IS_CTRL

0x00002300 = Byte Address

Description: Image Stabilization Control Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
is_en	0:0	rw	'h0	Image stabilization state 0: off ; 1: on

ISP_IS_RECENTER

0x00002304 = Byte Address

Description: Image Stabilization Re-center Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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is_recenter	2:0	rw	'h0	0: re-center feature switched off 1..7: re-centering by (cur_h/v_offs-h/v_offs)/2^recenter

ISP_IS_H_OFFS

0x00002308 = Byte Address

Description: Horizontal Offset of Output Window

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
is_h_offs	13:0	rw	'h0	Horizontal picture offset in pixels

ISP_IS_V_OFFS

0x0000230c = Byte Address

Description: Vertical Offset of Output Window

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
is_v_offs	13:0	rw	'h0	Vertical picture offset in lines

ISP_IS_H_SIZE

0x00002310 = Byte Address

Description: Horizontal Output Picture Size

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
is_h_size	13:0	rw	'h1000	Horizontal picture size in pixels if ISP_MODE is set to: 001: (ITU-R BT.656 YUV) 010: (ITU-R BT.601 YUV) 011: (ITU-R BT.601 Bayer RGB) 101: (ITU-R BT.656 Bayer RGB) Only even numbers are accepted, because complete quadruples of YUYV (YCbYCr) are needed for the following modules. If an odd size is programmed, the value will be truncated to even size.

ISP_IS_V_SIZE

0x00002314 = Byte Address

Description: Vertical Output Picture Size

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
is_v_size	13:0	rw	'hc00	Vertical picture size in lines

ISP_IS_MAX_DX

0x00002318 = Byte Address

Description: Maximum Horizontal Displacement

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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is_max_dx	12:0	rw	'h0	Maximum allowed accumulated horizontal displacement in pixels

ISP_IS_MAX_DY

0x0000231c = Byte Address

Description: Maximum Vertical Displacement

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
is_max_dy	12:0	rw	'h0	Maximum allowed accumulated vertical displacement in lines

ISP_IS_DISPLACE

0x00002320 = Byte Address

Description: Camera Displacement

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
is_dy	28:16	rw	'h0	ISP_IS will compensate for vertical camera displacement of DY lines in the next frame
is_dx	12:0	rw	'h0	ISP_IS will compensate for horizontal camera displacement of DX pixels in the next frame

ISP_IS_H_OFFS_SHD

0x00002324 = Byte Address

Description: Current Horizontal Offset of Output Window - Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
is_h_offs_shd	13:0	r	'h0	Current horizontal picture offset in lines

ISP_IS_V_OFFS_SHD

0x00002328 = Byte Address

Description: Current Vertical Offset of Output Window - Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
is_v_offs_shd	13:0	r	'h0	Current vertical picture offset in lines

ISP_IS_H_SIZE_SHD

0x0000232c = Byte Address

Description: Current Output Horizontal Picture Size - Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
is_h_size_shd	13:0	r	'h1000	Current horizontal picture size in pixels

ISP_IS_V_SIZE_SHD

0x00002330 = Byte Address

Description: Current Output Vertical Picture Size - Shadow Register

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Sub-Field Name	Bits	rw	Reset	Description
is_v_size_shd	13:0	r	'hc00	Current vertical picture size in pixels

ISP_EXPV2_CTRL

0x000026a0 = Byte Address

Description: raw exposure control

Sub-Field Name	Bits	rw	Reset	Description
expv2_input_select	3:2	rw	'h0	00: degamma output 01: awb_gain output 10: WDR3 output 11: lsc
expv2_autostop	1:1	rw	'h0	0: continous measurement 1: stop measuring after a complete frame
expv2_enable	0:0	rw	'h0	1: start measuring a frame The exp block will reset this bit and halt after completing one frame if bit "autostop" is set to 1

ISP_EXPV2_OFFSET

0x000026a4 = Byte Address

Description: offset for X and Y direction

Sub-Field Name	Bits	rw	Reset	Description
expv2_v_offset	28:16	rw	'h0	Vertical offset of first block in pixels
expv2_h_offset	12:0	rw	'h0	Horizontal offset of first block in pixels

ISP_EXPV2_SIZE

0x000026a8 = Byte Address

Description: xsize and ysize for each block

Sub-Field Name	Bits	rw	Reset	Description
expv2_v_size	24:16	rw	'h0	Vertical size in pixels of one block; should be multiple of 2
expv2_h_size	8:0	rw	'h0	Horizontal size in pixels of one block; should be multiple of 2

ISP_EXPV2_SIZE_INVERT

0x000026ac = Byte Address

Description: invert value

Sub-Field Name	Bits	rw	Reset	Description
expv2_v_size_invert	31:16	rw	'h0	value int[2^16/vsize+1/2]
expv2_h_size_invert	15:0	rw	'h0	value int[2^16/hsize+1/2]

ISP_EXPV2_OFFSET_SHD

0x000026b4 = Byte Address

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Description: offset for X and Y direction-shadow

Sub-Field Name	Bits	rw	Reset	Description
expv2_v_offset	28:16	r	'h0	Shadow vertical offset of first block in pixels
expv2_h_offset	12:0	r	'h0	Shadow Horizontal offset of first block in pixels

ISP_EXPV2_SIZE_SHD

0x000026b8 = Byte Address

Description: xsize and ysize for each block-shadow

Sub-Field Name	Bits	rw	Reset	Description
expv2_v_size	24:16	r	'h0	Shadow vertical size in pixels of one block; should be multiple of 2
expv2_h_size	8:0	r	'h0	Shadow Horizontal size in pixels of one block; should be multiple of 2

ISP_EXPV2_SIZE_INVERT_SHD

0x000026bc = Byte Address

Description: invert value -shadow register

Sub-Field Name	Bits	rw	Reset	Description
expv2_v_size_invert	31:16	r	'h0	Shadow value int[2^16/vsize+1/2]
expv2_h_size_invert	15:0	r	'h0	Shadow value int[2^16/hsize+1/2]

ISP_DPCC_MODE

0x00002900 = Byte Address

Description: Global control register

Sub-Field Name	Bits	rw	Reset	Description
stage1_enable	2:2	rw	'h1	0: bypass stage1 1: enable stage1 (Default)
grayscale_mode	1:1	rw	'h0	0: BAYER DATA INPUT (Default) 1: enable gray scale data input from black and white sensors (without color filter array)
dpcc_enable	0:0	rw	'h0	0: bypass DPCC (Default) 1: enable DPCC

ISP_DPCC_OUT_MODE

0x00002904 = Byte Address

Description: Interpolation Mode for Correction Unit

Sub-Field Name	Bits	rw	Reset	Description
stage1_rb_3x3	3:3	rw	'h0	0: Stage1 red/blue 4 or 5 pixel output median (Default) 1: Stage1 red/blue 9 pixel (3x3) output median

stage1_g_3x3	2:2	rw	'h0	0: Stage1 green 4 or 5 pixel output median (Default) 1: Stage1 green 9 pixel (3x3) output median
stage1_incl_rb_center	1:1	rw	'h0	0: Stage1 does not include center pixel for red/blue output median 2x2 1: Stage1 include center pixel for red/blue output median 2x2+1 (Default)
stage1_incl_gr_center	0:0	rw	'h0	0: Stage1 does not include center pixel for green output median 2x2 1: Stage1 include center pixel for green output median 2x2+1 (Default)

ISP_DPCC_SET_USE

0x00002908 = Byte Address

Description: DPCC Methods Set Usage for Detection

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stage1_use_fix_set	3:3	rw	'h0	0: Stage1 does not use hard coded methods set 1: Stage1 use hard coded methods set (Default)
stage1_use_set_3	2:2	rw	'h0	0: Stage1 does not use methods set 3 (Default) 1: Stage1 use methods set 3
stage1_use_set_2	1:1	rw	'h0	0: Stage1 does not use methods set 2 (Default) 1: Stage1 use methods set 2
stage1_use_set_1	0:0	rw	'h1	0: Stage1 does not use methods set 1 1: Stage1 use methods set 1 (Default)

ISP_DPCC_METHODS_SET1

0x0000290c = Byte Address

Description: DPCC Methods Enable Bits for SET_1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rg_red_blue1_enable	12:12	rw	'h1	0: Bypass Rank Gradient check for green 1: Enable Rank Gradient check for green (Default)
rnd_red_blue1_enable	11:11	rw	'h1	0: Bypass Rank Neighbor Difference check for green 1: Enable Rank Neighbor Difference check for green (Default)
ro_red_blue1_enable	10:10	rw	'h1	0: Bypass Rank Order check for green 1: Enable Rank Order check for green (Default)
lc_red_blue1_enable	9:9	rw	'h0	0: Bypass Line check for green 1: Enable Line check for green (Default)

pg_red_blue1_enable	8:8	rw	'h1	0: Bypass Peak Gradient check for red_blue 1: Enable Peak Gradient check for red_blue (Default)
rg_green1_enable	4:4	rw	'h1	0: Bypass Rank Gradient check for green 1: Enable Rank Gradient check for green (Default)
rnd_green1_enable	3:3	rw	'h1	0: Bypass Rank Neighbor Difference check for green 1: Enable Rank Neighbor Difference check for green (Default)
ro_green1_enable	2:2	rw	'h1	0: Bypass Rank Order check for green 1: Enable Rank Order check for green (Default)
lc_green1_enable	1:1	rw	'h0	0: Bypass Line check for green 1: Enable Line check for green (Default)
pg_green1_enable	0:0	rw	'h1	0: Bypass Peak Gradient check for green 1: Enable Peak Gradient check for green (Default)

ISP_DPCC_METHODS_SET2

0x00002910 = Byte Address

Description: DPCC Methods Enable Bits for SET_2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rg_red_blue2_enable	12:12	rw	'h0	0: Bypass Rank Gradient check for green 1: Enable Rank Gradient check for green (Default)
rnd_red_blue2_enable	11:11	rw	'h0	0: Bypass Rank Neighbor Difference check for green 1: Enable Rank Neighbor Difference check for green (Default)
ro_red_blue2_enable	10:10	rw	'h0	0: Bypass Rank Order check for green 1: Enable Rank Order check for green (Default)
lc_red_blue2_enable	9:9	rw	'h0	0: Bypass Line check for green 1: Enable Line check for green (Default)
pg_red_blue2_enable	8:8	rw	'h0	0: Bypass Peak Gradient check for red_blue 1: Enable Peak Gradient check for red_blue (Default)
rg_green2_enable	4:4	rw	'h0	0: Bypass Rank Gradient check for green 1: Enable Rank Gradient check for green (Default)
rnd_green2_enable	3:3	rw	'h0	0: Bypass Rank Neighbor Difference check for green 1: Enable Rank Neighbor Difference check for green (Default)
ro_green2_enable	2:2	rw	'h0	0: Bypass Rank Order check for green 1: Enable Rank Order check for green (Default)

lc_green2_enable	1:1	rw	'h0	0: Bypass Line check for green 1: Enable Line check for green (Default)
pg_green2_enable	0:0	rw	'h0	0: Bypass Peak Gradient check for green 1: Enable Peak Gradient check for green (Default)

ISP_DPCC_METHODS_SET3

0x00002914 = Byte Address

Description: DPCC Methods Enable Bits for SET_3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rg_red_blue3_enable	12:12	rw	'h0	0: Bypass Rank Gradient check for green 1: Enable Rank Gradient check for green (Default)
rnd_red_blue3_enable	11:11	rw	'h0	0: Bypass Rank Neighbor Difference check for green 1: Enable Rank Neighbor Difference check for green (Default)
ro_red_blue3_enable	10:10	rw	'h0	0: Bypass Rank Order check for green 1: Enable Rank Order check for green (Default)
lc_red_blue3_enable	9:9	rw	'h0	0: Bypass Line check for green 1: Enable Line check for green (Default)
pg_red_blue3_enable	8:8	rw	'h0	0: Bypass Peak Gradient check for red_blue 1: Enable Peak Gradient check for red_blue (Default)
rg_green3_enable	4:4	rw	'h0	0: Bypass Rank Gradient check for green 1: Enable Rank Gradient check for green (Default)
rnd_green3_enable	3:3	rw	'h0	0: Bypass Rank Neighbor Difference check for green 1: Enable Rank Neighbor Difference check for green (Default)
ro_green3_enable	2:2	rw	'h0	0: Bypass Rank Order check for green 1: Enable Rank Order check for green (Default)
lc_green3_enable	1:1	rw	'h0	0: Bypass Line check for green 1: Enable Line check for green (Default)
pg_green3_enable	0:0	rw	'h0	0: Bypass Peak Gradient check for green 1: Enable Peak Gradient check for green (Default)

ISP_DPCC_LINE_THRESH_1

0x00002918 = Byte Address

Description: Line Threshold for SET_1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
line_thr1_rb	15:8	rw	'h0	Line threshold for set 1 red/blue
line_thr1_g	7:0	rw	'h0	Line threshold for set 1 green

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ISP_DPCC_LINE_MAD_FAC_1

0x0000291c = Byte Address

Description: Mean Absolute Difference (MAD) Factor Line Check Set 1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
line_mad_fac_1_rb	13:8	rw	'h0	Line MAD factor for set 1 red/blue
line_mad_fac_1_g	5:0	rw	'h0	Line MAD factor for set 1 green

ISP_DPCC_PG_FAC_1

0x00002920 = Byte Address

Description: Peak Gradient Factor for Set 1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pg_fac_1_rb	13:8	rw	'h0	Peak gradient factor for set 1 red/blue
pg_fac_1_g	5:0	rw	'h0	Peak gradient factor for set 1 green

ISP_DPCC_RND_THRESH_1

0x00002924 = Byte Address

Description: Rank Neighbor Difference Threshold for Set 1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rnd_thr1_rb	15:8	rw	'h0	Rank Neighbor Difference threshold for set 1 red/blue
rnd_thr1_g	7:0	rw	'h0	Rank Neighbor Difference threshold for set 1 green

ISP_DPCC_RG_FAC_1

0x00002928 = Byte Address

Description: Rank Gradient Factor for Set 1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rg_fac_1_rb	13:8	rw	'h0	Rank gradient factor for set 1 red/blue
rg_fac_1_g	5:0	rw	'h0	Rank gradient factor for set 1 red/blue

ISP_DPCC_LINE_THRESH_2

0x0000292c = Byte Address

Description: Line Threshold for SET_2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
line_thr2_rb	15:8	rw	'h0	Line threshold for set 2 red/blue
line_thr2_g	7:0	rw	'h0	Line threshold for set 2 green

ISP_DPCC_LINE_MAD_FAC_2

0x00002930 = Byte Address

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Description: Mean Absolute Difference (MAD) Factor Line Check Set 2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
line_mad_fac_2_rb	13:8	rw	'h0	Line MAD factor for set 2 red/blue
line_mad_fac_2_g	5:0	rw	'h0	Line MAD factor for set 2 green

ISP_DPCC_PG_FAC_2

0x00002934 = Byte Address

Description: Peak Gradient Factor for Set 2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pg_fac_2_rb	13:8	rw	'h0	Peak gradient factor for set 2 red/blue
pg_fac_2_g	5:0	rw	'h0	Peak gradient factor for set 1 green

ISP_DPCC_RND_THRESH_2

0x00002938 = Byte Address

Description: Rank Neighbor Difference Threshold for Set 2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rnd_thr2_rb	15:8	rw	'h0	Rank Neighbor Difference threshold for set 2 red/blue
rnd_thr2_g	7:0	rw	'h0	Rank Neighbor Difference threshold for set 2 green

ISP_DPCC_RG_FAC_2

0x0000293c = Byte Address

Description: Rank Gradient Factor for Set 2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rg_fac_2_rb	13:8	rw	'h0	Rank gradient factor for set 2 red/blue
rg_fac_2_g	5:0	rw	'h0	Rank gradient factor for set 2 red/blue

ISP_DPCC_LINE_THRESH_3

0x00002940 = Byte Address

Description: Line Threshold for SET_3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
line_thr3_rb	15:8	rw	'h0	Line threshold for set 2 red/blue
line_thr3_g	7:0	rw	'h0	Line threshold for set 3 green

ISP_DPCC_LINE_MAD_FAC_3

0x00002944 = Byte Address

Description: Mean Absolute Difference (MAD) Factor Line Check Set 3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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line_mad_fac_3_rb	13:8	rw	'h0	Line MAD factor for set 3 red/blue
line_mad_fac_3_g	5:0	rw	'h0	Line MAD factor for set 3 green

ISP_DPCC_PG_FAC_3

0x00002948 = Byte Address

Description: Peak Gradient Factor for Set 3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pg_fac_3_rb	13:8	rw	'h0	Peak gradient factor for set 3 red/blue
pg_fac_3_g	5:0	rw	'h0	Peak gradient factor for set 3 green

ISP_DPCC_RND_THRESH_3

0x0000294c = Byte Address

Description: Rank Neighbor Difference Threshold for Set 3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rnd_thr3_rb	15:8	rw	'h0	Rank Neighbor Difference threshold for set 3 red/blue
rnd_thr3_g	7:0	rw	'h0	Rank Neighbor Difference threshold for set 3 green

ISP_DPCC_RG_FAC_3

0x00002950 = Byte Address

Description: Rank Gradient Factor for Set 3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rg_fac_3_rb	13:8	rw	'h0	Rank gradient factor for set 3 red/blue
rg_fac_3_g	5:0	rw	'h0	Rank gradient factor for set 3 red/blue

ISP_DPCC_RO_LIMITS

0x00002954 = Byte Address

Description: Rank Order Limits

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ro_3_rb	11:10	rw	'h0	Rank order limit for set 3 red/blue
ro_3_g	9:8	rw	'h0	Rank order limit for set 3 green
ro_2_rb	7:6	rw	'h0	Rank order limit for set 2 red/blue
ro_2_g	5:4	rw	'h0	Rank order limit for set 2 green
ro_1_rb	3:2	rw	'h0	Rank order limit for set 1 red/blue
ro_1_g	1:0	rw	'h0	Rank order limit for set 1 green

ISP_DPCC_RND_OFFS

0x00002958 = Byte Address

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Description: Differential Rank Offsets for Rank Neighbor Difference

Sub-Field Name	Bits	rw	Reset	Description
rnd_offs_3_rb	11:10	rw	'h0	Rank Offset to Neighbor for set 3 red/blue
rnd_offs_3_g	9:8	rw	'h0	Rank Offset to Neighbor for set 3 green
rnd_offs_2_rb	7:6	rw	'h0	Rank Offset to Neighbor for set 2 red/blue
rnd_offs_2_g	5:4	rw	'h0	Rank Offset to Neighbor for set 2 green
rnd_offs_1_rb	3:2	rw	'h0	Rank Offset to Neighbor for set 1 red/blue
rnd_offs_1_g	1:0	rw	'h0	Rank Offset to Neighbor for set 1 green

ISP_DPCC_BPT_CTRL

0x0000295c = Byte Address

Description: Bad Pixel Table Settings

Sub-Field Name	Bits	rw	Reset	Description
bpt_rb_3x3	11:11	rw	'h0	0: if BPT active red/blue 4 or 5 pixel output median (Default) 1: if BPT active red/blue 9 pixel (3x3) output median
bpt_g_3x3	10:10	rw	'h0	0: if BPT active green 4 or 5 pixel output median (Default) 1: if BPT active green 9 pixel (3x3) output median
bpt_incl_rb_center	9:9	rw	'h0	0: if BPT active do not include center pixel for red/blue output median 2x2 (Default) 1: if BPT active include center pixel for red/blue output median 2x2+1
bpt_incl_gr_center	8:8	rw	'h0	0: if BPT active do not include center pixel for green output median 2x2 (Default) 1: if BPT active include center pixel for green output median 2x2+1
bpt_use_fix_set	7:7	rw	'h0	0: for BPT write do not use hard coded methods set (Default) 1: for BPT write use hard coded methods set
bpt_use_set_3	6:6	rw	'h0	0: for BPT write do not use methods set 3 (Default) 1: for BPT write use methods set 3
bpt_use_set_2	5:5	rw	'h0	0: for BPT write do not use methods set 2 (Default) 1: for BPT write use methods set 2
bpt_use_set_1	4:4	rw	'h0	0: for BPT write do not use methods set 1 (Default) 1: for BPT write use methods set 1
bpt_cor_en	1:1	rw	'h0	Table based correction 0: Disabled; 1: Enabled
bpt_det_en	0:0	rw	'h0	Bad pixel detection write to memory 0: Disabled; 1: Enabled

ISP_DPCC_BPT_NUM

0x00002960 = Byte Address

Description: Number of Entries for Bad Pixel Table

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
bp_number	11:0	rw	'h0	Number of current Bad Pixel entries in bad pixel table (BPT)

ISP_DPCC_BPT_ADDR

0x00002964 = Byte Address

Description: TABLE Start Address for Table-Based Correction Algorithm

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
bp_table_addr	10:0	rw	'h0	Table RAM start address for read or write operations. The address counter is incremented at each read or write access to the data register (auto-increment mechanism).

ISP_DPCC_BPT_DATA

0x00002968 = Byte Address

Description: TABLE DATA Register for Read and Write Access of Table RAM

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
bpt_v_addr	27:16	rw	'h0	Bad Pixel vertical address (pixel position)
bpt_h_addr	12:0	rw	'h0	Bad Pixel horizontal address (pixel position)

ISP_COMPAND_CTRL

0x00003200 = Byte Address

Description: Compand Control Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
compand_ctrl_bls_enable	3:3	rw	'h0	Enable Black Level Subtraction in the Compand module
compand_ctrl_soft_reset_flag	2:2	rw	'h0	Soft reset of compand module
compand_ctrl_expand_enable	0:0	rw	'h0	Enable Expand in the Compand module

ISP_COMPAND_BLS_A_FIXED

0x00003204 = Byte Address

Description: BLS A value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
compand_bls_a_fixed	20:0	rw	'h0	BLS value

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ISP_COMPAND_BLS_B_FIXED

0x00003208 = Byte Address

Description: BLS B value

Sub-Field Name	Bits	rw	Reset	Description
compand_bls_b_fixed	20:0	rw	'h0	BLS value

ISP_COMPAND_BLS_C_FIXED

0x0000320c = Byte Address

Description: BLS C value

Sub-Field Name	Bits	rw	Reset	Description
compand_bls_c_fixed	20:0	rw	'h0	BLS value

ISP_COMPAND_BLS_D_FIXED

0x00003210 = Byte Address

Description: BLS D value

Sub-Field Name	Bits	rw	Reset	Description
compand_bls_d_fixed	20:0	rw	'h0	BLS value

ISP_COMPAND_EXPAND_PX_0

0x00003214 = Byte Address

Description: Distance of x0~x5 in expand curve

Sub-Field Name	Bits	rw	Reset	Description
compand_expand_px_5	29:25	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_4	24:20	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_3	19:15	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_2	14:10	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_1	9:5	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_0	4:0	rw	'h0	Distance of x in expand curve (distance_n=1<

ISP_COMPAND_EXPAND_PX_1

0x00003218 = Byte Address

Description: Distance of x6~x11 in expand curve

Sub-Field Name	Bits	rw	Reset	Description
compand_expand_px_11	29:25	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_10	24:20	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_9	19:15	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_8	14:10	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_7	9:5	rw	'h0	Distance of x in expand curve (distance_n=1<

compand_expand_px_6	4:0	rw	'h0	Distance of x in expand curve (distance_n=1<

ISP_COMPAND_EXPAND_PX_2

0x0000321c = Byte Address

Description: Distance of x12~x17 in expand curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
compand_expand_px_17	29:25	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_16	24:20	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_15	19:15	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_14	14:10	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_13	9:5	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_12	4:0	rw	'h0	Distance of x in expand curve (distance_n=1<

ISP_COMPAND_EXPAND_PX_3

0x00003220 = Byte Address

Description: Distance of x18~x23 in expand curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
compand_expand_px_23	29:25	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_22	24:20	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_21	19:15	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_20	14:10	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_19	9:5	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_18	4:0	rw	'h0	Distance of x in expand curve (distance_n=1<

ISP_COMPAND_EXPAND_PX_4

0x00003224 = Byte Address

Description: Distance of x24~x29 in expand curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
compand_expand_px_29	29:25	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_28	24:20	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_27	19:15	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_26	14:10	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_25	9:5	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_24	4:0	rw	'h0	Distance of x in expand curve (distance_n=1<

ISP_COMPAND_EXPAND_PX_5

0x00003228 = Byte Address

Description: Distance of x30~x35 in expand curve

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Sub-Field Name	Bits	rw	Reset	Description
compand_expand_px_35	29:25	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_34	24:20	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_33	19:15	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_32	14:10	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_31	9:5	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_30	4:0	rw	'h0	Distance of x in expand curve (distance_n=1<

ISP_COMPAND_EXPAND_PX_6

0x0000322c = Byte Address

Description: Distance of x36~x41 in expand curve

Sub-Field Name	Bits	rw	Reset	Description
compand_expand_px_41	29:25	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_40	24:20	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_39	19:15	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_38	14:10	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_37	9:5	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_36	4:0	rw	'h0	Distance of x in expand curve (distance_n=1<

ISP_COMPAND_EXPAND_PX_7

0x00003230 = Byte Address

Description: Distance of x42~x47 in expand curve

Sub-Field Name	Bits	rw	Reset	Description
compand_expand_px_47	29:25	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_46	24:20	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_45	19:15	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_44	14:10	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_43	9:5	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_42	4:0	rw	'h0	Distance of x in expand curve (distance_n=1<

ISP_COMPAND_EXPAND_PX_8

0x00003234 = Byte Address

Description: Distance of x48~x53 in expand curve

Sub-Field Name	Bits	rw	Reset	Description
compand_expand_px_53	29:25	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_52	24:20	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_51	19:15	rw	'h0	Distance of x in expand curve (distance_n=1<

compand_expand_px_50	14:10	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_49	9:5	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_48	4:0	rw	'h0	Distance of x in expand curve (distance_n=1<

ISP_COMPAND_EXPAND_PX_9

0x00003238 = Byte Address

Description: Distance of x54~x59 in expand curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
compand_expand_px_59	29:25	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_58	24:20	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_57	19:15	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_56	14:10	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_55	9:5	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_54	4:0	rw	'h0	Distance of x in expand curve (distance_n=1<

ISP_COMPAND_EXPAND_PX_10

0x0000323c = Byte Address

Description: Distance of x60~x64 in expand curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
compand_expand_px_63	19:15	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_62	14:10	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_61	9:5	rw	'h0	Distance of x in expand curve (distance_n=1<
compand_expand_px_60	4:0	rw	'h0	Distance of x in expand curve (distance_n=1<

ISP_COMPAND_EXPAND_Y_ADDR

0x0000326c = Byte Address

Description: Address of y memory in expand curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
compand_expand_y_addr	31:0	rw	'h0	address of y memory in expand curve

ISP_COMPAND_EXPAND_Y_WRITE_DATA

0x00003270 = Byte Address

Description: Y data in expand curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
compand_expand_y_write_data	31:0	rw	'h0	y data in expand curve

ISP_COMPAND_EXPAND_X_ADDR

0x0000327c = Byte Address

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Description: Address of x memory in expand curve

Sub-Field Name	Bits	rw	Reset	Description
compand_expand_x_addr	31:0	rw	'h0	address of x memory in expand curve

ISP_COMPAND_EXPAND_X_WRITE_DATA

0x00003280 = Byte Address

Description: X data in expand curve

Sub-Field Name	Bits	rw	Reset	Description
compand_expand_x_write_data	31:0	rw	'h0	x data in expand curve

ISP_COMPAND_CTRL_SHD

0x0000328c = Byte Address

Description: Compand Control Register Shadow

Sub-Field Name	Bits	rw	Reset	Description
compand_ctrl_bls_enable_shd	3:3	r	'h0	Enable Black Level Subtraction in the Compand module
compand_ctrl_expand_enable_shd	0:0	r	'h0	Enable Expand in the Compand module

ISP_COMPAND_BLS_A_FIXED_SHD

0x00003290 = Byte Address

Description: BLS A value Shadow

Sub-Field Name	Bits	rw	Reset	Description
compand_bls_a_fixed_shd	20:0	r	'h0	BLS value

ISP_COMPAND_BLS_B_FIXED_SHD

0x00003294 = Byte Address

Description: BLS B value Shadow

Sub-Field Name	Bits	rw	Reset	Description
compand_bls_b_fixed_shd	20:0	r	'h0	BLS value

ISP_COMPAND_BLS_C_FIXED_SHD

0x00003298 = Byte Address

Description: BLS C value Shadow

Sub-Field Name	Bits	rw	Reset	Description
compand_bls_c_fixed_shd	20:0	r	'h0	BLS value

ISP_COMPAND_BLS_D_FIXED_SHD

0x0000329c = Byte Address

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Description: BLS D value Shadow

Sub-Field Name	Bits	rw	Reset	Description
compand_bls_d_fixed_shd	20:0	r	'h0	BLS value

ISP_STITCHING0_CTRL

0x00003300 = Byte Address

Description: HDR Stitch Control Register

Sub-Field Name	Bits	rw	Reset	Description
deghost_en_lsvs	29:29	rw	'h0	Enable/Disable deghost between ls frame and very short frame 0: Disable the deghost 1: Enable the deghost
deghost_en_ls	28:28	rw	'h0	Enable/Disable deghost between long frame and short frame 0: Disable the deghost 1: Enable the deghost
mode3_sft_en	27:27	rw	'h0	Enable/Disable mode3 data shift 0: Disable mode3 data shift 1: Enable mode3 data shift
share_en	26:26	rw	'h0	Enable/Disable share of sram 0: Disable share sram 1: Enable share sram
digital_gain_en_2	25:25	rw	'h0	Enable/Disable dgain of very short frame 0: Disable the dgain 1: Enable the dgain
digital_gain_en_1	24:24	rw	'h0	Enable/Disable dgain of short frame 0: Disable the dgain 1: Enable the dgain
digital_gain_en_0	23:23	rw	'h0	Enable/Disable dgain of long frame 0: Disable the dgain 1: Enable the dgain
regs_inform_en	22:22	rw	'h0	Enable/Disable inform module 0: Disable the inform 1: Enable the inform
vsync_pol	21:21	rw	'h0	VSYNC polarity reversal
hsync_pol	20:20	rw	'h0	HSYNC polarity reversal
awb_gain_enable	19:19	rw	'h0	Enable/Disable AWB Gain module 0: Disable the AWB gain 1: Enable the AWB gain
cfg_upd	18:18	w	'h0	Configuration update
gen_cfg_upd	17:17	w	'h0	Generate configuration update
gen_cfg_upd_fix	16:16	rw	'h0	Auto configuration update in the end of frame
bypass_select	14:13	rw	'h0	Bypass frame select 0: long frame 1: short frame 2: very short frame 3: Exposure3 frame
linear_combine_enable	12:12	rw	'h0	linear combination enable
base_frame_selection	11:11	rw	'h0	base frame select
combination_mode	10:9	rw	'h0	combination_mode

channel_config_bit	8:8	rw	'h0	channel_config_bit
b10_enable_bit	7:7	rw	'h0	10 bit enable
lin_enable_bit	6:6	rw	'h0	reserved
vs_enable_bit	5:5	rw	'h0	very short frame enable
bayer_pattern	4:3	rw	'h0	bayer pattern
soft_reset_flag	2:2	rw	'h0	Activate/deactivate Stitch software reset 0: software reset un-active 1: software reset active
mono_input_flag	1:1	rw	'h0	reserved
combine_enable_bit	0:0	rw	'h0	Enable/Disable Stitch module 0: Disable the Stitch module 1: Enable the Stitch module

ISP_STITCHING0_FRAME_WIDTH

0x00003304 = Byte Address

Description: Stitching frame width

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_frame_width	13:0	rw	'h0	Frame width

ISP_STITCHING0_FRAME_HEIGHT

0x00003308 = Byte Address

Description: Stitching frame height

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_frame_height	13:0	rw	'h0	Frame height

ISP_STITCHING0_EXPOSURE_BIT

0x0000330c = Byte Address

Description: Stitching frame data bit depth

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_l_bit_depth	31:24	rw	'h5	Exposure bit for L; valid range: [0..255]
stitching_s_bit_depth	23:16	rw	'h5	Exposure bit for S; valid range: [0..255]
stitching_vs_bit_depth	15:8	rw	'h5	Exposure bit for VS; valid range: [0..255]
stitching_ls_bit_depth	7:0	rw	'h5	Exposure bit for LS/E3; valid range: [0..255]

ISP_STITCHING0_COLOR_WEIGHT

0x00003310 = Byte Address

Description: Channel color weight for gray computation

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_color_weight_2	23:16	rw	'h1d	B color weight for Y conversion Valid range: [0..255]

stitching_color_weight_1	15:8	rw	'h46	G color weight for Y conversion Valid range: [0..255]
stitching_color_weight_0	7:0	rw	'h4d	R color weight for Y conversion Valid range: [0..255]

ISP_STITCHING0_BLS_EXP_0_A

0x00003314 = Byte Address

Description: Black level for exposure 0 channel A

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_0_r	31:16	rw	'h100	Digital gain value for exposure 0 red pixel
stitching_bls_exp_0_a	11:0	rw	'h0	Black level value for exposure 0 channel A

ISP_STITCHING0_BLS_EXP_0_B

0x00003318 = Byte Address

Description: Black level for exposure 0 channel B

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_0_g	31:16	rw	'h100	Digital gain value for exposure 0 blue pixel
stitching_bls_exp_0_b	11:0	rw	'h0	Black level value for exposure 0 channel B

ISP_STITCHING0_BLS_EXP_0_C

0x0000331c = Byte Address

Description: Black level for exposure 0 channel C

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_0_gr	31:16	rw	'h100	Digital gain value for exposure 0 green red pixel
stitching_bls_exp_0_c	11:0	rw	'h0	Black level value for exposure 0 channel C

ISP_STITCHING0_BLS_EXP_0_D

0x00003320 = Byte Address

Description: Black level for exposure 0 channel D

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_digital_gain_exp_0_gb	31:16	rw	'h100	Digital gain value for exposure 0 green blue pixel
stitching_bls_exp_0_d	11:0	rw	'h0	Black level value for exposure 0 channel D

ISP_STITCHING0_BLS_EXP_1_A

0x00003324 = Byte Address

Description: Black level for exposure 1 channel A

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Sub-Field Name	Bits	rw	Reset	Description
stitching_digital_gain_exp_1_r	31:16	rw	'h100	Digital gain value for exposure 1 red pixel
stitching_bls_exp_1_a	11:0	rw	'h0	Black level value for exposure 1 channel A

ISP_STITCHING0_BLS_EXP_1_B

0x00003328 = Byte Address

Description: Black level for exposure 1 channel B

Sub-Field Name	Bits	rw	Reset	Description
stitching_digital_gain_exp_1_g	31:16	rw	'h100	Digital gain value for exposure 1 blue pixel
stitching_bls_exp_1_b	11:0	rw	'h0	Black level value for exposure 1 channel B

ISP_STITCHING0_BLS_EXP_1_C

0x0000332c = Byte Address

Description: Black level for exposure 1 channel C

Sub-Field Name	Bits	rw	Reset	Description
stitching_digital_gain_exp_1_gr	31:16	rw	'h100	Digital gain value for exposure 1 green red pixel
stitching_bls_exp_1_c	11:0	rw	'h0	Black level value for exposure 1 channel C

ISP_STITCHING0_BLS_EXP_1_D

0x00003330 = Byte Address

Description: Black level for exposure 1 channel D

Sub-Field Name	Bits	rw	Reset	Description
stitching_digital_gain_exp_1_gb	31:16	rw	'h100	Digital gain value for exposure 1 green blue pixel
stitching_bls_exp_1_d	11:0	rw	'h0	Black level value for exposure 1 channel D

ISP_STITCHING0_BLS_EXP_2_A

0x00003334 = Byte Address

Description: Black level for exposure 2 channel A

Sub-Field Name	Bits	rw	Reset	Description
stitching_digital_gain_exp_2_r	31:16	rw	'h100	Digital gain value for exposure 2 red pixel
stitching_bls_exp_2_a	11:0	rw	'h0	Black level value for exposure 2 channel A

ISP_STITCHING0_BLS_EXP_2_B

0x00003338 = Byte Address

Description: Black level for exposure 2 channel B

Sub-Field Name	Bits	rw	Reset	Description
stitching_digital_gain_exp_2_b	31:16	rw	'h100	Digital gain value for exposure 2 blue pixel
stitching_bls_exp_2_b	11:0	rw	'h0	Black level value for exposure 2 channel B

ISP_STITCHING0_BLS_EXP_2_C

0x0000333c = Byte Address

Description: Black level for exposure 2 channel C

Sub-Field Name	Bits	rw	Reset	Description
stitching_digital_gain_exp_2_gr	31:16	rw	'h100	Digital gain value for exposure 2 green red pixel
stitching_bls_exp_2_c	11:0	rw	'h0	Black level value for exposure 2 channel C

ISP_STITCHING0_BLS_EXP_2_D

0x00003340 = Byte Address

Description: Black level for exposure 2 channel D

Sub-Field Name	Bits	rw	Reset	Description
stitching_digital_gain_exp_2_gb	31:16	rw	'h100	Digital gain value for exposure 2 green blue pixel
stitching_bls_exp_2_d	11:0	rw	'h0	Black level value for exposure 2 channel D

ISP_STITCHING0_RATIO_LS

0x00003344 = Byte Address

Description: Exposure ratio between Long and Short

Sub-Field Name	Bits	rw	Reset	Description
stitching_ratio_long_short_1	23:12	rw	'h400	Exposure merge ratio from S to L (greater than 1)
stitching_ratio_long_short_0	11:0	rw	'h400	Exposure merge ratio from S to L (less than 1)

ISP_STITCHING0_RATIO_VS

0x00003348 = Byte Address

Description: Exposure ratio between Very short and Short

Sub-Field Name	Bits	rw	Reset	Description
stitching_ratio_veryshort_short_1	23:12	rw	'h400	Exposure merge ratio from VS to S (greater than 1)
stitching_ratio_veryshort_short_0	11:0	rw	'h400	Exposure merge ratio from VS to S (less than 1)

ISP_STITCHING0_RATIO_LS_SHD

0x0000334c = Byte Address

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Description: Exposure ratio between Long and Short shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_ratio_long_short_shd_1	23:12	r	'h400	Shadow of Exposure merge ratio from S to L (greater than 1)
stitching_ratio_long_short_shd_0	11:0	r	'h400	Shadow of Exposure merge ratio from S to L (less than 1)

ISP_STITCHING0_RATIO_VS_SHD

0x00003350 = Byte Address

Description: Exposure ratio between Very short and Short shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_ratio_veryshort_short_shd_1	23:12	r	'h400	Shadow of Exposure merge ratio from VS to S (greater than 1)
stitching_ratio_veryshort_short_shd_0	11:0	r	'h400	Shadow of Exposure merge ratio from VS to S (less than 1)

ISP_STITCHING0_TRANS_RANGE_LINEAR

0x00003354 = Byte Address

Description: Linear transformation range start and end

Sub-Field Name	Bits	rw	Reset	Description
stitching_trans_range_start_linear	27:16	rw	'h19a	Transmission range norm factor for linear mode
stitching_trans_range_norm_factor_mul_linear	15:0	rw	'he66	Transmission range start for linear mode

ISP_STITCHING0_TRANS_RANGE_NONLINEAR

0x00003358 = Byte Address

Description: Nonlinear transformation range start and end

Sub-Field Name	Bits	rw	Reset	Description
stitching_trans_range_start_nonlinear	27:16	rw	'h19a	Transmission range norm factor for nonlinear mode
stitching_trans_range_norm_factor_mul_nonlinear	15:0	rw	'he66	Transmission range start for nonlinear mode

ISP_STITCHING0_SAT_LEVEL

0x0000335c = Byte Address

Description: Stitching saturation level

Sub-Field Name	Bits	rw	Reset	Description
stitching_short_ls_extend_bit	31:28	rw	'h0	Short extend bit

stitching_short_vs_extend_bit	27:24	rw	'h0	Very Short extend bit
stitching_veryshort_valid_thresh	23:12	rw	'h7ff	Very short valid threshold
stitching_veryshort_offset_val	11:0	rw	'h0	Short extend bit

ISP_STITCHING0_LONG_EXPOSURE

0x00003360 = Byte Address

Description: Long exposure time

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_long_exposure_time	11:0	rw	'h40	Exposure time for long exposure frame

ISP_STITCHING0_SHORT_EXPOSURE

0x00003364 = Byte Address

Description: Short exposure time

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_short_exposure_time	11:0	rw	'h8	Exposure time for short exposure frame

ISP_STITCHING0_VERY_SHORT_EXPOSURE

0x00003368 = Byte Address

Description: Very short exposure time

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_very_short_exposure_time	11:0	rw	'h8	Exposure time for very short exposure frame

ISP_STITCHING0_HDR_MODE

0x0000336c = Byte Address

Description: HDR combination mode

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

stitching_hdr_input_format_type	3:0	rw	'h0	Input Sensor HDR mode 0: dual DCG mode 3x12-bit (default) 1: DOL 3 Frame 3x12-bit 2: 3x12-bit line by line without waiting 3: 16-bit compressed data + 12-bit RAW 4: 2x12-bit dual DCG without waiting 5: DOL2 frame or 1 CG+VS + 12-bit RAW 6: L+S 2x12-bit RAW 7: DOL 4 Frame 4x12-bit 8: DOL 3 Frame 3x12-bit 9: DOL 2 Frame 2x12-bit 10: 16-bit compressed data + 12-bit RAW

ISP_STITCHING0_OUT_HBLANK

0x00003370 = Byte Address

Description: HDR blanking in horizontal direction

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_dummy_s_hblank	31:16	rw	'h10	Dummy line horizontal blank for short exposure
stitching_out_hblank	15:0	rw	'h80	Output data horizontal blank time

ISP_STITCHING0_OUT_VBLANK

0x00003374 = Byte Address

Description: HDR blanking in vertical direction

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_dummy_vs_hblank	31:16	rw	'h10	Dummy line horizontal blank for very short exposure
stitching_out_vblank	15:0	rw	'h20	Output data vertical blank time

ISP_STITCHING0_INTERRUPT_STATUS

0x00003378 = Byte Address

Description: Stitching interrupt status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_interrupt_status	2:0	r	'h0	reserved

ISP_STITCHING0_COMPRESS_X0

0x0000337c = Byte Address

Description: Compression LUT x index

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
comress_lut_first_valid_x	9:0	rw	'h0	Compression LUT first valid x index

ISP_STITCHING0_COMPRESS_X0_SHD

0x00003380 = Byte Address

Description: Compression LUT x index shadow

Sub-Field Name	Bits	rw	Reset	Description
comress_lut_first_valid_shd_x	9:0	r	'h0	Compression LUT first valid x index shadow

ISP_STITCHING0_CTRL_SHD

0x00003384 = Byte Address

Description: HDR Stitch Control Register Shadow

Sub-Field Name	Bits	rw	Reset	Description
digital_gain_en_3_shd	30:30	r	'h0	Enable/Disable dgain of exposure3 frame 0: Disable the dgain 1: Enable the dgain
digital_gain_en_2_shd	25:25	r	'h0	Enable/Disable dgain of very short frame 0: Disable the dgain 1: Enable the dgain
digital_gain_en_1_shd	24:24	r	'h0	Enable/Disable dgain of short frame 0: Disable the dgain 1: Enable the dgain
digital_gain_en_0_shd	23:23	r	'h0	Enable/Disable dgain of long frame 0: Disable the dgain 1: Enable the dgain
regs_inform_en_shd	22:22	r	'h0	Enable/Disable inform module 0: Disable the inform 1: Enable the inform
bypass_select_shd	14:13	r	'h0	shadow of bypass data selection
soft_reset_shd	2:2	r	'h0	shadow of soft reset
combine_enable_shd	0:0	r	'h0	shadow of HDR combine enable

ISP_STITCHING0_EXPOSURE_MEAN_1

0x00003388 = Byte Address

Description: Exposure mean 1

Sub-Field Name	Bits	rw	Reset	Description
short_exposure_mean_1	31:16	r	'h0	reserved
long_exposure_mean_1	15:0	r	'h0	reserved

ISP_STITCHING0_COMPRESS_LUT_0

0x000033a0 = Byte Address

Description: Compression LUT data 0

Sub-Field Name	Bits	rw	Reset	Description
stitching_compress_lut_2	29:20	rw	'h0	VS compression LUT2
stitching_compress_lut_1	19:10	rw	'h0	VS compression LUT1
stitching_compress_lut_0	9:0	rw	'h0	VS compression LUT0

ISP_STITCHING0_COMPRESS_LUT_1

0x000033a4 = Byte Address

Description: Compression LUT data 1

Sub-Field Name	Bits	rw	Reset	Description
stitching_compress_lut_5	29:20	rw	'h0	VS compression LUT5
stitching_compress_lut_4	19:10	rw	'h0	VS compression LUT4
stitching_compress_lut_3	9:0	rw	'h0	VS compression LUT3

ISP_STITCHING0_COMPRESS_LUT_2

0x000033a8 = Byte Address

Description: Compression LUT data 2

Sub-Field Name	Bits	rw	Reset	Description
stitching_compress_lut_8	29:20	rw	'h0	VS compression LUT8
stitching_compress_lut_7	19:10	rw	'h0	VS compression LUT7
stitching_compress_lut_6	9:0	rw	'h0	VS compression LUT6

ISP_STITCHING0_COMPRESS_LUT_3

0x000033ac = Byte Address

Description: Compression LUT data 3

Sub-Field Name	Bits	rw	Reset	Description
stitching_compress_lut_11	29:20	rw	'h0	VS compression LUT11
stitching_compress_lut_10	19:10	rw	'h0	VS compression LUT10
stitching_compress_lut_9	9:0	rw	'h0	VS compression LUT9

ISP_STITCHING0_COMPRESS_LUT_4

0x000033b0 = Byte Address

Description: Compression LUT data 4

Sub-Field Name	Bits	rw	Reset	Description
stitching_compress_lut_14	29:20	rw	'h0	VS compression LUT14
stitching_compress_lut_13	19:10	rw	'h0	VS compression LUT13
stitching_compress_lut_12	9:0	rw	'h0	VS compression LUT12

ISP_STITCHING0_COMPRESS_LUT_0_SHD

0x000033b4 = Byte Address

Description: Compression LUT data 0 shadow

Sub-Field Name	Bits	rw	Reset	Description
stitching_compress_lut_shd_2	29:20	r	'h0	Shadow of VS compression LUT2
stitching_compress_lut_shd_1	19:10	r	'h0	Shadow of VS compression LUT1

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stitching_compress_lut_shd_0	9:0	r	'h0	Shadow of VS compression LUT0

ISP_STITCHING0_COMPRESS_LUT_1_SHD

0x000033b8 = Byte Address

Description: Compression LUT data 1 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_compress_lut_shd_5	29:20	r	'h0	Shadow of VS compression LUT5
stitching_compress_lut_shd_4	19:10	r	'h0	Shadow of VS compression LUT4
stitching_compress_lut_shd_3	9:0	r	'h0	Shadow of VS compression LUT3

ISP_STITCHING0_COMPRESS_LUT_2_SHD

0x000033bc = Byte Address

Description: Compression LUT data 2 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_compress_lut_shd_8	29:20	r	'h0	Shadow of VS compression LUT8
stitching_compress_lut_shd_7	19:10	r	'h0	Shadow of VS compression LUT7
stitching_compress_lut_shd_6	9:0	r	'h0	Shadow of VS compression LUT6

ISP_STITCHING0_COMPRESS_LUT_3_SHD

0x000033c0 = Byte Address

Description: Compression LUT data 3 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_compress_lut_shd_11	29:20	r	'h0	Shadow of VS compression LUT11
stitching_compress_lut_shd_10	19:10	r	'h0	Shadow of VS compression LUT10
stitching_compress_lut_shd_9	9:0	r	'h0	Shadow of VS compression LUT9

ISP_STITCHING0_COMPRESS_LUT_4_SHD

0x000033c4 = Byte Address

Description: Compression LUT data 4 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

stitching_compress_lut_shd_14	29:20	r	'h0	Shadow of VS compression LUT14
stitching_compress_lut_shd_13	19:10	r	'h0	Shadow of VS compression LUT13
stitching_compress_lut_shd_12	9:0	r	'h0	Shadow of VS compression LUT12

ISP_STITCHING0_EXP0_AWB_GAIN_G

0x000033c8 = Byte Address

Description: AWB gain for exposure 0 channel G

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_exp0_awb_gain_gr	25:16	rw	'h100	Stitching AWB GR gain for exposure 0
stitching_exp0_awb_gain_gb	9:0	rw	'h100	Stitching AWB GB gain for exposure 0

ISP_STITCHING0_EXP0_AWB_GAIN_RB

0x000033cc = Byte Address

Description: AWB gain for exposure 0 channel RB

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_exp0_awb_gain_r	25:16	rw	'h100	Stitching AWB R gain for exposure 0
stitching_exp0_awb_gain_b	9:0	rw	'h100	Stitching AWB B gain for exposure 0

ISP_STITCHING0_EXP1_AWB_GAIN_G

0x000033d0 = Byte Address

Description: AWB gain for exposure 1 channel G

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_exp1_awb_gain_gr	25:16	rw	'h100	Stitching AWB GR gain for exposure 1
stitching_exp1_awb_gain_gb	9:0	rw	'h100	Stitching AWB GB gain for exposure 1

ISP_STITCHING0_EXP1_AWB_GAIN_RB

0x000033d4 = Byte Address

Description: AWB gain for exposure 1 channel RB

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_exp1_awb_gain_r	25:16	rw	'h100	Stitching AWB R gain for exposure 1
stitching_exp1_awb_gain_b	9:0	rw	'h100	Stitching AWB B gain for exposure 1

ISP_STITCHING0_EXP2_AWB_GAIN_G

0x000033d8 = Byte Address

Description: AWB gain for exposure 2 channel G

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

stitching_exp2_awb_gain_gr	25:16	rw	'h100	Stitching AWB GR gain for exposure 2
stitching_exp2_awb_gain_gb	9:0	rw	'h100	Stitching AWB GB gain for exposure 2

ISP_STITCHING0_EXP2_AWB_GAIN_RB

0x000033dc = Byte Address

Description: AWB gain for exposure 2 channel RB

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_exp2_awb_gain_r	25:16	rw	'h100	Stitching AWB R gain for exposure 2
stitching_exp2_awb_gain_b	9:0	rw	'h100	Stitching AWB B gain for exposure 2

ISP_STITCHING0_LONG_SAT_PARAMS

0x000033e0 = Byte Address

Description: Stitching saturation params for long exposure

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_long_sat_thresh	23:12	rw	'h7ff	Stitching long saturation threshold
stitching_long_sat_combine_weight	8:0	rw	'h0	Stitching long combine weight

ISP_STITCHING0_IMSC

0x000033e4 = Byte Address

Description: IMSC Interrupt Mask

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_imsc_in_size_err_vs	13:13	rw	'h0	Stitching HDR very short frame input size error interrupt mask
stitching_imsc_in_size_err_s	12:12	rw	'h0	Stitching HDR short frame input size error interrupt mask
stitching_imsc_in_size_err_l	11:11	rw	'h0	Stitching HDR long frame input size error interrupt mask
stitching_imsc_retiming_int	10:10	rw	'h0	Stitching HDR retiming interrupt mask
stitching_imsc_frame_end	9:9	rw	'h0	Stitching HDR frame end interrupt mask
stitching_imsc_hdr_hist_irq_vs	8:8	rw	'h0	Stitching HDR hist VS interrupt mask
stitching_imsc_hdr_hist_irq_s	7:7	rw	'h0	Stitching HDR hist S interrupt mask
stitching_imsc_hdr_hist_irq_l	6:6	rw	'h0	Stitching HDR hist L interrupt mask
stitching_imsc_hdr_exp_complete_vs	5:5	rw	'h0	Stitching HDR exp statistic VS interrupt mask

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stitching_imsc_hdr_exp_complete_s	4:4	rw	'h0	Stitching HDR exp statistic S interrupt mask
stitching_imsc_hdr_exp_complete_l	3:3	rw	'h0	Stitching HDR exp statistic L interrupt mask
stitching_imsc_fifo_empty	2:2	rw	'h0	Stitching HDR fifo_empty interrupt mask
stitching_imsc_exp_err2	1:1	rw	'h0	Stitching HDR very short exp time error L interrupt mask
stitching_imsc_exp_err1	0:0	rw	'h0	Stitching HDR short exp time error interrupt mask

ISP_STITCHING0_RIS

0x000033e8 = Byte Address

Description: RIS Raw Interrupt Status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_ris_in_size_err_vs	13:13	r	'h0	Stitching HDR very short frame input size error interrupt status
stitching_ris_in_size_err_s	12:12	r	'h0	Stitching HDR short frame input size error interrupt status
stitching_ris_in_size_err_l	11:11	r	'h0	Stitching HDR long frame input size error interrupt status
stitching_ris_retiming_int	10:10	r	'h0	Stitching HDR retiming interrupt status
stitching_ris_frame_end	9:9	r	'h0	Stitching HDR frame_end L interrupt status
stitching_ris_hdr_hist_irq_vs	8:8	r	'h0	Stitching HDR hist VS interrupt status
stitching_ris_hdr_hist_irq_s	7:7	r	'h0	Stitching HDR hist S interrupt status
stitching_ris_hdr_hist_irq_l	6:6	r	'h0	Stitching HDR hist L interrupt status
stitching_ris_hdr_exp_complete_vs	5:5	r	'h0	Stitching HDR exp statistic VS interrupt status
stitching_ris_hdr_exp_complete_s	4:4	r	'h0	Stitching HDR exp statistic S interrupt status
stitching_ris_hdr_exp_complete_l	3:3	r	'h0	Stitching HDR exp statistic L interrupt status
stitching_ris_fifo_empty	2:2	r	'h0	Stitching HDR fifo_empty interrupt status
stitching_ris_exp_err2	1:1	r	'h0	Stitching HDR very short exp time error L interrupt status
stitching_ris_exp_err1	0:0	r	'h0	Stitching HDR short exp time error L interrupt status

ISP_STITCHING0_MIS

0x000033ec = Byte Address

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Description: MIS Masked Interrupt Status

Sub-Field Name	Bits	rw	Reset	Description
stitching_mis_in_size_err_vs	13:13	r	'h0	Stitching HDR very short frame input size error interrupt masked status
stitching_mis_in_size_err_s	12:12	r	'h0	Stitching HDR short frame input size error interrupt masked status
stitching_mis_in_size_err_l	11:11	r	'h0	Stitching HDR long frame input size error interrupt masked status
stitching_mis_retiming_int	10:10	r	'h0	Stitching HDR retiming interrupt masked status
stitching_mis_frame_end	9:9	r	'h0	Stitching HDR frame_end masked interrupt status
stitching_mis_hdr_hist_irq_vs	8:8	r	'h0	Stitching HDR hist VS interrupt masked status
stitching_mis_hdr_hist_irq_s	7:7	r	'h0	Stitching HDR hist S interrupt masked status
stitching_mis_hdr_hist_irq_l	6:6	r	'h0	Stitching HDR hist L interrupt masked status
stitching_mis_hdr_exp_complete_vs	5:5	r	'h0	Stitching HDR exp statistic VS interrupt masked status
stitching_mis_hdr_exp_complete_s	4:4	r	'h0	Stitching HDR exp statistic S interrupt masked status
stitching_mis_hdr_exp_complete_l	3:3	r	'h0	Stitching HDR exp statistic L interrupt masked status
stitching_mis_fifo_empty	2:2	r	'h0	Stitching HDR frame_end L interrupt masked status
stitching_mis_exp_err2	1:1	r	'h0	Stitching HDR very short exp time error L interrupt masked status
stitching_mis_exp_err1	0:0	r	'h0	Stitching HDR retiming interrupt Masked status

ISP_STITCHING0_ICR

0x000033f0 = Byte Address

Description: ICR Interrupt Clear Register

Sub-Field Name	Bits	rw	Reset	Description
stitching_icr_in_size_err_vs	13:13	w	'h0	Stitching HDR very short frame input size error interrupt clear
stitching_icr_in_size_err_s	12:12	w	'h0	Stitching HDR short frame input size error interrupt clear
stitching_icr_in_size_err_l	11:11	w	'h0	Stitching HDR long frame input size error interrupt clear

stitching_icr_retiming_int	10:10	w	'h0	Stitching HDR retiming interrupt clear
stitching_icr_frame_end	9:9	w	'h0	Stitching HDR frame_end L interrupt clear
stitching_icr_hdr_hist_irq_vs	8:8	w	'h0	Stitching HDR hist VS interrupt clear
stitching_icr_hdr_hist_irq_s	7:7	w	'h0	Stitching HDR hist S interrupt clear
stitching_icr_hdr_hist_irq_l	6:6	w	'h0	Stitching HDR hist L interrupt clear
stitching_icr_hdr_exp_complete_vs	5:5	w	'h0	Stitching HDR exp statistic VS interrupt clear
stitching_icr_hdr_exp_complete_s	4:4	w	'h0	Stitching HDR exp statistic S interrupt clear
stitching_icr_hdr_exp_complete_l	3:3	w	'h0	Stitching HDR exp statistic L interrupt clear
stitching_icr_fifo_empty	2:2	w	'h0	Stitching HDR fifo empty L interrupt clear
stitching_icr_exp_err2	1:1	w	'h0	Stitching HDR very short exp time error L interrupt clear
stitching_icr_exp_err1	0:0	w	'h0	Stitching HDR short exp time error interrupt clear

ISP_STITCHING0_ISR

0x000033f4 = Byte Address

Description: ISR Interrupt Set Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_isr_in_size_err_vs	13:13	w	'h0	Stitching HDR very short frame input size error interrupt set
stitching_isr_in_size_err_s	12:12	w	'h0	Stitching HDR short frame input size error interrupt set
stitching_isr_in_size_err_l	11:11	w	'h0	Stitching HDR long frame input size error interrupt set
stitching_isr_retiming_int	10:10	w	'h0	Stitching HDR retiming interrupt set
stitching_isr_frame_end	9:9	w	'h0	Stitching HDR frame_end L interrupt set
stitching_isr_hdr_hist_irq_vs	8:8	w	'h0	Stitching HDR hist VS interrupt set
stitching_isr_hdr_hist_irq_s	7:7	w	'h0	Stitching HDR hist S interrupt set
stitching_isr_hdr_hist_irq_l	6:6	w	'h0	Stitching HDR hist L interrupt set
stitching_isr_hdr_exp_complete_vs	5:5	w	'h0	Stitching HDR exp statistic VS interrupt set
stitching_isr_hdr_exp_complete_s	4:4	w	'h0	Stitching HDR exp statistic S interrupt set
stitching_isr_hdr_exp_complete_l	3:3	w	'h0	Stitching HDR exp statistic L interrupt set

stitching_isr_fifo_empty	2:2	w	'h0	Stitching HDR fifo empty L interrupt set
stitching_isr_exp_err2	1:1	w	'h0	Stitching HDR very short exp time error L interrupt set
stitching_isr_exp_err1	0:0	w	'h0	Stitching HDR short exp time error interrupt set

ISP_STITCHING0_BLS_EXP_OUT_0

0x000033f8 = Byte Address

Description: Stitching black level for output

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_bls_exp_out_a	27:16	rw	'h0	Black level value for Stitching output channel A
stitching_bls_exp_out_b	11:0	rw	'h0	Black level value for Stitching output channel B

ISP_STITCHING0_BLS_EXP_OUT_1

0x000033fc = Byte Address

Description: Stitching black level for output

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_bls_exp_out_c	27:16	rw	'h0	Black level value for Stitching output channel C
stitching_bls_exp_out_d	11:0	rw	'h0	Black level value for Stitching output channel D

ISP_CURVE_CTRL

0x0000391c = Byte Address

Description: curve control

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_mode	2:1	rw	'h0	2'b00: mode0 2'b01: mode1 2'b10: mode2 2'b11: Reserved
curve_enable	0:0	rw	'h0	1'b0: CA disable 1'b1: CA enable

ISP_CURVE_LUT_X_ADDR

0x00003920 = Byte Address

Description: Pointer to entry of X value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_x_addr	6:0	rw	'h0	Pointer to entry of X value LUT for Curve

ISP_CURVE_LUT_X_WRITE_DATA

0x00003924 = Byte Address

Description: Write Data of X LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_x_write_data	11:0	rw	'h0	Write Data of X LUT for Curve

ISP_CURVE_LUT_LUMA_ADDR

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0x00003928 = Byte Address

Description: Pointer to entry of LUMA value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_luma_addr	6:0	rw	'h0	Pointer to entry of LUMA value LUT for Curve

ISP_CURVE_LUT_LUMA_WRITE_DATA

0x0000392c = Byte Address

Description: Write Data of LUMA LUT value for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_luma_write_data	10:0	rw	'h0	Write Data of LUMA LUT value for Curve

ISP_CURVE_LUT_CHROMA_ADDR

0x00003930 = Byte Address

Description: Pointer to entry of CHROMA value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_chroma_addr	6:0	rw	'h0	Pointer to entry of CHROMA value LUT for Curve

ISP_CURVE_LUT_CHROMA_WRITE_DATA

0x00003934 = Byte Address

Description: Write Data of CHROMA LUT value for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_chroma_write_data	11:0	rw	'h0	Write Data of CHROMA LUT value for Curve

ISP_CURVE_LUT_SHIFT_ADDR

0x00003938 = Byte Address

Description: Pointer to entry of SHIFT value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_shift_addr	6:0	rw	'h0	Pointer to entry of SHIFT value LUT for Curve

ISP_CURVE_LUT_SHIFT_WRITE_DATA

0x0000393c = Byte Address

Description: Write Data of SHIFT LUT value for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_shift_write_data	4:0	rw	'h0	Write Data of SHIFT LUT value for Curve

ISP_CURVE_CTRL_SHD

0x00003940 = Byte Address

Description: curve control shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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curve_mode_shd	2:1	r	'h0	shadow bit
curve_enable_shd	0:0	r	'h0	shadow bit

ISP_CURVE_LUT_LUMA_ADDR_SHD

0x00003944 = Byte Address

Description: Pointer to entry of LUMA value LUT for Curve shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_luma_addr_shd	6:0	rw	'h0	shadow LUT

ISP_CURVE_LUT_LUMA_WRITE_DATA_SHD

0x00003948 = Byte Address

Description: Write Data of LUMA LUT value for Curve shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_luma_write_data_shd	10:0	r	'h0	Shadow LUT

ISP_CURVE_LUT_CHROMA_ADDR_SHD

0x0000394c = Byte Address

Description: Pointer to entry of CHROMA value LUT for Curve shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_chroma_addr_shd	6:0	rw	'h0	Shadow LUT

ISP_CURVE_LUT_CHROMA_WRITE_DATA_SHD

0x00003950 = Byte Address

Description: Write Data of CHROMA LUT value for Curve shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_chroma_write_data_shd	11:0	r	'h0	Shadow LUT

ISP_CURVE_LUT_SHIFT_ADDR_SHD

0x00003954 = Byte Address

Description: Pointer to entry of SHIFT value LUT for Curve shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
curve_lut_shift_addr_shd	6:0	rw	'h0	Shadow LUT

ISP_CURVE_LUT_SHIFT_WRITE_DATA_SHD

0x00003958 = Byte Address

Description: Write Data of SHIFT LUT value for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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curve_lut_shift_write_data_shd	4:0	r	'h0	Shadow LUT

ISP_DENOISE2D_CTRL

0x00003a00 = Byte Address

Description: Denoise2D Control Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise2d_lsc_gain_comp_enable	4:4	rw	'h0	Enable/Disable lsc gain compensation 0: Disable lsc gain compensation 1: Enable lsc gain compensation
denoise2d_luma_enable	3:3	rw	'h0	Enable/Disable luma control 0: Disable luma 1: Enable luma
denoise2d_motion_enable	2:2	rw	'h0	Enable/Disable motion 0: Disable motion 1: Enable motion
denoise2d_soft_reset	1:1	rw	'h0	Activate/deactivate Denoise_2d software reset 0: software reset un-active 1: software reset active

ISP_DENOISE2D_SIGMA_Y_0

0x00003a08 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_2_a	29:24	rw	'h0	Denoise filter sigma 2 [11:6]
denoise_sigma_y_1	23:12	rw	'h0	Denoise filter sigma 1
denoise_sigma_y_0	11:0	rw	'hfff	Denoise filter sigma 0

ISP_DENOISE2D_SIGMA_Y_1

0x00003a0c = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_2_b	29:24	rw	'h0	Denoise filter sigma 2 [5:0]
denoise_sigma_y_4	23:12	rw	'h0	Denoise filter sigma 4
denoise_sigma_y_3	11:0	rw	'h0	Denoise filter sigma 3

ISP_DENOISE2D_SIGMA_Y_2

0x00003a10 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_7_a	29:24	rw	'h0	Denoise filter sigma 7 [11:6]

denoise_sigma_y_6	23:12	rw	'h0	Denoise filter sigma 6
denoise_sigma_y_5	11:0	rw	'h0	Denoise filter sigma 5

ISP_DENOISE2D_SIGMA_Y_3

0x00003a14 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_7_b	29:24	rw	'h0	Denoise filter sigma 7 [5:0]
denoise_sigma_y_9	23:12	rw	'h0	Denoise filter sigma 9
denoise_sigma_y_8	11:0	rw	'h0	Denoise filter sigma 8

ISP_DENOISE2D_SIGMA_Y_4

0x00003a18 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_12_a	29:24	rw	'h0	Denoise filter sigma 12 [11:6]
denoise_sigma_y_11	23:12	rw	'h0	Denoise filter sigma 11
denoise_sigma_y_10	11:0	rw	'h0	Denoise filter sigma 10

ISP_DENOISE2D_SIGMA_Y_5

0x00003a1c = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_12_b	29:24	rw	'h0	Denoise filter sigma 12 [5:0]
denoise_sigma_y_14	23:12	rw	'h0	Denoise filter sigma 14
denoise_sigma_y_13	11:0	rw	'h0	Denoise filter sigma 13

ISP_DENOISE2D_SIGMA_Y_6

0x00003a20 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_17_a	29:24	rw	'h0	Denoise filter sigma 17 [11:6]
denoise_sigma_y_16	23:12	rw	'h0	Denoise filter sigma 16
denoise_sigma_y_15	11:0	rw	'h0	Denoise filter sigma 15

ISP_DENOISE2D_SIGMA_Y_7

0x00003a24 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

denoise_sigma_y_17_b	29:24	rw	'h0	Denoise filter sigma 17 [5:0]
denoise_sigma_y_19	23:12	rw	'h0	Denoise filter sigma 19
denoise_sigma_y_18	11:0	rw	'h0	Denoise filter sigma 18

ISP_DENOISE2D_SIGMA_Y_8

0x00003a28 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_22_a	29:24	rw	'h0	Denoise filter sigma 22 [11:6]
denoise_sigma_y_21	23:12	rw	'h0	Denoise filter sigma 21
denoise_sigma_y_20	11:0	rw	'h0	Denoise filter sigma 20

ISP_DENOISE2D_SIGMA_Y_9

0x00003a2c = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_22_b	29:24	rw	'h0	Denoise filter sigma 22 [5:0]
denoise_sigma_y_24	23:12	rw	'h0	Denoise filter sigma 24
denoise_sigma_y_23	11:0	rw	'h0	Denoise filter sigma 23

ISP_DENOISE2D_SIGMA_Y_10

0x00003a30 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_27_a	29:24	rw	'h0	Denoise filter sigma 27 [11:6]
denoise_sigma_y_26	23:12	rw	'h0	Denoise filter sigma 26
denoise_sigma_y_25	11:0	rw	'h0	Denoise filter sigma 25

ISP_DENOISE2D_SIGMA_Y_11

0x00003a34 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_27_b	29:24	rw	'h0	Denoise filter sigma 27 [5:0]
denoise_sigma_y_29	23:12	rw	'h0	Denoise filter sigma 29
denoise_sigma_y_28	11:0	rw	'h0	Denoise filter sigma 28

ISP_DENOISE2D_SIGMA_Y_12

0x00003a38 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

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Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_32_a	29:24	rw	'h0	Denoise filter sigma 32 [11:6]
denoise_sigma_y_31	23:12	rw	'h0	Denoise filter sigma 31
denoise_sigma_y_30	11:0	rw	'h0	Denoise filter sigma 30

ISP_DENOISE2D_SIGMA_Y_13

0x00003a3c = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_32_b	29:24	rw	'h0	Denoise filter sigma 32 [5:0]
denoise_sigma_y_34	23:12	rw	'h0	Denoise filter sigma 34
denoise_sigma_y_33	11:0	rw	'h0	Denoise filter sigma 33

ISP_DENOISE2D_SIGMA_Y_14

0x00003a40 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_37_a	29:24	rw	'h0	Denoise filter sigma 37 [11:6]
denoise_sigma_y_36	23:12	rw	'h0	Denoise filter sigma 36
denoise_sigma_y_35	11:0	rw	'h0	Denoise filter sigma 35

ISP_DENOISE2D_SIGMA_Y_15

0x00003a44 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_37_b	29:24	rw	'h0	Denoise filter sigma 37 [5:0]
denoise_sigma_y_39	23:12	rw	'h0	Denoise filter sigma 39
denoise_sigma_y_38	11:0	rw	'h0	Denoise filter sigma 38

ISP_DENOISE2D_SIGMA_Y_16

0x00003a48 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_42_a	29:24	rw	'h0	Denoise filter sigma 42 [11:6]
denoise_sigma_y_41	23:12	rw	'h0	Denoise filter sigma 41
denoise_sigma_y_40	11:0	rw	'h0	Denoise filter sigma 40

ISP_DENOISE2D_SIGMA_Y_17

0x00003a4c = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_42_b	29:24	rw	'h0	Denoise filter sigma 42 [5:0]
denoise_sigma_y_44	23:12	rw	'h0	Denoise filter sigma 44
denoise_sigma_y_43	11:0	rw	'h0	Denoise filter sigma 43

ISP_DENOISE2D_SIGMA_Y_18

0x00003a50 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_47_a	29:24	rw	'h0	Denoise filter sigma 47 [11:6]
denoise_sigma_y_46	23:12	rw	'h0	Denoise filter sigma 46
denoise_sigma_y_45	11:0	rw	'h0	Denoise filter sigma 45

ISP_DENOISE2D_SIGMA_Y_19

0x00003a54 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_47_b	29:24	rw	'h0	Denoise filter sigma 47 [5:0]
denoise_sigma_y_49	23:12	rw	'h0	Denoise filter sigma 49
denoise_sigma_y_48	11:0	rw	'h0	Denoise filter sigma 48

ISP_DENOISE2D_SIGMA_Y_20

0x00003a58 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_52_a	29:24	rw	'h0	Denoise filter sigma 52 [11:6]
denoise_sigma_y_51	23:12	rw	'h0	Denoise filter sigma 51
denoise_sigma_y_50	11:0	rw	'h0	Denoise filter sigma 50

ISP_DENOISE2D_SIGMA_Y_21

0x00003a5c = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_52_b	29:24	rw	'h0	Denoise filter sigma 52 [5:0]
denoise_sigma_y_54	23:12	rw	'h0	Denoise filter sigma 54
denoise_sigma_y_53	11:0	rw	'h0	Denoise filter sigma 53

ISP_DENOISE2D_SIGMA_Y_22

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0x00003a60 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_57_a	29:24	rw	'h0	Denoise filter sigma 57 [11:6]
denoise_sigma_y_56	23:12	rw	'h0	Denoise filter sigma 56
denoise_sigma_y_55	11:0	rw	'h0	Denoise filter sigma 55

ISP_DENOISE2D_SIGMA_Y_23

0x00003a64 = Byte Address

Description: Denoise2D Filter Sigma LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_57_b	29:24	rw	'h0	Denoise filter sigma 57 [5:0]
denoise_sigma_y_59	23:12	rw	'h0	Denoise filter sigma 59
denoise_sigma_y_58	11:0	rw	'h0	Denoise filter sigma 58

ISP_DENOISE2D_SIGMA_Y_0_SHD

0x00003a68 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_2_a_shd	29:24	r	'h0	Denoise filter sigma 2 [11:6] shadow
denoise_sigma_y_1_shd	23:12	r	'h0	Denoise filter sigma 1 shadow
denoise_sigma_y_0_shd	11:0	r	'hfff	Denoise filter sigma 0 shadow

ISP_DENOISE2D_SIGMA_Y_1_SHD

0x00003a6c = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_2_b_shd	29:24	r	'h0	Denoise filter sigma 2 [5:0] shadow
denoise_sigma_y_4_shd	23:12	r	'h0	Denoise filter sigma 4 shadow
denoise_sigma_y_3_shd	11:0	r	'h0	Denoise filter sigma 3 shadow

ISP_DENOISE2D_SIGMA_Y_2_SHD

0x00003a70 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_7_a_shd	29:24	r	'h0	Denoise filter sigma 7 [11:6] shadow
denoise_sigma_y_6_shd	23:12	r	'h0	Denoise filter sigma 6 shadow
denoise_sigma_y_5_shd	11:0	r	'h0	Denoise filter sigma 5 shadow

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ISP_DENOISE2D_SIGMA_Y_3_SHD

0x00003a74 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_7_b_shd	29:24	r	'h0	Denoise filter sigma 7 [5:0] shadow
denoise_sigma_y_9_shd	23:12	r	'h0	Denoise filter sigma 9 shadow
denoise_sigma_y_8_shd	11:0	r	'h0	Denoise filter sigma 8 shadow

ISP_DENOISE2D_SIGMA_Y_4_SHD

0x00003a78 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_12_a_shd	29:24	r	'h0	Denoise filter sigma 12 [11:6] shadow
denoise_sigma_y_11_shd	23:12	r	'h0	Denoise filter sigma 11 shadow
denoise_sigma_y_10_shd	11:0	r	'h0	Denoise filter sigma 10 shadow

ISP_DENOISE2D_SIGMA_Y_5_SHD

0x00003a7c = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_12_b_shd	29:24	r	'h0	Denoise filter sigma 12 [5:0] shadow
denoise_sigma_y_14_shd	23:12	r	'h0	Denoise filter sigma 14 shadow
denoise_sigma_y_13_shd	11:0	r	'h0	Denoise filter sigma 13 shadow

ISP_DENOISE2D_SIGMA_Y_6_SHD

0x00003a80 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_17_a_shd	29:24	r	'h0	Denoise filter sigma 17 [11:6] shadow
denoise_sigma_y_16_shd	23:12	r	'h0	Denoise filter sigma 16 shadow
denoise_sigma_y_15_shd	11:0	r	'h0	Denoise filter sigma 15 shadow

ISP_DENOISE2D_SIGMA_Y_7_SHD

0x00003a84 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_17_b_shd	29:24	r	'h0	Denoise filter sigma 17 [5:0] shadow
denoise_sigma_y_19_shd	23:12	r	'h0	Denoise filter sigma 19 shadow

denoise_sigma_y_18_shd	11:0	r	'h0	Denoise filter sigma 18 shadow

ISP_DENOISE2D_SIGMA_Y_8_SHD

0x00003a88 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_22_a_shd	29:24	r	'h0	Denoise filter sigma 22 [11:6] shadow
denoise_sigma_y_21_shd	23:12	r	'h0	Denoise filter sigma 21 shadow
denoise_sigma_y_20_shd	11:0	r	'h0	Denoise filter sigma 20 shadow

ISP_DENOISE2D_SIGMA_Y_9_SHD

0x00003a8c = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_22_b_shd	29:24	r	'h0	Denoise filter sigma 22 [5:0] shadow
denoise_sigma_y_24_shd	23:12	r	'h0	Denoise filter sigma 24 shadow
denoise_sigma_y_23_shd	11:0	r	'h0	Denoise filter sigma 23 shadow

ISP_DENOISE2D_SIGMA_Y_10_SHD

0x00003a90 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_27_a_shd	29:24	r	'h0	Denoise filter sigma 27 [11:6] shadow
denoise_sigma_y_26_shd	23:12	r	'h0	Denoise filter sigma 26 shadow
denoise_sigma_y_25_shd	11:0	r	'h0	Denoise filter sigma 25 shadow

ISP_DENOISE2D_SIGMA_Y_11_SHD

0x00003a94 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_27_b_shd	29:24	r	'h0	Denoise filter sigma 27 [5:0] shadow
denoise_sigma_y_29_shd	23:12	r	'h0	Denoise filter sigma 29 shadow
denoise_sigma_y_28_shd	11:0	r	'h0	Denoise filter sigma 28 shadow

ISP_DENOISE2D_SIGMA_Y_12_SHD

0x00003a98 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_32_a_shd	29:24	r	'h0	Denoise filter sigma 32 [11:6] shadow

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denoise_sigma_y_31_shd	23:12	r	'h0	Denoise filter sigma 31 shadow
denoise_sigma_y_30_shd	11:0	r	'h0	Denoise filter sigma 30 shadow

ISP_DENOISE2D_SIGMA_Y_13_SHD

0x00003a9c = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_32_b_shd	29:24	r	'h0	Denoise filter sigma 32 [5:0] shadow
denoise_sigma_y_34_shd	23:12	r	'h0	Denoise filter sigma 34 shadow
denoise_sigma_y_33_shd	11:0	r	'h0	Denoise filter sigma 33 shadow

ISP_DENOISE2D_SIGMA_Y_14_SHD

0x00003aa0 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_37_a_shd	29:24	r	'h0	Denoise filter sigma 37 [11:6] shadow
denoise_sigma_y_36_shd	23:12	r	'h0	Denoise filter sigma 36 shadow
denoise_sigma_y_35_shd	11:0	r	'h0	Denoise filter sigma 35 shadow

ISP_DENOISE2D_SIGMA_Y_15_SHD

0x00003aa4 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_37_b_shd	29:24	r	'h0	Denoise filter sigma 37 [5:0] shadow
denoise_sigma_y_39_shd	23:12	r	'h0	Denoise filter sigma 39 shadow
denoise_sigma_y_38_shd	11:0	r	'h0	Denoise filter sigma 38 shadow

ISP_DENOISE2D_SIGMA_Y_16_SHD

0x00003aa8 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_42_a_shd	29:24	r	'h0	Denoise filter sigma 42 [11:6] shadow
denoise_sigma_y_41_shd	23:12	r	'h0	Denoise filter sigma 41 shadow
denoise_sigma_y_40_shd	11:0	r	'h0	Denoise filter sigma 40 shadow

ISP_DENOISE2D_SIGMA_Y_17_SHD

0x00003aac = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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denoise_sigma_y_42_b_shd	29:24	r	'h0	Denoise filter sigma 42 [5:0] shadow
denoise_sigma_y_44_shd	23:12	r	'h0	Denoise filter sigma 44 shadow
denoise_sigma_y_43_shd	11:0	r	'h0	Denoise filter sigma 43 shadow

ISP_DENOISE2D_SIGMA_Y_18_SHD

0x00003ab0 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_47_a_shd	29:24	r	'h0	Denoise filter sigma 47 [11:6] shadow
denoise_sigma_y_46_shd	23:12	r	'h0	Denoise filter sigma 46 shadow
denoise_sigma_y_45_shd	11:0	r	'h0	Denoise filter sigma 45 shadow

ISP_DENOISE2D_SIGMA_Y_19_SHD

0x00003ab4 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_47_b_shd	29:24	r	'h0	Denoise filter sigma 47 [5:0] shadow
denoise_sigma_y_49_shd	23:12	r	'h0	Denoise filter sigma 49 shadow
denoise_sigma_y_48_shd	11:0	r	'h0	Denoise filter sigma 48 shadow

ISP_DENOISE2D_SIGMA_Y_20_SHD

0x00003ab8 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_52_a_shd	29:24	r	'h0	Denoise filter sigma 52 [11:6] shadow
denoise_sigma_y_51_shd	23:12	r	'h0	Denoise filter sigma 51 shadow
denoise_sigma_y_50_shd	11:0	r	'h0	Denoise filter sigma 50 shadow

ISP_DENOISE2D_SIGMA_Y_21_SHD

0x00003abc = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_y_52_b_shd	29:24	r	'h0	Denoise filter sigma 52 [5:0] shadow
denoise_sigma_y_54_shd	23:12	r	'h0	Denoise filter sigma 54 shadow
denoise_sigma_y_53_shd	11:0	r	'h0	Denoise filter sigma 53 shadow

ISP_DENOISE2D_SIGMA_Y_22_SHD

0x00003ac0 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_57_a_shd	29:24	r	'h0	Denoise filter sigma 57 [11:6] shadow
denoise_sigma_y_56_shd	23:12	r	'h0	Denoise filter sigma 56 shadow
denoise_sigma_y_55_shd	11:0	r	'h0	Denoise filter sigma 55 shadow

ISP_DENOISE2D_SIGMA_Y_23_SHD

0x00003ac4 = Byte Address

Description: Denoise2D Filter Sigma LUT Shadow Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_y_57_b_shd	29:24	r	'h0	Denoise filter sigma 57 [5:0] shadow
denoise_sigma_y_59_shd	23:12	r	'h0	Denoise filter sigma 59 shadow
denoise_sigma_y_58_shd	11:0	r	'h0	Denoise filter sigma 58 shadow

ISP_DENOISE2D_CTRL_SHD

0x00003ac8 = Byte Address

Description: Denoise2D Control Shadow Register

Sub-Field Name	Bits	rw	Reset	Description
denoise2d_lsc_gain_comp_enable_shd	4:4	r	'h0	Enable/Disable lsc gain compensation shadow 0: Disable lsc gain compensation 1: Enable lsc gain compensation
denoise2d_luma_enable_shd	3:3	r	'h0	Enable/Disable luma control shadow 0: Disable luma 1: Enable luma
denoise2d_motion_enable_shd	2:2	r	'h0	Enable/Disable motion shadow 0: Disable motion 1: Enable motion
denoise2d_enable_shd	0:0	r	'h0	Enable/Disable Denoise2D module shadow 0: Disable the 2DNR 1: Enable the 2DNR

ISP_DENOISE2D_SIGMA_SQR

0x00003ad0 = Byte Address

Description: Denoise2D Sigma square root Register

Sub-Field Name	Bits	rw	Reset	Description
denoise2d_sigma_sqr	11:0	rw	'h0	Sigma square root of filter

ISP_DENOISE2D_WEIGHT_MUL_FACTOR

0x00003ad4 = Byte Address

Description: Denoise2D sad Weight transfer factor Register

Sub-Field Name	Bits	rw	Reset	Description
denoise2d_weight_mul_factor	10:0	rw	'h400	SAD to weight calculation parameter

ISP_DENOISE2D_SIGMA_SQR_SHD

0x00003ad8 = Byte Address

Description: Denoise2D Sigma square root Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise2d_sigma_sqr_shd	11:0	r	'h0	Sigma square root of filter shadow

ISP_DENOISE2D_WEIGHT_MUL_FACTOR_SHD

0x00003adc = Byte Address

Description: Denoise2D sad Weight transfer factor Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise2d_weight_mul_factor_shd	10:0	r	'h400	SAD to weight calculation parameter shadow

ISP_DENOISE2D_DUMMY_HBLANK

0x00003ae0 = Byte Address

Description: Denoise2D Dummy hblank cycles Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise2d_dummy_hblank	15:0	rw	'h10	dummy lines hblank cycles

ISP_DENOISE3D2_BLENDING

0x00003aec = Byte Address

Description: Denoise3D2 blending params

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_nlm_strength_offset	29:19	rw	'h0	Denoise3D2: 2dnr strength offset
denoise3d_v20_nlm_strength_max	18:8	rw	'h0	Denoise3D2: 2dnr strength max value
denoise3d_v20_nlm_strength_slope	7:0	rw	'h0	Denoise3D2: 2dnr strength slope

ISP_DENOISE3D2_BLENDING_SHD

0x00003af0 = Byte Address

Description: Denoise3D2 blending params shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_nlm_strength_offset_shd	29:19	r	'h0	Denoise3D2: 2dnr strength offset shadow
denoise3d_v20_nlm_strength_max_shd	18:8	r	'h0	Denoise3D2: 2dnr strength max value shadow

denoise3d_v20_nlm_strength_slope_shd	7:0	r	'h0	Denoise3D2: 2dnr strength slope shadow

ISP_DENOISE2D_SIGMA_LUMA_LUT_0

0x00003f00 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_0	31:28	rw	'h0	Denoise filter sigma luma px0
denoise_sigma_luma_x_0	27:16	rw	'h0	Denoise filter sigma luma x0
denoise_sigma_luma_y_0_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y0
denoise_sigma_luma_y_0	13:0	rw	'h100	Denoise filter sigma luma y0

ISP_DENOISE2D_SIGMA_LUMA_LUT_1

0x00003f04 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_1	31:28	rw	'h8	Denoise filter sigma luma px1
denoise_sigma_luma_x_1	27:16	rw	'h0	Denoise filter sigma luma x1
denoise_sigma_luma_y_1_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y1
denoise_sigma_luma_y_1	13:0	rw	'h100	Denoise filter sigma luma y1

ISP_DENOISE2D_SIGMA_LUMA_LUT_2

0x00003f08 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_2	31:28	rw	'h0	Denoise filter sigma luma px2
denoise_sigma_luma_x_2	27:16	rw	'h0	Denoise filter sigma luma x2
denoise_sigma_luma_y_2_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y2
denoise_sigma_luma_y_2	13:0	rw	'h100	Denoise filter sigma luma y2

ISP_DENOISE2D_SIGMA_LUMA_LUT_3

0x00003f0c = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_3	31:28	rw	'h0	Denoise filter sigma luma px3
denoise_sigma_luma_x_3	27:16	rw	'h0	Denoise filter sigma luma x3

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denoise_sigma_luma_y_3_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y3
denoise_sigma_luma_y_3	13:0	rw	'h100	Denoise filter sigma luma y3

ISP_DENOISE2D_SIGMA_LUMA_LUT_4

0x00003f10 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_4	31:28	rw	'h8	Denoise filter sigma luma px4
denoise_sigma_luma_x_4	27:16	rw	'h0	Denoise filter sigma luma x4
denoise_sigma_luma_y_4_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y4
denoise_sigma_luma_y_4	13:0	rw	'h100	Denoise filter sigma luma y4

ISP_DENOISE2D_SIGMA_LUMA_LUT_5

0x00003f14 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_5	31:28	rw	'h0	Denoise filter sigma luma px5
denoise_sigma_luma_x_5	27:16	rw	'h0	Denoise filter sigma luma x5
denoise_sigma_luma_y_5_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y5
denoise_sigma_luma_y_5	13:0	rw	'h100	Denoise filter sigma luma y5

ISP_DENOISE2D_SIGMA_LUMA_LUT_6

0x00003f18 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_6	31:28	rw	'h0	Denoise filter sigma luma px6
denoise_sigma_luma_x_6	27:16	rw	'h0	Denoise filter sigma luma x6
denoise_sigma_luma_y_6_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y6
denoise_sigma_luma_y_6	13:0	rw	'h100	Denoise filter sigma luma y6

ISP_DENOISE2D_SIGMA_LUMA_LUT_7

0x00003f1c = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_7	31:28	rw	'h8	Denoise filter sigma luma px7

denoise_sigma_luma_x_7	27:16	rw	'h0	Denoise filter sigma luma x7
denoise_sigma_luma_y_7_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y7
denoise_sigma_luma_y_7	13:0	rw	'h100	Denoise filter sigma luma y7

ISP_DENOISE2D_SIGMA_LUMA_LUT_8

0x00003f20 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_8	31:28	rw	'h0	Denoise filter sigma luma px8
denoise_sigma_luma_x_8	27:16	rw	'h0	Denoise filter sigma luma x8
denoise_sigma_luma_y_8_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y8
denoise_sigma_luma_y_8	13:0	rw	'h100	Denoise filter sigma luma y8

ISP_DENOISE2D_SIGMA_LUMA_LUT_9

0x00003f24 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_9	31:28	rw	'h0	Denoise filter sigma luma px9
denoise_sigma_luma_x_9	27:16	rw	'h0	Denoise filter sigma luma x9
denoise_sigma_luma_y_9_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y9
denoise_sigma_luma_y_9	13:0	rw	'h100	Denoise filter sigma luma y9

ISP_DENOISE2D_SIGMA_LUMA_LUT_10

0x00003f28 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_10	31:28	rw	'h0	Denoise filter sigma luma px10
denoise_sigma_luma_x_10	27:16	rw	'h0	Denoise filter sigma luma x10
denoise_sigma_luma_y_10_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y10
denoise_sigma_luma_y_10	13:0	rw	'h100	Denoise filter sigma luma y10

ISP_DENOISE2D_SIGMA_LUMA_LUT_11

0x00003f2c = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

denoise_sigma_luma_px_11	31:28	rw	'h0	Denoise filter sigma luma px11
denoise_sigma_luma_x_11	27:16	rw	'h0	Denoise filter sigma luma x11
denoise_sigma_luma_y_11_h_fixed	15:14	rw	'h0	Denoise filter sigma luma y11
denoise_sigma_luma_y_11	13:0	rw	'h100	Denoise filter sigma luma y11

ISP_DENOISE2D_SIGMA_LUMA_LUT_0_SHD

0x00003f30 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_0	31:28	r	'h0	Denoise filter sigma luma px0
denoise_sigma_luma_x_0	27:16	r	'h0	Denoise filter sigma luma x0
denoise_sigma_luma_y_0_h_fixed	15:14	r	'h0	Denoise filter sigma luma y0
denoise_sigma_luma_y_0	13:0	r	'h100	Denoise filter sigma luma y0

ISP_DENOISE2D_SIGMA_LUMA_LUT_1_SHD

0x00003f34 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_1	31:28	r	'h8	Denoise filter sigma luma px1
denoise_sigma_luma_x_1	27:16	r	'h0	Denoise filter sigma luma x1
denoise_sigma_luma_y_1_h_fixed	15:14	r	'h0	Denoise filter sigma luma y1
denoise_sigma_luma_y_1	13:0	r	'h100	Denoise filter sigma luma y1

ISP_DENOISE2D_SIGMA_LUMA_LUT_2_SHD

0x00003f38 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_2	31:28	r	'h0	Denoise filter sigma luma px2
denoise_sigma_luma_x_2	27:16	r	'h0	Denoise filter sigma luma x2
denoise_sigma_luma_y_2_h_fixed	15:14	r	'h0	Denoise filter sigma luma y2
denoise_sigma_luma_y_2	13:0	r	'h100	Denoise filter sigma luma y2

ISP_DENOISE2D_SIGMA_LUMA_LUT_3_SHD

0x00003f3c = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

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Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_luma_px_3	31:28	r	'h0	Denoise filter sigma luma px3
denoise_sigma_luma_x_3	27:16	r	'h0	Denoise filter sigma luma x3
denoise_sigma_luma_y_3_h_fixed	15:14	r	'h0	Denoise filter sigma luma y3
denoise_sigma_luma_y_3	13:0	r	'h100	Denoise filter sigma luma y3

ISP_DENOISE2D_SIGMA_LUMA_LUT_4_SHD

0x00003f40 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_luma_px_4	31:28	r	'h8	Denoise filter sigma luma px4
denoise_sigma_luma_x_4	27:16	r	'h0	Denoise filter sigma luma x4
denoise_sigma_luma_y_4_h_fixed	15:14	r	'h0	Denoise filter sigma luma y4
denoise_sigma_luma_y_4	13:0	r	'h100	Denoise filter sigma luma y4

ISP_DENOISE2D_SIGMA_LUMA_LUT_5_SHD

0x00003f44 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_luma_px_5	31:28	r	'h0	Denoise filter sigma luma px5
denoise_sigma_luma_x_5	27:16	r	'h0	Denoise filter sigma luma x5
denoise_sigma_luma_y_5_h_fixed	15:14	r	'h0	Denoise filter sigma luma y5
denoise_sigma_luma_y_5	13:0	r	'h100	Denoise filter sigma luma y5

ISP_DENOISE2D_SIGMA_LUMA_LUT_6_SHD

0x00003f48 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

Sub-Field Name	Bits	rw	Reset	Description
denoise_sigma_luma_px_6	31:28	r	'h0	Denoise filter sigma luma px6
denoise_sigma_luma_x_6	27:16	r	'h0	Denoise filter sigma luma x6
denoise_sigma_luma_y_6_h_fixed	15:14	r	'h0	Denoise filter sigma luma y6
denoise_sigma_luma_y_6	13:0	r	'h100	Denoise filter sigma luma y6

ISP_DENOISE2D_SIGMA_LUMA_LUT_7_SHD

0x00003f4c = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_7	31:28	r	'h8	Denoise filter sigma luma px7
denoise_sigma_luma_x_7	27:16	r	'h0	Denoise filter sigma luma x7
denoise_sigma_luma_y_7_h_fixed	15:14	r	'h0	Denoise filter sigma luma y7
denoise_sigma_luma_y_7	13:0	r	'h100	Denoise filter sigma luma y7

ISP_DENOISE2D_SIGMA_LUMA_LUT_8_SHD

0x00003f50 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_8	31:28	r	'h0	Denoise filter sigma luma px8
denoise_sigma_luma_x_8	27:16	r	'h0	Denoise filter sigma luma x8
denoise_sigma_luma_y_8_h_fixed	15:14	r	'h0	Denoise filter sigma luma y8
denoise_sigma_luma_y_8	13:0	r	'h100	Denoise filter sigma luma y8

ISP_DENOISE2D_SIGMA_LUMA_LUT_9_SHD

0x00003f54 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_9	31:28	r	'h0	Denoise filter sigma luma px9
denoise_sigma_luma_x_9	27:16	r	'h0	Denoise filter sigma luma x9
denoise_sigma_luma_y_9_h_fixed	15:14	r	'h0	Denoise filter sigma luma y9
denoise_sigma_luma_y_9	13:0	r	'h100	Denoise filter sigma luma y9

ISP_DENOISE2D_SIGMA_LUMA_LUT_10_SHD

0x00003f58 = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_10	31:28	r	'h0	Denoise filter sigma luma px10
denoise_sigma_luma_x_10	27:16	r	'h0	Denoise filter sigma luma x10
denoise_sigma_luma_y_10_h_fixed	15:14	r	'h0	Denoise filter sigma luma y10
denoise_sigma_luma_y_10	13:0	r	'h100	Denoise filter sigma luma y10

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ISP_DENOISE2D_SIGMA_LUMA_LUT_11_SHD

0x00003f5c = Byte Address

Description: Denoise2D Filter Sigma Luma LUT Shadow Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise_sigma_luma_px_11	31:28	r	'h0	Denoise filter sigma luma px11
denoise_sigma_luma_x_11	27:16	r	'h0	Denoise filter sigma luma x11
denoise_sigma_luma_y_11_h_fixed	15:14	r	'h0	Denoise filter sigma luma y11
denoise_sigma_luma_y_11	13:0	r	'h100	Denoise filter sigma luma y11

ISP_HDR_HIST_PROP

0x00003b00 = Byte Address

Description: HDR Histogram pre-divider

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stepsize	9:3	rw	'h0	Histogram pre-divider Process every (step size)th pixel; all other pixels are skipped 0,1,2: not allowed 3: process every third input pixel 4: process every fourth input pixel ... 7FH: process every 127th pixel
hist_mode	2:0	rw	'h0	Histogram mode, luminance is taken at ISP output before output formatter, RGB is taken at xtalk output 0: disable, no measurements 1: RGB combined histogram 2: R histogram 3: G histogram 4: B histogram 5: Y (luminance) histogram 6, 7: must not be used

ISP_HDR_HIST_H_OFFS

0x00003b04 = Byte Address

Description: Histogram Window Horizontal Offset for First Window of 25 Sub-Windows

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_h_offs	12:0	rw	'h0	Horizontal offset of first window in pixels.

ISP_HDR_HIST_V_OFFS

0x00003b08 = Byte Address

Description: Histogram Window Vertical Offset for First Window of 25 Sub-Windows

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_v_offs	12:0	rw	'h0	Vertical offset of first window in pixels.

ISP_HDR_HIST_H_SIZE

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0x00003b0c = Byte Address

Description: Histogram Window Horizontal (Sub-) Window Size

Sub-Field Name	Bits	rw	Reset	Description
hist_h_size	10:0	rw	'h0	Horizontal size of one sub-window (in pixels)

ISP_HDR_HIST_V_SIZE

0x00003b10 = Byte Address

Description: Histogram Window Vertical (Sub-) Window Size

Sub-Field Name	Bits	rw	Reset	Description
hist_v_size	10:0	rw	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_0

0x00003b14 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 0

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_0_0	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_1

0x00003b18 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 1

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_0_1	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_2

0x00003b1c = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 2

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_0_2	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_3

0x00003b20 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 3

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_0_3	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

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ISP_HDR_HIST_0_BIN_4

0x00003b24 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 4

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_0_4	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_5

0x00003b28 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 5

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_0_5	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_6

0x00003b2c = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 6

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_0_6	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_7

0x00003b30 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 7

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_0_7	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_8

0x00003b34 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 8

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_0_8	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_9

0x00003b38 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 9

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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hist_bin_0_9	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_10

0x00003b3c = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 10

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_0_10	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_11

0x00003b40 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 11

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_0_11	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_12

0x00003b44 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 12

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_0_12	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_13

0x00003b48 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 13

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_0_13	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_14

0x00003b4c = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 14

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_0_14	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_0_BIN_15

0x00003b50 = Byte Address

Description: Exposure0 Histogram Measurement Result Bin 15

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_0_15	19:0	r	'h0	Exposure 0 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_0

0x00003b54 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 0

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_1_0	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_1

0x00003b58 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 1

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_1_1	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_2

0x00003b5c = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 2

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_1_2	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_3

0x00003b60 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 3

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_1_3	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_4

0x00003b64 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 4

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_1_4	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

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ISP_HDR_HIST_1_BIN_5

0x00003b68 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 5

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_1_5	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_6

0x00003b6c = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 6

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_1_6	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_7

0x00003b70 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 7

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_1_7	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_8

0x00003b74 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 8

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_1_8	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_9

0x00003b78 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 9

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_1_9	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_10

0x00003b7c = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 10

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

hist_bin_1_10	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_11

0x00003b80 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 11

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_1_11	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_12

0x00003b84 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 12

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_1_12	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_13

0x00003b88 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 13

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_1_13	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_14

0x00003b8c = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 14

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_1_14	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_1_BIN_15

0x00003b90 = Byte Address

Description: Exposure1 Histogram Measurement Result Bin 15

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_1_15	19:0	r	'h0	Exposure 1 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_0

0x00003b94 = Byte Address

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Description: Exposure2 Histogram Measurement Result Bin 0

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_2_0	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_1

0x00003b98 = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 1

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_2_1	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_2

0x00003b9c = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 2

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_2_2	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_3

0x00003ba0 = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 3

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_2_3	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_4

0x00003ba4 = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 4

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_2_4	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_5

0x00003ba8 = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 5

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_2_5	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

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ISP_HDR_HIST_2_BIN_6

0x00003bac = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 6

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_2_6	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_7

0x00003bb0 = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 7

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_2_7	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_8

0x00003bb4 = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 8

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_2_8	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_9

0x00003bb8 = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 9

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_2_9	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_10

0x00003bbc = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 10

Sub-Field Name	Bits	rw	Reset	Description
hist_bin_2_10	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_11

0x00003bc0 = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 11

Sub-Field Name	Bits	rw	Reset	Description

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hist_bin_2_11	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_12

0x00003bc4 = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 12

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_2_12	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_13

0x00003bc8 = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 13

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_2_13	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_14

0x00003bcc = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 14

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_2_14	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_HIST_2_BIN_15

0x00003bd0 = Byte Address

Description: Exposure2 Histogram Measurement Result Bin 15

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_2_15	19:0	r	'h0	Exposure 2 Measured bin count as 16-bit unsigned integer value plus 4 bit fractional part

ISP_HDR_EXP_CONF

0x00003c00 = Byte Address

Description: HDR Exposure Control

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_alt_mode	31:31	rw	'h0	0: luminance calculation according to Y=16+0.25R+0.5G+0.1094B 1: luminance calculation according to Y=(R+G+B) x 0.332
exp_src_select	2:2	rw	'h0	Data source select 0: data source is all 0 1: grey data

autostop	1:1	rw	'h0	0: continuous measurement 1: stop measuring after a complete frame
exp_ena	0:0	rw	'h0	1: start measuring a frame. If autostop is set to 1, the exposure block will reset this bit and halt after completing one frame.

ISP_HDR_EXP_H_OFFSET

0x00003c04 = Byte Address

Description: Horizontal Offset for First Block

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_h_offset	12:0	rw	'h0	Horizontal offset of first block in pixels.

ISP_HDR_EXP_V_OFFSET

0x00003c08 = Byte Address

Description: Vertical Offset for First Block

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_v_offset	12:0	rw	'h0	Vertical offset of first block in pixels.

ISP_HDR_EXP_H_SIZE

0x00003c0c = Byte Address

Description: Horizontal Size of One Block

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_h_size	10:0	rw	'h0	Horizontal size of one block in pixels.

ISP_HDR_EXP_V_SIZE

0x00003c10 = Byte Address

Description: Vertical Size of One Block

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_v_size	10:0	rw	'h0	Vertical size of one block in pixels (must be a multiple of 2). Note: The vertical size must be set in a way that after the last measurement window, At least two lines of the image will follow.

ISP_HDR_EXP_MEAN_0_00

0x00003c14 = Byte Address

Description: Exposure 0 Mean luminance value of block 00

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_00	7:0	r	'h0	Exposure 0 Mean luminance value of block 00 (x,y)

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ISP_HDR_EXP_MEAN_0_10

0x00003c18 = Byte Address

Description: Exposure 0 Mean luminance value of block 10 (x,y)

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_10	7:0	r	'h0	Exposure 0 Mean luminance value of block 10 (x,y)

ISP_HDR_EXP_MEAN_0_20

0x00003c1c = Byte Address

Description: Exposure 0 Mean luminance value of block 20

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_20	7:0	r	'h0	Exposure 0 Mean luminance value of block 20 (x,y)

ISP_HDR_EXP_MEAN_0_30

0x00003c20 = Byte Address

Description: Exposure 0 Mean luminance value of block 30

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_30	7:0	r	'h0	Exposure 0 Mean luminance value of block 30 (x,y)

ISP_HDR_EXP_MEAN_0_40

0x00003c24 = Byte Address

Description: Exposure 0 Mean luminance value of block 40

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_40	7:0	r	'h0	Exposure 0 Mean luminance value of block 40 (x,y)

ISP_HDR_EXP_MEAN_0_01

0x00003c28 = Byte Address

Description: Exposure 0 Mean luminance value of block 01

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_01	7:0	r	'h0	Exposure 0 Mean luminance value of block 01 (x,y)

ISP_HDR_EXP_MEAN_0_11

0x00003c2c = Byte Address

Description: Exposure 0 Mean luminance value of block 11

Sub-Field Name	Bits	rw	Reset	Description

exp_hdr_mean_11	7:0	r	'h0	Exposure 0 Mean luminance value of block 11 (x,y)

ISP_HDR_EXP_MEAN_0_21

0x00003c30 = Byte Address

Description: Exposure 0 Mean luminance value of block 21

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_21	7:0	r	'h0	Exposure 0 Mean luminance value of block 21 (x,y)

ISP_HDR_EXP_MEAN_0_31

0x00003c34 = Byte Address

Description: Exposure 0 Mean luminance value of block 31

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_31	7:0	r	'h0	Exposure 0 Mean luminance value of block 31 (x,y)

ISP_HDR_EXP_MEAN_0_41

0x00003c38 = Byte Address

Description: Exposure 0 Mean luminance value of block 41

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_41	7:0	r	'h0	Exposure 0 Mean luminance value of block 41 (x,y)

ISP_HDR_EXP_MEAN_0_02

0x00003c3c = Byte Address

Description: Exposure 0 Mean luminance value of block 02

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_02	7:0	r	'h0	Exposure 0 Mean luminance value of block 02 (x,y)

ISP_HDR_EXP_MEAN_0_12

0x00003c40 = Byte Address

Description: Exposure 0 Mean luminance value of block 12

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_12	7:0	r	'h0	Exposure 0 Mean luminance value of block 12 (x,y)

ISP_HDR_EXP_MEAN_0_22

0x00003c44 = Byte Address

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Description: Exposure 0 Mean luminance value of block 22

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_22	7:0	r	'h0	Exposure 0 Mean luminance value of block 22 (x,y)

ISP_HDR_EXP_MEAN_0_32

0x00003c48 = Byte Address

Description: Exposure 0 Mean luminance value of block 32

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_32	7:0	r	'h0	Exposure 0 Mean luminance value of block 32 (x,y)

ISP_HDR_EXP_MEAN_0_42

0x00003c4c = Byte Address

Description: Exposure 0 Mean luminance value of block 42

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_42	7:0	r	'h0	Exposure 0 Mean luminance value of block 42 (x,y)

ISP_HDR_EXP_MEAN_0_03

0x00003c50 = Byte Address

Description: Exposure 0 Mean luminance value of block 03

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_03	7:0	r	'h0	Exposure 0 Mean luminance value of block 03 (x,y)

ISP_HDR_EXP_MEAN_0_13

0x00003c54 = Byte Address

Description: Exposure 0 Mean luminance value of block 13

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_13	7:0	r	'h0	Exposure 0 Mean luminance value of block 13 (x,y)

ISP_HDR_EXP_MEAN_0_23

0x00003c58 = Byte Address

Description: Exposure 0 Mean luminance value of block 23

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_23	7:0	r	'h0	Exposure 0 Mean luminance value of block 23 (x,y)

ISP_HDR_EXP_MEAN_0_33

0x00003c5c = Byte Address

Description: Exposure 0 Mean luminance value of block 33

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_33	7:0	r	'h0	Exposure 0 Mean luminance value of block 33 (x,y)

ISP_HDR_EXP_MEAN_0_43

0x00003c60 = Byte Address

Description: Exposure 0 Mean luminance value of block 43

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_43	7:0	r	'h0	Exposure 0 Mean luminance value of block 43 (x,y)

ISP_HDR_EXP_MEAN_0_04

0x00003c64 = Byte Address

Description: Exposure 0 Mean luminance value of block 04

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_04	7:0	r	'h0	Exposure 0 Mean luminance value of block 04 (x,y)

ISP_HDR_EXP_MEAN_0_14

0x00003c68 = Byte Address

Description: Exposure 0 Mean luminance value of block 14

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_14	7:0	r	'h0	Exposure 0 Mean luminance value of block 14 (x,y)

ISP_HDR_EXP_MEAN_0_24

0x00003c6c = Byte Address

Description: Exposure 0 Mean luminance value of block 24

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_24	7:0	r	'h0	Exposure 0 Mean luminance value of block 24 (x,y)

ISP_HDR_EXP_MEAN_0_34

0x00003c70 = Byte Address

Description: Exposure 0 Mean luminance value of block 34

Sub-Field Name	Bits	rw	Reset	Description

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exp_hdr_mean_34	7:0	r	'h0	Exposure 0 Mean luminance value of block 34 (x,y)

ISP_HDR_EXP_MEAN_0_44

0x00003c74 = Byte Address

Description: Exposure 0 Mean luminance value of block 44

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_44	7:0	r	'h0	Exposure 0 Mean luminance value of block 44 (x,y)

ISP_HDR_EXP_MEAN_1_00

0x00003c78 = Byte Address

Description: Exposure 1 Mean luminance value of block 00

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_00	7:0	r	'h0	Exposure 1 Mean luminance value of block 00 (x,y)

ISP_HDR_EXP_MEAN_1_10

0x00003c7c = Byte Address

Description: Exposure 1 Mean luminance value of block 10

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_10	7:0	r	'h0	Exposure 1 Mean luminance value of block 10 (x,y)

ISP_HDR_EXP_MEAN_1_20

0x00003c80 = Byte Address

Description: Exposure 1 Mean luminance value of block 20

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_20	7:0	r	'h0	Exposure 1 Mean luminance value of block 20 (x,y)

ISP_HDR_EXP_MEAN_1_30

0x00003c84 = Byte Address

Description: Exposure 1 Mean luminance value of block 30

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_30	7:0	r	'h0	Exposure 1 Mean luminance value of block 30 (x,y)

ISP_HDR_EXP_MEAN_1_40

0x00003c88 = Byte Address

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Description: Exposure 1 Mean luminance value of block 40

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_40	7:0	r	'h0	Exposure 1 Mean luminance value of block 40 (x,y)

ISP_HDR_EXP_MEAN_1_01

0x00003c8c = Byte Address

Description: Exposure 1 Mean luminance value of block 01

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_01	7:0	r	'h0	Exposure 1 Mean luminance value of block 01 (x,y)

ISP_HDR_EXP_MEAN_1_11

0x00003c90 = Byte Address

Description: Exposure 1 Mean luminance value of block 11

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_11	7:0	r	'h0	Exposure 1 Mean luminance value of block 11 (x,y)

ISP_HDR_EXP_MEAN_1_21

0x00003c94 = Byte Address

Description: Exposure 1 Mean luminance value of block 21

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_21	7:0	r	'h0	Exposure 1 Mean luminance value of block 21 (x,y)

ISP_HDR_EXP_MEAN_1_31

0x00003c98 = Byte Address

Description: Exposure 1 Mean luminance value of block 31

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_31	7:0	r	'h0	Exposure 1 Mean luminance value of block 31 (x,y)

ISP_HDR_EXP_MEAN_1_41

0x00003c9c = Byte Address

Description: Exposure 1 Mean luminance value of block 41

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_41	7:0	r	'h0	Exposure 1 Mean luminance value of block 41 (x,y)

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ISP_HDR_EXP_MEAN_1_02

0x00003ca0 = Byte Address

Description: Exposure 1 Mean luminance value of block 02

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_02	7:0	r	'h0	Exposure 1 Mean luminance value of block 02 (x,y)

ISP_HDR_EXP_MEAN_1_12

0x00003ca4 = Byte Address

Description: Exposure 1 Mean luminance value of block 12

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_12	7:0	r	'h0	Exposure 1 Mean luminance value of block 12 (x,y)

ISP_HDR_EXP_MEAN_1_22

0x00003ca8 = Byte Address

Description: Exposure 1 Mean luminance value of block 22

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_22	7:0	r	'h0	Exposure 1 Mean luminance value of block 22 (x,y)

ISP_HDR_EXP_MEAN_1_32

0x00003cac = Byte Address

Description: Exposure 1 Mean luminance value of block 32

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_32	7:0	r	'h0	Exposure 1 Mean luminance value of block 32 (x,y)

ISP_HDR_EXP_MEAN_1_42

0x00003cb0 = Byte Address

Description: Exposure 1 Mean luminance value of block 42

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_42	7:0	r	'h0	Exposure 1 Mean luminance value of block 42 (x,y)

ISP_HDR_EXP_MEAN_1_03

0x00003cb4 = Byte Address

Description: Exposure 1 Mean luminance value of block 03

Sub-Field Name	Bits	rw	Reset	Description

exp_hdr_mean_03	7:0	r	'h0	Exposure 1 Mean luminance value of block 03 (x,y)

ISP_HDR_EXP_MEAN_1_13

0x00003cb8 = Byte Address

Description: Exposure 1 Mean luminance value of block 13

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_13	7:0	r	'h0	Exposure 1 Mean luminance value of block 13 (x,y)

ISP_HDR_EXP_MEAN_1_23

0x00003cbc = Byte Address

Description: Exposure 1 Mean luminance value of block 23

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_23	7:0	r	'h0	Exposure 1 Mean luminance value of block 23 (x,y)

ISP_HDR_EXP_MEAN_1_33

0x00003cc0 = Byte Address

Description: Exposure 1 Mean luminance value of block 33

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_33	7:0	r	'h0	Exposure 1 Mean luminance value of block 33 (x,y)

ISP_HDR_EXP_MEAN_1_43

0x00003cc4 = Byte Address

Description: Exposure 1 Mean luminance value of block 43

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_43	7:0	r	'h0	Exposure 1 Mean luminance value of block 43 (x,y)

ISP_HDR_EXP_MEAN_1_04

0x00003cc8 = Byte Address

Description: Exposure 1 Mean luminance value of block 04

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_04	7:0	r	'h0	Exposure 1 Mean luminance value of block 04 (x,y)

ISP_HDR_EXP_MEAN_1_14

0x00003ccc = Byte Address

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Description: Exposure 1 Mean luminance value of block 14

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_14	7:0	r	'h0	Exposure 1 Mean luminance value of block 14 (x,y)

ISP_HDR_EXP_MEAN_1_24

0x00003cd0 = Byte Address

Description: Exposure 1 Mean luminance value of block 24

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_24	7:0	r	'h0	Exposure 1 Mean luminance value of block 24 (x,y)

ISP_HDR_EXP_MEAN_1_34

0x00003cd4 = Byte Address

Description: Exposure 1 Mean luminance value of block 34

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_34	7:0	r	'h0	Exposure 1 Mean luminance value of block 34 (x,y)

ISP_HDR_EXP_MEAN_1_44

0x00003cd8 = Byte Address

Description: Exposure 1 Mean luminance value of block 44

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_44	7:0	r	'h0	Exposure 1 Mean luminance value of block 44 (x,y)

ISP_HDR_EXP_MEAN_2_00

0x00003cdc = Byte Address

Description: Exposure 2 Mean luminance value of block 00

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_00	7:0	r	'h0	Exposure 2 Mean luminance value of block 00 (x,y)

ISP_HDR_EXP_MEAN_2_10

0x00003ce0 = Byte Address

Description: Exposure 2 Mean luminance value of block 10

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_10	7:0	r	'h0	Exposure 2 Mean luminance value of block 10 (x,y)

ISP_HDR_EXP_MEAN_2_20

0x00003ce4 = Byte Address

Description: Exposure 2 Mean luminance value of block 20

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_20	7:0	r	'h0	Exposure 2 Mean luminance value of block 20 (x,y)

ISP_HDR_EXP_MEAN_2_30

0x00003ce8 = Byte Address

Description: Exposure 2 Mean luminance value of block 30

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_30	7:0	r	'h0	Exposure 2 Mean luminance value of block 30 (x,y)

ISP_HDR_EXP_MEAN_2_40

0x00003cec = Byte Address

Description: Exposure 2 Mean luminance value of block 40

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_40	7:0	r	'h0	Exposure 2 Mean luminance value of block 40 (x,y)

ISP_HDR_EXP_MEAN_2_01

0x00003cf0 = Byte Address

Description: Exposure 2 Mean luminance value of block 01

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_01	7:0	r	'h0	Exposure 2 Mean luminance value of block 01 (x,y)

ISP_HDR_EXP_MEAN_2_11

0x00003cf4 = Byte Address

Description: Exposure 2 Mean luminance value of block 11

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_11	7:0	r	'h0	Exposure 2 Mean luminance value of block 11 (x,y)

ISP_HDR_EXP_MEAN_2_21

0x00003cf8 = Byte Address

Description: Exposure 2 Mean luminance value of block 21

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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exp_hdr_mean_21	7:0	r	'h0	Exposure 2 Mean luminance value of block 21 (x,y)

ISP_HDR_EXP_MEAN_2_31

0x00003cfc = Byte Address

Description: Exposure 2 Mean luminance value of block 31

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_31	7:0	r	'h0	Exposure 2 Mean luminance value of block 31 (x,y)

ISP_HDR_EXP_MEAN_2_41

0x00003d00 = Byte Address

Description: Exposure 2 Mean luminance value of block 41

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_41	7:0	r	'h0	Exposure 2 Mean luminance value of block 41 (x,y)

ISP_HDR_EXP_MEAN_2_02

0x00003d04 = Byte Address

Description: Exposure 2 Mean luminance value of block 02

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_02	7:0	r	'h0	Exposure 2 Mean luminance value of block 02 (x,y)

ISP_HDR_EXP_MEAN_2_12

0x00003d08 = Byte Address

Description: Exposure 2 Mean luminance value of block 12

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_12	7:0	r	'h0	Exposure 2 Mean luminance value of block 12 (x,y)

ISP_HDR_EXP_MEAN_2_22

0x00003d0c = Byte Address

Description: Exposure 2 Mean luminance value of block 22

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
exp_hdr_mean_22	7:0	r	'h0	Exposure 2 Mean luminance value of block 22 (x,y)

ISP_HDR_EXP_MEAN_2_32

0x00003d10 = Byte Address

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Description: Exposure 2 Mean luminance value of block 32

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_32	7:0	r	'h0	Exposure 2 Mean luminance value of block 32 (x,y)

ISP_HDR_EXP_MEAN_2_42

0x00003d14 = Byte Address

Description: Exposure 2 Mean luminance value of block 42

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_42	7:0	r	'h0	Exposure 2 Mean luminance value of block 42 (x,y)

ISP_HDR_EXP_MEAN_2_03

0x00003d18 = Byte Address

Description: Exposure 2 Mean luminance value of block 03

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_03	7:0	r	'h0	Exposure 2 Mean luminance value of block 03

ISP_HDR_EXP_MEAN_2_13

0x00003d1c = Byte Address

Description: Exposure 2 Mean luminance value of block 13

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_13	7:0	r	'h0	Exposure 2 Mean luminance value of block 13 (x,y)

ISP_HDR_EXP_MEAN_2_23

0x00003d20 = Byte Address

Description: Exposure 2 Mean luminance value of block 23

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_23	7:0	r	'h0	Exposure 2 Mean luminance value of block 23 (x,y)

ISP_HDR_EXP_MEAN_2_33

0x00003d24 = Byte Address

Description: Exposure 2 Mean luminance value of block 33

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_33	7:0	r	'h0	Exposure 2 Mean luminance value of block 33 (x,y)

ISP_HDR_EXP_MEAN_2_43

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0x00003d28 = Byte Address

Description: Exposure 2 Mean luminance value of block 43

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_43	7:0	r	'h0	Exposure 2 Mean luminance value of block 43 (x,y)

ISP_HDR_EXP_MEAN_2_04

0x00003d2c = Byte Address

Description: Exposure 2 Mean luminance value of block 04

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_04	7:0	r	'h0	Exposure 2 Mean luminance value of block 04 (x,y)

ISP_HDR_EXP_MEAN_2_14

0x00003d30 = Byte Address

Description: Exposure 2 Mean luminance value of block 14

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_14	7:0	r	'h0	Exposure 2 Mean luminance value of block 14 (x,y)

ISP_HDR_EXP_MEAN_2_24

0x00003d34 = Byte Address

Description: Exposure 2 Mean luminance value of block 24

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_24	7:0	r	'h0	Exposure 2 Mean luminance value of block 24 (x,y)

ISP_HDR_EXP_MEAN_2_34

0x00003d38 = Byte Address

Description: Exposure 2 Mean luminance value of block 34

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_34	7:0	r	'h0	Exposure 2 Mean luminance value of block 34 (x,y)

ISP_HDR_EXP_MEAN_2_44

0x00003d3c = Byte Address

Description: Exposure 2 Mean luminance value of block 44

Sub-Field Name	Bits	rw	Reset	Description
exp_hdr_mean_44	7:0	r	'h0	Exposure 2 Mean luminance value of block 44 (x,y)

ISP_HDR_EXP_H_OFFSET_SHD

0x00003d40 = Byte Address

Description: Horizontal offset shadow

Sub-Field Name	Bits	rw	Reset	Description
exp_h_offset	12:0	r	'h0	Shadow Horizontal offset of first block in pixels.

ISP_HDR_EXP_V_OFFSET_SHD

0x00003d44 = Byte Address

Description: Vertical offset shadow

Sub-Field Name	Bits	rw	Reset	Description
exp_v_offset	12:0	r	'h0	Shadow Vertical offset of first block in pixels.

ISP_HDR_EXP_H_SIZE_SHD

0x00003d48 = Byte Address

Description: Horizontal size shadow

Sub-Field Name	Bits	rw	Reset	Description
exp_h_size	10:0	r	'h0	Shadow Horizontal size of one block in pixels.

ISP_HDR_EXP_V_SIZE_SHD

0x00003d4c = Byte Address

Description: Vertical size shadow

Sub-Field Name	Bits	rw	Reset	Description
exp_v_size	10:0	r	'h0	Shadow Vertical size of one block in pixels (must be a multiple of 2). Note: The vertical size must be set in a way that after the last measurement window, At least two lines of the image will follow.

ISP_REG_FE_CTRL

0x00003d60 = Byte Address

Description: Fe ctrl

Sub-Field Name	Bits	rw	Reset	Description
ahb_wr_with_fe_en	1:1	rw	'h0	Control whether ahb write is allowed during reg_fe_cfg_sel on 1'b0: disable 1'b1: enable
reg_fe_cfg_sel	0:0	rw	'h0	1'b0: use the legacy cfg bus, i.e., AHB bus. 1'b1: use command buffer. Note: this bit should be set to 1'b1 before start command buffer by isp_reg_fe_dma_start

ISP_REG_FE_DMA_START

0x00003d64 = Byte Address

Description: fe dma start

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_start	16:16	w	'h0	Set to 1' b1 to trigger fe(command buffer) to work
reg_fe_dma_count_qw	15:0	w	'h0	Indicate the count of buffer by unit of 8 bytes(QW)

ISP_REG_FE_DMA_AD

0x00003d68 = Byte Address

Description: fe dma address

Sub-Field Name	Bits	rw	Reset	Description
reg_fe_dma_ad	30:4	w	'h0	command buffer address base in memory range

ISP_REG_FE_IMSC

0x00003d6c = Byte Address

Description: fe interrupt mask enable

Sub-Field Name	Bits	rw	Reset	Description
reserved0	31:6	rw	'h0	reserved0
mcm_hw_batch_end	5:5	rw	'h0	mask bit for mcm_hw batch completion
reg_fe_frame_cfg_cmpl	4:4	rw	'h0	Mask bit for frame configuration completion interrupt
reg_fe_stall_wait	3:3	rw	'h0	Mask bit for stall command interrupt
reg_fe_addr_intervene	2:2	rw	'h0	Mask bit for unexpected AHB address access interrupt during FE configuration
reg_fe_isp_val_intervene	1:1	rw	'h0	Mask bit for unexpected valid data interrupt at ISP input during FE configuration
reg_fe_cfg_end	0:0	rw	'h0	mask bit for FE configuration completion

ISP_REG_FE_ISR

0x00003d70 = Byte Address

Description: fe interrupt set register

Sub-Field Name	Bits	rw	Reset	Description
mcm_hw_batch_end	5:5	w	'h0	set interrupt for mcm_hw batch completion
reg_fe_frame_cfg_cmpl	4:4	w	'h0	set interrupt for frame configuration completion interrupt
reg_fe_stall_wait	3:3	w	'h0	set interrupt for stall command interrupt

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reg_fe_addr_intervene	2:2	w	'h0	Set interrupt of unexpected AHB address access interrupt during FE configuration
reg_fe_isp_val_intervene	1:1	w	'h0	Set interrupt of unexpected valid data interrupt at ISP input during FE configuration
reg_fe_cfg_end	0:0	w	'h0	Set interrupt of FE configuration completion

ISP_REG_FE_MIS

0x00003d74 = Byte Address

Description: fe interrupt masked status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_hw_batch_end	5:5	r	'h0	masked status for mcm_hw batch completion
reg_fe_frame_cfg_cmpl	4:4	r	'h0	Masked status for frame configuration completion interrupt
reg_fe_stall_wait	3:3	r	'h0	Masked status for stall command interrupt
reg_fe_addr_intervene	2:2	r	'h0	Masked status of unexpected AHB address access interrupt during FE configuration
reg_fe_isp_val_intervene	1:1	r	'h0	Masked status of unexpected valid data interrupt at ISP input during FE configuration
reg_fe_cfg_end	0:0	r	'h0	Masked status of FE configuration completion

ISP_REG_FE_ICR

0x00003d78 = Byte Address

Description: fe interrupt clear register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_hw_batch_end	5:5	w	'h0	clear bit for mcm_hw batch completion
reg_fe_frame_cfg_cmpl	4:4	w	'h0	clear for frame configuration completion interrupt
reg_fe_stall_wait	3:3	w	'h0	clear bit for stall command interrupt
reg_fe_addr_intervene	2:2	w	'h0	Clear unexpected AHB address access interrupt during FE configuration
reg_fe_isp_val_intervene	1:1	w	'h0	Clear unexpected valid data interrupt at ISP input during FE configuration
reg_fe_cfg_end	0:0	w	'h0	Clear FE configuration completion

ISP_REG_FE_RIS

0x00003d7c = Byte Address

Description: fe interrupt raw status

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_hw_batch_end	5:5	r	'h0	raw status for mcm_hw batch completion

reg_fe_frame_cfg_cmpl	4:4	r	'h0	raw status for frame configuration completion interrupt
reg_fe_stall_wait	3:3	r	'h0	raw status for stall command interrupt
reg_fe_addr_intervene	2:2	r	'h0	RAW status of unexpected AHB address access interrupt during FE configuration
reg_fe_isp_val_intervene	1:1	r	'h0	RAW status of unexpected valid data interrupt at ISP input during FE configuration
reg_fe_cfg_end	0:0	r	'h0	RAW status of FE configuration completion

ISP_REG_FE_ADDR_INTERVENE

0x00003d80 = Byte Address

Description: fe abnormal intervene address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_addr_intervene	15:0	r	'h0	The address for unexpected AHB address access interrupt during FE configuration

ISP_REG_FE_DEBUG_SELECT

0x00003d84 = Byte Address

Description: fe debug signal

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_debug_select	7:0	rw	'h0	Select signal for debug signal within FE

ISP_REG_FE_DEBUG_SIGNALS

0x00003d88 = Byte Address

Description: fe debug signal

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
reg_fe_debug_signals	31:0	r	'h0	debug signal within FE, and default value is only for register bash test, without practical meaning

ISP_GCRGB_R_PX_0

0x00000900 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_r_px_5	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_4	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_3	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_2	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_1	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<

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rgb_gc_r_px_0	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_PX_1

0x00000904 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_r_px11	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_10	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_9	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_8	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_7	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_6	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_PX_2

0x00000908 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_r_px_17	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_16	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_15	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_14	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_13	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_12	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_PX_3

0x0000090c = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_r_px_23	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_22	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_21	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_20	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_19	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_18	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_PX_4

0x00000910 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

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Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_r_px_29	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_28	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_27	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_26	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_25	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_24	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_PX_5

0x00000914 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_r_px_35	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_34	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_33	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_32	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_31	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_30	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_PX_6

0x00000918 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_r_px_41	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_40	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_39	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_38	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_37	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_36	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_PX_7

0x0000091c = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_r_px_47	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_46	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_45	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<

rgb_gc_r_px_44	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_43	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_42	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_PX_8

0x00000920 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_r_px_53	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_52	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_51	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_50	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_49	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_48	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_PX_9

0x00000924 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_r_px_59	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_58	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_57	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_56	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_55	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_54	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_PX_10

0x00000928 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_r_px_63	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_62	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_61	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_r_px_60	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_Y_ADDR

0x0000092c = Byte Address

Description: Address of y memory in gamma curve

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_r_y_addr	5:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_R_Y_WRITE_DATA

0x00000930 = Byte Address

Description: Y data in gamma curve

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_r_y_write_data	31:0	rw	'h0	Y data in gamma curve

ISP_GCRGB_R_X_ADDR

0x00000934 = Byte Address

Description: Address of x memory in gamma curve

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_r_x_addr	5:0	rw	'h0	Address of x memory in gamma curve

ISP_GCRGB_R_X_WRITE_DATA

0x00000938 = Byte Address

Description: X data in gamma curve

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_r_x_write_data	31:0	rw	'h0	X data in gamma curve

ISP_GCRGB_G_PX_0

0x00000940 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_g_px_5	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_4	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_3	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_2	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_1	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_0	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_PX_1

0x00000944 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_g_px11	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_10	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_9	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<

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rgb_gc_g_px_8	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_7	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_6	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_PX_2

0x00000948 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_g_px_17	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_16	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_15	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_14	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_13	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_12	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_PX_3

0x0000094c = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_g_px_23	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_22	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_21	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_20	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_19	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_18	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_PX_4

0x00000950 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_g_px_29	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_28	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_27	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_26	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_25	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_24	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_PX_5

0x00000954 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_g_px_35	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_34	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_33	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_32	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_31	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_30	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_PX_6

0x00000958 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_g_px_41	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_40	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_39	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_38	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_37	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_36	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_PX_7

0x0000095c = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_g_px_47	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_46	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_45	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_44	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_43	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_42	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_PX_8

0x00000960 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_g_px_53	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_52	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<

rgb_gc_g_px_51	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_50	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_49	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_48	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_PX_9

0x00000964 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_g_px_59	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_58	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_57	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_56	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_55	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_54	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_PX_10

0x00000968 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_g_px_63	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_62	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_61	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_g_px_60	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_Y_ADDR

0x0000096c = Byte Address

Description: Address of y memory in gamma curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_g_y_addr	5:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_G_Y_WRITE_DATA

0x00000970 = Byte Address

Description: Y data in gamma curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_g_y_write_data	31:0	rw	'h0	Y data in gamma curve

ISP_GCRGB_G_X_ADDR

0x00000974 = Byte Address

Description: Address of x memory in gamma curve

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_g_x_addr	5:0	rw	'h0	Address of x memory in gamma curve

ISP_GCRGB_G_X_WRITE_DATA

0x00000978 = Byte Address

Description: X data in gamma curve

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_g_x_write_data	31:0	rw	'h0	X data in gamma curve

ISP_GCRGB_B_PX_0

0x00000980 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_b_px_5	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_4	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_3	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_2	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_1	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_0	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_B_PX_1

0x00000984 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_b_px11	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_10	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_9	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_8	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_7	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_6	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_B_PX_2

0x00000988 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_b_px_17	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<

rgb_gc_b_px_16	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_15	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_14	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_13	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_12	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_B_PX_3

0x0000098c = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_b_px_23	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_22	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_21	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_20	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_19	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_18	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_B_PX_4

0x00000990 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_b_px_29	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_28	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_27	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_26	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_25	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_24	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_B_PX_5

0x00000994 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_b_px_35	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_34	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_33	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_32	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_31	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<

rgb_gc_b_px_30	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_B_PX_6

0x00000998 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_b_px_41	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_40	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_39	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_38	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_37	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_36	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_B_PX_7

0x0000099c = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_b_px_47	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_46	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_45	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_44	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_43	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_42	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_B_PX_8

0x000009a0 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
rgb_gc_b_px_53	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_52	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_51	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_50	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_49	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_48	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_B_PX_9

0x000009a4 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

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Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_b_px_59	29:25	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_58	24:20	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_57	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_56	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_55	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_54	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_B_PX_10

0x000009a8 = Byte Address

Description: Distance of x in gamma curve (distance_n=1<

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_b_px_63	19:15	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_62	14:10	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_61	9:5	rw	'h0	Distance of x in gamma curve (distance_n=1<
rgb_gc_b_px_60	4:0	rw	'h0	Distance of x in gamma curve (distance_n=1<

ISP_GCRGB_B_Y_ADDR

0x000009ac = Byte Address

Description: Address of y memory in gamma curve

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_b_y_addr	5:0	rw	'h0	Address of y memory in gamma curve

ISP_GCRGB_B_Y_WRITE_DATA

0x000009b0 = Byte Address

Description: Y data in gamma curve

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_b_y_write_data	31:0	rw	'h0	Y data in gamma curve

ISP_GCRGB_B_X_ADDR

0x000009b4 = Byte Address

Description: Address of x memory in gamma curve

Sub-Field Name	Bits	rw	Reset	Description
rgb_gc_b_x_addr	5:0	rw	'h0	Address of x memory in gamma curve

ISP_GCRGB_B_X_WRITE_DATA

0x000009b8 = Byte Address

Description: X data in gamma curve

Sub-Field Name	Bits	rw	Reset	Description

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rgb_gc_b_x_write_data	31:0	rw	'h0	X data in gamma curve

ISP_HIST256_PROP

0x00002100 = Byte Address

Description: Histogram properties

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stepsize	9:3	rw	'h0	Histogram pre-divider Process every (step size)th pixel; all other pixels are skipped 0,1,2: not allowed 3: process every third input pixel 4: process every fourth input pixel ... 7FH: process every 127th pixel
hist_mode	2:0	rw	'h0	Histogram mode, luminance is taken at ISP output before output formatter, RGB is taken at xtalk output 0: disable, no measurements 1: reserved 2: R histogram 3: G histogram 4: B histogram 5: Y (luminance) histogram 6, 7: must not be used

ISP_HIST256_H_OFFS

0x00002104 = Byte Address

Description: Histogram window horizontal offset for first window

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_h_offs_1	15:13	rw	'h0	Horizontal offset of first window in pixels.
hist_h_offs	12:0	rw	'h0	Horizontal offset of first window in pixels.

ISP_HIST256_V_OFFS

0x00002108 = Byte Address

Description: Histogram window vertical offset for first window

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_v_offs_1	15:13	rw	'h0	Vertical offset of first window in pixels.
hist_v_offs	12:0	rw	'h0	Vertical offset of first window in pixels.

ISP_HIST256_H_SIZE

0x0000210c = Byte Address

Description: Histogram Window Horizontal (Sub-) Window Size

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_h_size_1	15:11	rw	'h0	Horizontal size of one sub-window (in pixels)
hist_h_size	10:0	rw	'h0	Horizontal size of one sub-window (in pixels)

ISP_HIST256_V_SIZE

0x00002110 = Byte Address

Description: Histogram Window Vertical (Sub-)Window Size

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_v_size_1	15:11	rw	'h0	Vertical size of one sub-window (in lines)
hist_v_size	10:0	rw	'h0	Vertical size of one sub-window (in lines)

ISP_HIST256_BIN_N

0x00002114 = Byte Address

Description: Histogram Measurement Result Bin data

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_bin_n	29:0	r	'h0	Measured bin count as 26-bit unsigned integer value plus 4 bit fractional part

ISP_HIST256_WEIGHT_00TO30

0x00002154 = Byte Address

Description: Weighting Factor for Sub-Windows 00/10/20/30

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_weight_00	4:0	rw	'h10	Weighting factor for sub-window 00
hist_weight_10	12:8	rw	'h10	Weighting factor for sub-window 10
hist_weight_20	20:16	rw	'h10	Weighting factor for sub-window 20
hist_weight_30	28:24	rw	'h10	Weighting factor for sub-window 30

ISP_HIST256_WEIGHT_40TO21

0x00002158 = Byte Address

Description: Weighting Factor for Sub-Windows 40/01/11/21

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_weight_40	4:0	rw	'h10	Weighting factor for sub-window 30
hist_weight_01	12:8	rw	'h10	Weighting factor for sub-window 01
hist_weight_11	20:16	rw	'h10	Weighting factor for sub-window 11
hist_weight_21	28:24	rw	'h10	Weighting factor for sub-window 21

ISP_HIST256_WEIGHT_31TO12

0x0000215c = Byte Address

Description: Weighting Factor for Sub-Windows 31/41/02/12

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_weight_31	4:0	rw	'h10	Weighting factor for sub-window 31
hist_weight_41	12:8	rw	'h10	Weighting factor for sub-window 41
hist_weight_02	20:16	rw	'h10	Weighting factor for sub-window 02

hist_weight_12	28:24	rw	'h10	Weighting factor for sub-window 12

ISP_HIST256_WEIGHT_22TO03

0x00002160 = Byte Address

Description: Weighting Factor for Sub-Windows 22/32/42/03

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_weight_22	4:0	rw	'h10	Weighting factor for sub-window 22
hist_weight_32	12:8	rw	'h10	Weighting factor for sub-window 32
hist_weight_42	20:16	rw	'h10	Weighting factor for sub-window 42
hist_weight_03	28:24	rw	'h10	Weighting factor for sub-window 03

ISP_HIST256_WEIGHT_13TO43

0x00002164 = Byte Address

Description: Weighting Factor for Sub-Windows 03/23/33/43

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_weight_13	4:0	rw	'h10	Weighting factor for sub-window 13
hist_weight_23	12:8	rw	'h10	Weighting factor for sub-window 23
hist_weight_33	20:16	rw	'h10	Weighting factor for sub-window 33
hist_weight_43	28:24	rw	'h10	Weighting factor for sub-window 43

ISP_HIST256_WEIGHT_04TO34

0x00002168 = Byte Address

Description: Weighting Factor for Sub-Windows 04/14/24/34

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_weight_04	4:0	rw	'h10	Weighting factor for sub-window 04
hist_weight_14	12:8	rw	'h10	Weighting factor for sub-window 14
hist_weight_24	20:16	rw	'h10	Weighting factor for sub-window 24
hist_weight_34	28:24	rw	'h10	Weighting factor for sub-window 34

ISP_HIST256_WEIGHT_44

0x0000216c = Byte Address

Description: Weighting Factor for Sub-Window 44

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hist_weight_44	4:0	rw	'h10	Weighting factor for sub-window 44

MI_MCM_G2_CTRL

0x00005000 = Byte Address

Description: MCM Path MI control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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mcm_wr1_cfg_upd	5:5	w	'h0	Follow mi_mcm_ctrl
mcm_init_offset_en	4:4	rw	'h0	Follow mi_mcm_ctrl
mcm_init_base_en	3:3	rw	'h0	Follow mi_mcm_ctrl
mcm_wr0_cfg_upd	2:2	w	'h0	Follow mi_mcm_ctrl
mcm_wr_skip	1:1	w	'h0	Follow mi_mcm_ctrl
mcm_wr_auto_update	0:0	rw	'h0	Follow mi_mcm_ctrl

MI_MCM_G2_FMT

0x00005004 = Byte Address

Description: MCM pixel format control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_wr1_fmt_aligned	17:16	rw	'h0	Follow mi_mcm_fmt
mcm_wr0_fmt_aligned	15:14	rw	'h0	Follow mi_mcm_fmt
mcm_wr1_raw_bit	11:8	rw	'h0	Follow mi_mcm_fmt
mcm_wr0_raw_bit	7:4	rw	'h0	Follow mi_mcm_fmt

MI_MCM_G2_BUS_CFG

0x00005008 = Byte Address

Description: MCM AXI Bus cfg

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_wr0_swap_raw	15:12	rw	'h0	Follow mi_mcm_bus_cfg
mcm_wr1_swap_raw	11:8	rw	'h0	Follow mi_mcm_bus_cfg
mcm_wr_issue_cap	5:4	rw	'h0	Follow mi_mcm_bus_cfg
mcm_wr_burst_len	1:0	rw	'h0	Follow mi_mcm_bus_cfg

MI_MCM_G2_RAW0_BASE_AD_INIT

0x0000500c = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_base_ad_init	31:4	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW0_SIZE_INIT

0x00005010 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_size_init	28:4	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW0_OFFS_CNT_INIT

0x00005014 = Byte Address

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Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw0_offs_cnt_init	28:4	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW0_LLENGTH

0x00005018 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw0_llengh	31:0	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW0_PIC_WIDTH

0x0000501c = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw0_pic_width	31:0	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW0_PIC_HEIGHT

0x00005020 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw0_pic_height	31:0	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW0_PIC_SIZE

0x00005024 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw0_pic_size	31:0	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW0_OFFS_CNT_START

0x00005028 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw0_offs_cnt_start	28:4	r	'h0	Follow mi_mcm

MI_MCM_G2_RAW0_BASE_AD_SHD

0x0000502c = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw0_base_ad	31:4	r	'h0	Follow mi_mcm

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MI_MCM_G2_RAW0_SIZE_SHD

0x00005030 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_size	28:4	r	'h0	Follow mi_mcm

MI_MCM_G2_RAW0_OFFS_CNT_SHD

0x00005034 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_offs_cnt	28:4	r	'h0	Follow mi_mcm

MI_MCM_G2_RAW1_BASE_AD_INIT

0x00005038 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_base_ad_init	31:4	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW1_SIZE_INIT

0x0000503c = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_size_init	28:4	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW1_OFFS_CNT_INIT

0x00005040 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_offs_cnt_init	28:4	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW1_LLENGTH

0x00005044 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_llengh	31:0	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW1_PIC_WIDTH

0x00005048 = Byte Address

Description: Follow Main Path RAW

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Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_pic_width	31:0	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW1_PIC_HEIGHT

0x0000504c = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_pic_height	31:0	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW1_PIC_SIZE

0x00005050 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_pic_size	31:0	rw	'h0	Follow mi_mcm

MI_MCM_G2_RAW1_OFFS_CNT_START

0x00005054 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_offs_cnt_start	28:4	r	'h0	Follow mi_mcm

MI_MCM_G2_RAW1_BASE_AD_SHD

0x00005058 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_base_ad	31:4	r	'h0	Follow mi_mcm

MI_MCM_G2_RAW1_SIZE_SHD

0x0000505c = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_size	28:4	r	'h0	Follow mi_mcm

MI_MCM_G2_RAW1_OFFS_CNT_SHD

0x00005060 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw1_offs_cnt	28:4	r	'h0	Follow mi_mcm

ISP_DENOISE3D2_CTRL

0x00005300 = Byte Address

Description: Denoise3D2 ctrl

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_dither_en	15:15	rw	'h0	NRV4 dithering enable bit
denoise3d_gauss_blur_en	14:14	rw	'h0	NRV4 gaussian blur enable bit
denoise3d_motion_smooth_lvl	13:13	rw	'h0	NRV4 motion smoothing level bit
denoise3d_v20_soft_rst	12:12	rw	'h0	Denoise3D2:soft reset bit
denoise3d_v20_invgamma_en	11:11	rw	'h0	Denoise3D2: inv gamma enable bit
denoise3d_v20_pregamma_en	10:10	rw	'h0	Denoise3D2: pregamma enable bit
denoise3d_v20_erode_en	9:9	rw	'h0	Denoise3D2: motion erosion enable bit
denoise3d_v20_motion_conv_en	8:8	rw	'h0	Denoise3D2: motion convergence enable bit
denoise3d_v20_inv_dgain_en	7:7	rw	'h0	Denoise3D2: inv digital gain enable bit
denoise3d_v20_inv_awb_gain_en	6:6	rw	'h0	Denoise3D2: inv awb gain enable bit
denoise3d_v20_comp_luma_en	5:5	rw	'h0	Denoise3D2: luma compensation enable bit
denoise3d_v20_ref_reset	4:4	rw	'h0	Denoise3D2: soft reset bit
denoise3d_v20_motion_dilate_enable	3:3	rw	'h0	Denoise3D2: motion dilation enable bit
denoise3d_v20_nlm_enable	2:2	rw	'h0	Denoise3D2: 2dnr enable bit
denoise3d_v20_tnr_enable	1:1	rw	'h0	Denoise3D2: tnr enable bit
denoise3d_v20_enable	0:0	rw	'h0	Denoise3D2: enable bit

ISP_DENOISE3D2_STRENGTH

0x00005304 = Byte Address

Description: Denoise3D2 strength

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v40_tnr_strength2	15:8	rw	'h0	Denoise3D2: tnr strength2
denoise3d_v20_tnr_strength	7:0	rw	'h0	Denoise3D2: tnr strength

ISP_DENOISE3D2_NOISE

0x00005308 = Byte Address

Description: Denoise3D2 noise params

Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_noise_level	31:16	rw	'h0	Denoise3D2: noise level
denoise3d_v20_noise_mean	15:0	rw	'h0	Denoise3D2: noise mean

ISP_DENOISE3D2_MOTION

0x0000530c = Byte Address

Description: Denoise3D2 motion params

Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_noise_thresh	31:16	rw	'h0	Denoise3D2: noise threshold
denoise3d_v20_motion_mean	15:0	rw	'h0	Denoise3D2: motion mean

ISP_DENOISE3D2_RANGE

0x00005310 = Byte Address

Description: Denoise3D2 range params

Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_tnr_range_h	15:12	rw	'h0	Denoise3D2: tnr horizontal range value
denoise3d_v20_tnr_range_v	11:8	rw	'h0	Denoise3D2: tnr vertical range value
denoise3d_v20_tnr_dilate_range_h	7:4	rw	'h0	Denoise3D2: tnr dilation horizontal range value
denoise3d_v20_tnr_dilate_range_v	3:0	rw	'h0	Denoise3D2: tnr dilation vertical range value

ISP_DENOISE3D2_MOTION_INV

0x00005314 = Byte Address

Description: Denoise3D2 motion inv value

Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_motion_inv	19:0	rw	'h0	Denoise3D2: motion inv factor

ISP_DENOISE3D2_UPDATE

0x00005318 = Byte Address

Description: Denoise3D2 update params

Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_thr_update	26:16	rw	'h0	Denoise3D2: tnr update factor
denoise3d_v20_motion_thr_update	10:0	rw	'h0	Denoise3D2: motion tnr update factor

ISP_DENOISE3D2_TNR

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0x0000531c = Byte Address

Description: Denoise3D2 motion detection params

Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_motion_pre_weight	21:16	rw	'h0	Denoise3D2: pre motion weight
denoise3d_v20_motion_slope	10:0	rw	'h0	Denoise3D2: motion slope

ISP_DENOISE3D2_SHIFT

0x00005320 = Byte Address

Description: Denoise3D2 shift params

Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_motion_conv_shift	31:28	rw	'h0	Denoise3D2: motion convergence shift bit
denoise3d_v20_motion_conv_max	27:17	rw	'h0	Denoise3D2: motion convergence max value
denoise3d_v20_tnr_val_shift_bit	16:11	rw	'h0	Denoise3D2: tnr value shift bit
denoise3d_v20_tnr_diff_norm_shift_bit	10:5	rw	'h0	Denoise3D2: tnr diff shift bit
denoise3d_v20_sad_weight	4:0	rw	'h0	Denoise3D2: tnr sad weight

ISP_DENOISE3D2_LUMA

0x00005324 = Byte Address

Description: Denoise3D2 luma params

Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_pixel_val_shift	30:20	rw	'h0	Denoise3D2: luma compensation pixel value shift bit
denoise3d_v20_pixel_slop	19:12	rw	'h0	Denoise3D2: luma compensation pixel slop value
denoise3d_v20_pixel_slop_shift	11:8	rw	'h0	Denoise3D2: luma compensation slope shift value
denoise3d_v20_pixel_slop_min	7:0	rw	'h0	Denoise3D2: luma compensation pixel slop min value

ISP_DENOISE3D2_CTRL_SHD

0x00005328 = Byte Address

Description: Denoise3D2 ctrl shadow

Sub-Field Name	Bits	rw	Reset	Description

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denoise3d_v20_dither_en	15:15	r	'h0	NRV4 dithering enable bit
denoise3d_gauss_blur_en	14:14	r	'h0	NRV4 gaussian blur enable bit
denoise3d_motion_smooth_lvl	13:13	r	'h0	NRV4 motion smoothing level bit
denoise3d_v20_soft_rst_shd	12:12	r	'h0	Denoise3D2: soft reset bit shadow
denoise3d_v20_invgamma_en_shd	11:11	r	'h0	Denoise3D2: inv gamma enable bit shadow
denoise3d_v20_pregamma_en_shd	10:10	r	'h0	Denoise3D2: pregamma enable bit shadow
denoise3d_v20_erode_en_shd	9:9	r	'h0	Denoise3D2: motion erosion enable bit shadow
denoise3d_v20_motion_conv_en_shd	8:8	r	'h0	Denoise3D2: motion convergence enable bit shadow
denoise3d_v20_inv_dgain_en_shd	7:7	r	'h0	Denoise3D2: inv digital gain enable bit shadow
denoise3d_v20_inv_awb_gain_en_shd	6:6	r	'h0	Denoise3D2: inv awb gain enable bit shadow
denoise3d_v20_comp_luma_en_shd	5:5	r	'h0	Denoise3D2: luma compensation enable bit shadow
denoise3d_v20_ref_reset_shd	4:4	r	'h0	Denoise3D2: soft reset bit shadow
denoise3d_v20_motion_dilate_enable_shd	3:3	r	'h0	Denoise3D2: motion dilation enable bit shadow
denoise3d_v20_nlm_enable_shd	2:2	r	'h0	Denoise3D2: 2dnr enable bit shadow
denoise3d_v20_tnr_enable_shd	1:1	r	'h0	Denoise3D2: tnr enable bit shadow
denoise3d_v20_enable_shd	0:0	r	'h0	Denoise3D2: enable bit shadow

ISP_DENOISE3D2_STRENGTH_SHD

0x0000532c = Byte Address

Description: Denoise3D2 strength shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v40_tnr_strength2_shd	15:8	r	'h0	Denoise3D2: tnr strength2 shadow
denoise3d_v40_tnr_strength_shd	7:0	r	'h0	Denoise3D2: tnr strength shadow

ISP_DENOISE3D2_NOISE_SHD

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0x00005330 = Byte Address

Description: Denoise3D2 noise params shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_noise_level_shd	29:16	r	'h0	Denoise3D2: noise level shadow
denoise3d_v20_noise_mean_shd	13:0	r	'h0	Denoise3D2: noise mean shadow

ISP_DENOISE3D2_MOTION_SHD

0x00005334 = Byte Address

Description: Denoise3D2 motion params shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_nosie_thresh_shd	29:16	r	'h0	Denoise3D2: noise threshold shadow
denoise3d_v20_motion_mean_shd	13:0	r	'h0	Denoise3D2: motion mean shadow

ISP_DENOISE3D2_RANGE_SHD

0x00005338 = Byte Address

Description: Denoise3D2 range params shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_tnr_range_h_shd	9:8	r	'h0	Denoise3D2: tnr horizontal range value shadow
denoise3d_v20_tnr_range_v_shd	7:6	r	'h0	Denoise3D2: tnr vertical range value shadow
denoise3d_v20_tnr_dilate_range_h_shd	5:3	r	'h0	Denoise3D2: tnr dilation horizontal range value shadow
denoise3d_v20_tnr_dilate_range_v_shd	2:0	r	'h0	Denoise3D2: tnr dilation vertical range value shadow

ISP_DENOISE3D2_MOTION_INV_SHD

0x0000533c = Byte Address

Description: Denoise3D2 motion inv value shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_motion_inv_shd	19:0	r	'h0	Denoise3D2: motion inv factor shadow

ISP_DENOISE3D2_UPDATE_SHD

0x00005340 = Byte Address

Description: Denoise3D2 update params shadow

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Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_thr_update_shd	26:16	r	'h0	Denoise3D2: tnr update factor shadow
denoise3d_v20_motion_thr_update_shd	10:0	r	'h0	Denoise3D2: motion tnr update factor shadow

ISP_DENOISE3D2_TNR_SHD

0x00005344 = Byte Address

Description: Denoise3D2 motion detection params shadow

Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_motion_pre_weight_shd	26:16	r	'h0	Denoise3D2: pre motion weight shadow
denoise3d_v20_motion_slope_shd	10:0	r	'h0	Denoise3D2: motion slope shadow

ISP_DENOISE3D2_SHIFT_SHD

0x00005348 = Byte Address

Description: Denoise3D2 shift params shadow

Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_motion_conv_shift_shd	31:28	r	'h0	Denoise3D2: motion convergence shift bit shadow
denoise3d_v20_motion_conv_max_shd	27:17	r	'h0	Denoise3D2: motion convergence max value shadow
denoise3d_v20_tnr_val_shift_bit_shd	16:11	r	'h0	Denoise3D2: tnr value shift bit shadow
denoise3d_v20_tnr_diff_norm_shift_bit_shd	10:5	r	'h0	Denoise3D2: tnr diff shift bit shadow
denoise3d_v20_sad_weight_shd	4:0	r	'h0	Denoise3D2: tnr sad weight shadow

ISP_DENOISE3D2_LUMA_SHD

0x0000534c = Byte Address

Description: Denoise3D2 luma params shadow

Sub-Field Name	Bits	rw	Reset	Description
denoise3d_v20_pixel_val_shift_shd	30:20	r	'h0	Denoise3D2: luma compensation pixel value shift bit shadow
denoise3d_v20_pixel_slop_min_shd	19:12	r	'h0	Denoise3D2: luma compensation pixel slop value shadow

denoise3d_v20_pixel_slop_shift_shd	11:8	r	'h0	Denoise3D2: luma compensation slope shift value shadow

ISP_DENOISE3D2_BG_VAL_SUM

0x00005350 = Byte Address

Description: Denoise3D2 background value sum stats

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_bg_val_sum	31:0	r	'h0	Denoise3D2: background value sum stats

ISP_DENOISE3D2_MO_VAL_SUM

0x00005354 = Byte Address

Description: Denoise3D2 foreground value sum stats

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_mo_val_sum	31:0	r	'h0	Denoise3D2: foreground value sum stats

ISP_DENOISE3D2_BG_CNT

0x00005358 = Byte Address

Description: Denoise3D2 background pixel count stats

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_bg_pix_cnt	31:0	r	'h0	Denoise3D2: background pixel count stats

ISP_DENOISE3D2_MO_CNT

0x0000535c = Byte Address

Description: Denoise3D2 foreground pixel count stats

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_mo_pix_cnt	31:0	r	'h0	Denoise3D2: foreground pixel count stats

ISP_DENOISE3D2_FRAME_AVG

0x00005360 = Byte Address

Description: Denoise3D2 frame average

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_frame_avg	31:0	r	'h0	Denoise3D2: frame average stats

ISP_DENOISE3D2_DUMMY_HBLANK

0x00005364 = Byte Address

Description: Denoise3D2 dummy hblank

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d_v20_dummy_hblank	15:0	rw	'h0	Denoise3D2: dummy hblank

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ISP_DENOISE3D2_PREGAMMA_Y_0

0x00005400 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_2_a	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 2 [11:6]
denoise3d2_pregamma_y_1	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 1
denoise3d2_pregamma_y_0	11:0	rw	'hfff	Denoise 3DV2 Pregamma Y 0

ISP_DENOISE3D2_PREGAMMA_Y_1

0x00005404 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_2_b	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 2 [5:0]
denoise3d2_pregamma_y_4	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 4
denoise3d2_pregamma_y_3	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 3

ISP_DENOISE3D2_PREGAMMA_Y_2

0x00005408 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_7_a	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 7 [11:6]
denoise3d2_pregamma_y_6	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 6
denoise3d2_pregamma_y_5	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 5

ISP_DENOISE3D2_PREGAMMA_Y_3

0x0000540c = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_7_b	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 7 [5:0]
denoise3d2_pregamma_y_9	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 9
denoise3d2_pregamma_y_8	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 8

ISP_DENOISE3D2_PREGAMMA_Y_4

0x00005410 = Byte Address

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Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise3d2_pregamma_y_12_a	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 12 [11:6]
denoise3d2_pregamma_y_11	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 11
denoise3d2_pregamma_y_10	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 10

ISP_DENOISE3D2_PREGAMMA_Y_5

0x00005414 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise3d2_pregamma_y_12_b	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 12 [5:0]
denoise3d2_pregamma_y_14	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 14
denoise3d2_pregamma_y_13	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 13

ISP_DENOISE3D2_PREGAMMA_Y_6

0x00005418 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise3d2_pregamma_y_17_a	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 17 [11:6]
denoise3d2_pregamma_y_16	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 16
denoise3d2_pregamma_y_15	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 15

ISP_DENOISE3D2_PREGAMMA_Y_7

0x0000541c = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	Bits	rw	Reset	Description
denoise3d2_pregamma_y_17_b	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 17 [5:0]
denoise3d2_pregamma_y_19	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 19

denoise3d2_pregamma_y_18	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 18

ISP_DENOISE3D2_PREGAMMA_Y_8

0x00005420 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_22_a	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 22 [11:6]
denoise3d2_pregamma_y_21	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 21
denoise3d2_pregamma_y_20	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 20

ISP_DENOISE3D2_PREGAMMA_Y_9

0x00005424 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_22_b	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 22 [5:0]
denoise3d2_pregamma_y_24	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 24
denoise3d2_pregamma_y_23	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 23

ISP_DENOISE3D2_PREGAMMA_Y_10

0x00005428 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_27_a	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 27 [11:6]
denoise3d2_pregamma_y_26	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 26
denoise3d2_pregamma_y_25	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 25

ISP_DENOISE3D2_PREGAMMA_Y_11

0x0000542c = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

denoise3d2_pregamma_y_27_b	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 27 [5:0]
denoise3d2_pregamma_y_29	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 29
denoise3d2_pregamma_y_28	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 28

ISP_DENOISE3D2_PREGAMMA_Y_12

0x00005430 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_32_a	29:24	rw	'h0	Denoise 3DV2 Pregamma Y 32 [11:6]
denoise3d2_pregamma_y_31	23:12	rw	'h0	Denoise 3DV2 Pregamma Y 31
denoise3d2_pregamma_y_30	11:0	rw	'h0	Denoise 3DV2 Pregamma Y 30

ISP_DENOISE3D2_PREGAMMA_Y_13

0x00005434 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_32_b	5:0	rw	'h0	Denoise 3DV2 Pregamma Y 32 [5:0]

ISP_DENOISE3D2_PREGAMMA_Y_0_SHD

0x00005438 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_2_a_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 2 [11:6]
denoise3d2_pregamma_y_1_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 1
denoise3d2_pregamma_y_0_shd	11:0	r	'hfff	Denoise 3DV2 Pregamma Y Shadow 0

ISP_DENOISE3D2_PREGAMMA_Y_1_SHD

0x0000543c = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

denoise3d2_pregamma_y_2_b_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 2 [5:0]
denoise3d2_pregamma_y_4_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 4
denoise3d2_pregamma_y_3_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 3

ISP_DENOISE3D2_PREGAMMA_Y_2_SHD

0x00005440 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_7_a_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 7 [11:6]
denoise3d2_pregamma_y_6_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 6
denoise3d2_pregamma_y_5_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 5

ISP_DENOISE3D2_PREGAMMA_Y_3_SHD

0x00005444 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_7_b_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 7 [5:0]
denoise3d2_pregamma_y_9_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 9
denoise3d2_pregamma_y_8_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 8

ISP_DENOISE3D2_PREGAMMA_Y_4_SHD

0x00005448 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_12_a_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 12 [11:6]
denoise3d2_pregamma_y_11_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 11
denoise3d2_pregamma_y_10_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 10

ISP_DENOISE3D2_PREGAMMA_Y_5_SHD

0x0000544c = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_12_b_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 12 [5:0]
denoise3d2_pregamma_y_14_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 14
denoise3d2_pregamma_y_13_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 13

ISP_DENOISE3D2_PREGAMMA_Y_6_SHD

0x00005450 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_17_a_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 17 [11:6]
denoise3d2_pregamma_y_16_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 16
denoise3d2_pregamma_y_15_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 15

ISP_DENOISE3D2_PREGAMMA_Y_7_SHD

0x00005454 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_17_b_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 17 [5:0]
denoise3d2_pregamma_y_19_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 19
denoise3d2_pregamma_y_18_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 18

ISP_DENOISE3D2_PREGAMMA_Y_8_SHD

0x00005458 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_22_a_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 22 [11:6]

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denoise3d2_pregamma_y_21_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 21
denoise3d2_pregamma_y_20_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 20

ISP_DENOISE3D2_PREGAMMA_Y_9_SHD

0x0000545c = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_22_b_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 22 [5:0]
denoise3d2_pregamma_y_24_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 24
denoise3d2_pregamma_y_23_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 23

ISP_DENOISE3D2_PREGAMMA_Y_10_SHD

0x00005460 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_27_a_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 27 [11:6]
denoise3d2_pregamma_y_26_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 26
denoise3d2_pregamma_y_25_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 25

ISP_DENOISE3D2_PREGAMMA_Y_11_SHD

0x00005464 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
denoise3d2_pregamma_y_27_b_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 27 [5:0]
denoise3d2_pregamma_y_29_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 29
denoise3d2_pregamma_y_28_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 28

ISP_DENOISE3D2_PREGAMMA_Y_12_SHD

0x00005468 = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	Bits	rw	Reset	Description
denoise3d2_pregamma_y_32_a_shd	29:24	r	'h0	Denoise 3DV2 Pregamma Y Shadow 32 [11:6]
denoise3d2_pregamma_y_31_shd	23:12	r	'h0	Denoise 3DV2 Pregamma Y Shadow 31
denoise3d2_pregamma_y_30_shd	11:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 30

ISP_DENOISE3D2_PREGAMMA_Y_13_SHD

0x0000546c = Byte Address

Description: Denoise3D2 Pregamma Y LUT Register Shadow

Sub-Field Name	Bits	rw	Reset	Description
denoise3d2_pregamma_y_32_b_shd	5:0	r	'h0	Denoise 3DV2 Pregamma Y Shadow 32 [5:0]

MI_PP_CTRL

0x00005500 = Byte Address

Description: pp control register

Sub-Field Name	Bits	rw	Reset	Description
pp_rd_yuv_cfg_update	7:7	w	'h0	Forced configuration update which leads to an immediate update of the shadow registers for this read path.
pp_rd_yuv_auto_update	6:6	rw	'h0	Automatic update of configuration registers for post path at frame end of read. 0: disabled; 1: enabled

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pp_init_offset_en	5:5	rw	'h0	Enables updating of the offset counters shadow registers for self path to the programmed register init values. MI_PP_Y/CB/CR_OFFS_CNT_INIT-> MI_PP_Y/CB/CR_OFFS_CNT_SHD The update will be executed either when a forced software update occurs (in register MI_PP_CTRL bit cfg_upd = 1) or when an automatic configuration update signal arrives at the MI input port. The latter is split into main and self picture. So only the corresponding main/self shadow registers are affected. After a picture skip has been performed init_offset_en selects between skip restart and skip init mode (see bit skip in register MI_MP_CTRL).
pp_init_base_en	4:4	rw	'h0	Enables updating of the base address and buffer size shadow registers for self picture to the programmed register init values. MI_PP_Y/CB/CR_BASE_AD_INIT-> MI_PP_Y/CB/CR_BASE_AD_SHD MI_PP_Y/CB/CR_SIZE_INIT-> MI_PP_Y/CB/CR_SIZE_SHD The update will be executed either when a forced software update occurs (in register MI_PP_CTRL bit cfg_upd = 1) or when an automatic configuration update signal arrives at the MI input port. The latter is split into main and self picture. Only the corresponding main/self shadow registers are affected.
pp_mi_cfg_upd	3:3	w	'h0	Forced configuration update which leads to an immediate update of the shadow registers. Depending on the init_offset_en and init_base_en enable bits in the MI_PP_CTRL register, the offset counter, base address and buffer size shadow registers are also updated.
pp_mi_skip	2:2	w	'h0	Skip of current or next starting main picture.
pp_auto_update	1:1	rw	'h0	Automatic update of configuration registers for post path at frame end. 0: disabled; 1: enabled
pp_pingpong_enable	0:0	rw	'h0	Ping-pong mode of configuration registers for post path at frame end. 0: disabled; 1: enabled

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MI_PP_FMT

0x00005504 = Byte Address

Description: PP pixel format control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pp_rd_raw_sel	31:31	rw	'h0	1'b0: YUV read is selected 1'b1: RAW read is selected
pp_rd_raw_aligned	30:29	rw	'h0	00: Unaligned 01: Aligned mode0 10: Aligned mode1 11: Reserved
pp_rd_raw_bit	28:26	rw	'h0	000: RAW8 001: RAW10 010: RAW12 011: RAW14 100: RAW16 101: RAW20, or Reserved
pp_rd_yuv_nvy	25:24	rw	'h0	00: Y channel in the first position 01: Y in the second 10: Y in the third 11: Reserved When in YUV444 or RGB888 interleave: 00: Y channel in the first and third position 01: Y channel in the second and fourth position
pp_rd_yuv_nv21	23:23	rw	'h0	0: No swap 1: Swap Cr and Cb channel for semi-planar and YUV444 or RGB interleave and YUV422 interleave
pp_rd_yuv_aligned	22:22	rw	'h0	0: Unaligned; 1: Aligned
pp_rd_yuv_str	21:20	rw	'h0	00: YUV/RGB Semi-Planar 01: YUV Interleave, RGB Raster 10: YUV/RGB Planar 11: Reserved
pp_rd_yuv_fmt	19:18	rw	'h0	00: YUV420 01: YUV422 10: YUV444 11: Reserved
pp_rd_yuv_bit	17:17	rw	'h0	0: 8 bit YUV/RGB 1: 10 bit YUV/RGB
pp_wr_raw_sel	16:16	rw	'h0	0: write yuv 1: write raw
pp_wr_raw_aligned	15:14	rw	'h0	00: Unaligned 01: Aligned mode0 10: Aligned mode1 11: Reserved
pp_wr_raw_bit	13:11	rw	'h0	000: RAW8 001: RAW10 010: RAW12 011: RAW14 100: RAW16 101: RAW20, or Reserved
pp_wr_yuv_word_aligned_msb_en	10:10	rw	'h0	0: 10bit yuv in LSB of 16bit; 1: 10bit yuv in MSB of 16bit;
pp_wr_yuv_word_aligned	9:9	rw	'h0	0: disable; 1:enable

pp_wr_yuv_nvy	8:7	rw	'h0	00: Y channel in the first position 01: Y in the second 10: Y in the third 11: Reserved When in YUV444 or RGB888 interleave: 00: Y channel in the first and third position 01: Y channel in the second and forth position
pp_wr_yuv_nv21	6:6	rw	'h0	0: No swap 1: Swap Cr and Cb channel for semi-planar and YUV444 or RGB interleave and YUV422 interleave
pp_wr_yuv_aligned	5:5	rw	'h0	0: Unaligned; 1: Aligned
pp_wr_yuv_str	4:3	rw	'h0	00: YUV/RGB SemiPlanar 01: YUV Interleave, RGB Raster 10: YUV/RGB Planar 11: Y Only
pp_wr_yuv_fmt	2:1	rw	'h0	00: YUV420 01: YUV422 10: YUV444 11: Reserved
pp_wr_yuv_bit	0:0	rw	'h0	0: 8 bit YUV/RGB 1: 10 bit YUV/RGB

MI_PP_DATA_SWAP_CTRL

0x00005508 = Byte Address

Description: PP data swap control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pp_rd_swap_v	27:24	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
pp_rd_swap_u	23:20	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
pp_rd_swap_y	19:16	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
pp_wr_swap_v	11:8	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words

pp_wr_swap_u	7:4	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
pp_wr_swap_y	3:0	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words

MI_PP_Y_BASE_AD_INIT

0x00005520 = Byte Address

Description: Base address for self path 1 picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pp_y_base_ad_init	31:4	rw	'h0	Base address of Self picture Y component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_PP_Y_SIZE_INIT

0x00005524 = Byte Address

Description: Size of self path 1 picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pp_y_size_init	28:4	rw	'h0	Size of Self Path 1 picture Y component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_PP_Y_OFFS_CNT_INIT

0x00005528 = Byte Address

Description: Offset counter init value for self path 1 picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

pp_y_offs_cnt_init	28:4	rw	'h0	Offset counter initial value of Self Path 1 picture Y component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update. Note: Set control bit init_base_en before updating so that a forced or automatic update can take effect.

MI_PP_Y_LLENGTH

0x0000552c = Byte Address

Description: Line length of self path 1 picture Y component

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pp_y_llengh	31:0	rw	'h0	Line length of Self Path 1 picture Y component or RGB picture in bytes, also known as line stride. This value should be aligned to bus width (16-byte), and greater than or equal to the valid data length (lval), except when mode is 4:2:x planar, where the value should be aligned to 2 * (bus width) (32 bytes). In planar mode, the line length of the Cb and Cr component is assumed according to the YCbCr format, i.e. half for 4:2:x and the same size for 4:4:4. In semi planar 4:2:x mode, the line length of the Cb and Cr component is assumed to be the same size. Note: Line length always refers to the line length of the output image. This is particularly important when rotating. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 1. Note: Refer to the Hardware Integration document for a detailed calculation of line length.

MI_PP_Y_PIC_WIDTH

0x00005530 = Byte Address

Description: Y component image width

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

pp_y_pic_width	31:0	rw	'h0	Image width of the Self Path 1 picture Y component or RGB picture in pixels. For YCbCr 4:2:x and RGB 565, the image width must be a multiple of 2. If no line stride is used but flipping required, the image width must be a multiple of 8 for 4:2:x planar or 4 for 4:4:4 planar/4:2:x semi planar. There are no restrictions for RGB 888/666. In planar mode, the image width of the Cb and Cr component is assumed according to the YCbCr format, i.e. half for 4:2:x and the same size for 4:4:4. In semi planar 4:2:x mode, the image width of the Cb component (which includes Cr) is assumed the same size. In interleave mode, no Cb/Cr image width is used. Note: Image width always refers to the picture width of the output image. This is particularly important when rotating. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 1.

MI_PP_Y_PIC_HEIGHT

0x00005534 = Byte Address

Description: Y component image height

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pp_y_pic_height	31:0	rw	'h0	Image height of the Self Path 1 picture Y component or RGB picture in pixels. In planar and semi planar mode, the image height of the Cb and Cr component is assumed according to the YCbCr format, i.e. half for 4:2:0 and the same for 4:2:2 and 4:4:4. Note: Image height always refers to the picture height of the output image. This is particularly important when rotating. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 1.

MI_PP_Y_PIC_SIZE

0x00005538 = Byte Address

Description: Y component image memory size in bytes

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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pp_y_pic_size	31:0	rw	'h0	Image size of the Self Path 1 picture Y component or RGB picture in bytes which has to be the Y line length multiplied by the Y image height (PP_y_llength * PP_y_pic_height). In planar mode the image size of the Cb and Cr component is assumed according to the YCbCr format, i.e. a quarter for 4:2:0, half for 4:2:2 and the same for 4:4:4. In semi planar mode the image size of the Cb and Cr component is assumed half for 4:2:0 and the same size for 4:2:2. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 1.

MI_PP_Y_OFFS_CNT_START

0x00005560 = Byte Address

Description: Offset counter start value for self path 1 picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pp_y_offs_cnt_start	28:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (Self Path 1 picture Y component). Updated at frame end. Note: Soft reset will reset the contents to the reset value.

MI_PP_Y_BASE_AD_SHD

0x0000556c = Byte Address

Description: Base address shadow register for self path 1 picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pp_y_base_ad	31:4	r	'h0	Base address of Self Path 1 picture Y component ring buffer.

MI_PP_Y_SIZE_SHD

0x00005570 = Byte Address

Description: Size shadow register of self path 1 picture Y component ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
pp_y_size	28:4	r	'h0	Size of Self Path 1 picture Y component ring buffer.

MI_PP_Y_OFFS_CNT_SHD

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0x00005574 = Byte Address

Description: Current offset counter of self path 1 picture Y component ring buffer

Sub-Field Name	Bits	rw	Reset	Description
pp_y_offs_cnt	28:4	r	'h0	Current offset counter of Self Path 1 picture Y component ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_PP_Y_LVAL_BYTES

0x000055cc = Byte Address

Description: PPW RAW0 Component line valid bytes number

Sub-Field Name	Bits	rw	Reset	Description
pp_y_width_bytes	15:0	rw	'h0	Sizes of bytes occupied by valid data of a line.

MI_SP1_YCBCR_ENTRY_Y_CNT

0x000055e8 = Byte Address

Description: the line count of y

Sub-Field Name	Bits	rw	Reset	Description
sp1_ycbcr_entry_y_cnt	15:0	r	'h0	the line count of y

MI_SP1_YCBCR_ENTRY_CB_CNT

0x000055ec = Byte Address

Description: the line count of cb

Sub-Field Name	Bits	rw	Reset	Description
sp1_ycbcr_entry_cb_cnt	15:0	r	'h0	the line count of cb

MI_SP1_YCBCR_ENTRY_CR_CNT

0x000055f0 = Byte Address

Description: the line count of cr

Sub-Field Name	Bits	rw	Reset	Description
sp1_ycbcr_entry_cr_cnt	15:0	r	'h0	the line count of cr

MI_SP2_YCBCR_ENTRY_Y_CNT

0x000055f4 = Byte Address

Description: the line count of y

Sub-Field Name	Bits	rw	Reset	Description
sp2_ycbcr_entry_y_cnt	15:0	r	'h0	the line count of y

MI_SP2_YCBCR_ENTRY_CB_CNT

0x000055f8 = Byte Address

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Description: the line count of cb

Sub-Field Name	Bits	rw	Reset	Description
sp2_ycbcr_entry_cb_cnt	15:0	r	'h0	the line count of cb

MI_SP2_YCBCR_ENTRY_CR_CNT

0x000055fc = Byte Address

Description: the line count of cr

Sub-Field Name	Bits	rw	Reset	Description
sp2_ycbcr_entry_cr_cnt	15:0	r	'h0	the line count of cr

MI_SP2_RAW2_CTRL

0x00005600 = Byte Address

Description: SP2 RAW2 control register

Sub-Field Name	Bits	rw	Reset	Description
sp2_rd_raw_cfg_update	9:9	w	'h0	Forced configuration update which leads to an immediate update of the shadow registers for this RAW read path.
sp2_rd_raw_auto_update	8:8	rw	'h0	Automatic update of configuration registers for raw2 read at frame end. 0: disabled; 1: enabled
sp2_init_offset_en	5:5	rw	'h0	Enables updating of the offset counters shadow registers for self path to the programmed register init values.
sp2_init_base_en	4:4	rw	'h0	Enables updating of the base address and buffer size shadow registers for self picture to the programmed register init values.
sp2_mi_cfg_upd	3:3	w	'h0	Forced configuration update which leads to an immediate update of the shadow registers. Depending on the init_offset_en and init_base_en enable bits in the MI_SP2_CTRL register, the offset counter, base address and buffer size shadow registers are also updated.
sp2_mi_skip	2:2	w	'h0	Skip of current or next starting main picture.
sp2_auto_update	1:1	rw	'h0	Automatic update of configuration registers for Self Path 2 at frame end. 0: disabled; 1: enabled

MI_SP2_RAW2_FMT

0x00005604 = Byte Address

Description: SP2 RAW2 pixel format control register

Sub-Field Name	Bits	rw	Reset	Description

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sp2_rd_raw_aligned	24:23	rw	'h0	00: Unaligned 01: Aligned mode0 10: Aligned mode1 11: Reserved
sp2_rd_raw_bit	21:19	rw	'h0	000: RAW8 001: RAW10 010: RAW12 011: RAW14 100: RAW16 101: RAW20, or Reserved
sp2_wr_raw_aligned	10:9	rw	'h0	00: Unaligned 01: Aligned mode0 10: Aligned mode1 11: Reserved
sp2_wr_raw_bit	7:5	rw	'h0	000: RAW8 001: RAW10 010: RAW12 011: RAW14 100: RAW16 101: RAW20, or Reserved

MI_SP2_RAW2_DATA_SWAP_CTRL

0x00005608 = Byte Address

Description: sp2 raw2 data swap control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_rd_swap_raw	7:4	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
sp2_wr_swap_raw	3:0	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words

MI_SP2_RAW2_BASE_AD_INIT

0x00005620 = Byte Address

Description: Follow main path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_raw_base_ad_init	31:4	rw	'h0	Base address of Self Path 2 picture RAW component ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP2_RAW2_SIZE_INIT

0x00005624 = Byte Address

Description: Follow main path RAW

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Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_size_init	28:4	rw	'h0	Size of Self Path 2 picture RAW data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP2_RAW2_OFFS_CNT_INIT

0x00005628 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_offs_cnt_init	28:4	rw	'h0	Offset counter initial value of Self Path 2 picture RAW data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_SP2_RAW2_LLENGTH

0x0000562c = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_llengh	31:0	rw	'h0	Line length of Self Path 2 RAW data in bytes, also known as line stride. If no line stride is used, line length must match image width. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

MI_SP2_RAW2_PIC_WIDTH

0x00005630 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_pic_width	31:0	rw	'h0	Image width of the RAW component picture in pixels. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

MI_SP2_RAW2_PIC_HEIGHT

0x00005634 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_pic_height	31:0	rw	'h0	Image height of RAW component in pixels. Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

MI_SP2_RAW2_PIC_SIZE

0x00005638 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_pic_size	31:0	rw	'h0	Image size of the RAW component in pixels which has to be the Y line length multiplied by the RAW image height (sp2_raw_llength * sp2_raw_pic_height). Note: Programmed value becomes effective immediately; therefore, write to the register only if no picture data is sent to the Self Path 2.

MI_SP2_RAW2_OFFS_CNT_START

0x0000563c = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_offs_cnt_start	28:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (Self Path 2 picture RAW data). Updated at frame end. Note: A soft reset resets the contents to the reset value.

MI_SP2_RAW2_BASE_AD_SHD

0x00005640 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
sp2_raw_base_ad	31:4	r	'h0	Base address of Self Path 2 picture RAW data ring buffer.

MI_SP2_RAW2_SIZE_SHD

0x00005644 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description

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sp2_raw_size	28:4	r	'h0	Size of Self Path 2 picture RAW data ring buffer.

MI_SP2_RAW2_OFFS_CNT_SHD

0x00005648 = Byte Address

Description: Follow main path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_raw_offs_cnt	28:4	r	'h0	Current offset counter of Self Path 2 picture RAW data ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_SP2_DMA_RAW2_PIC_START_AD

0x00005660 = Byte Address

Description: RAW component image start address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_pic_start_ad	31:4	rw	'h0	Image start address of the RAW component.

MI_SP2_DMA_RAW2_PIC_WIDTH

0x00005664 = Byte Address

Description: RAW component image width

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_pic_width	14:0	rw	'h0	Image width of the RAW component in pixels.

MI_SP2_DMA_RAW2_PIC_LLENGTH

0x00005668 = Byte Address

Description: RAW component original line length

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_pic_llength	31:0	rw	'h0	Line length of the RAW component in bytes of the original image in memory. For an uncropped image, where lines follow each other without offset (no line stride), line length must match image width.

MI_SP2_DMA_RAW2_PIC_SIZE

0x0000566c = Byte Address

Description: RAW component image memory size in bytes

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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sp2_dma_raw_pic_size	27:0	rw	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (dma_raw_llength * dma_raw_pic_height).

MI_SP2_DMA_RAW2_PIC_START_AD_SHD

0x00005670 = Byte Address

Description: Shadow register of RAW component image start address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_pic_start_ad	31:4	r	'h0	Image start address of the RAW component.

MI_SP2_DMA_RAW2_PIC_WIDTH_SHD

0x00005674 = Byte Address

Description: RAW component image width

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_pic_width	14:0	r	'h0	Image width of the RAW component in pixels.

MI_SP2_DMA_RAW2_PIC_LLENGTH_SHD

0x00005678 = Byte Address

Description: RAW component original line length

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_pic_llength	31:0	r	'h0	Line length of the RAW component in bytes of the original image in memory. For an uncropped image, where lines follow each other without offset (no line stride), line length must match image width.

MI_SP2_DMA_RAW2_PIC_SIZE_SHD

0x0000567c = Byte Address

Description: RAW component image memory size in bytes

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_pic_size	27:0	r	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (dma_raw_llength * dma_raw_pic_height).

MI_SP2_DMA_RAW2_PIC_LVAL

0x00005680 = Byte Address

Description: SP2_DMA RAW Component line valid bytes number, 128 bit Alignment.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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sp2_dma_raw_width_bytes	15:0	rw	'h0	Sizes occupied by valid data of a line; value should be a multiple of 16 (bus width = 16 bytes).

MI_SP2_DMA_RAW2_PIC_LVAL_SHD

0x00005684 = Byte Address

Description: SP2_DMA RAW Component line valid bytes number, 128 bit Alignment.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_dma_raw_width_bytes	15:0	r	'h0	Sizes occupied by valid data of a line; value should be a multiple of 16 (bus width = 16 bytes).

MI_MP_YCBCR_ENTRY_LINE_NUM

0x00005690 = Byte Address

Description: the line count in each entry

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_ycbcr_entry_line_num	15:0	rw	'h0	the line count in each entriy

MI_MP_YCBCR_ENTRY_PIC_CNT

0x00005694 = Byte Address

Description: the line count of picture

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_ycbcr_entry_pic_cnt	15:0	r	'h0	the line count of picture

MI_MP_YCBCR_ENTRY_Y_CNT

0x00005698 = Byte Address

Description: the line count of y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_ycbcr_entry_y_cnt	15:0	r	'h0	the line count of y

MI_MP_YCBCR_ENTRY_CB_CNT

0x0000569c = Byte Address

Description: the line count of cb

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_ycbcr_entry_cb_cnt	15:0	r	'h0	the line count of cb

MI_MP_YCBCR_ENTRY_CR_CNT

0x000056a0 = Byte Address

Description: the line count of cr

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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mp_ycbcr_entry_cr_cnt	15:0	r	'h0	the line count of cr

MI_MP_RAW_ENTRY_LINE_NUM

0x000056a4 = Byte Address

Description: the line count in each entry

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_raw_entry_line_num	15:0	rw	'h0	the line count in each entriy

MI_MP_RAW_ENTRY_PIC_CNT

0x000056a8 = Byte Address

Description: the line count of picture

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_raw_entry_pic_cnt	15:0	r	'h0	the line count of picture

MI_SP1_PPW_YCBCR_ENTRY_LINE_NUM

0x000056ac = Byte Address

Description: the line count in each entry

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp1_ycbcr_entry_line_num	15:0	rw	'h0	the line count in each entriy

MI_SP1_PPW_YCBCR_ENTRY_PIC_CNT

0x000056b0 = Byte Address

Description: the line count of picture

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp1_ycbcr_entry_pic_cnt	15:0	r	'h0	the line count of picture

MI_MCM_RAW0_ENTRY_LINE_NUM

0x000056b4 = Byte Address

Description: the line count in each entry

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_entry_line_num	15:0	rw	'h0	the line count in each entriy

MI_MCM_RAW0_ENTRY_PIC_CNT

0x000056b8 = Byte Address

Description: the line count of picture

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw0_entry_pic_cnt	15:0	r	'h0	the line count of picture

MI_IMSC3

0x000056d0 = Byte Address

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Description: Interrupt Mask

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw3_entry_cmpl	22:22	rw	'h0	Mask bit for MCM RAW3 entry cmpl interrupt
mcm_raw2_entry_cmpl	21:21	rw	'h0	Mask bit for MCM RAW2 entry cmpl interrupt
mcm_raw1_entry_cmpl	20:20	rw	'h0	Mask bit for MCM RAW1 entry cmpl interrupt
mcm_g2_raw1_buf_full	19:19	rw	'h0	Mask bit for MCM RAW1 buffer full interrupt
mcm_g2_raw0_buf_full	18:18	rw	'h0	Mask bit for MCM RAW0 buffer full interrupt
wrap_mcm_g2_raw1	17:17	rw	'h0	Mask bit for MCM RAW1 address wrap interrupt
wrap_mcm_g2_raw0	16:16	rw	'h0	Mask bit for MCM RAW0 address wrap interrupt
mcm_g2_raw1_frame_end	15:15	rw	'h0	Mask bit for MCM RAW1 end of frame interrupt
mcm_g2_raw0_frame_end	14:14	rw	'h0	Mask bit for MCM RAW0 end of frame interrupt
sp2_ycbcr_entry_cmpl	13:13	rw	'h0	Mask bit for mp ycbcr entry cmpl interrupt
sp1_ycbcr_entry_cmpl	12:12	rw	'h0	Mask bit for mp ycbcr entry cmpl interrupt
mp_cr_pr_outsync	11:11	rw	'h0	Mask bit for mp cr pr outsync interrupt
mp_cr_pr_timeout	10:10	rw	'h0	Mask bit for mp cr pr timeout interrupt
mp_cb_pr_outsync	9:9	rw	'h0	Mask bit for mp cb pr outsync interrupt
mp_cb_pr_timeout	8:8	rw	'h0	Mask bit for mp cb pr timeout interrupt
mp_y_pr_outsync	7:7	rw	'h0	Mask bit for mp y pr outsync interrupt
mp_y_pr_timeout	6:6	rw	'h0	Mask bit for mp y pr timeout interrupt
bus_idle_done	5:5	rw	'h0	Mask bit for bus idle operation
mcm_raw0_entry_cmpl	4:4	rw	'h0	Mask bit for MCM RAW0 entry cmpl interrupt
sp1_ppw_entry_cmpl	3:3	rw	'h0	Mask bit for sp1 ppw entry cmpl interrupt
mp_raw_entry_cmpl	2:2	rw	'h0	Mask bit formp raw entry cmpl interrupt
mp_ycbcr_entry_cmpl	1:1	rw	'h0	Mask bit for mp ycbcr entry cmpl interrupt
pp_dma_y_entry_cmpl	0:0	rw	'h0	Mask bit for pp dma y entry cmpl interrupt

MI_ISR3

0x000056d4 = Byte Address

Description: Interrupt Mask

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw3_entry_cmpl	22:22	w	'h0	Set bit for MCM RAW3 entry cmpl interrupt
mcm_raw2_entry_cmpl	21:21	w	'h0	set bit for MCM RAW2 entry cmpl interrupt
mcm_raw1_entry_cmpl	20:20	w	'h0	Set bit for MCM RAW1 entry cmpl interrupt
mcm_g2_raw1_buf_full	19:19	w	'h0	Set bit for MCM RAW1 buffer full interrupt

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mcm_g2_raw0_buf_full	18:18	w	'h0	Set bit for MCM RAW0 buffer full interrupt
wrap_mcm_g2_raw1	17:17	w	'h0	Set bit for MCM RAW1 address wrap interrupt
wrap_mcm_g2_raw0	16:16	w	'h0	Set bit for MCM RAW0 address wrap interrupt
mcm_g2_raw1_frame_end	15:15	w	'h0	Set bit for MCM RAW1 end of frame interrupt
mcm_g2_raw0_frame_end	14:14	w	'h0	Set bit for MCM RAW0 end of frame interrupt
sp2_ycbcr_entry_cmpl	13:13	w	'h0	Set bit for mp ycbcr entry cmpl interrupt
sp1_ycbcr_entry_cmpl	12:12	w	'h0	Set bit for mp ycbcr entry cmpl interrupt
mp_cr_pr_outsync	11:11	w	'h0	Set bit for mp cr pr outsync interrupt
mp_cr_pr_timeout	10:10	w	'h0	Set bit for mp cr pr timeout interrupt
mp_cb_pr_outsync	9:9	w	'h0	Set bit for mp cb pr outsync interrupt
mp_cb_pr_timeout	8:8	w	'h0	Set bit for mp cb pr timeout interrupt
mp_y_pr_outsync	7:7	w	'h0	Set bit for mp y pr outsync interrupt
mp_y_pr_timeout	6:6	w	'h0	Set bit for mp y pr timeout interrupt
bus_idle_done	5:5	w	'h0	Set bit for bus idle operation
mcm_raw0_entry_cmpl	4:4	w	'h0	Set bit for MCM RAW0 entry cmpl interrupt
sp1_ppw_entry_cmpl	3:3	w	'h0	Set bit for sp1 ppw entry cmpl interrupt
mp_raw_entry_cmpl	2:2	w	'h0	Set bit formp raw entry cmpl interrupt
mp_ycbcr_entry_cmpl	1:1	w	'h0	Set bit for mp ycbcr entry cmpl interrupt
pp_dma_y_entry_cmpl	0:0	w	'h0	Set bit for pp dma y entry cmpl interrupt

MI_MIS3

0x000056d8 = Byte Address

Description: Interrupt Mask

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw3_entry_cmpl	22:22	r	'h0	Masked bit for MCM RAW3 entry cmpl interrupt
mcm_raw2_entry_cmpl	21:21	r	'h0	Masked bit for MCM RAW2 entry cmpl interrupt
mcm_raw1_entry_cmpl	20:20	r	'h0	Masked bit for MCM RAW1 entry cmpl interrupt
mcm_g2_raw1_buf_full	19:19	r	'h0	Masked bit for MCM RAW1 buffer full interrupt
mcm_g2_raw0_buf_full	18:18	r	'h0	Masked bit for MCM RAW0 buffer full interrupt
wrap_mcm_g2_raw1	17:17	r	'h0	Masked bit for MCM RAW1 address wrap interrupt
wrap_mcm_g2_raw0	16:16	r	'h0	Masked bit for MCM RAW0 address wrap interrupt

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mcm_g2_raw1_frame_end	15:15	r	'h0	Masked bit for MCM RAW1 end of frame interrupt
mcm_g2_raw0_frame_end	14:14	r	'h0	Masked bit for MCM RAW0 end of frame interrupt
sp2_ycbcr_entry_cmpl	13:13	r	'h0	Masked bit for mp ycbcr entry cmpl interrupt
sp1_ycbcr_entry_cmpl	12:12	r	'h0	Masked bit for mp ycbcr entry cmpl interrupt
mp_cr_pr_outsync	11:11	r	'h0	Masked bit for mp cr pr outsync interrupt
mp_cr_pr_timeout	10:10	r	'h0	Masked bit for mp cr pr timeout interrupt
mp_cb_pr_outsync	9:9	r	'h0	Masked bit for mp cb pr outsync interrupt
mp_cb_pr_timeout	8:8	r	'h0	Masked bit for mp cb pr timeout interrupt
mp_y_pr_outsync	7:7	r	'h0	Masked bit for mp y pr outsync interrupt
mp_y_pr_timeout	6:6	r	'h0	Masked bit for mp y pr timeout interrupt
bus_idle_done	5:5	r	'h0	Masked bit for bus idle operation
mcm_raw0_entry_cmpl	4:4	r	'h0	Masked bit for MCM RAW0 entry cmpl interrupt
sp1_ppw_entry_cmpl	3:3	r	'h0	Masked bit for sp1 ppw entry cmpl interrupt
mp_raw_entry_cmpl	2:2	r	'h0	Masked bit formp raw entry cmpl interrupt
mp_ycbcr_entry_cmpl	1:1	r	'h0	Masked bit for mp ycbcr entry cmpl interrupt
pp_dma_y_entry_cmpl	0:0	r	'h0	Masked bit for pp dma y entry cmpl interrupt

MI_ICR3

0x000056dc = Byte Address

Description: Interrupt Mask

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw3_entry_cmpl	22:22	w	'h0	Clear bit for MCM RAW3 entry cmpl interrupt
mcm_raw2_entry_cmpl	21:21	w	'h0	Clear bit for MCM RAW2 entry cmpl interrupt
mcm_raw1_entry_cmpl	20:20	w	'h0	Clear bit for MCM RAW1 entry cmpl interrupt
mcm_g2_raw1_buf_full	19:19	w	'h0	Clear bit for MCM RAW1 buffer full interrupt
mcm_g2_raw0_buf_full	18:18	w	'h0	Clear bit for MCM RAW0 buffer full interrupt
wrap_mcm_g2_raw1	17:17	w	'h0	Clear bit for MCM RAW1 address wrap interrupt
wrap_mcm_g2_raw0	16:16	w	'h0	Clear bit for MCM RAW0 address wrap interrupt
mcm_g2_raw1_frame_end	15:15	w	'h0	Clear bit for MCM RAW1 end of frame interrupt
mcm_g2_raw0_frame_end	14:14	w	'h0	Clear bit for MCM RAW0 end of frame interrupt
sp2_ycbcr_entry_cmpl	13:13	w	'h0	Clear bit for mp ycbcr entry cmpl interrupt
sp1_ycbcr_entry_cmpl	12:12	w	'h0	Clear bit for mp ycbcr entry cmpl interrupt

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mp_cr_pr_outsync	11:11	w	'h0	Clear bit for mp cr pr outsync interrupt
mp_cr_pr_timeout	10:10	w	'h0	Clear bit for mp cr pr timeout interrupt
mp_cb_pr_outsync	9:9	w	'h0	Clear bit for mp cb pr outsync interrupt
mp_cb_pr_timeout	8:8	w	'h0	Clear bit for mp cb pr timeout interrupt
mp_y_pr_outsync	7:7	w	'h0	Clear bit for mp y pr outsync interrupt
mp_y_pr_timeout	6:6	w	'h0	Clear bit for mp y pr timeout interrupt
bus_idle_done	5:5	w	'h0	Clear bit for bus idle operation
mcm_raw0_entry_cmpl	4:4	w	'h0	Clear bit for MCM RAW0 entry cmpl interrupt
sp1_ppw_entry_cmpl	3:3	w	'h0	Clear bit for sp1 ppw entry cmpl interrupt
mp_raw_entry_cmpl	2:2	w	'h0	Clear bit formp raw entry cmpl interrupt
mp_ycbcr_entry_cmpl	1:1	w	'h0	Clear bit for mp ycbcr entry cmpl interrupt
pp_dma_y_entry_cmpl	0:0	w	'h0	Clear bit for pp dma y entry cmpl interrupt

MI_RIS3

0x000056e0 = Byte Address

Description: Interrupt Mask

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw3_entry_cmpl	22:22	r	'h0	Raw status for MCM RAW3 entry cmpl interrupt
mcm_raw2_entry_cmpl	21:21	r	'h0	Raw status for MCM RAW2 entry cmpl interrupt
mcm_raw1_entry_cmpl	20:20	r	'h0	Raw status for MCM RAW1 entry cmpl interrupt
mcm_g2_raw1_buf_full	19:19	r	'h0	RAW status of MCM RAW1 buffer full interrupt
mcm_g2_raw0_buf_full	18:18	r	'h0	RAW status of MCM RAW0 buffer full interrupt
wrap_mcm_g2_raw1	17:17	r	'h0	RAW status of MCM RAW1 address wrap interrupt
wrap_mcm_g2_raw0	16:16	r	'h0	RAW status of MCM RAW0 address wrap interrupt
mcm_g2_raw1_frame_end	15:15	r	'h0	RAW status of MCM RAW1 end of frame interrupt
mcm_g2_raw0_frame_end	14:14	r	'h0	RAW status of MCM RAW0 end of frame interrupt
sp2_ycbcr_entry_cmpl	13:13	r	'h0	Mask bit for mp ycbcr entry cmpl interrupt
sp1_ycbcr_entry_cmpl	12:12	r	'h0	Mask bit for mp ycbcr entry cmpl interrupt
mp_cr_pr_outsync	11:11	r	'h0	Mask bit for mp cr pr outsync interrupt
mp_cr_pr_timeout	10:10	r	'h0	Mask bit for mp cr pr timeout interrupt

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mp_cb_pr_outsync	9:9	r	'h0	Mask bit for mp cb pr outsync interrupt
mp_cb_pr_timeout	8:8	r	'h0	Mask bit for mp cb pr timeout interrupt
mp_y_pr_outsync	7:7	r	'h0	Mask bit for mp y pr outsync interrupt
mp_y_pr_timeout	6:6	r	'h0	Mask bit for mp y pr timeout interrupt
bus_idle_done	5:5	r	'h0	Mask bit for bus idle operation
mcm_raw0_entry_cmpl	4:4	r	'h0	Mask bit for MCM RAW0 entry cmpl interrupt
sp1_ppw_entry_cmpl	3:3	r	'h0	Mask bit for sp1 ppw entry cmpl interrupt
mp_raw_entry_cmpl	2:2	r	'h0	Mask bit formp raw entry cmpl interrupt
mp_ycbcr_entry_cmpl	1:1	r	'h0	Mask bit for mp ycbcr entry cmpl interrupt
pp_dma_y_entry_cmpl	0:0	r	'h0	Mask bit for pp dma y entry cmpl interrupt

MI_HDR_CTRL

0x00005700 = Byte Address

Description: MI_HDR control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_dma_start_continuous	8:8	rw	'h0	Enables continuous mode. If set to 1, the same frame is read back over and over. A start pulse on dma_start is needed only for the first time. To stop continuous mode reset this bit (takes effect after the next frame end) or execute a soft reset. 0: Disabled; 1: enabled
hdr_dma_start	7:7	w	'h0	Enables DMA access 0: Disabled; 1: enabled
hdr_wr_enable	6:6	rw	'h0	1b'0: write disable 1b'1: write enable
hdr_rd_raw_cfg_update	5:5	w	'h0	Forced configuration update which leads to an immediate update of the shadow registers for this RAW read path.
hdr_rd_raw_auto_update	4:4	rw	'h0	Frame end update which leads to an immediate update of the shadow registers for this RAW read path.
hdr_init_offset_en	3:3	rw	'h0	Enables updating of the offset counters shadow registers for hdr path to the programmed register init values.
hdr_init_base_en	2:2	rw	'h0	Enables updating of the base address and buffer size shadow registers for hdr picture to the programmed register init values.

hdr_mi_cfg_upd	1:1	w	'h0	Forced configuration update which leads to an immediate update of the shadow registers. Depending on the init_offset_en and init_base_en enable bits in the MI_HDR_CTRL register, the offset counter, base address and buffer size shadow registers are also updated.
hdr_auto_update	0:0	rw	'h0	Automatic update of configuration registers forhdr path at frame end. 0: disabled; 1: enabled

MI_HDR_FMT

0x00005704 = Byte Address

Description: MI_HDR format register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_rd_xs_bit	29:28	rw	'h0	
hdr_rd_str1	27:26	rw	'h0	{hdr_rd_str1,hdr_rd_str} combination for HDRv3 indicate, in HDR mode Bit0: there's L Bit1: there's S Bit2: there's VS Bit3: there's XS
hdr_rd_raw_aligned	25:24	rw	'h0	00: Unaligned 01: Aligned mode0 10: Aligned mode1 11: Reserved
hdr_rd_vs_bit	23:22	rw	'h0	2'b00: 12bit 2'b01: 16bit 2'b1x: reserved
hdr_rd_s_bit	21:20	rw	'h0	2'b00: 12bit 2'b01: 16bit 2'b1x: reserved
hdr_rd_l_bit	19:18	rw	'h0	2'b00: 12bit 2'b01: 16bit 2'b1x: reserved
hdr_rd_str	17:16	rw	'h0	For HDRv2 2'b00: L 2'b01: L,S 2'b10: L,S,VS 2'b11: L,VS For HDRv3, refer to field hdr_rd_str1
hdr_wr_xs_bit	13:12	rw	'h0	2'b00: 12bit 2'b01: 16bit 2'b1x: reserved
hdr_wr_str1	11:10	rw	'h0	{hdr_wr_str1,hdr_wr_str} combination for HDRv3 indicate, in HDR mode Bit0: there's L Bit1: there's S Bit2: there's VS Bit3: there's XS
hdr_wr_raw_aligned	9:8	rw	'h0	00: Unaligned 01: Aligned mode0 10: Aligned mode1 11: Reserved
hdr_wr_vs_bit	7:6	rw	'h0	2'b00: 12bit 2'b01: 16bit 2'b1x: reserved
hdr_wr_s_bit	5:4	rw	'h0	2'b00: 12bit 2'b01: 16bit 2'b1x: reserved
hdr_wr_l_bit	3:2	rw	'h0	2'b00: 12bit 2'b01: 16bit 2'b1x: reserved
hdr_wr_str	1:0	rw	'h0	For HDRv2 2'b00: L 2'b01: L,S 2'b10: L,S,VS 2'b11: L,VS For HDRv3, refer to field hdr_wr_str1

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MI_HDR_DMA_START_BY_LINES

0x00005708 = Byte Address

Description: MI_HDR dma register

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_start_by_lines	15:0	rw	'h0	the written lines count of the last appearing frame after which start dma read

MI_HDR_RET_BUS_CFG

0x0000570c = Byte Address

Description: HDR AXI Bus cfg

Sub-Field Name	Bits	rw	Reset	Description
hdr_rd_swap_raw	19:16	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words
hdr_wr_swap_raw	3:0	rw	'h0	Swap bits control for 16 byte data width on AXI bus; YUV interleave and RGB raster modes share the same bits. Bit 0 to swap bytes Bit 1 to swap words Bit 2 to swap double words Bit 3 to swap four words

MI_HDR_RET_BUS_ID

0x00005710 = Byte Address

Description: HDR AXI Bus ID configuration

Sub-Field Name	Bits	rw	Reset	Description
hdr_bus_sw_en	23:23	rw	'h0	Control whether BUS works, and set to 1'b1 before ISP works
hdr_rd_burst_len	22:21	rw	'h0	Maximum burst length. 00: 4 01: 8 10:16 11: Reserved
hdr_wr_burst_len	20:19	rw	'h0	Maximum burst length. 00: 4 01: 8 10:16 11: Reserved
hdr_rd_id_en	18:18	rw	'h0	Hdr Path AXI ports read ID configuration enable
hdr_rd_id_cfg	17:10	rw	'h0	Hdr Path AXI ports read ID configuration
hdr_wr_id_en	8:8	rw	'h0	Hdr Path AXI ports write ID configuration enable
hdr_wr_id_cfg	7:0	rw	'h0	Hdr Path AXI ports write ID configuration

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MI_HDR_RET_BUS_TIMEO

0x00005714 = Byte Address

Description: HDR AXI Bus ctrol register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_bus_timeo_en	31:31	rw	'h0	Hdr Path AXI bus time out waiting enable
hdr_bus_timeo	30:1	rw	'h0	Hdr Path AXI bus time out waiting clock cycle numbers
hdr_bus_timeo_interrupt_disable	0:0	rw	'h0	This interrupt function switch: 1'b0: enable 1'b1: disable

MI_HDR_PIC_WIDTH

0x00005720 = Byte Address

Description: raw data width

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_l_pic_width	31:0	rw	'h0	L component picture width

MI_HDR_PIC_HEIGHT

0x00005724 = Byte Address

Description: raw data height

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_l_pic_height	31:0	rw	'h0	L component picture height

MI_HDR_L_BASE_AD_INIT

0x00005740 = Byte Address

Description: Base address for hdr path raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_l_base_ad_init	31:4	rw	'h0	Base address of L component ring buffer

MI_HDR_L_SIZE_INIT

0x00005744 = Byte Address

Description: Size of hdr path picture raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_l_size_init	28:4	rw	'h0	size of L component ring buffer

MI_HDR_L_OFFS_CNT_INIT

0x00005748 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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hdr_l_offs_cnt_init	28:4	rw	'h0	Offset counter initial value of MCM Path picture RAW data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_HDR_L_LLENGTH

0x0000574c = Byte Address

Description: Line length of hdr raw data

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_l_llength	31:0	rw	'h0	stride of L component

MI_HDR_L_PIC_LVAL

0x00005750 = Byte Address

Description: hdr raw data line valid bytes number, 128 bit Alignment.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_l_pic_lval	15:0	rw	'h0	Sizes occupied by valid data of a line; value should be a multiple of 16 (bus width = 16 bytes).

MI_HDR_L_IRQ_OFFS_INIT

0x00005754 = Byte Address

Description: Fill level interrupt offset value for hdr raw data

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_l_irq_offs_init	31:4	rw	'h0	fill level interrupt offset value of L component

MI_HDR_L_OFFS_CNT_START

0x00005758 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_l_offs_cnt_start	28:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (MCM Path picture RAW data). Updated at frame end. Note: A soft reset resets the contents to the reset value.

MI_HDR_L_BASE_AD_SHD

0x0000575c = Byte Address

Description: Base address for hdr raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

hdr_l_base_ad_shd	31:4	r	'h0	Base address of L component ring buffer

MI_HDR_L_SIZE_SHD

0x00005760 = Byte Address

Description: Size address for hdr raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_l_size_shd	28:4	r	'h0	size of L component ring buffer

MI_HDR_L_OFFS_CNT_SHD

0x00005764 = Byte Address

Description: Follow main path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_l_offs_cnt	28:4	r	'h0	Current offset counter of Self Path 2 picture RAW data ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_HDR_L_IRQ_OFFS_SHD

0x00005768 = Byte Address

Description: Fill level interrupt offset value for hdr raw data

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_l_irq_offs_shd	31:4	r	'h0	fill level interrupt offset value of L component

MI_HDR_S_BASE_AD_INIT

0x00005780 = Byte Address

Description: Base address for hdr path raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_s_base_ad_init	31:4	rw	'h0	Base address of S component ring buffer

MI_HDR_S_SIZE_INIT

0x00005784 = Byte Address

Description: Size of hdr path picture raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_s_size_init	28:4	rw	'h0	size of S component ring buffer

MI_HDR_S_OFFS_CNT_INIT

0x00005788 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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hdr_s_offs_cnt_init	28:4	rw	'h0	Offset counter initial value of MCM Path picture RAW data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_HDR_S_LLENGTH

0x0000578c = Byte Address

Description: Line length of hdr raw data

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_s_llength	31:0	rw	'h0	stride of S component

MI_HDR_S_PIC_LVAL

0x00005790 = Byte Address

Description: hdr raw data line valid bytes number, 128 bit Alignment.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_s_pic_lval	15:0	rw	'h0	Sizes occupied by valid data of a line; value should be a multiple of 16 (bus width = 16 bytes).

MI_HDR_S_IRQ_OFFS_INIT

0x00005794 = Byte Address

Description: Fill level interrupt offset value for hdr raw data

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_s_irq_offs_init	31:4	rw	'h0	fill level interrupt offset value of S component

MI_HDR_S_OFFS_CNT_START

0x00005798 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_s_offs_cnt_start	28:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (MCM Path picture RAW data). Updated at frame end. Note: A soft reset resets the contents to the reset value.

MI_HDR_S_BASE_AD_SHD

0x0000579c = Byte Address

Description: Base address for hdr raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

hdr_s_base_ad_shd	31:4	r	'h0	Base address of S component ring buffer

MI_HDR_S_SIZE_SHD

0x000057a0 = Byte Address

Description: Size address for hdr raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_s_size_shd	28:4	r	'h0	size of S component ring buffer

MI_HDR_S_OFFS_CNT_SHD

0x000057a4 = Byte Address

Description: Follow main path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_s_offs_cnt	28:4	r	'h0	Current offset counter of Self Path 2 picture RAW data ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_HDR_S_IRQ_OFFS_SHD

0x000057a8 = Byte Address

Description: Fill level interrupt offset value for hdr raw data

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_s_irq_offs_shd	31:0	r	'h0	fill level interrupt offset value of S component

MI_HDR_VS_BASE_AD_INIT

0x000057c0 = Byte Address

Description: Base address for hdr path raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_vs_base_ad_init	31:4	rw	'h0	Base address of VS component ring buffer

MI_HDR_VS_SIZE_INIT

0x000057c4 = Byte Address

Description: Size of hdr path picture raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_vs_size_init	28:4	rw	'h0	size of VS component ring buffer

MI_HDR_VS_OFFS_CNT_INIT

0x000057c8 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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hdr_vs_offs_cnt_init	28:4	rw	'h0	Offset counter initial value of MCM Path picture RAW data ring buffer. Programmed value becomes effective (visible in corresponding shadow register) after a soft reset, a forced software update or an automatic configuration update.

MI_HDR_VS_LLENGTH

0x000057cc = Byte Address

Description: Line length of hdr raw data

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_vs_llength	31:0	rw	'h0	stride of VS component

MI_HDR_VS_PIC_LVAL

0x000057d0 = Byte Address

Description: hdr raw data line valid bytes number, 128 bit Alignment.

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_vs_pic_lval	15:0	rw	'h0	Sizes occupied by valid data of a line; value should be a multiple of 16 (bus width = 16 bytes).

MI_HDR_VS_IRQ_OFFS_INIT

0x000057d4 = Byte Address

Description: Fill level interrupt offset value for hdr raw data

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_vs_irq_offs_init	31:4	rw	'h0	fill level interrupt offset value of VS component

MI_HDR_VS_OFFS_CNT_START

0x000057d8 = Byte Address

Description: Follow Main Path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_vs_offs_cnt_start	28:4	r	'h0	Offset counter value which points to the start address of the previously processed picture (MCM Path picture RAW data). Updated at frame end. Note: A soft reset resets the contents to the reset value.

MI_HDR_VS_BASE_AD_SHD

0x000057dc = Byte Address

Description: Base address for hdr raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

hdr_vs_base_ad_shd	31:4	r	'h0	Base address of VS component ring buffer

MI_HDR_VS_SIZE_SHD

0x000057e0 = Byte Address

Description: Size address for hdr raw data ring buffer

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_vs_size_shd	28:4	r	'h0	size of VS component ring buffer

MI_HDR_VS_OFFS_CNT_SHD

0x000057e4 = Byte Address

Description: Follow main path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_vs_offs_cnt	28:4	r	'h0	Current offset counter of Self Path 2 picture RAW data ring buffer for address generation. Note: Soft reset will reset the contents to the reset value.

MI_HDR_VS_IRQ_OFFS_SHD

0x000057e8 = Byte Address

Description: Fill level interrupt offset value for hdr raw data

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_vs_irq_offs_shd	31:0	r	'h0	fill level interrupt offset value of VS component

MI_HDR_DMA_PIC_WIDTH

0x00005800 = Byte Address

Description: Follow main path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_dma_pic_width	14:0	rw	'h0	Image width of the RAW component picture in pixels.

MI_HDR_DMA_PIC_HEIGHT

0x00005804 = Byte Address

Description: Follow main path RAW

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_dma_pic_height	14:0	rw	'h0	Image height of the RAW component picture in pixels.

MI_HDR_DMA_L_BASE_AD_INIT

0x00005820 = Byte Address

Description: RAW component image start address

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Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_l_base_ad_init	31:4	rw	'h0	RAW component image start address

MI_HDR_DMA_L_SIZE_INIT

0x00005824 = Byte Address

Description: RAW component image memory size in bytes

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_l_size_init	31:4	rw	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (dma_raw_llength * dma_raw_pic_height).

MI_HDR_DMA_L_LLENGTH

0x00005828 = Byte Address

Description: RAW component original line length

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_l_pic_llength	31:0	rw	'h0	Line length of the RAW component in bytes of the original image in memory. For an uncropped image, where lines follow each other without offset (no line stride), line length must match image width.

MI_HDR_DMA_L_LVAL

0x0000582c = Byte Address

Description: HDR_DMA L RAW Component line valid bytes number, 128 bit Alignment.

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_l_width_bytes	15:0	rw	'h0	Sizes occupied by valid data of a line; value should be a multiple of 16 (bus width = 16 bytes).

MI_HDR_DMA_L_BASE_AD_SHD

0x00005830 = Byte Address

Description: RAW component image start address

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_l_base_ad_init	31:4	r	'h0	RAW component image start address

MI_HDR_DMA_L_SIZE_SHD

0x00005834 = Byte Address

Description: RAW component image memory size in bytes

Sub-Field Name	Bits	rw	Reset	Description

hdr_dma_l_size_init	31:4	r	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (dma_raw_llength * dma_raw_pic_height).

MI_HDR_DMA_L_SIZE_BUF_INIT

0x00005838 = Byte Address

Description: RAW component image memory size in bytes in frame buffer of memory

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_dma_l_size_buf_init	31:4	rw	'h0	Image size of the RAW component in bytes which has to be the multiple of llength. This value indicate the memory isze in system sram or ddr. And it may be less than the size register

MI_HDR_DMA_L_SIZE_BUF_SHD

0x0000583c = Byte Address

Description: RAW component image memory size in bytes in frame buffer of memory

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_dma_l_size_buf_init	31:4	r	'h0	Image size of the RAW component in bytes which has to be the multiple of llength. This value indicate the memory isze in system sram or ddr. And it may be less than the size register

MI_HDR_DMA_S_BASE_AD_INIT

0x00005850 = Byte Address

Description: RAW component image start address

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_dma_s_base_ad_init	31:4	rw	'h0	RAW component image start address

MI_HDR_DMA_S_SIZE_INIT

0x00005854 = Byte Address

Description: RAW component image memory size in bytes

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_dma_s_size_init	31:4	rw	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (dma_raw_llength * dma_raw_pic_height).

MI_HDR_DMA_S_LLENGTH

0x00005858 = Byte Address

Description: RAW component original line length

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Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_s_pic_llength	31:0	rw	'h0	Line length of the RAW component in bytes of the original image in memory. For an uncropped image, where lines follow each other without offset (no line stride), line length must match image width.

MI_HDR_DMA_S_LVAL

0x0000585c = Byte Address

Description: HDR_DMA L RAW Component line valid bytes number, 128 bit Alignment.

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_s_width_bytes	15:0	rw	'h0	Sizes occupied by valid data of a line; value should be a multiple of 16 (bus width = 16 bytes).

MI_HDR_DMA_S_BASE_AD_SHD

0x00005860 = Byte Address

Description: RAW component image start address

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_s_base_ad_init	31:4	r	'h0	RAW component image start address

MI_HDR_DMA_S_SIZE_SHD

0x00005864 = Byte Address

Description: RAW component image memory size in bytes

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_s_size_init	31:4	r	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (dma_raw_llength * dma_raw_pic_height).

MI_HDR_DMA_S_SIZE_BUF_INIT

0x00005868 = Byte Address

Description: RAW component image memory size in bytes in frame buffer of memory

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_s_size_buf_init	31:4	rw	'h0	Image size of the RAW component in bytes which has to be the multiple of llength. This value indicate the memory isze in system sram or ddr. And it may be less than the size register

MI_HDR_DMA_S_SIZE_BUF_SHD

0x0000586c = Byte Address

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Description: RAW component image memory size in bytes in frame buffer of memory

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_s_size_buf_init	31:4	r	'h0	Image size of the RAW component in bytes which has to be the multiple of llength. This value indicate the memory isze in system sram or ddr. And it may be less than the size register

MI_HDR_DMA_VS_BASE_AD_INIT

0x00005880 = Byte Address

Description: RAW component image start address

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_vs_base_ad_init	31:4	rw	'h0	RAW component image start address

MI_HDR_DMA_VS_SIZE_INIT

0x00005884 = Byte Address

Description: RAW component image memory size in bytes

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_vs_size_init	31:4	rw	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (dma_raw_llength * dma_raw_pic_height).

MI_HDR_DMA_VS_LLENGTH

0x00005888 = Byte Address

Description: RAW component original line length

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_vs_pic_llength	31:0	rw	'h0	Line length of the RAW component in bytes of the original image in memory. For an uncropped image, where lines follow each other without offset (no line stride), line length must match image width.

MI_HDR_DMA_VS_LVAL

0x0000588c = Byte Address

Description: HDR_DMA L RAW Component line valid bytes number, 128 bit Alignment.

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_vs_width_bytes	15:0	rw	'h0	Sizes occupied by valid data of a line; value should be a multiple of 16 (bus width = 16 bytes).

MI_HDR_DMA_VS_BASE_AD_SHD

0x00005890 = Byte Address

Description: RAW component image start address

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_vs_base_ad_init	31:4	r	'h0	RAW component image start address

MI_HDR_DMA_VS_SIZE_SHD

0x00005894 = Byte Address

Description: RAW component image memory size in bytes

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_vs_size_init	31:4	r	'h0	Image size of the RAW component in bytes which has to be the RAW line length multiplied by the RAW image height (dma_raw_llength * dma_raw_pic_height).

MI_HDR_DMA_VS_SIZE_BUF_INIT

0x00005898 = Byte Address

Description: RAW component image memory size in bytes in frame buffer of memory

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_vs_size_buf_init	31:4	rw	'h0	Image size of the RAW component in bytes which has to be the multiple of llength. This value indicate the memory isze in system sram or ddr. And it may be less than the size register

MI_HDR_DMA_VS_SIZE_BUF_SHD

0x0000589c = Byte Address

Description: RAW component image memory size in bytes in frame buffer of memory

Sub-Field Name	Bits	rw	Reset	Description
hdr_dma_vs_size_buf_init	31:4	r	'h0	Image size of the RAW component in bytes which has to be the multiple of llength. This value indicate the memory isze in system sram or ddr. And it may be less than the size register

HDR_RT_CTRL

0x00005900 = Byte Address

Description: HDR_RT control register

Sub-Field Name	Bits	rw	Reset	Description
rt_inform_enable	17:17	rw	'h0	rt_inform_enable
vs_isnt_sram_stored	16:16	rw	'h0	1'b0: use internal line SRAM 1'b1: use external line memory

s_isnt_sram_stored	15:15	rw	'h0	1‘b0: use internal line SRAM 1'b1: use external line memory
l_isnt_sram_stored	14:14	rw	'h0	1‘b0: use internal line SRAM 1'b1: use external line memory
vs_dump	13:13	rw	'h0	vs_dump
s_dump	12:12	rw	'h0	s_dump
l_dump	11:11	rw	'h0	l_dump
mem_boundary_upd	10:10	w	'h0	mem_boundary_upd, auto clear after setting
hdr_rt_pixel_in_num	9:9	rw	'h0	hdr_rt_pixel_in_num: 0: D1(one pixel in one clk) 1: D2(two pixels in one clk)
hdr_rt_vsync_pol	8:8	rw	'h0	Vertical sync polarity 0: high active; 1: low active
hdr_rt_hsync_pol	7:7	rw	'h0	Horizontal sync polarity 0: high active; 1: low active
hdr_retiming_enable	6:6	rw	'h0	1b'0:disabled 1b'1:enabled
long_use_ddr_only_en	5:5	rw	'h0	enable only use long addr
short_use_ddr_only_en	4:4	rw	'h0	enable only use short addr
dump_mode_en	3:3	rw	'h0	all channel(L,S,VS go into DDR): 0: dump mode disable 1: dump mode enable
l_vs_combining_enable	2:2	rw	'h0	0: not l, vs for 2 exposure 1: l, vs for 2 exposure
exposure_count	1:0	rw	'h0	

HDR_RT_INTERVAL

0x00005904 = Byte Address

Description: HDR_RT interval

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_rt_interval1	31:16	r	'h0	the interval line count from L to VS
hdr_rt_interval0	15:0	r	'h0	the interval line count from L to S

HDR_RT_H_BLANK

0x00005908 = Byte Address

Description: HDR_RT_H_BLANK

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
hdr_rt_h_blank	15:0	rw	'h0	the blank internal between lines, by cycle of clock

HDR_RT_V_BLANK

0x0000590c = Byte Address

Description: HDR_RT_V_BLANK

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Sub-Field Name	Bits	rw	Reset	Description
hdr_rt_v_blank	23:0	rw	'h0	the blank internal between frames, by cycle of clock

HDR_RT_ACQ_H_SIZE

0x00005910 = Byte Address

Description: HDR_RT_ACQ_H_SIZE

Sub-Field Name	Bits	rw	Reset	Description
hdr_rt_acq_h_size	13:0	rw	'h0	the input sensor size of width

HDR_RT_ACQ_V_SIZE

0x00005914 = Byte Address

Description: HDR_RT_ACQ_V_SIZE

Sub-Field Name	Bits	rw	Reset	Description
hdr_rt_acq_v_size	13:0	rw	'h0	the input sensor size of height

HDR_RT_MEAS_HBLANK_L_L

0x00005918 = Byte Address

Description: HDR_RT_MEAS_HBLANK_L_L

Sub-Field Name	Bits	rw	Reset	Description
hdr_rt_meas_hblank_l_l	23:0	r	'h0	the h blank of L-to-L component

HDR_RT_MEAS_HBLANK_L_S

0x0000591c = Byte Address

Description: HDR_RT_MEAS_HBLANK_L_S

Sub-Field Name	Bits	rw	Reset	Description
hdr_rt_meas_hblank_l_s	23:0	r	'h0	the h blank of L-to-S component

HDR_RT_MEAS_HBLANK_S_VS

0x00005920 = Byte Address

Description: HDR_RT_MEAS_HBLANK_S_VS

Sub-Field Name	Bits	rw	Reset	Description
hdr_rt_meas_hblank_s_vs	23:0	r	'h0	the h blank of S-to-VS component

HDR_RT_MEAS_HBLANK_L_VS

0x00005924 = Byte Address

Description: HDR_RT_MEAS_HBLANK_L_VS

Sub-Field Name	Bits	rw	Reset	Description
hdr_rt_meas_hblank_l_vs	23:0	r	'h0	the h blank of L-to-VS component

HDR_RT_STATUS

0x00005928 = Byte Address

Description: HDR_RT_STATUS

Sub-Field Name	Bits	rw	Reset	Description
vs_pix_loss	2:2	r	'h0	input side backpress leading to pixel loss
s_pix_loss	1:1	r	'h0	input side backpress leading to pixel loss
l_pix_loss	0:0	r	'h0	input side backpress leading to pixel loss

MI_SEGMENT_CTRL

0x00005a00 = Byte Address

Description: MI_SEGMENT_CTRL

Sub-Field Name	Bits	rw	Reset	Description
mi_reset_offs_u_sp2	25:25	w	'h0	offset reset for sp2 u
mi_reset_offs_y_sp2	24:24	w	'h0	offset reset for sp2 y
mi_reset_offs_u_sp1	22:22	w	'h0	offset reset for sp1 u
mi_reset_offs_y_sp1	21:21	w	'h0	offset reset for sp1 y
mi_reset_offs_u_mp	19:19	w	'h0	offset reset for mp u
mi_reset_offs_y_mp	18:18	w	'h0	offset reset for mp y
mi_sp2_cr_segment_start	17:17	w	'h0	sp2 cb segment producer start, for SBI
mi_sp2_cb_segment_start	16:16	w	'h0	sp2 cb segment producer start, for SBI
mi_sp2_y_segment_start	15:15	w	'h0	sp2 y segment producer start, for SBI
mi_sp2_cr_segment_enable	14:14	rw	'h0	sp2_cr_segment_enable, for SBI
mi_sp2_cb_segment_enable	13:13	rw	'h0	sp2_cb_segment_enable, for SBI
mi_sp2_y_segment_enable	12:12	rw	'h0	sp2_y_segment_enable, for SBI
mi_sp1_cr_segment_start	11:11	w	'h0	sp1 cb segment producer start, for SBI
mi_sp1_cb_segment_start	10:10	w	'h0	sp1 cb segment producer start, for SBI
mi_sp1_y_segment_start	9:9	w	'h0	sp1 y segment producer start, for SBI
mi_sp1_cr_segment_enable	8:8	rw	'h0	sp1_cr_segment_enable, for SBI
mi_sp1_cb_segment_enable	7:7	rw	'h0	sp1_cb_segment_enable, for SBI
mi_sp1_y_segment_enable	6:6	rw	'h0	sp1_y_segment_enable, for SBI
mi_mp_cr_segment_start	5:5	w	'h0	mp cb segment producer start, for SBI
mi_mp_cb_segment_start	4:4	w	'h0	mp cb segment producer start, for SBI
mi_mp_y_segment_start	3:3	w	'h0	mp y segment producer start, for SBI
mi_mp_cr_segment_enable	2:2	rw	'h0	mp_cr_segment_enable, for SBI
mi_mp_cb_segment_enable	1:1	rw	'h0	mp_cb_segment_enable, for SBI
mi_mp_y_segment_enable	0:0	rw	'h0	mp_y_segment_enable, for SBI

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MI_MP_YCBCR_STREAM_ID

0x00005a04 = Byte Address

Description: MI_MP_YCBCR_STREAM_ID

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mi_mp_cr_stream_id	23:16	rw	'h0	mp_cr_stream_id, for SBI
mi_mp_cb_stream_id	15:8	rw	'h0	mp_cb_stream_id, for SBI
mi_mp_y_stream_id	7:0	rw	'h0	mp_y_stream_id, for SBI

MI_MP_Y_SEGMENT_ENTRY_SIZE

0x00005a08 = Byte Address

Description: MI_MP_Y_SEGMENT_ENTRY_SIZE

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_segment_entry_size	15:0	rw	'h0	mp_y_segment_entry_size, for SBI

MI_MP_Y_SEGMENT_ENTRY_COUNT

0x00005a0c = Byte Address

Description: MI_MP_Y_SEGMENT_ENTRY_COUNT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_y_segment_entry_count	15:0	rw	'h0	mp_y_segment_entry_count, for SBI

MI_MP_CB_SEGMENT_ENTRY_SIZE

0x00005a10 = Byte Address

Description: MI_MP_CB_SEGMENT_ENTRY_SIZE

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_cb_segment_entry_size	15:0	rw	'h0	mp_cb_segment_entry_size, for SBI

MI_MP_CB_SEGMENT_ENTRY_COUNT

0x00005a14 = Byte Address

Description: MI_MP_CB_SEGMENT_ENTRY_COUNT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_cb_segment_entry_count	15:0	rw	'h0	mp_cb_segment_entry_count, for SBI

MI_MP_CR_SEGMENT_ENTRY_SIZE

0x00005a18 = Byte Address

Description: MI_MP_CR_SEGMENT_ENTRY_SIZE

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_cr_segment_entry_size	15:0	rw	'h0	mp_cr_segment_entry_size, for SBI

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MI_MP_CR_SEGMENT_ENTRY_COUNT

0x00005a1c = Byte Address

Description: MI_MP_CR_SEGMENT_ENTRY_COUNT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mp_cr_segment_entry_count	15:0	rw	'h0	mp_cr_segment_entry_count, for SBI

MI_SP1_YCBCR_STREAM_ID

0x00005a20 = Byte Address

Description: MI_SP1_YCBCR_STREAM_ID

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mi_sp1_cr_stream_id	23:16	rw	'h0	sp1_cr_stream_id, for SBI
mi_sp1_cb_stream_id	15:8	rw	'h0	sp1_cb_stream_id, for SBI
mi_sp1_y_stream_id	7:0	rw	'h0	sp1_y_stream_id, for SBI

MI_SP1_Y_SEGMENT_ENTRY_SIZE

0x00005a24 = Byte Address

Description: MI_SP1_Y_SEGMENT_ENTRY_SIZE

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp1_y_segment_entry_size	15:0	rw	'h0	sp1_y_segment_entry_size, for SBI

MI_SP1_Y_SEGMENT_ENTRY_COUNT

0x00005a28 = Byte Address

Description: MI_SP1_Y_SEGMENT_ENTRY_COUNT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp1_y_segment_entry_count	15:0	rw	'h0	sp1_y_segment_entry_count, for SBI

MI_SP1_CB_SEGMENT_ENTRY_SIZE

0x00005a2c = Byte Address

Description: MI_SP1_CB_SEGMENT_ENTRY_SIZE

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp1_cb_segment_entry_size	15:0	rw	'h0	sp1_cb_segment_entry_size, for SBI

MI_SP1_CB_SEGMENT_ENTRY_COUNT

0x00005a30 = Byte Address

Description: MI_SP1_CB_SEGMENT_ENTRY_COUNT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp1_cb_segment_entry_count	15:0	rw	'h0	sp1_cb_segment_entry_count, for SBI

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MI_SP1_CR_SEGMENT_ENTRY_SIZE

0x00005a34 = Byte Address

Description: MI_SP1_CR_SEGMENT_ENTRY_SIZE

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp1_cr_segment_entry_size	15:0	rw	'h0	sp1_cr_segment_entry_size, for SBI

MI_SP1_CR_SEGMENT_ENTRY_COUNT

0x00005a38 = Byte Address

Description: MI_SP1_CR_SEGMENT_ENTRY_COUNT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp1_cr_segment_entry_count	15:0	rw	'h0	sp1_cr_segment_entry_count, for SBI

MI_SP2_YCBCR_STREAM_ID

0x00005a3c = Byte Address

Description: MI_SP2_YCBCR_STREAM_ID

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mi_sp2_cr_stream_id	23:16	rw	'h0	sp2_cr_stream_id, for SBI
mi_sp2_cb_stream_id	15:8	rw	'h0	sp2_cb_stream_id, for SBI
mi_sp2_y_stream_id	7:0	rw	'h0	sp2_y_stream_id, for SBI

MI_SP2_Y_SEGMENT_ENTRY_SIZE

0x00005a40 = Byte Address

Description: MI_SP2_Y_SEGMENT_ENTRY_SIZE

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_y_segment_entry_size	15:0	rw	'h0	sp2_y_segment_entry_size, for SBI

MI_SP2_Y_SEGMENT_ENTRY_COUNT

0x00005a44 = Byte Address

Description: MI_SP2_Y_SEGMENT_ENTRY_COUNT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_y_segment_entry_count	15:0	rw	'h0	sp2_y_segment_entry_count, for SBI

MI_SP2_CB_SEGMENT_ENTRY_SIZE

0x00005a48 = Byte Address

Description: MI_SP2_CB_SEGMENT_ENTRY_SIZE

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cb_segment_entry_size	15:0	rw	'h0	sp2_cb_segment_entry_size, for SBI

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MI_SP2_CB_SEGMENT_ENTRY_COUNT

0x00005a4c = Byte Address

Description: MI_SP2_CB_SEGMENT_ENTRY_COUNT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cb_segment_entry_count	15:0	rw	'h0	sp2_cb_segment_entry_count, for SBI

MI_SP2_CR_SEGMENT_ENTRY_SIZE

0x00005a50 = Byte Address

Description: MI_SP2_CR_SEGMENT_ENTRY_SIZE

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cr_segment_entry_size	15:0	rw	'h0	sp2_cr_segment_entry_size, for SBI

MI_SP2_CR_SEGMENT_ENTRY_COUNT

0x00005a54 = Byte Address

Description: MI_SP2_CR_SEGMENT_ENTRY_COUNT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
sp2_cr_segment_entry_count	15:0	rw	'h0	sp2_cr_segment_entry_count, for SBI

MI_MCM_RAW1_ENTRY_LINE_NUM

0x00005a88 = Byte Address

Description: the line count in each entry

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_entry_line_num	15:0	rw	'h0	the line count in each entriy

MI_MCM_RAW1_ENTRY_PIC_CNT

0x00005a8c = Byte Address

Description: the line count of picture

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw1_entry_pic_cnt	15:0	r	'h0	the line count of picture

MI_MCM_RAW2_ENTRY_LINE_NUM

0x00005a90 = Byte Address

Description: the line count in each entry

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
mcm_raw2_entry_line_num	15:0	rw	'h0	the line count in each entriy

MI_MCM_RAW2_ENTRY_PIC_CNT

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0x00005a94 = Byte Address

Description: the line count of picture

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw2_entry_pic_cnt	15:0	r	'h0	the line count of picture

MI_MCM_RAW3_ENTRY_LINE_NUM

0x00005a98 = Byte Address

Description: the line count in each entry

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw3_entry_line_num	15:0	rw	'h0	the line count in each entriy

MI_MCM_RAW3_ENTRY_PIC_CNT

0x00005a9c = Byte Address

Description: the line count of picture

Sub-Field Name	Bits	rw	Reset	Description
mcm_raw3_entry_pic_cnt	15:0	r	'h0	the line count of picture

ISP_GAMMA_FE_CTRL

0x00005f00 = Byte Address

Description: Gamma FE control register

Sub-Field Name	Bits	rw	Reset	Description
isp_gamma_fe_bls_en	1:1	rw	'h0	Enable Black Level Subtraction in the Gamma FE module
isp_gamma_fe_ctrl_en	0:0	rw	'h0	Enable Gamma FE module

ISP_GAMMA_FE_PRE_BLS_A_FIXED

0x00005f04 = Byte Address

Description: Gamma FE pre BLS A value

Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_pre_bls_a_fixed	20:0	rw	'h0	Gamma FE pre BLS A value

ISP_GAMMA_FE_PRE_BLS_B_FIXED

0x00005f08 = Byte Address

Description: Gamma FE pre BLS B value

Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_pre_bls_b_fixed	20:0	rw	'h0	Gamma FE pre BLS B value

ISP_GAMMA_FE_PRE_BLS_C_FIXED

0x00005f0c = Byte Address

Description: Gamma FE pre BLS C value

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Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_pre_bls_c_fixed	20:0	rw	'h0	Gamma FE pre BLS C value

ISP_GAMMA_FE_PRE_BLS_D_FIXED

0x00005f10 = Byte Address

Description: Gamma FE pre BLS D value

Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_pre_bls_d_fixed	20:0	rw	'h0	Gamma FE pre BLS D value

ISP_GAMMA_FE_BLS_A_FIXED

0x00005f14 = Byte Address

Description: Gamma FE BLS A value

Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_bls_a_fixed	20:0	rw	'h0	Gamma FE BLS A value

ISP_GAMMA_FE_BLS_B_FIXED

0x00005f18 = Byte Address

Description: Gamma FE BLS B value

Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_bls_b_fixed	20:0	rw	'h0	Gamma FE BLS B value

ISP_GAMMA_FE_BLS_C_FIXED

0x00005f1c = Byte Address

Description: Gamma FE BLS C value

Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_bls_c_fixed	20:0	rw	'h0	Gamma FE BLS C value

ISP_GAMMA_FE_BLS_D_FIXED

0x00005f20 = Byte Address

Description: Gamma FE BLS D value

Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_bls_d_fixed	20:0	rw	'h0	Gamma FE BLS D value

ISP_GAMMA_FE_Y_ADDR

0x00005f24 = Byte Address

Description: address of y memory in gamme fe curve

Sub-Field Name	Bits	rw	Reset	Description
compand_expand_y_addr	31:0	rw	'h0	address of y memory in gamme fe curve

ISP_GAMMA_FE_Y_WRITE_DATA

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0x00005f28 = Byte Address

Description: px and y data in gamma fe curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
compand_expand_y_write_data	31:0	rw	'h0	px and y data in gamma fe curve

ISP_GAMMA_FE_CTRL_SHD

0x00005f2c = Byte Address

Description: Shadow for Gamma Fe control register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_gamma_fe_bls_en_shd	1:1	r	'h0	Shadow for Enable Black Level Subtraction in the Gamma FE module
isp_gamma_fe_ctrl_en_shd	0:0	r	'h0	Shadow for Enable Gamma FE module

ISP_GAMMA_FE_PRE_BLS_A_FIXED_SHD

0x00005f30 = Byte Address

Description: Shadow for Gamma FE pre BLS A value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_fe_pre_bls_a_fixed_shd	20:0	r	'h0	Shadow for Gamma FE pre BLS A value

ISP_GAMMA_FE_PRE_BLS_B_FIXED_SHD

0x00005f34 = Byte Address

Description: Shadow for Gamma FE pre BLS B value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_fe_pre_bls_b_fixed_shd	20:0	r	'h0	Shadow for Gamma FE pre BLS B value

ISP_GAMMA_FE_PRE_BLS_C_FIXED_SHD

0x00005f38 = Byte Address

Description: Shadow for Gamma FE pre BLS C value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_fe_pre_bls_c_fixed_shd	20:0	r	'h0	Shadow for Gamma FE pre BLS C value

ISP_GAMMA_FE_PRE_BLS_D_FIXED_SHD

0x00005f3c = Byte Address

Description: Shadow for Gamma FE pre BLS D value

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Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_pre_bls_d_fixed_shd	20:0	r	'h0	Shadow for Gamma FE pre BLS D value

ISP_GAMMA_FE_BLS_A_FIXED_SHD

0x00005f40 = Byte Address

Description: Shadow for Gamma FE BLS A value

Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_bls_a_fixed_shd	20:0	r	'h0	Shadow for Gamma FE BLS A value

ISP_GAMMA_FE_BLS_B_FIXED_SHD

0x00005f44 = Byte Address

Description: Shadow for Gamma FE BLS B value

Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_bls_b_fixed_shd	20:0	r	'h0	Shadow for Gamma FE BLS B value

ISP_GAMMA_FE_BLS_C_FIXED_SHD

0x00005f48 = Byte Address

Description: Shadow for Gamma FE BLS C value

Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_bls_c_fixed_shd	20:0	r	'h0	Shadow for Gamma FE BLS C value

ISP_GAMMA_FE_BLS_D_FIXED_SHD

0x00005f4c = Byte Address

Description: Shadow for Gamma FE BLS D value

Sub-Field Name	Bits	rw	Reset	Description
gamma_fe_bls_d_fixed_shd	20:0	r	'h0	Shadow for Gamma FE BLS D value

ISP_GAMMA_BE_CTRL

0x00006000 = Byte Address

Description: Gamma BE control register

Sub-Field Name	Bits	rw	Reset	Description
isp_gamma_be_bls_en	1:1	rw	'h0	Enable Black Level Subtraction in the Gamma BE module
isp_gamma_be_ctrl_en	0:0	rw	'h0	Enable Gamma BE module

ISP_GAMMA_BE_PRE_BLS_A_FIXED

0x00006004 = Byte Address

Description: Gamma BE pre BLS A value

Sub-Field Name	Bits	rw	Reset	Description

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gamma_be_pre_bls_a_fixed	20:0	rw	'h0	Gamma BE pre BLS A value

ISP_GAMMA_BE_PRE_BLS_B_FIXED

0x00006008 = Byte Address

Description: Gamma BE pre BLS B value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_pre_bls_b_fixed	20:0	rw	'h0	Gamma BE pre BLS B value

ISP_GAMMA_BE_PRE_BLS_C_FIXED

0x0000600c = Byte Address

Description: Gamma BE pre BLS C value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_pre_bls_c_fixed	20:0	rw	'h0	Gamma BE pre BLS C value

ISP_GAMMA_BE_PRE_BLS_D_FIXED

0x00006010 = Byte Address

Description: Gamma BE pre BLS D value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_pre_bls_d_fixed	20:0	rw	'h0	Gamma BE pre BLS D value

ISP_GAMMA_BE_BLS_A_FIXED

0x00006014 = Byte Address

Description: Gamma BE BLS A value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_bls_a_fixed	20:0	rw	'h0	Gamma BE BLS A value

ISP_GAMMA_BE_BLS_B_FIXED

0x00006018 = Byte Address

Description: Gamma BE BLS B value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_bls_b_fixed	20:0	rw	'h0	Gamma BE BLS B value

ISP_GAMMA_BE_BLS_C_FIXED

0x0000601c = Byte Address

Description: Gamma BE BLS C value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_bls_c_fixed	20:0	rw	'h0	Gamma BE BLS C value

ISP_GAMMA_BE_BLS_D_FIXED

0x00006020 = Byte Address

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Description: Gamma BE BLS D value

Sub-Field Name	Bits	rw	Reset	Description
gamma_be_bls_d_fixed	20:0	rw	'h0	Gamma BE BLS D value

ISP_GAMMA_BE_Y_ADDR

0x00006024 = Byte Address

Description: address of y memory in Gamma BE curve

Sub-Field Name	Bits	rw	Reset	Description
compand_expand_y_addr	31:0	rw	'h0	address of y memory in gamme be curve

ISP_GAMMA_BE_Y_WRITE_DATA

0x00006028 = Byte Address

Description: px and y data in Gamma BE curve

Sub-Field Name	Bits	rw	Reset	Description
compand_expand_y_write_data	31:0	rw	'h0	px and y data in gamma be curve

ISP_GAMMA_BE_CTRL_SHD

0x0000602c = Byte Address

Description: Shadow for Gamma BE control register

Sub-Field Name	Bits	rw	Reset	Description
isp_gamma_be_bls_en_shd	1:1	r	'h0	Shadow for Enable Black Level Subtraction in the Gamma BE module
isp_gamma_be_ctrl_en_shd	0:0	r	'h0	Shadow for Enable Gamma BE module

ISP_GAMMA_BE_PRE_BLS_A_FIXED_SHD

0x00006030 = Byte Address

Description: Shadow for Gamma BE pre BLS A value

Sub-Field Name	Bits	rw	Reset	Description
gamma_be_pre_bls_a_fixed_shd	20:0	r	'h0	Shadow for Gamma BE pre BLS A value

ISP_GAMMA_BE_PRE_BLS_B_FIXED_SHD

0x00006034 = Byte Address

Description: Shadow for Gamma BE pre BLS B value

Sub-Field Name	Bits	rw	Reset	Description
gamma_be_pre_bls_b_fixed_shd	20:0	r	'h0	Shadow for Gamma BE pre BLS B value

ISP_GAMMA_BE_PRE_BLS_C_FIXED_SHD

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0x00006038 = Byte Address

Description: Shadow for Gamma BE pre BLS C value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_pre_bls_c_fixed_shd	20:0	r	'h0	Shadow for Gamma BE pre BLS C value

ISP_GAMMA_BE_PRE_BLS_D_FIXED_SHD

0x0000603c = Byte Address

Description: Shadow for Gamma BE pre BLS D value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_pre_bls_d_fixed_shd	20:0	r	'h0	Shadow for Gamma BE pre BLS D value

ISP_GAMMA_BE_BLS_A_FIXED_SHD

0x00006040 = Byte Address

Description: Shadow for Gamma BE BLS A value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_bls_a_fixed_shd	20:0	r	'h0	Shadow for Gamma BE BLS A value

ISP_GAMMA_BE_BLS_B_FIXED_SHD

0x00006044 = Byte Address

Description: Shadow for Gamma BE BLS B value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_bls_b_fixed_shd	20:0	r	'h0	Shadow for Gamma BE BLS B value

ISP_GAMMA_BE_BLS_C_FIXED_SHD

0x00006048 = Byte Address

Description: Shadow for Gamma BE BLS C value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_bls_c_fixed_shd	20:0	r	'h0	Shadow for Gamma BE BLS C value

ISP_GAMMA_BE_BLS_D_FIXED_SHD

0x0000604c = Byte Address

Description: Shadow for Gamma BE BLS D value

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
gamma_be_bls_d_fixed_shd	20:0	r	'h0	Shadow for Gamma BE BLS D value

ISP_WDR5_CTRL

0x00006100 = Byte Address

Description: WDR 5.0 Control Register

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Sub-Field Name	Bits	rw	Reset	Description
wdr5_dummy_blk	16:1	rw	'h0	WDR5:dummy_blk
wdr5_enable	0:0	rw	'h0	WDR5: set enable

ISP_WDR5_BLOCK_SIZE

0x00006104 = Byte Address

Description: WDR 5.0 block size

Sub-Field Name	Bits	rw	Reset	Description
wdr5_block_height	19:10	rw	'h0	WDR5: set block height; this is equal to floor(width/16)
wdr5_block_width	9:0	rw	'h0	WDR5: set block width; this is equal to floor(height/16)

ISP_WDR5_BLOCK_AREA_FACTOR

0x00006108 = Byte Address

Description: WDR 5.0 block area

Sub-Field Name	Bits	rw	Reset	Description
wdr5_block_area_inverse	19:0	rw	'h0	WDR5: set the inverse of block area; this is equal to floor(1024*1024/(block_height*block_width))

ISP_WDR5_VALUE_WEIGHT

0x0000610c = Byte Address

Description: WDR 5.0 value weight

Sub-Field Name	Bits	rw	Reset	Description
wdr5_value_weight_3	19:15	rw	'h0	WDR5: set weight for gray level calculation, default(h8)
wdr5_value_weight_2	14:10	rw	'h0	WDR5: set weight for gray level calculation, default(h8)
wdr5_value_weight_1	9:5	rw	'h0	WDR5: set weight for gray level calculation, default(h8)
wdr5_value_weight_0	4:0	rw	'h0	WDR5: set weight for gray level calculation, default(h8)

ISP_WDR5_STRENGTH

0x00006110 = Byte Address

Description: WDR 5.0 effect control

Sub-Field Name	Bits	rw	Reset	Description
wdr5_strength	24:17	rw	'h0	WDR5: set strength, default(h80)

wdr5_low_strength	16:8	rw	'h0	WDR5: set low bit strength ,default(h100)
wdr5_high_strength	7:0	rw	'h0	WDR5:set high bit strength, default(h00)

ISP_WDR5_CONTRAST

0x00006114 = Byte Address

Description: WDR 5.0 contrast control

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_drc_bayer_ratio	19:11	rw	'h0	WDR5:equal ratio in stitching,default(h01)
wdr5_contrast	10:0	rw	'h0	WDR5: set contrast, default(h100)

ISP_WDR5_PIXEL_ENTROPY

0x00006118 = Byte Address

Description: WDR 5.0 entropy control

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_pixel_base_entropy	19:10	rw	'h0	WDR5: set factor for all pixel based on image entropy, default(h00)
wdr5_pixel_slope_entropy	9:0	rw	'h0	WDR5: set factor for all pixel based on image entropy, default(h00)

ISP_WDR5_SIGMA_HEIGHT

0x0000611c = Byte Address

Description: WDR 5.0 vertical factor

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_biliteral_height_sigma	19:0	rw	'h0	WDR5: set sigma height for internal biliteral interpolation, default(h2222)

ISP_WDR5_SIGMA_WIDTH

0x00006120 = Byte Address

Description: WDR 5.0 horizontal factor

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_biliteral_width_sigma	19:0	rw	'h0	WDR5: set sigma width for internal biliteral interpolation, default(h3D22)

ISP_WDR5_SIGMA_VALUE

0x00006124 = Byte Address

Description: WDR 5.0 gray factor

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_biliteral_value_sigma	19:0	rw	'h0	WDR5: set sigma value for internal biliteral interpolation, default(h400)

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ISP_WDR5_BLOCK_FLAG

0x00006128 = Byte Address

Description: WDR 5.0 block adjust

Sub-Field Name	Bits	rw	Reset	Description
wdr5_block_row_flag	31:16	rw	'h0	WDR5: set whether its block height should +1 for 16x16 block.
wdr5_block_col_flag	15:0	rw	'h0	WDR5: set whether its block width should +1 for 16x16 block.

ISP_WDR5_PIXEL_ADJUST

0x0000612c = Byte Address

Description: WDR 5.0 pixel control

Sub-Field Name	Bits	rw	Reset	Description
wdr5_pixel_base_adjust	15:8	rw	'h0	WDR5: set factor for all pixel based on external preference, default(h0)
wdr5_pixel_slope_adjust	7:0	rw	'h0	WDR5: set factor for all pixel based on external preference, default(h80)

ISP_WDR5_FLAT

0x00006130 = Byte Address

Description: WDR 5.0 flat control

Sub-Field Name	Bits	rw	Reset	Description
wdr5_local_flat_mode	19:19	rw	'h0	WDR5: set local flat mode, default(h00)
wdr5_local_flat_level	18:15	rw	'h0	WDR5: set local flat level, default(h05)
wdr5_global_curve_mode	14:14	rw	'h0	WDR5: set global curve mode, default(h00)
wdr5_global_flat_level	13:10	rw	'h0	WDR5: set global flat level, default(h05)
wdr5_global_fixed_weight	9:0	rw	'h0	WDR5: set global fixed weight, default(h3ff)

ISP_WDR5_FRAME_AVERAGE

0x00006134 = Byte Address

Description: WDR 5.0 frame average output

Sub-Field Name	Bits	rw	Reset	Description
wdr5_frame_average	31:0	r	'h0	WDR5: ouput image average

ISP_WDR5_FRAME_STD

0x00006138 = Byte Address

Description: WDR 5.0 frame std output

Sub-Field Name	Bits	rw	Reset	Description

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wdr5_frame_std	31:0	r	'h0	WDR5: ouput image STD

ISP_WDR5_STRENGTH_SHD

0x0000613c = Byte Address

Description: WDR 5.0 effect control shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_enable_shd	25:25	r	'h0	WDR5: set enable shadow
wdr5_strength_shd	24:17	r	'h0	WDR5: set strength shadow
wdr5_low_strength_shd	16:8	r	'h0	WDR5: set low bit strength shadow
wdr5_high_strength_shd	7:0	r	'h0	WDR5:set high bit strength shadow

ISP_WDR5_CONTRAST_SHD

0x00006140 = Byte Address

Description: WDR 5.0 contrast control shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_drc_bayer_ratio_shd	19:11	r	'h0	WDR5:equal ratio in stitching shadow
wdr5_contrast_shd	10:0	r	'h0	WDR5: set contrast shadow

ISP_WDR5_PIXEL_ENTROPY_SHD

0x00006144 = Byte Address

Description: WDR 5.0 entropy control shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_pixel_base_entropy_shd	19:10	r	'h0	WDR5: set factor for all pixel based on image entropy shadow
wdr5_pixel_slope_entropy_shd	9:0	r	'h0	WDR5: set factor for all pixel based on image entropy shadow

ISP_WDR5_PIXEL_ADJUST_SHD

0x00006148 = Byte Address

Description: WDR 5.0 pixel control shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_pixel_base_adjust_shd	15:8	r	'h0	WDR5: set factor for all pixel based on external preference shadow
wdr5_pixel_slope_adjust_shd	7:0	r	'h0	WDR5: set factor for all pixel based on external preference shadow

ISP_WDR5_FLAT_SHD

0x0000614c = Byte Address

Description: WDR 5.0 flat control shadow

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Sub-Field Name	Bits	rw	Reset	Description
wdr5_local_flat_mode_shd	19:19	r	'h0	WDR5: set local flat mode shadow
wdr5_local_flat_level_shd	18:15	r	'h0	WDR5: set local flat leve lshadow
wdr5_global_curve_mode_shd	14:14	r	'h0	WDR5: set global curve mode shadow
wdr5_global_flat_level_shd	13:10	r	'h0	WDR5: set global flat level shadow
wdr5_global_fixed_weight_shd	9:0	r	'h0	WDR5: set global fixed weight shadow

ISP_WDR5_LUT_HISTOGRAM_ADDR

0x00006150 = Byte Address

Description: Pointer to entry of histogram value LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_histogram_addr	5:0	rw	'h0	Pointer to entry of histogram value LUT for Curve

ISP_WDR5_LUT_HISTOGRAM_WRITE_DATA

0x00006154 = Byte Address

Description: Write data of histogram LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_histogram_write_data	29:0	rw	'h0	Write data of histogram LUT for Curve

ISP_WDR5_LUT_SHIFT_ADDR

0x00006158 = Byte Address

Description: Pointer to entry of shift value LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_shift_addr	3:0	rw	'h0	Pointer to entry of shift value LUT for Curve

ISP_WDR5_LUT_SHIFT_WRITE_DATA

0x0000615c = Byte Address

Description: Write data of shift LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_shift_write_data	29:0	rw	'h0	Write data of shift LUT for Curve

ISP_WDR5_LUT_SHIFT0_ADDR

0x00006160 = Byte Address

Description: Pointer to entry of shift0 value LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description

wdr5_lut_shift0_addr	3:0	rw	'h0	Pointer to entry of shift0 value LUT for Curve

ISP_WDR5_LUT_SHIFT0_WRITE_DATA

0x00006164 = Byte Address

Description: Write data of shift0 LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_shift0_write_data	31:0	rw	'h0	Write data of shift0 LUT for Curve

ISP_WDR5_LUT_GAMMAPRE_ADDR

0x00006168 = Byte Address

Description: Pointer to entry of gamma pre value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_gammapre_addr	5:0	rw	'h0	Pointer to entry of gamma pre value LUT for Curve

ISP_WDR5_LUT_GAMMAPRE_WRITE_DATA

0x0000616c = Byte Address

Description: Write data of gamma pre LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_gammapre_write_data	29:0	rw	'h0	Write data of gamma pre LUT for Curve

ISP_WDR5_LUT_GAMMAUP_ADDR

0x00006170 = Byte Address

Description: Pointer to entry of gamma up value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_gammaup_addr	5:0	rw	'h0	Pointer to entry of gamma up value LUT for Curve

ISP_WDR5_LUT_GAMMAUP_WRITE_DATA

0x00006174 = Byte Address

Description: Write data of gamma up LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_gammaup_write_data	29:0	rw	'h0	Write data of gamma up LUT for Curve

ISP_WDR5_LUT_GAMMADOWN_ADDR

0x00006178 = Byte Address

Description: Pointer to entry of gamma down value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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wdr5_lut_gammadown_addr	4:0	rw	'h0	Pointer to entry of gamma down value LUT for Curve

ISP_WDR5_LUT_GAMMADOWN_WRITE_DATA

0x0000617c = Byte Address

Description: Write data of gamma down LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_gammadown_write_data	29:0	rw	'h0	Write data of gamma down LUT for Curve

ISP_WDR5_LUT_ENTROPY_ADDR

0x00006180 = Byte Address

Description: Pointer to entry of entropy value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_entropy_addr	4:0	rw	'h0	Pointer to entry of entropy value LUT for Curve

ISP_WDR5_LUT_ENTROPY_WRITE_DATA

0x00006184 = Byte Address

Description: Write data of entropy LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_entropy_write_data	26:0	rw	'h0	Write data of entropy LUT for Curve

ISP_WDR5_LUT_DISTANCE_WEIGHT_ADDR

0x00006188 = Byte Address

Description: Pointer to entry of distance weight value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_distance_weight_addr	4:0	rw	'h0	Pointer to entry of distance weight value LUT for Curve

ISP_WDR5_LUT_DISTANCE_WEIGHT_WRITE_DATA

0x0000618c = Byte Address

Description: Write data ofdistance weight LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_distance_weight_write_data	27:0	rw	'h0	Write data ofdistance weight LUT for Curve

ISP_WDR5_LUT_DIFFERENCE_WEIGHT_ADDR

0x00006190 = Byte Address

Description: Pointer to entry of difference weight value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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wdr5_lut_difference_weight_addr	4:0	rw	'h0	Pointer to entry of difference weight value LUT for Curve

ISP_WDR5_LUT_DIFFERENCE_WEIGHT_WRITE_DATA

0x00006194 = Byte Address

Description: Write data of difference weight LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_difference_weight_write_data	27:0	rw	'h0	Write data of difference weight LUT for Curve

ISP_WDR5_LUT_WEIGHT_RAW_ADDR

0x00006198 = Byte Address

Description: Pointer to entry of raw weight value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_weight_raw_addr	4:0	rw	'h0	Pointer to entry of raw weight value LUT for Curve

ISP_WDR5_LUT_WEIGHT_RAW_WRITE_DATA

0x0000619c = Byte Address

Description: Write data of raw weight LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_weight_raw_write_data	29:0	rw	'h0	Write data of raw weight LUT for Curve

ISP_WDR5_LUT_MAXGAIN_ADDR

0x000061a0 = Byte Address

Description: Pointer to entry of maxgain value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_maxgain_addr	5:0	rw	'h0	Pointer to entry of maxgain value LUT for Curve;The number of combined bins for WDR5 is 44; The number of combined bins for WDR5.1 is 33

ISP_WDR5_LUT_MAXGAIN_WRITE_DATA

0x000061a4 = Byte Address

Description: Write data of maxgain LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_maxgain_write_data_h	31:30	rw	'h0	Write data of maxgain LUT for Curve;bin_combine_0[31:0]=bin_0+bin_1(inWDR5.1)

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wdr5_lut_maxgain_write_data	29:0	rw	'h0	Write data of maxgain LUT for Curve;bin_combine_0[29:0]=bin_0+bin_1[9:0]; bin_combine_1[29:0]=bin_1[19:10]+bin_2(in WDR5)

ISP_WDR5_LUT_MINGAIN_ADDR

0x000061a8 = Byte Address

Description: Pointer to entry of mingain value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_mingain_addr	5:0	rw	'h0	Pointer to entry of mingain value LUT for Curve;The number of combined bins for WDR5 is 44; The number of combined bins for WDR5.1 is 33

ISP_WDR5_LUT_MINGAIN_WRITE_DATA

0x000061ac = Byte Address

Description: Write data of mingain LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_mingain_write_data_h	31:30	rw	'h0	Write data of mingain LUT for Curve;bin_combine_0[31:0]=bin_0+bin_1(inWDR5.1)
wdr5_lut_mingain_write_data	29:0	rw	'h0	Write data of mingain LUT for Curve;The number of combined bins for WDR5 is 44; The number of combined bins for WDR5.1 is 33

ISP_WDR5_LUT_FLAT_FACTOR_ADDR

0x000061b0 = Byte Address

Description: Pointer to entry of flat factor value LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_flat_factor_addr	6:0	rw	'h0	Pointer to entry of flat factor value LUT for Curve

ISP_WDR5_LUT_FLAT_FACTOR_WRITE_DATA

0x000061b4 = Byte Address

Description: Write data of flat factor LUT for Curve

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_flat_factor_write_data	29:0	rw	'h0	Write data of flat factor LUT for Curve

ISP_WDR5_LUT_FLAT_LEVEL_ADDR

0x000061b8 = Byte Address

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Description: Pointer to entry of flat level value LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_flat_level_addr	3:0	rw	'h0	Pointer to entry of flat level value LUT for Curve

ISP_WDR5_LUT_FLAT_LEVEL_WRITE_DATA

0x000061bc = Byte Address

Description: Write data of flat level LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_flat_level_write_data	31:0	rw	'h0	Write data of flat level LUT for Curve

ISP_WDR5_LUT_SAT_X_ADDR

0x000061c0 = Byte Address

Description: Pointer to entry of sat x value LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_sat_x_addr	2:0	rw	'h0	Pointer to entry of sat x value LUT for Curve

ISP_WDR5_LUT_SAT_X_WRITE_DATA

0x000061c4 = Byte Address

Description: Write data of sat x LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_sat_x_write_data	26:0	rw	'h0	Write data of sat x LUT for Curve

ISP_WDR5_LUT_SAT_SHIFT_ADDR

0x000061c8 = Byte Address

Description: Pointer to entry of sat shift value LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_sat_shift_addr	1:0	rw	'h0	Pointer to entry of sat shift value LUT for Curve

ISP_WDR5_LUT_SAT_SHIFT_WRITE_DATA

0x000061cc = Byte Address

Description: Write data of sat shift LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_sat_shift_write_data	31:0	rw	'h0	Write data of sat shift LUT for Curve

ISP_WDR5_LUT_SAT_ADDR

0x000061d0 = Byte Address

Description: Pointer to entry of sat value LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_sat_addr	2:0	rw	'h0	Pointer to entry of sat value LUT for Curve

ISP_WDR5_LUT_SAT_WRITE_DATA

0x000061d4 = Byte Address

Description: Write data of sat LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_sat_write_data	29:0	rw	'h0	Write data of sat LUT for Curve

ISP_WDR5_LUT_DEGAMMA_ADDR

0x000061d8 = Byte Address

Description: Pointer to entry of degamma value LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_degamma_addr	5:0	rw	'h0	Pointer to entry of degamma value LUT for Curve;The number of combined bins for WDR5 is 44; The number of combined bins for WDR5.1 is 33

ISP_WDR5_LUT_DEGAMMA_WRITE_DATA

0x000061dc = Byte Address

Description: Write data of degamma LUT for Curve

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_degamma_write_data_h	31:30	rw	'h0	Write data of degamma LUT for Curve;bin_combine_0[31:0]=bin_0+bin_1(inWDR5.1)
wdr5_lut_degamma_write_data	29:0	rw	'h0	Write data of degamma LUT for Curve;bin_combine_0[29:0]=bin_0+bin_1[9:0]; bin_combine_1[29:0]=bin_1[19:10]+bin_2(inWDR5)

ISP_WDR5_LUT_GAMMAUP_ADDR_SHD

0x000061e0 = Byte Address

Description: Pointer to entry of gamma up value LUT for Curve shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_gammaup_addr_shd	5:0	rw	'h0	Pointer to entry of gamma up value LUT for Curve shadow

ISP_WDR5_LUT_GAMMAUP_WRITE_DATA_SHD

0x000061e4 = Byte Address

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Description: Write data of gamma up LUT for Curve shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_gammaup_write_data_shd	29:0	r	'h0	Write data of gamma up LUT for Curve shadow

ISP_WDR5_LUT_WEIGHT_RAW_ADDR_SHD

0x000061e8 = Byte Address

Description: Pointer to entry of raw weight value LUT for Curve shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_weight_raw_addr_shd	4:0	rw	'h0	Pointer to entry of raw weight value LUT for Curve shadow

ISP_WDR5_LUT_WEIGHT_RAW_WRITE_DATA_SHD

0x000061ec = Byte Address

Description: Write data of raw weight LUT for Curve shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_weight_raw_write_data_shd	29:0	r	'h0	Write data of raw weight LUT for Curve shadow

ISP_WDR5_LUT_MAXGAIN_ADDR_SHD

0x000061f0 = Byte Address

Description: Pointer to entry of maxgain value LUT for Curve shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_maxgain_addr_shd	5:0	rw	'h0	Pointer to entry of maxgain value LUT for Curve shadow;The number of combined bins for WDR5 is 44; The number of combined bins for WDR5.1 is 33

ISP_WDR5_LUT_MAXGAIN_WRITE_DATA_SHD

0x000061f4 = Byte Address

Description: Write data of maxgain LUT for Curve shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_maxgain_write_data_shd_h	31:30	r	'h0	Write data of maxgain LUT for Curve shadow;bin_combine_0[31:0]=bin_0+bin_1(in WDR5.1)
wdr5_lut_maxgain_write_data_shd	29:0	r	'h0	Write data of maxgain LUT for Curve shadow;;bin_combine_0[29:0]=bin_0+bin_1[9:0];bin_combine_1[29:0]=bin_1[19:10]+bin_2(in WDR5)

ISP_WDR5_LUT_MINGAIN_ADDR_SHD

0x000061f8 = Byte Address

Description: Pointer to entry of mingain value LUT for Curve shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_mingain_addr_shd	5:0	rw	'h0	Pointer to entry of mingain value LUT for Curve shadow;The number of combined bins for WDR5 is 44; The number of combined bins for WDR5.1 is 33

ISP_WDR5_LUT_MINGAIN_WRITE_DATA_SHD

0x000061fc = Byte Address

Description: Write data of mingain LUT for Curve shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_mingain_write_data_shd_h	31:30	r	'h0	Write data of mingain LUT for Curve shadow;bin_combine_0[31:0]=bin_0+bin_1(in WDR5.1)
wdr5_lut_mingain_write_data_shd	29:0	r	'h0	Write data of mingain LUT for Curve shadow;bin_combine_0[29:0]=bin_0+bin_1[9:0];bin_combine_1[29:0]=bin_1[19:10]+bin_2(in WDR5)

ISP_WDR5_LUT_SAT_ADDR_SHD

0x00006200 = Byte Address

Description: Pointer to entry of sat value LUT for Curve shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_sat_addr_shd	2:0	rw	'h0	Pointer to entry of sat value LUT for Curve shadow

ISP_WDR5_LUT_SAT_WRITE_DATA_SHD

0x00006204 = Byte Address

Description: Write data of sat LUT for Curve shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_lut_sat_write_data_shd	29:0	r	'h0	Write data of sat LUT for Curve shadow

ISP_WDR5_LUT_DEGAMMA_ADDR_SHD

0x00006208 = Byte Address

Description: Pointer to entry of degamma value LUT for Curve shadow

Sub-Field Name	Bits	rw	Reset	Description

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wdr5_lut_degamma_addr_shd	5:0	rw	'h0	Pointer to entry of degamma value LUT for Curve shadow;The number of combined bins for WDR5 is 44; The number of combined bins for WDR5.1 is 33

ISP_WDR5_LUT_DEGAMMA_WRITE_DATA_SHD

0x0000620c = Byte Address

Description: Write data of degamma LUT for Curve shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lut_degamma_write_data_shd_h	31:30	r	'h0	Write data of degamma LUT for Curve shadow;bin_combine_0[31:0]=bin_0+bin_1(in WDR5.1)
wdr5_lut_degamma_write_data_shd	29:0	r	'h0	Write data of degamma LUT for Curve shadow;bin_combine_0[29:0]=bin_0+bin_1[9:0];bin_combine_1[29:0]=bin_1[19:10]+bin_2(in WDR5)

ISP_WDR5_LUMA_WEIGHT_1

0x00006210 = Byte Address

Description: WDR5.1 luma weight 1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_rgb_coef_2	23:16	rw	'h0	WDR5.1 rgb coef 2
wdr5_rgb_coef_1	15:8	rw	'h0	WDR5.1 rgb coef 1
wdr5_rgb_coef_0	7:0	rw	'h0	WDR5.1 rgb coef 0

ISP_WDR5_LUMA_WEIGHT_2

0x00006214 = Byte Address

Description: WDR5.1 luma weight 2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lightness_weight	31:24	rw	'h0	WDR5.1 lightness weight
wdr5_color_weight_2	23:16	rw	'h0	WDR5.1 color weight 2
wdr5_color_weight_1	15:8	rw	'h0	WDR5.1 color weight 1
wdr5_color_weight_0	7:0	rw	'h0	WDR5.1 color weight 0

ISP_WDR5_LIGHT_CTRL

0x00006218 = Byte Address

Description: WDR5.1 halo color adjustment control

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_light_sat_hithr_factor	23:9	rw	'h0	WDR5.1 light sat hithr factor

wdr5_light_sat_lothr	8:1	rw	'h0	WDR5.1 light sat lothr
wdr5_light_enable	0:0	rw	'h0	WDR5.1 light enable

ISP_WDR5_RED_THR_0

0x0000621c = Byte Address

Description: WDR5.1 red component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_red_factor_01_low_bit	29:20	rw	'h0	WDR5.1 red factor 01 low bit
wdr5_red_thr_0	19:0	rw	'h0	WDR5.1 red thr 0

ISP_WDR5_RED_THR_1

0x00006220 = Byte Address

Description: WDR5.1 red component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_red_factor_01_high_bit	29:20	rw	'h0	WDR5.1 red factor 01 high bit
wdr5_red_thr_1	19:0	rw	'h0	WDR5.1 red thr 1

ISP_WDR5_RED_THR_2

0x00006224 = Byte Address

Description: WDR5.1 red component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_red_factor_23_low_bit	29:20	rw	'h0	WDR5.1 red factor 23 low bit
wdr5_red_thr_2	19:0	rw	'h0	WDR5.1 red thr 2

ISP_WDR5_RED_THR_3

0x00006228 = Byte Address

Description: WDR5.1 red component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_red_factor_23_high_bit	29:20	rw	'h0	WDR5.1 red factor 23 high bit
wdr5_red_thr_3	19:0	rw	'h0	WDR5.1 red thr 3

ISP_WDR5_YELLOW_THR_0

0x0000622c = Byte Address

Description: WDR5.1 yellow component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_yellow_factor_01_low_bit	29:20	rw	'h0	WDR5.1 yellow factor 01 low bit
wdr5_yellow_thr_0	19:0	rw	'h0	WDR5.1 yellow thr 0

ISP_WDR5_YELLOW_THR_1

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0x00006230 = Byte Address

Description: WDR5.1 yellow component threshold of halo

Sub-Field Name	Bits	rw	Reset	Description
wdr5_yellow_factor_01_high_bit	29:20	rw	'h0	WDR5.1 yellow factor 01 high bit
wdr5_yellow_thr_1	19:0	rw	'h0	WDR5.1 yellow thr 1

ISP_WDR5_YELLOW_THR_2

0x00006234 = Byte Address

Description: WDR5.1 yellow component threshold of halo

Sub-Field Name	Bits	rw	Reset	Description
wdr5_yellow_factor_23_low_bit	29:20	rw	'h0	WDR5.1 yellow factor 23 low bit
wdr5_yellow_thr_2	19:0	rw	'h0	WDR5.1 yellow thr 2

ISP_WDR5_YELLOW_THR_3

0x00006238 = Byte Address

Description: WDR5.1 yellow component threshold of halo

Sub-Field Name	Bits	rw	Reset	Description
wdr5_yellow_factor_23_high_bit	29:20	rw	'h0	WDR5.1 yellow factor 23 high bit
wdr5_yellow_thr_3	19:0	rw	'h0	WDR5.1 yellow thr 3

ISP_WDR5_GREEN_THR_0

0x0000623c = Byte Address

Description: WDR5.1 green component threshold of halo

Sub-Field Name	Bits	rw	Reset	Description
wdr5_green_factor_01_low_bit	29:20	rw	'h0	WDR5.1 green factor 01 low bit
wdr5_green_thr_0	19:0	rw	'h0	WDR5.1 green thr 0

ISP_WDR5_GREEN_THR_1

0x00006240 = Byte Address

Description: WDR5.1 green component threshold of halo

Sub-Field Name	Bits	rw	Reset	Description
wdr5_green_factor_01_high_bit	29:20	rw	'h0	WDR5.1 green factor 01 high bit
wdr5_green_thr_1	19:0	rw	'h0	WDR5.1 green thr 1

ISP_WDR5_GREEN_THR_2

0x00006244 = Byte Address

Description: WDR5.1 green component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_green_factor_23_low_bit	29:20	rw	'h0	WDR5.1 green factor 23 low bit
wdr5_green_thr_2	19:0	rw	'h0	WDR5.1 green thr 2

ISP_WDR5_GREEN_THR_3

0x00006248 = Byte Address

Description: WDR5.1 green component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_green_factor_23_high_bit	29:20	rw	'h0	WDR5.1 green factor 23 high bit
wdr5_green_thr_3	19:0	rw	'h0	WDR5.1 green thr 3

ISP_WDR5_CYAN_THR_0

0x0000624c = Byte Address

Description: WDR5.1 cyan component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_cyan_factor_01_low_bit	29:20	rw	'h0	WDR5.1 cyan factor 01 low bit
wdr5_cyan_thr_0	19:0	rw	'h0	WDR5.1 cyan thr 0

ISP_WDR5_CYAN_THR_1

0x00006250 = Byte Address

Description: WDR5.1 cyan component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_cyan_factor_01_high_bit	29:20	rw	'h0	WDR5.1 cyan factor 01 high bit
wdr5_cyan_thr_1	19:0	rw	'h0	WDR5.1 cyan thr 1

ISP_WDR5_CYAN_THR_2

0x00006254 = Byte Address

Description: WDR5.1 cyan component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_cyan_factor_23_low_bit	29:20	rw	'h0	WDR5.1 cyan factor 23 low bit
wdr5_cyan_thr_2	19:0	rw	'h0	WDR5.1 cyan thr 2

ISP_WDR5_CYAN_THR_3

0x00006258 = Byte Address

Description: WDR5.1 cyan component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_cyan_factor_23_high_bit	29:20	rw	'h0	WDR5.1 cyan factor 23 high bit
wdr5_cyan_thr_3	19:0	rw	'h0	WDR5.1 cyan thr 3

ISP_WDR5_BLUE_THR_0

0x0000625c = Byte Address

Description: WDR5.1 blue component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_blue_factor_01_low_bit	29:20	rw	'h0	WDR5.1 blue factor 01 low bit
wdr5_blue_thr_0	19:0	rw	'h0	WDR5.1 blue thr 0

ISP_WDR5_BLUE_THR_1

0x00006260 = Byte Address

Description: WDR5.1 blue component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_blue_factor_01_high_bit	29:20	rw	'h0	WDR5.1 blue factor 01 high bit
wdr5_blue_thr_1	19:0	rw	'h0	WDR5.1 blue thr 1

ISP_WDR5_BLUE_THR_2

0x00006264 = Byte Address

Description: WDR5.1 blue component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_blue_factor_23_low_bit	29:20	rw	'h0	WDR5.1 blue factor 23 low bit
wdr5_blue_thr_2	19:0	rw	'h0	WDR5.1 blue thr 2

ISP_WDR5_BLUE_THR_3

0x00006268 = Byte Address

Description: WDR5.1 blue component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_blue_factor_23_high_bit	29:20	rw	'h0	WDR5.1 blue factor 23 high bit
wdr5_blue_thr_3	19:0	rw	'h0	WDR5.1 blue thr 3

ISP_WDR5_MAGENTA_THR_0

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0x0000626c = Byte Address

Description: WDR5.1 magenta component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_magenta_factor_01_low_bit	29:20	rw	'h0	WDR5.1 magenta factor 01 low bit
wdr5_magenta_thr_0	19:0	rw	'h0	WDR5.1 magenta thr 0

ISP_WDR5_MAGENTA_THR_1

0x00006270 = Byte Address

Description: WDR5.1 magenta component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_magenta_factor_01_high_bit	29:20	rw	'h0	WDR5.1 magenta factor 01 high bit
wdr5_magenta_thr_1	19:0	rw	'h0	WDR5.1 magenta thr 1

ISP_WDR5_MAGENTA_THR_2

0x00006274 = Byte Address

Description: WDR5.1 magenta component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_magenta_factor_23_low_bit	29:20	rw	'h0	WDR5.1 magenta factor 23 low bit
wdr5_magenta_thr_2	19:0	rw	'h0	WDR5.1 magenta thr 2

ISP_WDR5_MAGENTA_THR_3

0x00006278 = Byte Address

Description: WDR5.1 magenta component threshold of halo

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_magenta_factor_23_high_bit	29:20	rw	'h0	WDR5.1 magenta factor 23 high bit
wdr5_magenta_thr_3	19:0	rw	'h0	WDR5.1 magenta thr 3

ISP_WDR5_HLC_CTRL

0x0000627c = Byte Address

Description: WDR5.1 Highlight compensation control

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_hlc_slope	31:24	rw	'h0	WDR5.1 hlc slope
wdr5_hlc_base	23:5	rw	'h0	WDR5.1 hlc base
wdr5_hist_range_thr	4:0	rw	'h0	WDR5.1 hist range thr

ISP_WDR5_DAMP_CTRL

0x00006280 = Byte Address

Description: WDR5.1 damping function control

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_damp_avg_coef	19:13	rw	'h0	WDR5.1 damp avg coef
wdr5_damp_curve_coef	12:6	rw	'h0	WDR5.1 damp curve coef
wdr5_damp_filter_size	5:1	rw	'h0	WDR5.1 damp filter size
wdr5_damp_mode	0:0	rw	'h0	WDR5.1 damp mode

ISP_WDR5_DAMP_THR_LIMIT_1

0x00006284 = Byte Address

Description: WDR5.1 damping threshold

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_damp_lothr	26:7	rw	'h0	WDR5.1 damp lothr
wdr5_damp_coef_inc_limit	6:0	rw	'h0	WDR5.1 damp coef inc limit

ISP_WDR5_DAMP_THR_LIMIT_2

0x00006288 = Byte Address

Description: WDR5.1 damping threshold

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_damp_thr_factor	26:7	rw	'h0	WDR5.1 damp thr factor
wdr5_damp_coef_dec_limit	6:0	rw	'h0	WDR5.1 damp coef dec limit

ISP_WDR5_DAMP_MAX_MIN

0x0000628c = Byte Address

Description: WDR5.1 damping limitation

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_damp_curve_max	27:21	rw	'h0	WDR5.1 damp curve max
wdr5_damp_curve_min	20:14	rw	'h0	WDR5.1 damp curve min
wdr5_damp_avg_max	13:7	rw	'h0	WDR5.1 damp avg max
wdr5_damp_avg_min	6:0	rw	'h0	WDR5.1 damp avg min

ISP_WDR5_LUMA_WEIGHT_2_SHD

0x00006290 = Byte Address

Description: WDR5.1 luma weight 2 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_lightness_weight_shd	31:24	r	'h0	WDR5.1 lightness weight shd
wdr5_color_weight_2_shd	23:16	r	'h0	WDR5.1 color weight 2 shd

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wdr5_color_weight_1_shd	15:8	r	'h0	WDR5.1 color weight 1 shd
wdr5_color_weight_0_shd	7:0	r	'h0	WDR5.1 color weight 0 shd

ISP_WDR5_LIGHT_CTRL_SHD

0x00006294 = Byte Address

Description: WDR5.1 halo color adjustment control shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_light_sat_hithr_factor_shd	23:9	r	'h0	WDR5.1 light sat hithr factor shd
wdr5_light_sat_lothr_shd	8:1	r	'h0	WDR5.1 light sat lothr shd
wdr5_light_enable_shd	0:0	r	'h0	WDR5.1 light enable shd

ISP_WDR5_RED_THR_0_SHD

0x00006298 = Byte Address

Description: WDR5.1 red component threshold of halo shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_red_factor_01_low_bit_shd	29:20	r	'h0	WDR5.1 red factor 01 low bit shd
wdr5_red_thr_0_shd	19:0	r	'h0	WDR5.1 red thr 0 shd

ISP_WDR5_RED_THR_1_SHD

0x0000629c = Byte Address

Description: WDR5.1 red component threshold of halo shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_red_factor_01_high_bit_shd	29:20	r	'h0	WDR5.1 red factor 01 high bit shd
wdr5_red_thr_1_shd	19:0	r	'h0	WDR5.1 red thr 1 shd

ISP_WDR5_RED_THR_2_SHD

0x000062a0 = Byte Address

Description: WDR5.1 red component threshold of halo shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
wdr5_red_factor_23_low_bit_shd	29:20	r	'h0	WDR5.1 red factor 23 low bit shd
wdr5_red_thr_2_shd	19:0	r	'h0	WDR5.1 red thr 2 shd

ISP_WDR5_RED_THR_3_SHD

0x000062a4 = Byte Address

Description: WDR5.1 red component threshold of halo shadow

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Sub-Field Name	Bits	rw	Reset	Description
wdr5_red_factor_23_high_bit_shd	29:20	r	'h0	WDR5.1 red factor 23 high bit shd
wdr5_red_thr_3_shd	19:0	r	'h0	WDR5.1 red thr 3 shd

ISP_WDR5_YELLOW_THR_0_SHD

0x000062a8 = Byte Address

Description: WDR5.1 yellow component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_yellow_factor_01_low_bit_shd	29:20	r	'h0	WDR5.1 yellow factor 01 low bit shd
wdr5_yellow_thr_0_shd	19:0	r	'h0	WDR5.1 yellow thr 0 shd

ISP_WDR5_YELLOW_THR_1_SHD

0x000062ac = Byte Address

Description: WDR5.1 yellow component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_yellow_factor_01_high_bit_shd	29:20	r	'h0	WDR5.1 yellow factor 01 high bit shd
wdr5_yellow_thr_1_shd	19:0	r	'h0	WDR5.1 yellow thr 1 shd

ISP_WDR5_YELLOW_THR_2_SHD

0x000062b0 = Byte Address

Description: WDR5.1 yellow component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_yellow_factor_23_low_bit_shd	29:20	r	'h0	WDR5.1 yellow factor 23 low bit shd
wdr5_yellow_thr_2_shd	19:0	r	'h0	WDR5.1 yellow thr 2 shd

ISP_WDR5_YELLOW_THR_3_SHD

0x000062b4 = Byte Address

Description: WDR5.1 yellow component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_yellow_factor_23_high_bit_shd	29:20	r	'h0	WDR5.1 yellow factor 23 high bit shd
wdr5_yellow_thr_3_shd	19:0	r	'h0	WDR5.1 yellow thr 3 shd

ISP_WDR5_GREEN_THR_0_SHD

0x000062b8 = Byte Address

Description: WDR5.1 green component threshold of halo shadow

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Sub-Field Name	Bits	rw	Reset	Description
wdr5_green_factor_01_low_bit_shd	29:20	r	'h0	WDR5.1 green factor 01 low bit shd
wdr5_green_thr_0_shd	19:0	r	'h0	WDR5.1 green thr 0 shd

ISP_WDR5_GREEN_THR_1_SHD

0x000062bc = Byte Address

Description: WDR5.1 green component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_green_factor_01_high_bit_shd	29:20	r	'h0	WDR5.1 green factor 01 high bit shd
wdr5_green_thr_1_shd	19:0	r	'h0	WDR5.1 green thr 1 shd

ISP_WDR5_GREEN_THR_2_SHD

0x000062c0 = Byte Address

Description: WDR5.1 green component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_green_factor_23_low_bit_shd	29:20	r	'h0	WDR5.1 green factor 23 low bit shd
wdr5_green_thr_2_shd	19:0	r	'h0	WDR5.1 green thr 2 shd

ISP_WDR5_GREEN_THR_3_SHD

0x000062c4 = Byte Address

Description: WDR5.1 green component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_green_factor_23_high_bit_shd	29:20	r	'h0	WDR5.1 green factor 23 high bit shd
wdr5_green_thr_3_shd	19:0	r	'h0	WDR5.1 green thr 3 shd

ISP_WDR5_CYAN_THR_0_SHD

0x000062c8 = Byte Address

Description: WDR5.1 cyan component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_cyan_factor_01_low_bit_shd	29:20	r	'h0	WDR5.1 cyan factor 01 low bit shd
wdr5_cyan_thr_0_shd	19:0	r	'h0	WDR5.1 cyan thr 0 shd

ISP_WDR5_CYAN_THR_1_SHD

0x000062cc = Byte Address

Description: WDR5.1 cyan component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_cyan_factor_01_high_bit_shd	29:20	r	'h0	WDR5.1 cyan factor 01 high bit shd
wdr5_cyan_thr_1_shd	19:0	r	'h0	WDR5.1 cyan thr 1 shd

ISP_WDR5_CYAN_THR_2_SHD

0x000062d0 = Byte Address

Description: WDR5.1 cyan component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_cyan_factor_23_low_bit_shd	29:20	r	'h0	WDR5.1 cyan factor 23 low bit shd
wdr5_cyan_thr_2_shd	19:0	r	'h0	WDR5.1 cyan thr 2 shd

ISP_WDR5_CYAN_THR_3_SHD

0x000062d4 = Byte Address

Description: WDR5.1 cyan component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_cyan_factor_23_high_bit_shd	29:20	r	'h0	WDR5.1 cyan factor 23 high bit shd
wdr5_cyan_thr_3_shd	19:0	r	'h0	WDR5.1 cyan thr 3 shd

ISP_WDR5_BLUE_THR_0_SHD

0x000062d8 = Byte Address

Description: WDR5.1 blue component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_blue_factor_01_low_bit_shd	29:20	r	'h0	WDR5.1 blue factor 01 low bit shd
wdr5_blue_thr_0_shd	19:0	r	'h0	WDR5.1 blue thr 0 shd

ISP_WDR5_BLUE_THR_1_SHD

0x000062dc = Byte Address

Description: WDR5.1 blue component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_blue_factor_01_high_bit_shd	29:20	r	'h0	WDR5.1 blue factor 01 high bit shd
wdr5_blue_thr_1_shd	19:0	r	'h0	WDR5.1 blue thr 1 shd

ISP_WDR5_BLUE_THR_2_SHD

0x000062e0 = Byte Address

Description: WDR5.1 blue component threshold of halo shadow

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Sub-Field Name	Bits	rw	Reset	Description
wdr5_blue_factor_23_low_bit_shd	29:20	r	'h0	WDR5.1 blue factor 23 low bit shd
wdr5_blue_thr_2_shd	19:0	r	'h0	WDR5.1 blue thr 2 shd

ISP_WDR5_BLUE_THR_3_SHD

0x000062e4 = Byte Address

Description: WDR5.1 blue component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_blue_factor_23_high_bit_shd	29:20	r	'h0	WDR5.1 blue factor 23 high bit shd
wdr5_blue_thr_3_shd	19:0	r	'h0	WDR5.1 blue thr 3 shd

ISP_WDR5_MAGENTA_THR_0_SHD

0x000062e8 = Byte Address

Description: WDR5.1 magenta component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_magenta_factor_01_low_bit_shd	29:20	r	'h0	WDR5.1 magenta factor 01 low bit shd
wdr5_magenta_thr_0_shd	19:0	r	'h0	WDR5.1 magenta thr 0 shd

ISP_WDR5_MAGENTA_THR_1_SHD

0x000062ec = Byte Address

Description: WDR5.1 magenta component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_magenta_factor_01_high_bit_shd	29:20	r	'h0	WDR5.1 magenta factor 01 high bit shd
wdr5_magenta_thr_1_shd	19:0	r	'h0	WDR5.1 magenta thr 1 shd

ISP_WDR5_MAGENTA_THR_2_SHD

0x000062f0 = Byte Address

Description: WDR5.1 magenta component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_magenta_factor_23_low_bit_shd	29:20	r	'h0	WDR5.1 magenta factor 23 low bit shd
wdr5_magenta_thr_2_shd	19:0	r	'h0	WDR5.1 magenta thr 2 shd

ISP_WDR5_MAGENTA_THR_3_SHD

0x000062f4 = Byte Address

Description: WDR5.1 magenta component threshold of halo shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_magenta_factor_23_high_bit_shd	29:20	r	'h0	WDR5.1 magenta factor 23 high bit shd
wdr5_magenta_thr_3_shd	19:0	r	'h0	WDR5.1 magenta thr 3 shd

ISP_WDR5_HLC_CTRL_SHD

0x000062f8 = Byte Address

Description: WDR5.1 Highlight compensation control shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_hlc_slope_shd	31:24	r	'h0	WDR5.1 hlc slope shd
wdr5_hlc_base_shd	23:5	r	'h0	WDR5.1 hlc base shd
wdr5_hist_range_thr_shd	4:0	r	'h0	WDR5.1 hist range thr shd

ISP_WDR5_DAMP_CTRL_SHD

0x000062fc = Byte Address

Description: WDR5.1 damping function control shadow

Sub-Field Name	Bits	rw	Reset	Description
wdr5_damp_avg_coef_shd	19:13	r	'h0	WDR5.1 damp avg coef shd
wdr5_damp_curve_coef_shd	12:6	r	'h0	WDR5.1 damp curve coef shd
wdr5_damp_filter_size_shd	5:1	r	'h0	WDR5.1 damp filter size shd
wdr5_damp_mode_shd	0:0	r	'h0	WDR5.1 damp mode shd

ISP_YUVNR_CNR_CTRL

0x00006300 = Byte Address

Description: CNR control

Sub-Field Name	Bits	rw	Reset	Description
cnr_strength2	19:12	rw	'h0	Denoise strength of UV channel controlled by bright, the large it is, the denoise strength of bright area is weaker; Range:[0, 128]
cnr_strength1	11:4	rw	'h80	Denoise strength of UV channel; Range:[0, 128]
cnr_auto_cg_enable	3:3	rw	'h1	0: cnr internal auto clock gating disable 1: cnr internal auto clock gating enable.
cbcr_full_range	2:2	rw	'h0	0: Pixel value range according to BT.601 1: YCbCr full range 0...255
neighbor_pix_consider	1:1	rw	'h1	Whether neighbor pixels are considered when UV image is reconstructed; Range:[0, 1]
cnr_enable	0:0	rw	'h0	Enable Denoise of UV channel; Range:[0, 1]

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ISP_YUVNR_CNR_PYD_TFILT_MASK0

0x00006308 = Byte Address

Description: Texture filter mask0 of CNR

Sub-Field Name	Bits	rw	Reset	Description
texture_filter8	31:28	rw	'h6	Texture filter coefficent8 in UV channel; Range: [0, 15]
texture_filter7	27:24	rw	'h8	Texture filter coefficent7 in UV channel; Range: [0, 15]
texture_filter6	23:20	rw	'h6	Texture filter coefficent6 in UV channel; Range: [0, 15]
texture_filter5	19:16	rw	'h8	Texture filter coefficent5 in UV channel; Range: [0, 15]
texture_filter3	15:12	rw	'h8	Texture filter coefficent3 in UV channel; Range: [0, 15]
texture_filter2	11:8	rw	'h6	Texture filter coefficent2 in UV channel; Range: [0, 15]
texture_filter1	7:4	rw	'h8	Texture filter coefficent1 in UV channel; Range: [0, 15]
texture_filter0	3:0	rw	'h6	Texture filter coefficent0 in UV channel; Range: [0, 15]

ISP_YUVNR_CNR_PYD_TFILT_MASK1

0x0000630c = Byte Address

Description: Texture filter mask1 of CNR

Sub-Field Name	Bits	rw	Reset	Description
texture_filter4	6:0	rw	'h8	Texture filter coefficent4 in UV channel; Range: [0, 127]

ISP_YUVNR_CNR_PYD_TFILT_SHIFT

0x00006310 = Byte Address

Description: Texture filter shift of CNR

Sub-Field Name	Bits	rw	Reset	Description
texture_filter_shift	2:0	rw	'h6	Texture filter shift in UV channel; Range:[0, 7]; Constraints: the sum of all texture_filter is equal to the power of texture _filter _shift to the base 2

ISP_YUVNR_CNR_WEIGHT_TABLE_FILTER0_ADDR

0x00006314 = Byte Address

Description: Address of layer0 filter weight table in UV channel

Sub-Field Name	Bits	rw	Reset	Description
cnr_weight_lut_filter0_addr	6:0	rw	'h0	Address of layer0 filter weight table in UV channel; Range:[0, 63]

ISP_YUVNR_CNR_WEIGHT_TABLE_FILTER0_DATA

0x00006318 = Byte Address

Description: Data of layer0 filter weight table in UV channel

Sub-Field Name	Bits	rw	Reset	Description
cnr_weight_lut_filter0_data	9:0	rw	'h0	Data of layer0 filter weight table in UV channel; Range:[0, 255]; Constraints: Y0>=Y1>=Y2...>=Y63

ISP_YUVNR_CNR_WEIGHT_TABLE_FILTER1_ADDR

0x0000631c = Byte Address

Description: Address of layer1 filter weight table in UV channel

Sub-Field Name	Bits	rw	Reset	Description
cnr_weight_lut_filter1_addr	6:0	rw	'h0	Address of layer1 filter weight table in UV channel; Range:[0, 63]

ISP_YUVNR_CNR_WEIGHT_TABLE_FILTER1_DATA

0x00006320 = Byte Address

Description: Data of layer1 filter weight table in UV channel

Sub-Field Name	Bits	rw	Reset	Description
cnr_weight_lut_filter1_data	9:0	rw	'h0	Data of layer1 filter weight table in UV channel; Range:[0, 255]; Constraints: Y0>=Y1>=Y2...>=Y63

ISP_YUVNR_CNR_WEIGHT_TABLE_FILTER2_ADDR

0x00006324 = Byte Address

Description: Address of layer2 filter weight table in UV channel

Sub-Field Name	Bits	rw	Reset	Description
cnr_weight_lut_filter2_addr	6:0	rw	'h0	Address of layer2 filter weight table in UV channel; Range:[0, 63]

ISP_YUVNR_CNR_WEIGHT_TABLE_FILTER2_DATA

0x00006328 = Byte Address

Description: Data of layer2 filter weight table in UV channel

Sub-Field Name	Bits	rw	Reset	Description

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cnr_weight_lut_filter2_data	9:0	rw	'h0	Data of layer2 filter weight table in UV channel; Range:[0, 255]; Constraints: Y0>=Y1>=Y2...>=Y63

ISP_YUVNR_CNR_CTRL_SHD

0x00006354 = Byte Address

Description: Shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cnr_strength2_shd	19:12	r	'h0	Shadow register
cnr_strength1_shd	11:4	r	'h80	Shadow register
cnr_auto_cg_enable_shd	3:3	r	'h1	Shadow register
cbcr_full_range_shd	2:2	r	'h0	Shadow register
neighbor_pix_consider_shd	1:1	r	'h1	Shadow register
cnr_enable_shd	0:0	r	'h0	Shadow register

ISP_YUVNR_CNR_PYD_TFILT_MASK0_SHD

0x0000635c = Byte Address

Description: Shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
texture_filter8_shd	31:28	r	'h6	Shadow register
texture_filter7_shd	27:24	r	'h8	Shadow register
texture_filter6_shd	23:20	r	'h6	Shadow register
texture_filter5_shd	19:16	r	'h8	Shadow register
texture_filter3_shd	15:12	r	'h8	Shadow register
texture_filter2_shd	11:8	r	'h6	Shadow register
texture_filter1_shd	7:4	r	'h8	Shadow register
texture_filter0_shd	3:0	r	'h6	Shadow register

ISP_YUVNR_CNR_PYD_TFILT_MASK1_SHD

0x00006360 = Byte Address

Description: Shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
texture_filter4_shd	6:0	r	'h8	Shadow register

ISP_YUVNR_CNR_PYD_TFILT_SHIFT_SHD

0x00006364 = Byte Address

Description: Shadow register

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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texture_filter_shift_shd	2:0	r	'h6	Shadow register

ISP_YUVNR_DUMMY_HBLANK

0x000063a8 = Byte Address

Description: hblank control registers

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
cnr_dmy_hb	15:0	rw	'h10	control of hblank time for dummy lines. The unit is cycle.

ISP_STITCHING0_MOTION_WEIGHT

0x00006580 = Byte Address

Description: weight of shorter frame in the moving area

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dark_weight_shorter	31:22	rw	'h0	weight of shorter frame for dark area calculation
motion_weight_lsvs	21:11	rw	'h0	weight of very short frame in the moving area
motion_weight_ls	10:0	rw	'h0	weight of short frame in the moving area

ISP_STITCHING0_MOTION_THRESHOLD

0x00006584 = Byte Address

Description: threshold for deghost

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
motion_sat_threshold	23:11	rw	'h0	saturation threshold for motion detection
weight_update_threshold	10:0	rw	'h0	weight update threshold

ISP_STITCHING0_MOTION_LOWER_THRESHOLD_G_0

0x00006588 = Byte Address

Description: lower motion threshold for deghost between l-s frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
motion_lower_threshold_ls_gr	23:12	rw	'h0	lower GR channel motion threshold for deghost between long frame and short frame
motion_lower_threshold_ls_gb	11:0	rw	'h0	

ISP_STITCHING0_MOTION_LOWER_THRESHOLD_RB_0

0x0000658c = Byte Address

Description: lower motion threshold for deghost between l-s frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
motion_lower_threshold_ls_r	23:12	rw	'h0	lower R channel motion threshold for deghost between long frame and short frame

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motion_lower_threshold_ls_b	11:0	rw	'h0	lower B channel motion threshold for deghost between long frame and short frame

ISP_STITCHING0_MOTION_FACTOR_THRESHOLD_G_0

0x00006590 = Byte Address

Description: motion threshold factor for deghost between l-s frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
motion_factor_threshold_ls_gr	31:16	rw	'h0	GR channel motion factorthreshold for deghost between long frame and short frame
motion_factor_threshold_ls_gb	15:0	rw	'h0	GB channel motion factorthreshold for deghost between long frame and short frame

ISP_STITCHING0_MOTION_FACTOR_THRESHOLD_RB_0

0x00006594 = Byte Address

Description: motion threshold factor for deghost between l-s frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
motion_factor_threshold_ls_r	31:16	rw	'h0	R channel motion factorthreshold for deghost between long frame and short frame
motion_factor_threshold_ls_b	15:0	rw	'h0	B channel motion factorthreshold for deghost between long frame and short frame

ISP_STITCHING0_MOTION_LOWER_THRESHOLD_G_1

0x00006598 = Byte Address

Description: lower motion threshold for deghost between ls-vs frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
motion_lower_threshold_lsvs_gr	23:12	rw	'h0	lower GR channel motion threshold for deghost between ls frame and very short frame
motion_lower_threshold_lsvs_gb	11:0	rw	'h0	lower GB channel motion threshold for deghost between ls frame and very short frame

ISP_STITCHING0_MOTION_LOWER_THRESHOLD_RB_1

0x0000659c = Byte Address

Description: lower motion threshold for deghost between ls-vs frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
motion_lower_threshold_lsvs_r	23:12	rw	'h0	lower R channel motion threshold for deghost between ls frame and very short frame
motion_lower_threshold_lsvs_b	11:0	rw	'h0	lower B channel motion threshold for deghost between ls frame and very short frame

ISP_STITCHING0_MOTION_FACTOR_THRESHOLD_G_1

0x000065a0 = Byte Address

Description: motion threshold factor for deghost between ls-vs frame

Sub-Field Name	Bits	rw	Reset	Description
motion_factor_threshold_lsvs_gr	31:16	rw	'h0	GR channel motion factorthreshold for deghost between ls frame and very short frame
motion_factor_threshold_lsvs_gb	15:0	rw	'h0	GB channel motion factorthreshold for deghost between ls frame and very short frame

ISP_STITCHING0_MOTION_FACTOR_THRESHOLD_RB_1

0x000065a4 = Byte Address

Description: motion threshold factor for deghost between ls-vs frame

Sub-Field Name	Bits	rw	Reset	Description
motion_factor_threshold_lsvs_r	31:16	rw	'h0	R channel motion factorthreshold for deghost between ls frame and very short frame
motion_factor_threshold_lsvs_b	15:0	rw	'h0	B channel motion factorthreshold for deghost between ls frame and very short frame

ISP_STITCHING0_DARK_LOWER_THRESHOLD_G_0

0x000065a8 = Byte Address

Description: lower dark threshold for deghost between l-s frame

Sub-Field Name	Bits	rw	Reset	Description
dark_lower_threshold_ls_gr	23:12	rw	'h0	lower GR channel dark threshold for deghost between long frame and short frame
dark_lower_threshold_ls_gb	11:0	rw	'h0	lower GB channel dark threshold for deghost between long frame and short frame

ISP_STITCHING0_DARK_LOWER_THRESHOLD_RB_0

0x000065ac = Byte Address

Description: lower dark threshold for deghost between l-s frame

Sub-Field Name	Bits	rw	Reset	Description
dark_lower_threshold_ls_r	23:12	rw	'h0	lower R channel dark threshold for deghost between long frame and short frame
dark_lower_threshold_ls_b	11:0	rw	'h0	lower B channel dark threshold for deghost between long frame and short frame

ISP_STITCHING0_DARK_FACTOR_THRESHOLD_G_0

0x000065b0 = Byte Address

Description: dark threshold factor for deghost between l-s frame

Sub-Field Name	Bits	rw	Reset	Description

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dark_factor_threshold_ls_gr	31:16	rw	'h0	GR channel dark threshold factor for deghost between long frame and short frame
dark_factor_threshold_ls_gb	15:0	rw	'h0	GB channel dark threshold factor for deghost between long frame and short frame

ISP_STITCHING0_DARK_FACTOR_THRESHOLD_RB_0

0x000065b4 = Byte Address

Description: dark threshold factor for deghost between l-s frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dark_factor_threshold_ls_r	31:16	rw	'h0	R channel dark threshold factor for deghost between long frame and short frame
dark_factor_threshold_ls_b	15:0	rw	'h0	B channel dark threshold factor for deghost between long frame and short frame

ISP_STITCHING0_DARK_LOWER_THRESHOLD_G_1

0x000065b8 = Byte Address

Description: lower dark threshold for deghost between ls-vs frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dark_lower_threshold_lsvs_gr	23:12	rw	'h0	lower GR channel dark threshold for deghost between ls frame and very short frame
dark_lower_threshold_lsvs_gb	11:0	rw	'h0	lower GB channel dark threshold for deghost between ls frame and very short frame

ISP_STITCHING0_DARK_LOWER_THRESHOLD_RB_1

0x000065bc = Byte Address

Description: lower dark threshold for deghost between ls-vs frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dark_lower_threshold_lsvs_r	23:12	rw	'h0	lower R channel dark threshold for deghost between ls frame and very short frame
dark_lower_threshold_lsvs_b	11:0	rw	'h0	lower B channel dark threshold for deghost between ls frame and very short frame

ISP_STITCHING0_DARK_FACTOR_THRESHOLD_G_1

0x000065c0 = Byte Address

Description: dark threshold factor for deghost between ls-vs frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dark_factor_threshold_lsvs_gr	31:16	rw	'h0	GR channel dark threshold factor for deghost between ls frame and very short frame
dark_factor_threshold_lsvs_gb	15:0	rw	'h0	GB channel dark threshold factor for deghost between ls frame and very short frame

ISP_STITCHING0_DARK_FACTOR_THRESHOLD_RB_1

0x000065c4 = Byte Address

Description: dark threshold factor for deghost between ls-vs frame

Sub-Field Name	Bits	rw	Reset	Description
dark_factor_threshold_lsvs_r	31:16	rw	'h0	R channel dark threshold factor for deghost between ls frame and very short frame
dark_factor_threshold_lsvs_b	15:0	rw	'h0	B channel dark threshold factor for deghost between ls frame and very short frame

ISP_STITCHING0_MOTION_THRESHOLD_SHD

0x000065c8 = Byte Address

Description: shadow threshold for deghost

Sub-Field Name	Bits	rw	Reset	Description
motion_sat_threshold_shd	23:11	r	'h0	saturation threshold for motion detection
weight_update_threshold_shd	10:0	r	'h0	weight update threshold

ISP_STITCHING0_MOTION_LOWER_THRESHOLD_G_0_SHD

0x000065cc = Byte Address

Description: lower motion shadow threshold for deghost between l-s frame

Sub-Field Name	Bits	rw	Reset	Description
motion_lower_threshold_ls_gr_shd	23:12	r	'h0	lower GR channel motion threshold for deghost between long frame and short frame
motion_lower_threshold_ls_gb_shd	11:0	r	'h0	lower GB channel motion threshold for deghost between long frame and short frame

ISP_STITCHING0_MOTION_LOWER_THRESHOLD_RB_0_SHD

0x000065d0 = Byte Address

Description: lower motion shadow threshold for deghost between l-s frame

Sub-Field Name	Bits	rw	Reset	Description
motion_lower_threshold_ls_r_shd	23:12	r	'h0	lower R channel motion threshold for deghost between long frame and short frame
motion_lower_threshold_ls_b_shd	11:0	r	'h0	lower B channel motion threshold for deghost between long frame and short frame

ISP_STITCHING0_MOTION_FACTOR_THRESHOLD_G_0_SHD

0x000065d4 = Byte Address

Description: motion shadow threshold factor for deghost between l-s frame

Sub-Field Name	Bits	rw	Reset	Description

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motion_factor_threshold_ls_gr_shd	31:16	r	'h0	GR channel motion factorthreshold for deghost between long frame and short frame
motion_factor_threshold_ls_gb_shd	15:0	r	'h0	GB channel motion factorthreshold for deghost between long frame and short frame

ISP_STITCHING0_MOTION_FACTOR_THRESHOLD_RB_0_SHD

0x000065d8 = Byte Address

Description: motion shadow threshold factor for deghost between l-s frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
motion_factor_threshold_ls_r_shd	31:16	r	'h0	R channel motion factorthreshold for deghost between long frame and short frame
motion_factor_threshold_ls_b_shd	15:0	r	'h0	B channel motion factorthreshold for deghost between long frame and short frame

ISP_STITCHING0_MOTION_LOWER_THRESHOLD_G_1_SHD

0x000065dc = Byte Address

Description: lower motion shadow threshold for deghost between ls-vs frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
motion_lower_threshold_lsvs_gr_shd	23:12	r	'h0	lower GR channel motion threshold for deghost between ls frame and very short frame
motion_lower_threshold_lsvs_gb_shd	11:0	r	'h0	lower GB channel motion threshold for deghost between ls frame and very short frame

ISP_STITCHING0_MOTION_LOWER_THRESHOLD_RB_1_SHD

0x000065e0 = Byte Address

Description: lower motion shadow threshold for deghost between ls-vs frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
motion_lower_threshold_lsvs_r_shd	23:12	r	'h0	lower R channel motion threshold for deghost between ls frame and very short frame
motion_lower_threshold_lsvs_b_shd	11:0	r	'h0	lower B channel motion threshold for deghost between ls frame and very short frame

ISP_STITCHING0_MOTION_FACTOR_THRESHOLD_G_1_SHD

0x000065e4 = Byte Address

Description: motion shadow threshold factor for deghost between ls-vs frame

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
motion_factor_threshold_lsvs_gr_shd	31:16	r	'h0	GR channel motion factorthreshold for deghost between ls frame and very short frame
motion_factor_threshold_lsvs_gb_shd	15:0	r	'h0	GB channel motion factorthreshold for deghost between ls frame and very short frame

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ISP_STITCHING0_MOTION_FACTOR_THRESHOLD_RB_1_SHD

0x000065e8 = Byte Address

Description: motion shadow threshold factor for deghost between ls-vs frame

Sub-Field Name	Bits	rw	Reset	Description
motion_factor_threshold_lsvs_r_shd	31:16	r	'h0	R channel motion factorthreshold for deghost between ls frame and very short frame
motion_factor_threshold_lsvs_b_shd	15:0	r	'h0	B channel motion factorthreshold for deghost between ls frame and very short frame

ISP_STITCHING0_DARK_LOWER_THRESHOLD_G_0_SHD

0x000065ec = Byte Address

Description: lower dark shadow threshold for deghost between l-s frame

Sub-Field Name	Bits	rw	Reset	Description
dark_lower_threshold_ls_gr_shd	23:12	r	'h0	lower GR channel dark threshold for deghost between long frame and short frame
dark_lower_threshold_ls_gb_shd	11:0	r	'h0	lower GB channel dark threshold for deghost between long frame and short frame

ISP_STITCHING0_DARK_LOWER_THRESHOLD_RB_0_SHD

0x000065f0 = Byte Address

Description: lower dark shadow threshold for deghost between l-s frame

Sub-Field Name	Bits	rw	Reset	Description
dark_lower_threshold_ls_r_shd	23:12	r	'h0	lower R channel dark threshold for deghost between long frame and short frame
dark_lower_threshold_ls_b_shd	11:0	r	'h0	lower B channel dark threshold for deghost between long frame and short frame

ISP_STITCHING0_DARK_FACTOR_THRESHOLD_G_0_SHD

0x000065f4 = Byte Address

Description: dark shadow threshold factor for deghost between l-s frame

Sub-Field Name	Bits	rw	Reset	Description
dark_factor_threshold_ls_gr_shd	31:16	r	'h0	GR channel dark threshold factor for deghost between long frame and short frame
dark_factor_threshold_ls_gb_shd	15:0	r	'h0	GB channel dark threshold factor for deghost between long frame and short frame

ISP_STITCHING0_DARK_FACTOR_THRESHOLD_RB_0_SHD

0x000065f8 = Byte Address

Description: dark shadow threshold factor for deghost between l-s frame

Sub-Field Name	Bits	rw	Reset	Description
dark_factor_threshold_ls_r_shd	31:16	r	'h0	R channel dark threshold factor for deghost between long frame and short frame
dark_factor_threshold_ls_b_shd	15:0	r	'h0	B channel dark threshold factor for deghost between long frame and short frame

ISP_STITCHING0_DARK_LOWER_THRESHOLD_G_1_SHD

0x000065fc = Byte Address

Description: lower dark shadow threshold for deghost between ls-vs frame

Sub-Field Name	Bits	rw	Reset	Description
dark_lower_threshold_lsvs_gr_shd	23:12	r	'h0	lower GR channel dark threshold for deghost between ls frame and very short frame
dark_lower_threshold_lsvs_gb_shd	11:0	r	'h0	lower GB channel dark threshold for deghost between ls frame and very short frame

ISP_STITCHING0_DARK_LOWER_THRESHOLD_RB_1_SHD

0x00006600 = Byte Address

Description: lower dark shadow threshold for deghost between ls-vs frame

Sub-Field Name	Bits	rw	Reset	Description
dark_lower_threshold_lsvs_r_shd	23:12	r	'h0	lower R channel dark threshold for deghost between ls frame and very short frame
dark_lower_threshold_lsvs_b_shd	11:0	r	'h0	lower B channel dark threshold for deghost between ls frame and very short frame

ISP_STITCHING0_DARK_FACTOR_THRESHOLD_G_1_SHD

0x00006604 = Byte Address

Description: dark shadow threshold factor for deghost between ls-vs frame

Sub-Field Name	Bits	rw	Reset	Description
dark_factor_threshold_lsvs_gr_shd	31:16	r	'h0	GR channel dark threshold factor for deghost between ls frame and very short frame
dark_factor_threshold_lsvs_gb_shd	15:0	r	'h0	GB channel dark threshold factor for deghost between ls frame and very short frame

ISP_STITCHING0_DARK_FACTOR_THRESHOLD_RB_1_SHD

0x00006608 = Byte Address

Description: dark shadow threshold factor for deghost between ls-vs frame

Sub-Field Name	Bits	rw	Reset	Description
dark_factor_threshold_lsvs_r_shd	31:16	r	'h0	R channel dark threshold factor for deghost between ls frame and very short frame

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dark_factor_threshold_lsvs_b_shd	15:0	r	'h0	B channel dark threshold factor for deghost between ls frame and very short frame

ISP_STITCHING0_TRANS_RANGE_SHORT_FRAME

0x0000660c = Byte Address

Description: short frame transformation range start and end for linear mode

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_trans_range_start_linear	27:16	rw	'h0	short frame transmission range start for linear mode
stitching_trans_range_norm_factor_mul_linear	15:0	rw	'h0	short frame transmission range norm factor for linear mode

ISP_STITCHING0_TRANS_RANGE_VERY_SHORT_FRAME

0x00006610 = Byte Address

Description: very short frame transformation range start and end for linear mode

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_trans_range_start_nonlinear	27:16	rw	'h0	very short frame transmission range start for linear mode
stitching_trans_range_norm_factor_mul_nonlinear	15:0	rw	'h0	very short frame transmission range norm factor for linear mode

ISP_STITCHING0_TRANS_RANGE_SHORT_FRAME_SHD

0x00006614 = Byte Address

Description: short frame transformation range start and end for linear mode shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_trans_range_start_linear_shd	27:16	r	'h0	short frame transmission range start for linear mode
stitching_trans_range_norm_factor_mul_linear_shd	15:0	r	'h0	short frame transmission range norm factor for linear mode

ISP_STITCHING0_TRANS_RANGE_VERY_SHORT_FRAME_SHD

0x00006618 = Byte Address

Description: very short frame transformation range start and end for linear mode shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
stitching_trans_range_start_nonlinear_shd	27:16	r	'h0	very short frame transmission range start for linear mode
stitching_trans_range_norm_factor_mul_nonlinear_shd	15:0	r	'h0	very short frame transmission range norm factor for linear mode

MI_LT_RAW3_CTRL

0x000072f0 = Byte Address

Description: Control register

Sub-Field Name	Bits	rw	Reset	Description
mi_lt_raw3_path_enable	5:5	rw	'h0	write channel path enable
mi_lt_raw3_dma_start_id	4:1	rw	'h0	start id
mi_lt_raw3_dma_start	0:0	w	'h0	start trigger signal, auto clear after that

MI_LT_RAW3_BASE_AD

0x000072f4 = Byte Address

Description: Base address for hdr path raw data ring buffer

Sub-Field Name	Bits	rw	Reset	Description
mi_lt_raw3_base_ad_init	31:4	rw	'h0	Base address of raw component ring buffer

MI_LT_RAW3_LLENGTH

0x000072f8 = Byte Address

Description: Follow main path RAW

Sub-Field Name	Bits	rw	Reset	Description
mi_lt_raw3_llengh	31:0	rw	'h0	Line length of RAW data in bytes, also known as line stride. If no line stride is used, line length must match image width.

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_0

0x00007800 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_0

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_lsc_comp_lut_px0	31:28	rw	'h0	Denoise filter lsc gain compensation lut px0
isp_denoise2d_sigma_lsc_comp_lut_x0	27:16	rw	'h0	Denoise filter lsc gain compensation lut x0
isp_denoise2d_sigma_lsc_comp_lut_y0	15:0	rw	'h400	Denoise filter lsc gain compensation lut y0

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_1

0x00007804 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_1

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_lsc_comp_lut_px1	31:28	rw	'h8	Denoise filter lsc gain compensation lut px1

isp_denoise2d_sigma_lsc_comp_lut_x1	27:16	rw	'h0	Denoise filter lsc gain compensation lut x1
isp_denoise2d_sigma_lsc_comp_lut_y1	15:0	rw	'h400	Denoise filter lsc gain compensation lut y1

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_2

0x00007808 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px2	31:28	rw	'h0	Denoise filter lsc gain compensation lut px2
isp_denoise2d_sigma_lsc_comp_lut_x2	27:16	rw	'h0	Denoise filter lsc gain compensation lut x2
isp_denoise2d_sigma_lsc_comp_lut_y2	15:0	rw	'h400	Denoise filter lsc gain compensation lut y2

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_3

0x0000780c = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px3	31:28	rw	'h0	Denoise filter lsc gain compensation lut px3
isp_denoise2d_sigma_lsc_comp_lut_x3	27:16	rw	'h0	Denoise filter lsc gain compensation lut x3
isp_denoise2d_sigma_lsc_comp_lut_y3	15:0	rw	'h400	Denoise filter lsc gain compensation lut y3

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_4

0x00007810 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_4

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px4	31:28	rw	'h8	Denoise filter lsc gain compensation lut px4
isp_denoise2d_sigma_lsc_comp_lut_x4	27:16	rw	'h0	Denoise filter lsc gain compensation lut x4
isp_denoise2d_sigma_lsc_comp_lut_y4	15:0	rw	'h400	Denoise filter lsc gain compensation lut y4

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_5

0x00007814 = Byte Address

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Description: isp_denoise2d_sigma_lsc_comp_lut_5

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_lsc_comp_lut_px5	31:28	rw	'h0	Denoise filter lsc gain compensation lut px5
isp_denoise2d_sigma_lsc_comp_lut_x5	27:16	rw	'h0	Denoise filter lsc gain compensation lut x5
isp_denoise2d_sigma_lsc_comp_lut_y5	15:0	rw	'h400	Denoise filter lsc gain compensation lut y5

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_6

0x00007818 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_6

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_lsc_comp_lut_px6	31:28	rw	'h0	Denoise filter lsc gain compensation lut px6
isp_denoise2d_sigma_lsc_comp_lut_x6	27:16	rw	'h0	Denoise filter lsc gain compensation lut x6
isp_denoise2d_sigma_lsc_comp_lut_y6	15:0	rw	'h400	Denoise filter lsc gain compensation lut y6

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_7

0x0000781c = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_7

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_lsc_comp_lut_px7	31:28	rw	'h8	Denoise filter lsc gain compensation lut px7
isp_denoise2d_sigma_lsc_comp_lut_x7	27:16	rw	'h0	Denoise filter lsc gain compensation lut x7
isp_denoise2d_sigma_lsc_comp_lut_y7	15:0	rw	'h400	Denoise filter lsc gain compensation lut y7

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_8

0x00007820 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_8

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_lsc_comp_lut_px8	31:28	rw	'h0	Denoise filter lsc gain compensation lut px8
isp_denoise2d_sigma_lsc_comp_lut_x8	27:16	rw	'h0	Denoise filter lsc gain compensation lut x8

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isp_denoise2d_sigma_lsc_comp_lut_y8	15:0	rw	'h400	Denoise filter lsc gain compensation lut y8

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_9

0x00007824 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_9

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px9	31:28	rw	'h0	Denoise filter lsc gain compensation lut px9
isp_denoise2d_sigma_lsc_comp_lut_x9	27:16	rw	'h0	Denoise filter lsc gain compensation lut x9
isp_denoise2d_sigma_lsc_comp_lut_y9	15:0	rw	'h400	Denoise filter lsc gain compensation lut y9

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_10

0x00007828 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_10

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px10	31:28	rw	'h0	Denoise filter lsc gain compensation lut px10
isp_denoise2d_sigma_lsc_comp_lut_x10	27:16	rw	'h0	Denoise filter lsc gain compensation lut x10
isp_denoise2d_sigma_lsc_comp_lut_y10	15:0	rw	'h400	Denoise filter lsc gain compensation lut y10

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_11

0x0000782c = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_11

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px11	31:28	rw	'h0	Denoise filter lsc gain compensation lut px11
isp_denoise2d_sigma_lsc_comp_lut_x11	27:16	rw	'h0	Denoise filter lsc gain compensation lut x11
isp_denoise2d_sigma_lsc_comp_lut_y11	15:0	rw	'h400	Denoise filter lsc gain compensation lut y11

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_0_SHD

0x00007830 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_0_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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isp_denoise2d_sigma_lsc_comp_lut_px0_shd	31:28	r	'h0	Denoise filter lsc gain compensation lut px0
isp_denoise2d_sigma_lsc_comp_lut_x0_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x0
isp_denoise2d_sigma_lsc_comp_lut_y0_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y0

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_1_SHD

0x00007834 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_1_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px1_shd	31:28	r	'h8	Denoise filter lsc gain compensation lut px1
isp_denoise2d_sigma_lsc_comp_lut_x1_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x1
isp_denoise2d_sigma_lsc_comp_lut_y1_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y1

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_2_SHD

0x00007838 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_2_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px2_shd	31:28	r	'h0	Denoise filter lsc gain compensation lut px2
isp_denoise2d_sigma_lsc_comp_lut_x2_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x2
isp_denoise2d_sigma_lsc_comp_lut_y2_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y2

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_3_SHD

0x0000783c = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_3_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px3_shd	31:28	r	'h0	Denoise filter lsc gain compensation lut px3
isp_denoise2d_sigma_lsc_comp_lut_x3_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x3
isp_denoise2d_sigma_lsc_comp_lut_y3_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y3

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_4_SHD

0x00007840 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_4_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_lsc_comp_lut_px4_shd	31:28	r	'h8	Denoise filter lsc gain compensation lut px4
isp_denoise2d_sigma_lsc_comp_lut_x4_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x4
isp_denoise2d_sigma_lsc_comp_lut_y4_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y4

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_5_SHD

0x00007844 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_5_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_lsc_comp_lut_px5_shd	31:28	r	'h0	Denoise filter lsc gain compensation lut px5
isp_denoise2d_sigma_lsc_comp_lut_x5_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x5
isp_denoise2d_sigma_lsc_comp_lut_y5_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y5

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_6_SHD

0x00007848 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_6_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_lsc_comp_lut_px6_shd	31:28	r	'h0	Denoise filter lsc gain compensation lut px6
isp_denoise2d_sigma_lsc_comp_lut_x6_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x6
isp_denoise2d_sigma_lsc_comp_lut_y6_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y6

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_7_SHD

0x0000784c = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_7_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_lsc_comp_lut_px7_shd	31:28	r	'h8	Denoise filter lsc gain compensation lut px7

isp_denoise2d_sigma_lsc_comp_lut_x7_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x7
isp_denoise2d_sigma_lsc_comp_lut_y7_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y7

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_8_SHD

0x00007850 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_8_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px8_shd	31:28	r	'h0	Denoise filter lsc gain compensation lut px8
isp_denoise2d_sigma_lsc_comp_lut_x8_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x8
isp_denoise2d_sigma_lsc_comp_lut_y8_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y8

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_9_SHD

0x00007854 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_9_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px9_shd	31:28	r	'h0	Denoise filter lsc gain compensation lut px9
isp_denoise2d_sigma_lsc_comp_lut_x9_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x9
isp_denoise2d_sigma_lsc_comp_lut_y9_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y9

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_10_SHD

0x00007858 = Byte Address

Description: isp_denoise2d_sigma_lsc_comp_lut_10_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_lsc_comp_lut_px10_shd	31:28	r	'h0	Denoise filter lsc gain compensation lut px10
isp_denoise2d_sigma_lsc_comp_lut_x10_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x10
isp_denoise2d_sigma_lsc_comp_lut_y10_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y10

ISP_DENOISE2D_SIGMA_LSC_COMP_LUT_11_SHD

0x0000785c = Byte Address

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Description: isp_denoise2d_sigma_lsc_comp_lut_11_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_lsc_comp_lut_px11_shd	31:28	r	'h0	Denoise filter lsc gain compensation lut px11
isp_denoise2d_sigma_lsc_comp_lut_x11_shd	27:16	r	'h0	Denoise filter lsc gain compensation lut x11
isp_denoise2d_sigma_lsc_comp_lut_y11_shd	15:0	r	'h400	Denoise filter lsc gain compensation lut y11

ISP_DENOISE2D_MOTION_FAC_LUT_0

0x00007860 = Byte Address

Description: isp_denoise2d_motion_fac_lut_0

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_lut_px0	31:28	rw	'h0	Denoise filter motion factor lut px0
isp_denoise2d_motion_fac_lut_x0	27:16	rw	'h0	Denoise filter motion factor lut x0
isp_denoise2d_motion_fac_lut_y0	15:0	rw	'h40	Denoise filter motion factor lut y0

ISP_DENOISE2D_MOTION_FAC_LUT_1

0x00007864 = Byte Address

Description: isp_denoise2d_motion_fac_lut_1

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_lut_px1	31:28	rw	'h6	Denoise filter motion factor lut px1
isp_denoise2d_motion_fac_lut_x1	27:16	rw	'h0	Denoise filter motion factor lut x1
isp_denoise2d_motion_fac_lut_y1	15:0	rw	'h40	Denoise filter motion factor lut y1

ISP_DENOISE2D_MOTION_FAC_LUT_2

0x00007868 = Byte Address

Description: isp_denoise2d_motion_fac_lut_2

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_lut_px2	31:28	rw	'h6	Denoise filter motion factor lut px2
isp_denoise2d_motion_fac_lut_x2	27:16	rw	'h0	Denoise filter motion factor lut x2

isp_denoise2d_motion_fac_lut_y2	15:0	rw	'h40	Denoise filter motion factor lut y2

ISP_DENOISE2D_MOTION_FAC_LUT_3

0x0000786c = Byte Address

Description: isp_denoise2d_motion_fac_lut_3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px3	31:28	rw	'h6	Denoise filter motion factor lut px3
isp_denoise2d_motion_fac_lut_x3	27:16	rw	'h0	Denoise filter motion factor lut x3
isp_denoise2d_motion_fac_lut_y3	15:0	rw	'h40	Denoise filter motion factor lut y3

ISP_DENOISE2D_MOTION_FAC_LUT_4

0x00007870 = Byte Address

Description: isp_denoise2d_motion_fac_lut_4

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px4	31:28	rw	'h6	Denoise filter motion factor lut px4
isp_denoise2d_motion_fac_lut_x4	27:16	rw	'h0	Denoise filter motion factor lut x4
isp_denoise2d_motion_fac_lut_y4	15:0	rw	'h40	Denoise filter motion factor lut y4

ISP_DENOISE2D_MOTION_FAC_LUT_5

0x00007874 = Byte Address

Description: isp_denoise2d_motion_fac_lut_5

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px5	31:28	rw	'h6	Denoise filter motion factor lut px5
isp_denoise2d_motion_fac_lut_x5	27:16	rw	'h0	Denoise filter motion factor lut x5
isp_denoise2d_motion_fac_lut_y5	15:0	rw	'h40	Denoise filter motion factor lut y5

ISP_DENOISE2D_MOTION_FAC_LUT_6

0x00007878 = Byte Address

Description: isp_denoise2d_motion_fac_lut_6

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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isp_denoise2d_motion_fac_lut_px6	31:28	rw	'h6	Denoise filter motion factor lut px6
isp_denoise2d_motion_fac_lut_x6	27:16	rw	'h0	Denoise filter motion factor lut x6
isp_denoise2d_motion_fac_lut_y6	15:0	rw	'h40	Denoise filter motion factor lut y6

ISP_DENOISE2D_MOTION_FAC_LUT_7

0x0000787c = Byte Address

Description: isp_denoise2d_motion_fac_lut_7

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px7	31:28	rw	'h6	Denoise filter motion factor lut px7
isp_denoise2d_motion_fac_lut_x7	27:16	rw	'h0	Denoise filter motion factor lut x7
isp_denoise2d_motion_fac_lut_y7	15:0	rw	'h40	Denoise filter motion factor lut y7

ISP_DENOISE2D_MOTION_FAC_LUT_8

0x00007880 = Byte Address

Description: isp_denoise2d_motion_fac_lut_8

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px8	31:28	rw	'h6	Denoise filter motion factor lut px8
isp_denoise2d_motion_fac_lut_x8	27:16	rw	'h0	Denoise filter motion factor lut x8
isp_denoise2d_motion_fac_lut_y8	15:0	rw	'h40	Denoise filter motion factor lut y8

ISP_DENOISE2D_MOTION_FAC_LUT_9

0x00007884 = Byte Address

Description: isp_denoise2d_motion_fac_lut_9

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px9	31:28	rw	'h6	Denoise filter motion factor lut px9
isp_denoise2d_motion_fac_lut_x9	27:16	rw	'h0	Denoise filter motion factor lut x9
isp_denoise2d_motion_fac_lut_y9	15:0	rw	'h40	Denoise filter motion factor lut y9

ISP_DENOISE2D_MOTION_FAC_LUT_10

0x00007888 = Byte Address

Description: isp_denoise2d_motion_fac_lut_10

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_lut_px10	31:28	rw	'h6	Denoise filter motion factor lut px10
isp_denoise2d_motion_fac_lut_x10	27:16	rw	'h0	Denoise filter motion factor lut x10
isp_denoise2d_motion_fac_lut_y10	15:0	rw	'h40	Denoise filter motion factor lut y10

ISP_DENOISE2D_MOTION_FAC_LUT_11

0x0000788c = Byte Address

Description: isp_denoise2d_motion_fac_lut_11

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_lut_px11	31:28	rw	'h6	Denoise filter motion factor lut px11
isp_denoise2d_motion_fac_lut_x11	27:16	rw	'h0	Denoise filter motion factor lut x11
isp_denoise2d_motion_fac_lut_y11	15:0	rw	'h40	Denoise filter motion factor lut y11

ISP_DENOISE2D_MOTION_FAC_LUT_0_SHD

0x00007890 = Byte Address

Description: isp_denoise2d_motion_fac_lut_0_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_lut_px0_shd	31:28	r	'h0	Denoise filter motion factor lut px0
isp_denoise2d_motion_fac_lut_x0_shd	27:16	r	'h0	Denoise filter motion factor lut x0
isp_denoise2d_motion_fac_lut_y0_shd	15:0	r	'h40	Denoise filter motion factor lut y0

ISP_DENOISE2D_MOTION_FAC_LUT_1_SHD

0x00007894 = Byte Address

Description: isp_denoise2d_motion_fac_lut_1_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_lut_px1_shd	31:28	r	'h6	Denoise filter motion factor lut px1

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isp_denoise2d_motion_fac_lut_x1_shd	27:16	r	'h0	Denoise filter motion factor lut x1
isp_denoise2d_motion_fac_lut_y1_shd	15:0	r	'h40	Denoise filter motion factor lut y1

ISP_DENOISE2D_MOTION_FAC_LUT_2_SHD

0x00007898 = Byte Address

Description: isp_denoise2d_motion_fac_lut_2_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px2_shd	31:28	r	'h6	Denoise filter motion factor lut px2
isp_denoise2d_motion_fac_lut_x2_shd	27:16	r	'h0	Denoise filter motion factor lut x2
isp_denoise2d_motion_fac_lut_y2_shd	15:0	r	'h40	Denoise filter motion factor lut y2

ISP_DENOISE2D_MOTION_FAC_LUT_3_SHD

0x0000789c = Byte Address

Description: isp_denoise2d_motion_fac_lut_3_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px3_shd	31:28	r	'h6	Denoise filter motion factor lut px3
isp_denoise2d_motion_fac_lut_x3_shd	27:16	r	'h0	Denoise filter motion factor lut x3
isp_denoise2d_motion_fac_lut_y3_shd	15:0	r	'h40	Denoise filter motion factor lut y3

ISP_DENOISE2D_MOTION_FAC_LUT_4_SHD

0x000078a0 = Byte Address

Description: isp_denoise2d_motion_fac_lut_4_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px4_shd	31:28	r	'h6	Denoise filter motion factor lut px4
isp_denoise2d_motion_fac_lut_x4_shd	27:16	r	'h0	Denoise filter motion factor lut x4
isp_denoise2d_motion_fac_lut_y4_shd	15:0	r	'h40	Denoise filter motion factor lut y4

ISP_DENOISE2D_MOTION_FAC_LUT_5_SHD

0x000078a4 = Byte Address

Description: isp_denoise2d_motion_fac_lut_5_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_lut_px5_shd	31:28	r	'h6	Denoise filter motion factor lut px5
isp_denoise2d_motion_fac_lut_x5_shd	27:16	r	'h0	Denoise filter motion factor lut x5
isp_denoise2d_motion_fac_lut_y5_shd	15:0	r	'h40	Denoise filter motion factor lut y5

ISP_DENOISE2D_MOTION_FAC_LUT_6_SHD

0x000078a8 = Byte Address

Description: isp_denoise2d_motion_fac_lut_6_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_lut_px6_shd	31:28	r	'h6	Denoise filter motion factor lut px6
isp_denoise2d_motion_fac_lut_x6_shd	27:16	r	'h0	Denoise filter motion factor lut x6
isp_denoise2d_motion_fac_lut_y6_shd	15:0	r	'h40	Denoise filter motion factor lut y6

ISP_DENOISE2D_MOTION_FAC_LUT_7_SHD

0x000078ac = Byte Address

Description: isp_denoise2d_motion_fac_lut_7_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_lut_px7_shd	31:28	r	'h6	Denoise filter motion factor lut px7
isp_denoise2d_motion_fac_lut_x7_shd	27:16	r	'h0	Denoise filter motion factor lut x7
isp_denoise2d_motion_fac_lut_y7_shd	15:0	r	'h40	Denoise filter motion factor lut y7

ISP_DENOISE2D_MOTION_FAC_LUT_8_SHD

0x000078b0 = Byte Address

Description: isp_denoise2d_motion_fac_lut_8_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_lut_px8_shd	31:28	r	'h6	Denoise filter motion factor lut px8
isp_denoise2d_motion_fac_lut_x8_shd	27:16	r	'h0	Denoise filter motion factor lut x8

isp_denoise2d_motion_fac_lut_y8_shd	15:0	r	'h40	Denoise filter motion factor lut y8

ISP_DENOISE2D_MOTION_FAC_LUT_9_SHD

0x000078b4 = Byte Address

Description: isp_denoise2d_motion_fac_lut_9_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px9_shd	31:28	r	'h6	Denoise filter motion factor lut px9
isp_denoise2d_motion_fac_lut_x9_shd	27:16	r	'h0	Denoise filter motion factor lut x9
isp_denoise2d_motion_fac_lut_y9_shd	15:0	r	'h40	Denoise filter motion factor lut y9

ISP_DENOISE2D_MOTION_FAC_LUT_10_SHD

0x000078b8 = Byte Address

Description: isp_denoise2d_motion_fac_lut_10_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px10_shd	31:28	r	'h6	Denoise filter motion factor lut px10
isp_denoise2d_motion_fac_lut_x10_shd	27:16	r	'h0	Denoise filter motion factor lut x10
isp_denoise2d_motion_fac_lut_y10_shd	15:0	r	'h40	Denoise filter motion factor lut y10

ISP_DENOISE2D_MOTION_FAC_LUT_11_SHD

0x000078bc = Byte Address

Description: isp_denoise2d_motion_fac_lut_11_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_lut_px11_shd	31:28	r	'h6	Denoise filter motion factor lut px11
isp_denoise2d_motion_fac_lut_x11_shd	27:16	r	'h0	Denoise filter motion factor lut x11
isp_denoise2d_motion_fac_lut_y11_shd	15:0	r	'h40	Denoise filter motion factor lut y11

ISP_DENOISE2D_MOTION_ANCHOR_X

0x000078c0 = Byte Address

Description: isp_denoise2d_motion_anchor_x

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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isp_denoise2d_motion_anchor_x1	19:16	rw	'hc	Denoise2D Filter Motion Anchor x1 for detail boosting
isp_denoise2d_motion_anchor_x0	10:0	rw	'h0	Denoise2D Filter Motion Anchor x0 for detail boosting

ISP_DENOISE2D_STATIC_BOOST_L0_0

0x000078c4 = Byte Address

Description: isp_denoise2d_static_boost_l0_0

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_static_boost_l0_c1_t1	27:16	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 1 static region
isp_denoise2d_static_boost_l0_c0_t1	11:0	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 0 static region

ISP_DENOISE2D_STATIC_BOOST_L0_1

0x000078c8 = Byte Address

Description: isp_denoise2d_static_boost_l0_1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_static_boost_l0_c0_t2	27:16	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 0 static region
isp_denoise2d_static_boost_l0_c2_t1	11:0	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 2 static region

ISP_DENOISE2D_STATIC_BOOST_L0_2

0x000078cc = Byte Address

Description: isp_denoise2d_static_boost_l0_2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_static_boost_l0_c2_t2	27:16	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 2 static region
isp_denoise2d_static_boost_l0_c1_t2	11:0	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 1 static region

ISP_DENOISE2D_STATIC_BOOST_L0_3

0x000078d0 = Byte Address

Description: isp_denoise2d_static_boost_l0_3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_static_boost_fac_l0_c1	27:16	rw	'h0	frequency synthesis detail boosting factor for layer 0 channel 1 static region
isp_denoise2d_static_boost_fac_l0_c0	11:0	rw	'h0	frequency synthesis detail boosting factor for layer 0 channel 0 static region

ISP_DENOISE2D_STATIC_BOOST_L0_4

0x000078d4 = Byte Address

Description: isp_denoise2d_static_boost_l0_4

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_clip_val_l0_c0	27:16	rw	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 0 static region
isp_denoise2d_static_boost_fac_l0_c2	11:0	rw	'h0	frequency synthesis detail boosting factor for layer 0 channel 2 static region

ISP_DENOISE2D_STATIC_BOOST_L0_5

0x000078d8 = Byte Address

Description: isp_denoise2d_static_boost_l0_5

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_clip_val_l0_c2	27:16	rw	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 2 static region
isp_denoise2d_static_boost_clip_val_l0_c1	11:0	rw	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 1 static region

ISP_DENOISE2D_STATIC_BOOST_L1_0

0x000078dc = Byte Address

Description: isp_denoise2d_static_boost_l1_0

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_l1_c1_t1	27:16	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 1 static region
isp_denoise2d_static_boost_l1_c0_t1	11:0	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 0 static region

ISP_DENOISE2D_STATIC_BOOST_L1_1

0x000078e0 = Byte Address

Description: isp_denoise2d_static_boost_l1_1

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_l1_c0_t2	27:16	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 0 static region
isp_denoise2d_static_boost_l1_c2_t1	11:0	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 2 static region

ISP_DENOISE2D_STATIC_BOOST_L1_2

0x000078e4 = Byte Address

Description: isp_denoise2d_static_boost_l1_2

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Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_l1_c2_t2	27:16	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 2 static region
isp_denoise2d_static_boost_l1_c1_t2	11:0	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 1 static region

ISP_DENOISE2D_STATIC_BOOST_L1_3

0x000078e8 = Byte Address

Description: isp_denoise2d_static_boost_l1_3

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_fac_l1_c1	27:16	rw	'h0	frequency synthesis detail boosting factor for layer 1 channel 1 static region
isp_denoise2d_static_boost_fac_l1_c0	11:0	rw	'h0	frequency synthesis detail boosting factor for layer 1 channel 0 static region

ISP_DENOISE2D_STATIC_BOOST_L1_4

0x000078ec = Byte Address

Description: isp_denoise2d_static_boost_l1_4

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_clip_val_l1_c0	27:16	rw	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 0 static region
isp_denoise2d_static_boost_fac_l1_c2	11:0	rw	'h0	frequency synthesis detail boosting factor for layer 1 channel 2 static region

ISP_DENOISE2D_STATIC_BOOST_L1_5

0x000078f0 = Byte Address

Description: isp_denoise2d_static_boost_l1_5

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_clip_val_l1_c2	27:16	rw	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 2 static region
isp_denoise2d_static_boost_clip_val_l1_c1	11:0	rw	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 1 static region

ISP_DENOISE2D_MOVING_BOOST_L0_0

0x000078f4 = Byte Address

Description: isp_denoise2d_moving_boost_l0_0

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_l0_c1_t1	27:16	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 1 moving region

isp_denoise2d_moving_boost_l0_c0_t1	11:0	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 0 moving region

ISP_DENOISE2D_MOVING_BOOST_L0_1

0x000078f8 = Byte Address

Description: isp_denoise2d_moving_boost_l0_1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_moving_boost_l0_c0_t2	27:16	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 0 moving region
isp_denoise2d_moving_boost_l0_c2_t1	11:0	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 2 moving region

ISP_DENOISE2D_MOVING_BOOST_L0_2

0x000078fc = Byte Address

Description: isp_denoise2d_moving_boost_l0_2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_moving_boost_l0_c2_t2	27:16	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 2 moving region
isp_denoise2d_moving_boost_l0_c1_t2	11:0	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 1 moving region

ISP_DENOISE2D_MOVING_BOOST_L0_3

0x00007900 = Byte Address

Description: isp_denoise2d_moving_boost_l0_3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_moving_boost_fac_l0_c1	27:16	rw	'h0	frequency synthesis detail boosting factor for layer 0 channel 1 moving region
isp_denoise2d_moving_boost_fac_l0_c0	11:0	rw	'h0	frequency synthesis detail boosting factor for layer 0 channel 0 moving region

ISP_DENOISE2D_MOVING_BOOST_L0_4

0x00007904 = Byte Address

Description: isp_denoise2d_moving_boost_l0_4

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_moving_boost_clip_val_l0_c0	27:16	rw	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 0 moving region
isp_denoise2d_moving_boost_fac_l0_c2	11:0	rw	'h0	frequency synthesis detail boosting factor for layer 0 channel 2 moving region

ISP_DENOISE2D_MOVING_BOOST_L0_5

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0x00007908 = Byte Address

Description: isp_denoise2d_moving_boost_l0_5

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_clip_val_l0_c2	27:16	rw	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 2 moving region
isp_denoise2d_moving_boost_clip_val_l0_c1	11:0	rw	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 1 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_0

0x0000790c = Byte Address

Description: isp_denoise2d_moving_boost_l1_0

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_l1_c1_t1	27:16	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 1 moving region
isp_denoise2d_moving_boost_l1_c0_t1	11:0	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 0 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_1

0x00007910 = Byte Address

Description: isp_denoise2d_moving_boost_l1_1

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_l1_c0_t2	27:16	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 0 moving region
isp_denoise2d_moving_boost_l1_c2_t1	11:0	rw	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 2 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_2

0x00007914 = Byte Address

Description: isp_denoise2d_moving_boost_l1_2

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_l1_c2_t2	27:16	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 2 moving region
isp_denoise2d_moving_boost_l1_c1_t2	11:0	rw	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 1 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_3

0x00007918 = Byte Address

Description: isp_denoise2d_moving_boost_l1_3

Sub-Field Name	Bits	rw	Reset	Description

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isp_denoise2d_moving_boost_fac_l1_c1	27:16	rw	'h0	frequency synthesis detail boosting factor for layer 1 channel 1 moving region
isp_denoise2d_moving_boost_fac_l1_c0	11:0	rw	'h0	frequency synthesis detail boosting factor for layer 1 channel 0 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_4

0x0000791c = Byte Address

Description: isp_denoise2d_moving_boost_l1_4

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_moving_boost_clip_val_l1_c0	27:16	rw	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 0 moving region
isp_denoise2d_moving_boost_fac_l1_c2	11:0	rw	'h0	frequency synthesis detail boosting factor for layer 1 channel 2 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_5

0x00007920 = Byte Address

Description: isp_denoise2d_moving_boost_l1_5

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_moving_boost_clip_val_l1_c2	27:16	rw	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 2 moving region
isp_denoise2d_moving_boost_clip_val_l1_c1	11:0	rw	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 1 moving region

ISP_DENOISE2D_MOTION_FAC_PARAM_0

0x00007924 = Byte Address

Description: isp_denoise2d_motion_fac_param_0

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_motion_fac_mul_l1	31:16	rw	'h40	Denoise2D motion factor lut mul factor for .layer 1
isp_denoise2d_motion_fac_mul_l0	15:0	rw	'h40	Denoise2D motion factor lut mul factor for .layer 0

ISP_DENOISE2D_MOTION_FAC_PARAM_1

0x00007928 = Byte Address

Description: isp_denoise2d_motion_fac_param_1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_static_sigma_factor_l0	31:16	rw	'h2000	Denoise2D sigma factor for layer 0 static region
isp_denoise2d_motion_fac_mul_l2	15:0	rw	'h40	Denoise2D motion factor lut mul factor for .layer 2

ISP_DENOISE2D_MOTION_FAC_PARAM_2

0x0000792c = Byte Address

Description: isp_denoise2d_motion_fac_param_2

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_sigma_factor_l2	31:16	rw	'h2000	Denoise2D sigma factor for layer 2 static region
isp_denoise2d_static_sigma_factor_l1	15:0	rw	'h2000	Denoise2D sigma factor for layer 1 static region

ISP_DENOISE2D_MOTION_FAC_PARAM_3

0x00007930 = Byte Address

Description: isp_denoise2d_motion_fac_param_3

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_motion_max_val_mul	30:22	rw	'h100	Denoise2D sigma motion factor max mul value
isp_denoise2d_sigma_motion_max_val	21:11	rw	'h400	Denoise2D sigma motion factor max value
isp_denoise2d_sigma_motion_min_val	10:0	rw	'h0	Denoise2D sigma motion factor min value

ISP_DENOISE2D_SIGMA_SCALE_0

0x00007934 = Byte Address

Description: isp_denoise2d_sigma_scale_0

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_scale_l1	31:16	rw	'h4000	Denoise2D Filter sigma scale value for layer 1
isp_denoise2d_sigma_scale_l0	15:0	rw	'h4000	Denoise2D Filter sigma scale value for layer 0

ISP_DENOISE2D_SIGMA_SCALE_1

0x00007938 = Byte Address

Description: isp_denoise2d_sigma_scale_1

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_sigma_scale_l2	15:0	rw	'h4000	Denoise2D Filter sigma scale value for layer 2

ISP_DENOISE2D_MOTION_ANCHOR_X_SHD

0x0000793c = Byte Address

Description: isp_denoise2d_motion_anchor_x_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_anchor_x1_shd	19:16	r	'hc	Denoise2D Filter Motion Anchor x1 for detail boosting
isp_denoise2d_motion_anchor_x0_shd	10:0	r	'h0	Denoise2D Filter Motion Anchor x0 for detail boosting

ISP_DENOISE2D_STATIC_BOOST_L0_0_SHD

0x00007940 = Byte Address

Description: isp_denoise2d_static_boost_l0_0_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_l0_c1_t1_shd	27:16	r	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 1 static region
isp_denoise2d_static_boost_l0_c0_t1_shd	11:0	r	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 0 static region

ISP_DENOISE2D_STATIC_BOOST_L0_1_SHD

0x00007944 = Byte Address

Description: isp_denoise2d_static_boost_l0_1_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_l0_c0_t2_shd	27:16	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 0 static region
isp_denoise2d_static_boost_l0_c2_t1_shd	11:0	r	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 2 static region

ISP_DENOISE2D_STATIC_BOOST_L0_2_SHD

0x00007948 = Byte Address

Description: isp_denoise2d_static_boost_l0_2_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_l0_c2_t2_shd	27:16	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 2 static region
isp_denoise2d_static_boost_l0_c1_t2_shd	11:0	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 1 static region

ISP_DENOISE2D_STATIC_BOOST_L0_3_SHD

0x0000794c = Byte Address

Description: isp_denoise2d_static_boost_l0_3_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_fac_l0_c1_shd	27:16	r	'h0	frequency synthesis detail boosting factor for layer 0 channel 1 static region

isp_denoise2d_static_boost_fac_l0_c0_shd	11:0	r	'h0	frequency synthesis detail boosting factor for layer 0 channel 0 static region

ISP_DENOISE2D_STATIC_BOOST_L0_4_SHD

0x00007950 = Byte Address

Description: isp_denoise2d_static_boost_l0_4_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_static_boost_clip_val_l0_c0_shd	27:16	r	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 0 static region
isp_denoise2d_static_boost_fac_l0_c2_shd	11:0	r	'h0	frequency synthesis detail boosting factor for layer 0 channel 2 static region

ISP_DENOISE2D_STATIC_BOOST_L0_5_SHD

0x00007954 = Byte Address

Description: isp_denoise2d_static_boost_l0_5_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_static_boost_clip_val_l0_c2_shd	27:16	r	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 2 static region
isp_denoise2d_static_boost_clip_val_l0_c1_shd	11:0	r	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 1 static region

ISP_DENOISE2D_STATIC_BOOST_L1_0_SHD

0x00007958 = Byte Address

Description: isp_denoise2d_static_boost_l1_0_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_static_boost_l1_c1_t1_shd	27:16	r	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 1 static region
isp_denoise2d_static_boost_l1_c0_t1_shd	11:0	r	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 0 static region

ISP_DENOISE2D_STATIC_BOOST_L1_1_SHD

0x0000795c = Byte Address

Description: isp_denoise2d_static_boost_l1_1_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_static_boost_l1_c0_t2_shd	27:16	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 0 static region
isp_denoise2d_static_boost_l1_c2_t1_shd	11:0	r	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 2 static region

ISP_DENOISE2D_STATIC_BOOST_L1_2_SHD

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0x00007960 = Byte Address

Description: isp_denoise2d_static_boost_l1_2_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_l1_c2_t2_shd	27:16	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 2 static region
isp_denoise2d_static_boost_l1_c1_t2_shd	11:0	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 1 static region

ISP_DENOISE2D_STATIC_BOOST_L1_3_SHD

0x00007964 = Byte Address

Description: isp_denoise2d_static_boost_l1_3_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_fac_l1_c1_shd	27:16	r	'h0	frequency synthesis detail boosting factor for layer 1 channel 1 static region
isp_denoise2d_static_boost_fac_l1_c0_shd	11:0	r	'h0	frequency synthesis detail boosting factor for layer 1 channel 0 static region

ISP_DENOISE2D_STATIC_BOOST_L1_4_SHD

0x00007968 = Byte Address

Description: isp_denoise2d_static_boost_l1_4_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_clip_val_l1_c0_shd	27:16	r	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 0 static region
isp_denoise2d_static_boost_fac_l1_c2_shd	11:0	r	'h0	frequency synthesis detail boosting factor for layer 1 channel 2 static region

ISP_DENOISE2D_STATIC_BOOST_L1_5_SHD

0x0000796c = Byte Address

Description: isp_denoise2d_static_boost_l1_5_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_boost_clip_val_l1_c2_shd	27:16	r	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 2 static region
isp_denoise2d_static_boost_clip_val_l1_c1_shd	11:0	r	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 1 static region

ISP_DENOISE2D_MOVING_BOOST_L0_0_SHD

0x00007970 = Byte Address

Description: isp_denoise2d_moving_boost_l0_0_shd

Sub-Field Name	Bits	rw	Reset	Description

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isp_denoise2d_moving_boost_l0_c1_t1_shd	27:16	r	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 1 moving region
isp_denoise2d_moving_boost_l0_c0_t1_shd	11:0	r	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 0 moving region

ISP_DENOISE2D_MOVING_BOOST_L0_1_SHD

0x00007974 = Byte Address

Description: isp_denoise2d_moving_boost_l0_1_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_moving_boost_l0_c0_t2_shd	27:16	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 0 moving region
isp_denoise2d_moving_boost_l0_c2_t1_shd	11:0	r	'h0	frequency synthesis detail boosting treshold 1 for layer 0 channel 2 moving region

ISP_DENOISE2D_MOVING_BOOST_L0_2_SHD

0x00007978 = Byte Address

Description: isp_denoise2d_moving_boost_l0_2_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_moving_boost_l0_c2_t2_shd	27:16	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 2 moving region
isp_denoise2d_moving_boost_l0_c1_t2_shd	11:0	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 0 channel 1 moving region

ISP_DENOISE2D_MOVING_BOOST_L0_3_SHD

0x0000797c = Byte Address

Description: isp_denoise2d_moving_boost_l0_3_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_moving_boost_fac_l0_c1_shd	27:16	r	'h0	frequency synthesis detail boosting factor for layer 0 channel 1 moving region
isp_denoise2d_moving_boost_fac_l0_c0_shd	11:0	r	'h0	frequency synthesis detail boosting factor for layer 0 channel 0 moving region

ISP_DENOISE2D_MOVING_BOOST_L0_4_SHD

0x00007980 = Byte Address

Description: isp_denoise2d_moving_boost_l0_4_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_moving_boost_clip_val_l0_c0_shd	27:16	r	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 0 moving region
isp_denoise2d_moving_boost_fac_l0_c2_shd	11:0	r	'h0	frequency synthesis detail boosting factor for layer 0 channel 2 moving region

ISP_DENOISE2D_MOVING_BOOST_L0_5_SHD

0x00007984 = Byte Address

Description: isp_denoise2d_moving_boost_l0_5_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_clip_val_l0_c2_shd	27:16	r	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 2 moving region
isp_denoise2d_moving_boost_clip_val_l0_c1_shd	11:0	r	'hfff	frequency synthesis detail boosting clipping value for layer 0 channel 1 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_0_SHD

0x00007988 = Byte Address

Description: isp_denoise2d_moving_boost_l1_0_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_l1_c1_t1_shd	27:16	r	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 1 moving region
isp_denoise2d_moving_boost_l1_c0_t1_shd	11:0	r	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 0 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_1_SHD

0x0000798c = Byte Address

Description: isp_denoise2d_moving_boost_l1_1_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_l1_c0_t2_shd	27:16	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 0 moving region
isp_denoise2d_moving_boost_l1_c2_t1_shd	11:0	r	'h0	frequency synthesis detail boosting treshold 1 for layer 1 channel 2 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_2_SHD

0x00007990 = Byte Address

Description: isp_denoise2d_moving_boost_l1_2_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_l1_c2_t2_shd	27:16	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 2 moving region
isp_denoise2d_moving_boost_l1_c1_t2_shd	11:0	r	'hfff	frequency synthesis detail boosting treshold 2 for layer 1 channel 1 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_3_SHD

0x00007994 = Byte Address

Description: isp_denoise2d_moving_boost_l1_3_shd

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Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_fac_l1_c1_shd	27:16	r	'h0	frequency synthesis detail boosting factor for layer 1 channel 1 moving region
isp_denoise2d_moving_boost_fac_l1_c0_shd	11:0	r	'h0	frequency synthesis detail boosting factor for layer 1 channel 0 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_4_SHD

0x00007998 = Byte Address

Description: isp_denoise2d_moving_boost_l1_4_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_clip_val_l1_c0_shd	27:16	r	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 0 moving region
isp_denoise2d_moving_boost_fac_l1_c2_shd	11:0	r	'h0	frequency synthesis detail boosting factor for layer 1 channel 2 moving region

ISP_DENOISE2D_MOVING_BOOST_L1_5_SHD

0x0000799c = Byte Address

Description: isp_denoise2d_moving_boost_l1_5_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_moving_boost_clip_val_l1_c2_shd	27:16	r	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 2 moving region
isp_denoise2d_moving_boost_clip_val_l1_c1_shd	11:0	r	'hfff	frequency synthesis detail boosting clipping value for layer 1 channel 1 moving region

ISP_DENOISE2D_MOTION_FAC_PARAM_0_SHD

0x000079a0 = Byte Address

Description: isp_denoise2d_motion_fac_param_0_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_motion_fac_mul_l1_shd	31:16	r	'h40	Denoise2D motion factor lut mul factor for .layer 1
isp_denoise2d_motion_fac_mul_l0_shd	15:0	r	'h40	Denoise2D motion factor lut mul factor for .layer 0

ISP_DENOISE2D_MOTION_FAC_PARAM_1_SHD

0x000079a4 = Byte Address

Description: isp_denoise2d_motion_fac_param_1_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise2d_static_sigma_factor_l0_shd	31:16	r	'h2000	Denoise2D sigma factor for layer 0 static region

isp_denoise2d_motion_fac_mul_l2_shd	15:0	r	'h40	Denoise2D motion factor lut mul factor for .layer 2

ISP_DENOISE2D_MOTION_FAC_PARAM_2_SHD

0x000079a8 = Byte Address

Description: isp_denoise2d_motion_fac_param_2_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_static_sigma_factor_l2_shd	31:16	r	'h2000	Denoise2D sigma factor for layer 2 static region
isp_denoise2d_static_sigma_factor_l1_shd	15:0	r	'h2000	Denoise2D sigma factor for layer 1 static region

ISP_DENOISE2D_MOTION_FAC_PARAM_3_SHD

0x000079ac = Byte Address

Description: isp_denoise2d_motion_fac_param_3_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_motion_max_val_mul_shd	30:22	r	'h100	Denoise2D sigma motion factor max mul value
isp_denoise2d_sigma_motion_max_val_shd	21:11	r	'h400	Denoise2D sigma motion factor max value
isp_denoise2d_sigma_motion_min_val_shd	10:0	r	'h0	Denoise2D sigma motion factor min value

ISP_DENOISE2D_SIGMA_SCALE_0_SHD

0x000079b0 = Byte Address

Description: isp_denoise2d_sigma_scale_0_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_scale_l1_shd	31:16	r	'h4000	Denoise2D Filter sigma scale value for layer 1
isp_denoise2d_sigma_scale_l0_shd	15:0	r	'h4000	Denoise2D Filter sigma scale value for layer 0

ISP_DENOISE2D_SIGMA_SCALE_1_SHD

0x000079b4 = Byte Address

Description: isp_denoise2d_sigma_scale_1_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise2d_sigma_scale_l2_shd	15:0	r	'h4000	Denoise2D Filter sigma scale value for layer 2

ISP_DENOISE3D_MOTION_DIFF

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0x00007a00 = Byte Address

Description: isp_denoise3d_motion_diff

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_motion_diff_type	10:10	rw	'h1	Denoise3D Motion detection diff type
isp_denoise3d_motion_diff_shift_bit	9:5	rw	'ha	Denoise3D Motion detection sqr diff shift bit
isp_denoise3d_motion_dither_fac	4:0	rw	'hc	Denoise3D TNR dither factor

ISP_DENOISE3D_DECOMPOSE_KERNEL_0

0x00007a04 = Byte Address

Description: isp_denoise3d_decompose_kernel_0

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_decompose_kernel_v4	29:24	rw	'h1	Denoise3D Frequency Decomposition Kernel v4
isp_denoise3d_decompose_kernel_v3	23:18	rw	'h4	Denoise3D Frequency Decomposition Kernel v3
isp_denoise3d_decompose_kernel_v2	17:12	rw	'h6	Denoise3D Frequency Decomposition Kernel v2
isp_denoise3d_decompose_kernel_v1	11:6	rw	'h4	Denoise3D Frequency Decomposition Kernel v1
isp_denoise3d_decompose_kernel_v0	5:0	rw	'h1	Denoise3D Frequency Decomposition Kernel v0

ISP_DENOISE3D_DECOMPOSE_KERNEL_1

0x00007a08 = Byte Address

Description: isp_denoise3d_decompose_kernel_1

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_decompose_kernel_v9	29:24	rw	'h4	Denoise3D Frequency Decomposition Kernel v9
isp_denoise3d_decompose_kernel_v8	23:18	rw	'h10	Denoise3D Frequency Decomposition Kernel v8
isp_denoise3d_decompose_kernel_v7	17:12	rw	'h18	Denoise3D Frequency Decomposition Kernel v7
isp_denoise3d_decompose_kernel_v6	11:6	rw	'h10	Denoise3D Frequency Decomposition Kernel v6
isp_denoise3d_decompose_kernel_v5	5:0	rw	'h4	Denoise3D Frequency Decomposition Kernel v5

ISP_DENOISE3D_DECOMPOSE_KERNEL_2

0x00007a0c = Byte Address

Description: isp_denoise3d_decompose_kernel_2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_decompose_kernel_v14	29:24	rw	'h6	Denoise3D Frequency Decomposition Kernel v14
isp_denoise3d_decompose_kernel_v13	23:18	rw	'h18	Denoise3D Frequency Decomposition Kernel v13
isp_denoise3d_decompose_kernel_v12	17:12	rw	'h24	Denoise3D Frequency Decomposition Kernel v12
isp_denoise3d_decompose_kernel_v11	11:6	rw	'h18	Denoise3D Frequency Decomposition Kernel v11
isp_denoise3d_decompose_kernel_v10	5:0	rw	'h6	Denoise3D Frequency Decomposition Kernel v10

ISP_DENOISE3D_DECOMPOSE_KERNEL_3

0x00007a10 = Byte Address

Description: isp_denoise3d_decompose_kernel_3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_decompose_kernel_v19	29:24	rw	'h4	Denoise3D Frequency Decomposition Kernel v19
isp_denoise3d_decompose_kernel_v18	23:18	rw	'h10	Denoise3D Frequency Decomposition Kernel v18
isp_denoise3d_decompose_kernel_v17	17:12	rw	'h18	Denoise3D Frequency Decomposition Kernel v17
isp_denoise3d_decompose_kernel_v16	11:6	rw	'h10	Denoise3D Frequency Decomposition Kernel v16
isp_denoise3d_decompose_kernel_v15	5:0	rw	'h4	Denoise3D Frequency Decomposition Kernel v15

ISP_DENOISE3D_DECOMPOSE_KERNEL_4

0x00007a14 = Byte Address

Description: isp_denoise3d_decompose_kernel_4

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_decompose_kernel_v24	29:24	rw	'h1	Denoise3D Frequency Decomposition Kernel v24
isp_denoise3d_decompose_kernel_v23	23:18	rw	'h4	Denoise3D Frequency Decomposition Kernel v23

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isp_denoise3d_decompose_kernel_v22	17:12	rw	'h6	Denoise3D Frequency Decomposition Kernel v22
isp_denoise3d_decompose_kernel_v21	11:6	rw	'h4	Denoise3D Frequency Decomposition Kernel v21
isp_denoise3d_decompose_kernel_v20	5:0	rw	'h1	Denoise3D Frequency Decomposition Kernel v20

ISP_DENOISE3D_INV_SIGMA_L0

0x00007a18 = Byte Address

Description: isp_denoise3d_inv_sigma_l0

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_inv_sigma_l0	11:0	rw	'hfff	Denoise3D Inverse sigma for layer 1

ISP_DENOISE3D_INV_SIGMA_L1

0x00007a1c = Byte Address

Description: isp_denoise3d_inv_sigma_l1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_inv_sigma_l1	11:0	rw	'hfff	Denoise3D Inverse sigma for layer 0

ISP_DENOISE3D_OFFSET

0x00007a20 = Byte Address

Description: isp_denoise3d_offset

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_offset_l1	29:16	rw	'h2000	Denoise3D decompose offset value fpr layer 1
isp_denoise3d_offset_l0	13:0	rw	'h2000	Denoise3D decompose offset value fpr layer 0

ISP_DENOISE3D_MOTION_GAUSS_KERNEL

0x00007a24 = Byte Address

Description: isp_denoise3d_motion_gauss_kernel

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_motion_gauss_kernel_v4	24:20	rw	'h1	Denoise3D Motion detection smoothing gaussian kernel v4
isp_denoise3d_motion_gauss_kernel_v3	19:15	rw	'h4	Denoise3D Motion detection smoothing gaussian kernel v3
isp_denoise3d_motion_gauss_kernel_v2	14:10	rw	'h6	Denoise3D Motion detection smoothing gaussian kernel v2
isp_denoise3d_motion_gauss_kernel_v1	9:5	rw	'h4	Denoise3D Motion detection smoothing gaussian kernel v1

isp_denoise3d_motion_gauss_kernel_v0	4:0	rw	'h1	Denoise3D Motion detection smoothing gaussian kernel v0

ISP_DENOISE3D_TNR_LUMA_LUT_0

0x00007a28 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_0

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p0	31:28	rw	'h0	Denoise3D TNR luma LUT p0
isp_denoise3d_tnr_luma_lut_y0	27:12	rw	'h0	Denoise3D TNR luma LUT y0
isp_denoise3d_tnr_luma_lut_motion_slope0	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v0

ISP_DENOISE3D_TNR_LUMA_LUT_1

0x00007a2c = Byte Address

Description: isp_denoise3d_tnr_luma_lut_1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p1	31:28	rw	'h6	Denoise3D TNR luma LUT p1
isp_denoise3d_tnr_luma_lut_y1	27:12	rw	'h0	Denoise3D TNR luma LUT y1
isp_denoise3d_tnr_luma_lut_motion_slope1	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v1

ISP_DENOISE3D_TNR_LUMA_LUT_2

0x00007a30 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p2	31:28	rw	'h6	Denoise3D TNR luma LUT p2
isp_denoise3d_tnr_luma_lut_y2	27:12	rw	'h0	Denoise3D TNR luma LUT y2
isp_denoise3d_tnr_luma_lut_motion_slope2	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v2

ISP_DENOISE3D_TNR_LUMA_LUT_3

0x00007a34 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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isp_denoise3d_tnr_luma_lut_p3	31:28	rw	'h6	Denoise3D TNR luma LUT p3
isp_denoise3d_tnr_luma_lut_y3	27:12	rw	'h0	Denoise3D TNR luma LUT y3
isp_denoise3d_tnr_luma_lut_motion_slope3	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v3

ISP_DENOISE3D_TNR_LUMA_LUT_4

0x00007a38 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_4

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p4	31:28	rw	'h6	Denoise3D TNR luma LUT p4
isp_denoise3d_tnr_luma_lut_y4	27:12	rw	'h0	Denoise3D TNR luma LUT y4
isp_denoise3d_tnr_luma_lut_motion_slope4	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v4

ISP_DENOISE3D_TNR_LUMA_LUT_5

0x00007a3c = Byte Address

Description: isp_denoise3d_tnr_luma_lut_5

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p5	31:28	rw	'h6	Denoise3D TNR luma LUT p5
isp_denoise3d_tnr_luma_lut_y5	27:12	rw	'h0	Denoise3D TNR luma LUT y5
isp_denoise3d_tnr_luma_lut_motion_slope5	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v5

ISP_DENOISE3D_TNR_LUMA_LUT_6

0x00007a40 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_6

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p6	31:28	rw	'h6	Denoise3D TNR luma LUT p6
isp_denoise3d_tnr_luma_lut_y6	27:12	rw	'h0	Denoise3D TNR luma LUT y6
isp_denoise3d_tnr_luma_lut_motion_slope6	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v6

ISP_DENOISE3D_TNR_LUMA_LUT_7

0x00007a44 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_7

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_luma_lut_p7	31:28	rw	'h6	Denoise3D TNR luma LUT p7
isp_denoise3d_tnr_luma_lut_y7	27:12	rw	'h0	Denoise3D TNR luma LUT y7
isp_denoise3d_tnr_luma_lut_motion_slope7	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v7

ISP_DENOISE3D_TNR_LUMA_LUT_8

0x00007a48 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_8

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_luma_lut_p8	31:28	rw	'h6	Denoise3D TNR luma LUT p8
isp_denoise3d_tnr_luma_lut_y8	27:12	rw	'h0	Denoise3D TNR luma LUT y8
isp_denoise3d_tnr_luma_lut_motion_slope8	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v8

ISP_DENOISE3D_TNR_LUMA_LUT_9

0x00007a4c = Byte Address

Description: isp_denoise3d_tnr_luma_lut_9

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_luma_lut_p9	31:28	rw	'h6	Denoise3D TNR luma LUT p9
isp_denoise3d_tnr_luma_lut_y9	27:12	rw	'h0	Denoise3D TNR luma LUT y9
isp_denoise3d_tnr_luma_lut_motion_slope9	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v9

ISP_DENOISE3D_TNR_LUMA_LUT_10

0x00007a50 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_10

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_luma_lut_p10	31:28	rw	'h6	Denoise3D TNR luma LUT p10

isp_denoise3d_tnr_luma_lut_y10	27:12	rw	'h0	Denoise3D TNR luma LUT y10
isp_denoise3d_tnr_luma_lut_motion_slope10	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v10

ISP_DENOISE3D_TNR_LUMA_LUT_11

0x00007a54 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_11

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p11	31:28	rw	'h6	Denoise3D TNR luma LUT p11
isp_denoise3d_tnr_luma_lut_y11	27:12	rw	'h0	Denoise3D TNR luma LUT y11
isp_denoise3d_tnr_luma_lut_motion_slope11	11:0	rw	'h40	Denoise3D TNR luma LUT motion slope v11

ISP_DENOISE3D_TNR_BOOST_0

0x00007a58 = Byte Address

Description: isp_denoise3d_tnr_boost_0

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_boost_l0_c1_t1	27:16	rw	'h0	frequency synthesis detail boosting treshold 1 channel 1 in tnr
isp_denoise3d_tnr_boost_l0_c0_t1	11:0	rw	'h0	frequency synthesis detail boosting treshold 1 channel 0 in tnr

ISP_DENOISE3D_TNR_BOOST_1

0x00007a5c = Byte Address

Description: isp_denoise3d_tnr_boost_1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_boost_l0_c0_t2	27:16	rw	'hfff	frequency synthesis detail boosting treshold 2 channel 0 in tnr
isp_denoise3d_tnr_boost_l0_c2_t1	11:0	rw	'h0	frequency synthesis detail boosting treshold 1 channel 2 in tnr

ISP_DENOISE3D_TNR_BOOST_2

0x00007a60 = Byte Address

Description: isp_denoise3d_tnr_boost_2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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isp_denoise3d_tnr_boost_l0_c2_t2	27:16	rw	'hfff	frequency synthesis detail boosting treshold 2 channel 2 in tnr
isp_denoise3d_tnr_boost_l0_c1_t2	11:0	rw	'hfff	frequency synthesis detail boosting treshold 2 channel 1 in tnr

ISP_DENOISE3D_TNR_BOOST_3

0x00007a64 = Byte Address

Description: isp_denoise3d_tnr_boost_3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_boost_fac_l0_c1	27:16	rw	'h0	frequency synthesis detail boosting factor channel 1 in tnr
isp_denoise3d_tnr_boost_fac_l0_c0	11:0	rw	'h0	frequency synthesis detail boosting factor channel 0 in tnr

ISP_DENOISE3D_TNR_BOOST_4

0x00007a68 = Byte Address

Description: isp_denoise3d_tnr_boost_4

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_boost_clip_val_l0_c0	27:16	rw	'hfff	frequency synthesis detail boosting clipping value channel 0 in tnr
isp_denoise3d_tnr_boost_fac_l0_c2	11:0	rw	'h0	frequency synthesis detail boosting factor channel 2 in tnr

ISP_DENOISE3D_TNR_BOOST_5

0x00007a6c = Byte Address

Description: isp_denoise3d_tnr_boost_5

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_boost_clip_val_l0_c2	27:16	rw	'hfff	frequency synthesis detail boosting clipping value channel 2 in tnr
isp_denoise3d_tnr_boost_clip_val_l0_c1	11:0	rw	'hfff	frequency synthesis detail boosting clipping value channel 1 in tnr

ISP_DENOISE3D_MOTION_DIFF_SHD

0x00007a70 = Byte Address

Description: isp_denoise3d_motion_diff_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_motion_diff_type_shd	10:10	r	'h1	Denoise3D Motion detection diff type
isp_denoise3d_motion_diff_shift_bit_shd	9:5	r	'ha	Denoise3D Motion detection sqr diff shift bit

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isp_denoise3d_motion_dither_fac_shd	4:0	r	'hc	Denoise3D TNR dither factor

ISP_DENOISE3D_DECOMPOSE_KERNEL_0_SHD

0x00007a74 = Byte Address

Description: isp_denoise3d_decompose_kernel_0_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_decompose_kernel_v4_shd	29:24	r	'h1	Denoise3D Frequency Decomposition Kernel v4
isp_denoise3d_decompose_kernel_v3_shd	23:18	r	'h4	Denoise3D Frequency Decomposition Kernel v3
isp_denoise3d_decompose_kernel_v2_shd	17:12	r	'h6	Denoise3D Frequency Decomposition Kernel v2
isp_denoise3d_decompose_kernel_v1_shd	11:6	r	'h4	Denoise3D Frequency Decomposition Kernel v1
isp_denoise3d_decompose_kernel_v0_shd	5:0	r	'h1	Denoise3D Frequency Decomposition Kernel v0

ISP_DENOISE3D_DECOMPOSE_KERNEL_1_SHD

0x00007a78 = Byte Address

Description: isp_denoise3d_decompose_kernel_1_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_decompose_kernel_v9_shd	29:24	r	'h4	Denoise3D Frequency Decomposition Kernel v9
isp_denoise3d_decompose_kernel_v8_shd	23:18	r	'h10	Denoise3D Frequency Decomposition Kernel v8
isp_denoise3d_decompose_kernel_v7_shd	17:12	r	'h18	Denoise3D Frequency Decomposition Kernel v7
isp_denoise3d_decompose_kernel_v6_shd	11:6	r	'h10	Denoise3D Frequency Decomposition Kernel v6
isp_denoise3d_decompose_kernel_v5_shd	5:0	r	'h4	Denoise3D Frequency Decomposition Kernel v5

ISP_DENOISE3D_DECOMPOSE_KERNEL_2_SHD

0x00007a7c = Byte Address

Description: isp_denoise3d_decompose_kernel_2_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_decompose_kernel_v14_shd	29:24	r	'h6	Denoise3D Frequency Decomposition Kernel v14

isp_denoise3d_decompose_kernel_v13_shd	23:18	r	'h18	Denoise3D Frequency Decomposition Kernel v13
isp_denoise3d_decompose_kernel_v12_shd	17:12	r	'h24	Denoise3D Frequency Decomposition Kernel v12
isp_denoise3d_decompose_kernel_v11_shd	11:6	r	'h18	Denoise3D Frequency Decomposition Kernel v11
isp_denoise3d_decompose_kernel_v10_shd	5:0	r	'h6	Denoise3D Frequency Decomposition Kernel v10

ISP_DENOISE3D_DECOMPOSE_KERNEL_3_SHD

0x00007a80 = Byte Address

Description: isp_denoise3d_decompose_kernel_3_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_decompose_kernel_v19_shd	29:24	r	'h4	Denoise3D Frequency Decomposition Kernel v19
isp_denoise3d_decompose_kernel_v18_shd	23:18	r	'h10	Denoise3D Frequency Decomposition Kernel v18
isp_denoise3d_decompose_kernel_v17_shd	17:12	r	'h18	Denoise3D Frequency Decomposition Kernel v17
isp_denoise3d_decompose_kernel_v16_shd	11:6	r	'h10	Denoise3D Frequency Decomposition Kernel v16
isp_denoise3d_decompose_kernel_v15_shd	5:0	r	'h4	Denoise3D Frequency Decomposition Kernel v15

ISP_DENOISE3D_DECOMPOSE_KERNEL_4_SHD

0x00007a84 = Byte Address

Description: isp_denoise3d_decompose_kernel_4_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_decompose_kernel_v24_shd	29:24	r	'h1	Denoise3D Frequency Decomposition Kernel v24
isp_denoise3d_decompose_kernel_v23_shd	23:18	r	'h4	Denoise3D Frequency Decomposition Kernel v23
isp_denoise3d_decompose_kernel_v22_shd	17:12	r	'h6	Denoise3D Frequency Decomposition Kernel v22
isp_denoise3d_decompose_kernel_v21_shd	11:6	r	'h4	Denoise3D Frequency Decomposition Kernel v21
isp_denoise3d_decompose_kernel_v20_shd	5:0	r	'h1	Denoise3D Frequency Decomposition Kernel v20

ISP_DENOISE3D_INV_SIGMA_L0_SHD

0x00007a88 = Byte Address

Description: isp_denoise3d_inv_sigma_l0_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_inv_sigma_l0_shd	11:0	r	'hfff	Denoise3D Inverse sigma for layer 1

ISP_DENOISE3D_INV_SIGMA_L1_SHD

0x00007a8c = Byte Address

Description: isp_denoise3d_inv_sigma_l1_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_inv_sigma_l1_shd	11:0	r	'hfff	Denoise3D Inverse sigma for layer 0

ISP_DENOISE3D_OFFSET_SHD

0x00007a90 = Byte Address

Description: isp_denoise3d_offset_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_offset_l1_shd	29:16	r	'h2000	Denoise3D decompose offset value fpr layer 1
isp_denoise3d_offset_l0_shd	13:0	r	'h2000	Denoise3D decompose offset value fpr layer 0

ISP_DENOISE3D_MOTION_GAUSS_KERNEL_SHD

0x00007a94 = Byte Address

Description: isp_denoise3d_motion_gauss_kernel_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_motion_gauss_kernel_v4_shd	24:20	r	'h1	Denoise3D Motion detection smoothing gaussian kernel v4
isp_denoise3d_motion_gauss_kernel_v3_shd	19:15	r	'h4	Denoise3D Motion detection smoothing gaussian kernel v3
isp_denoise3d_motion_gauss_kernel_v2_shd	14:10	r	'h6	Denoise3D Motion detection smoothing gaussian kernel v2
isp_denoise3d_motion_gauss_kernel_v1_shd	9:5	r	'h4	Denoise3D Motion detection smoothing gaussian kernel v1
isp_denoise3d_motion_gauss_kernel_v0_shd	4:0	r	'h1	Denoise3D Motion detection smoothing gaussian kernel v0

ISP_DENOISE3D_TNR_LUMA_LUT_0_SHD

0x00007a98 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_0_shd

Sub-Field Name	Bits	rw	Reset	Description

isp_denoise3d_tnr_luma_lut_p0_shd	31:28	r	'h0	Denoise3D TNR luma LUT p0
isp_denoise3d_tnr_luma_lut_y0_shd	27:12	r	'h0	Denoise3D TNR luma LUT y0
isp_denoise3d_tnr_luma_lut_motion_slope0_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v0

ISP_DENOISE3D_TNR_LUMA_LUT_1_SHD

0x00007a9c = Byte Address

Description: isp_denoise3d_tnr_luma_lut_1_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p1_shd	31:28	r	'h6	Denoise3D TNR luma LUT p1
isp_denoise3d_tnr_luma_lut_y1_shd	27:12	r	'h0	Denoise3D TNR luma LUT y1
isp_denoise3d_tnr_luma_lut_motion_slope1_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v1

ISP_DENOISE3D_TNR_LUMA_LUT_2_SHD

0x00007aa0 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_2_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p2_shd	31:28	r	'h6	Denoise3D TNR luma LUT p2
isp_denoise3d_tnr_luma_lut_y2_shd	27:12	r	'h0	Denoise3D TNR luma LUT y2
isp_denoise3d_tnr_luma_lut_motion_slope2_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v2

ISP_DENOISE3D_TNR_LUMA_LUT_3_SHD

0x00007aa4 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_3_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p3_shd	31:28	r	'h6	Denoise3D TNR luma LUT p3
isp_denoise3d_tnr_luma_lut_y3_shd	27:12	r	'h0	Denoise3D TNR luma LUT y3
isp_denoise3d_tnr_luma_lut_motion_slope3_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v3

ISP_DENOISE3D_TNR_LUMA_LUT_4_SHD

0x00007aa8 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_4_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_luma_lut_p4_shd	31:28	r	'h6	Denoise3D TNR luma LUT p4
isp_denoise3d_tnr_luma_lut_y4_shd	27:12	r	'h0	Denoise3D TNR luma LUT y4
isp_denoise3d_tnr_luma_lut_motion_slope4_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v4

ISP_DENOISE3D_TNR_LUMA_LUT_5_SHD

0x00007aac = Byte Address

Description: isp_denoise3d_tnr_luma_lut_5_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_luma_lut_p5_shd	31:28	r	'h6	Denoise3D TNR luma LUT p5
isp_denoise3d_tnr_luma_lut_y5_shd	27:12	r	'h0	Denoise3D TNR luma LUT y5
isp_denoise3d_tnr_luma_lut_motion_slope5_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v5

ISP_DENOISE3D_TNR_LUMA_LUT_6_SHD

0x00007ab0 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_6_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_luma_lut_p6_shd	31:28	r	'h6	Denoise3D TNR luma LUT p6
isp_denoise3d_tnr_luma_lut_y6_shd	27:12	r	'h0	Denoise3D TNR luma LUT y6
isp_denoise3d_tnr_luma_lut_motion_slope6_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v6

ISP_DENOISE3D_TNR_LUMA_LUT_7_SHD

0x00007ab4 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_7_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_luma_lut_p7_shd	31:28	r	'h6	Denoise3D TNR luma LUT p7

isp_denoise3d_tnr_luma_lut_y7_shd	27:12	r	'h0	Denoise3D TNR luma LUT y7
isp_denoise3d_tnr_luma_lut_motion_slope7_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v7

ISP_DENOISE3D_TNR_LUMA_LUT_8_SHD

0x00007ab8 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_8_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p8_shd	31:28	r	'h6	Denoise3D TNR luma LUT p8
isp_denoise3d_tnr_luma_lut_y8_shd	27:12	r	'h0	Denoise3D TNR luma LUT y8
isp_denoise3d_tnr_luma_lut_motion_slope8_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v8

ISP_DENOISE3D_TNR_LUMA_LUT_9_SHD

0x00007abc = Byte Address

Description: isp_denoise3d_tnr_luma_lut_9_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p9_shd	31:28	r	'h6	Denoise3D TNR luma LUT p9
isp_denoise3d_tnr_luma_lut_y9_shd	27:12	r	'h0	Denoise3D TNR luma LUT y9
isp_denoise3d_tnr_luma_lut_motion_slope9_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v9

ISP_DENOISE3D_TNR_LUMA_LUT_10_SHD

0x00007ac0 = Byte Address

Description: isp_denoise3d_tnr_luma_lut_10_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_luma_lut_p10_shd	31:28	r	'h6	Denoise3D TNR luma LUT p10
isp_denoise3d_tnr_luma_lut_y10_shd	27:12	r	'h0	Denoise3D TNR luma LUT y10
isp_denoise3d_tnr_luma_lut_motion_slope10_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v10

ISP_DENOISE3D_TNR_LUMA_LUT_11_SHD

0x00007ac4 = Byte Address

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Description: isp_denoise3d_tnr_luma_lut_11_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_luma_lut_p11_shd	31:28	r	'h6	Denoise3D TNR luma LUT p11
isp_denoise3d_tnr_luma_lut_y11_shd	27:12	r	'h0	Denoise3D TNR luma LUT y11
isp_denoise3d_tnr_luma_lut_motion_slope11_shd	11:0	r	'h40	Denoise3D TNR luma LUT motion slope v11

ISP_DENOISE3D_TNR_BOOST_0_SHD

0x00007ac8 = Byte Address

Description: isp_denoise3d_tnr_boost_0_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_boost_l0_c1_t1_shd	27:16	r	'h0	frequency synthesis detail boosting treshold 1 channel 1 in tnr
isp_denoise3d_tnr_boost_l0_c0_t1_shd	11:0	r	'h0	frequency synthesis detail boosting treshold 1 channel 0 in tnr

ISP_DENOISE3D_TNR_BOOST_1_SHD

0x00007acc = Byte Address

Description: isp_denoise3d_tnr_boost_1_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_boost_l0_c0_t2_shd	27:16	r	'hfff	frequency synthesis detail boosting treshold 2 channel 0 in tnr
isp_denoise3d_tnr_boost_l0_c2_t1_shd	11:0	r	'h0	frequency synthesis detail boosting treshold 1 channel 2 in tnr

ISP_DENOISE3D_TNR_BOOST_2_SHD

0x00007ad0 = Byte Address

Description: isp_denoise3d_tnr_boost_2_shd

Sub-Field Name	Bits	rw	Reset	Description
isp_denoise3d_tnr_boost_l0_c2_t2_shd	27:16	r	'hfff	frequency synthesis detail boosting treshold 2 channel 2 in tnr
isp_denoise3d_tnr_boost_l0_c1_t2_shd	11:0	r	'hfff	frequency synthesis detail boosting treshold 2 channel 1 in tnr

ISP_DENOISE3D_TNR_BOOST_3_SHD

0x00007ad4 = Byte Address

Description: isp_denoise3d_tnr_boost_3_shd

Sub-Field Name	Bits	rw	Reset	Description

isp_denoise3d_tnr_boost_fac_l0_c1_shd	27:16	r	'h0	frequency synthesis detail boosting factor channel 1 in tnr
isp_denoise3d_tnr_boost_fac_l0_c0_shd	11:0	r	'h0	frequency synthesis detail boosting factor channel 0 in tnr

ISP_DENOISE3D_TNR_BOOST_4_SHD

0x00007ad8 = Byte Address

Description: isp_denoise3d_tnr_boost_4_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_boost_clip_val_l0_c0_shd	27:16	r	'hfff	frequency synthesis detail boosting clipping value channel 0 in tnr
isp_denoise3d_tnr_boost_fac_l0_c2_shd	11:0	r	'h0	frequency synthesis detail boosting factor channel 2 in tnr

ISP_DENOISE3D_TNR_BOOST_5_SHD

0x00007adc = Byte Address

Description: isp_denoise3d_tnr_boost_5_shd

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_denoise3d_tnr_boost_clip_val_l0_c2_shd	27:16	r	'hfff	frequency synthesis detail boosting clipping value channel 2 in tnr
isp_denoise3d_tnr_boost_clip_val_l0_c1_shd	11:0	r	'hfff	frequency synthesis detail boosting clipping value channel 1 in tnr

ISP_AFMV3_CTRL

0x00007b00 = Byte Address

Description: afm iir module control

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_hist_thr	31:24	rw	'he6	histogram threshold for afm iir
isp_afm_iir_fdata_shift	23:20	rw	'h8	FIR filter data shift
isp_afm_iir_idata_shift	19:16	rw	'h8	IIR filter data shift
isp_afm_iir_blklevel_raw	15:4	rw	'h0	black level on 12 bit raw
isp_afm_iir_gamma_en	1:1	rw	'h0	gamma enable signal
isp_afm_iir_en	0:0	rw	'h0	afm iir enable signal

ISP_AFMV3_BLK_START

0x00007b04 = Byte Address

Description: block position of total roi

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_blk_ypos	31:16	rw	'h0	y position of total roi

isp_afm_iir_blk_xpos	15:0	rw	'h0	x position of total roi

ISP_AFMV3_BLK_SIZE

0x00007b08 = Byte Address

Description: block size of total roi

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_blk_height	31:16	rw	'h3c	block height total
isp_afm_iir_blk_width	15:0	rw	'h3c	block width total

ISP_AFMV3_FILTER_WEIGHT

0x00007b0c = Byte Address

Description: filter weight for different filters

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_lshift_weight	31:24	rw	'h10	FIR low pass filter weight
isp_afm_fir_hshift_weight	23:16	rw	'h10	FIR high pass filter weight
isp_afm_iir_lshift_weight	15:8	rw	'h10	IIR low pass filter weight
isp_afm_iir_hshift_weight	7:0	rw	'h10	IIR high pass filter weight

ISP_AFMV3_IIR_H_FILTER1_COEFF01

0x00007b10 = Byte Address

Description: IIR high pass first firstiir coeff0/1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_hfilt1_coeff1	25:13	rw	'h0	IIR high pass filter first firstiir coeff1
isp_afm_iir_hfilt1_coeff0	12:0	rw	'h0	IIR high pass filter first firstiir coeff0

ISP_AFMV3_IIR_H_FILTER1_COEFF23

0x00007b14 = Byte Address

Description: IIR high pass first firstiir coeff2/3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_hfilt1_coeff3	25:13	rw	'h0	IIR high pass filter first firstiir coeff3
isp_afm_iir_hfilt1_coeff2	12:0	rw	'h0	IIR high pass filter first firstiir coeff2

ISP_AFMV3_IIR_H_FILTER2_COEFF01

0x00007b18 = Byte Address

Description: IIR high pass second firstiir coeff0/1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_hfilt2_coeff1	25:13	rw	'h0	IIR high pass filter second firstiir coeff1
isp_afm_iir_hfilt2_coeff0	12:0	rw	'h0	IIR high pass filter second firstiir coeff0

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ISP_AFMV3_IIR_H_FILTER2_COEFF23

0x00007b1c = Byte Address

Description: IIR high pass second firstiir coeff2/3

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_hfilt2_coeff3	25:13	rw	'h0	IIR high pass filter second firstiir coeff3
isp_afm_iir_hfilt2_coeff2	12:0	rw	'h0	IIR high pass filter second firstiir coeff2

ISP_AFMV3_IIR_H_FILTER3_COEFF01

0x00007b20 = Byte Address

Description: IIR high pass lastiir coeff0/1

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_hfilt3_coeff1	25:13	rw	'h0	IIR high pass filter lasttiir coeff1
isp_afm_iir_hfilt3_coeff0	12:0	rw	'h0	IIR high pass filter lasttiir coeff0

ISP_AFMV3_IIR_H_FILTER3_COEFF23

0x00007b24 = Byte Address

Description: IIR high pass lastiir coeff2/3

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_hfilt3_coeff3	25:13	rw	'h0	IIR high pass filter lasttiir coeff3
isp_afm_iir_hfilt3_coeff2	12:0	rw	'h0	IIR high pass filter lasttiir coeff2

ISP_AFMV3_IIR_L_FILTER1_COEFF01

0x00007b28 = Byte Address

Description: IIR low pass first firstiir coeff0/1

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_lfilt1_coeff1	25:13	rw	'h0	IIR low pass filter first firstiir coeff1
isp_afm_iir_lfilt1_coeff0	12:0	rw	'h0	IIR low pass filter first firstiir coeff0

ISP_AFMV3_IIR_L_FILTER1_COEFF23

0x00007b2c = Byte Address

Description: IIR low pass first firstiir coeff2/3

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_lfilt1_coeff3	25:13	rw	'h0	IIR low pass filter first firstiir coeff3
isp_afm_iir_lfilt1_coeff2	12:0	rw	'h0	IIR low pass filter first firstiir coeff2

ISP_AFMV3_IIR_L_FILTER2_COEFF01

0x00007b30 = Byte Address

Description: IIR low pass second firstiir coeff0/1

Sub-Field Name	Bits	rw	Reset	Description

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isp_afm_iir_lfilt2_coeff1	25:13	rw	'h0	IIR low pass filter second firstiir coeff1
isp_afm_iir_lfilt2_coeff0	12:0	rw	'h0	IIR low pass filter second firstiir coeff0

ISP_AFMV3_IIR_L_FILTER2_COEFF23

0x00007b34 = Byte Address

Description: IIR low pass second firstiir coeff2/3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lfilt2_coeff3	25:13	rw	'h0	IIR low pass filter second firstiir coeff3
isp_afm_iir_lfilt2_coeff2	12:0	rw	'h0	IIR low pass filter second firstiir coeff2

ISP_AFMV3_IIR_L_FILTER3_COEFF01

0x00007b38 = Byte Address

Description: IIR low pass lastiir coeff0/1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lfilt3_coeff1	25:13	rw	'h0	IIR low pass filter lastiir coeff1
isp_afm_iir_lfilt3_coeff0	12:0	rw	'h0	IIR low pass filter lastiir coeff0

ISP_AFMV3_IIR_L_FILTER3_COEFF23

0x00007b3c = Byte Address

Description: IIR low pass lastiir coeff2/3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lfilt3_coeff3	25:13	rw	'h0	IIR low pass filter lastiir coeff3
isp_afm_iir_lfilt3_coeff2	12:0	rw	'h0	IIR low pass filter lastiir coeff2

ISP_AFMV3_FIR_H_FILTER_COEFF

0x00007b40 = Byte Address

Description: FIR high pass filter coeff

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_hfilt_vb1	31:24	rw	'h10	FIR high pass filter coeff vb1
isp_afm_fir_hfilt_va1	23:16	rw	'h20	FIR high pass filter coeff va1
isp_afm_fir_hfilt_hb1	15:8	rw	'h10	FIR high pass filter coeff hb1
isp_afm_fir_hfilt_ha1	7:0	rw	'h20	FIR high pass filter coeff ha1

ISP_AFMV3_FIR_L_FILTER_COEFF_HOR

0x00007b44 = Byte Address

Description: FIR low pass filter horizontal coeff

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_lfilt_hd1	31:24	rw	'h07	FIR low pass filter coeff hd1

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isp_afm_fir_lfilt_hc1	23:16	rw	'h07	FIR low pass filter coeff hc1
isp_afm_fir_lfilt_hb1	15:8	rw	'h07	FIR low pass filter coeff hb1
isp_afm_fir_lfilt_ha1	7:0	rw	'h09	FIR low pass filter coeff ha1

ISP_AFMV3_FIR_L_FILTER_COEFF_VER

0x00007b48 = Byte Address

Description: FIR low pass filter vertical coeff

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_lfilt_vd1	31:24	rw	'h09	FIR low pass filter coeff vd1
isp_afm_fir_lfilt_vc1	23:16	rw	'h09	FIR low pass filter coeff vc1
isp_afm_fir_lfilt_vb1	15:8	rw	'h09	FIR low pass filter coeff vb1
isp_afm_fir_lfilt_va1	7:0	rw	'h0a	FIR low pass filter coeff va1

ISP_AFMV3_GAMMA_PX_0

0x00007b4c = Byte Address

Description: distance of x in gamma curve px0~5

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_gamma_px_5	29:25	rw	'h06	gamma px_5
isp_afm_iir_gamma_px_4	24:20	rw	'h06	gamma px_4
isp_afm_iir_gamma_px_3	19:15	rw	'h06	gamma px_3
isp_afm_iir_gamma_px_2	14:10	rw	'h06	gamma px_2
isp_afm_iir_gamma_px_1	9:5	rw	'h06	gamma px_1
isp_afm_iir_gamma_px_0	4:0	rw	'h06	gamma px_0

ISP_AFMV3_GAMMA_PX_1

0x00007b50 = Byte Address

Description: distance of x in gamma curve px6~11

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_gamma_px_11	29:25	rw	'h06	gamma px_11
isp_afm_iir_gamma_px_10	24:20	rw	'h06	gamma px_10
isp_afm_iir_gamma_px_9	19:15	rw	'h06	gamma px_9
isp_afm_iir_gamma_px_8	14:10	rw	'h06	gamma px_8
isp_afm_iir_gamma_px_7	9:5	rw	'h06	gamma px_7
isp_afm_iir_gamma_px_6	4:0	rw	'h06	gamma px_6

ISP_AFMV3_GAMMA_PX_2

0x00007b54 = Byte Address

Description: distance of x in gamma curve px12~17

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Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_17	29:25	rw	'h06	gamma px_17
isp_afm_iir_gamma_px_16	24:20	rw	'h06	gamma px_16
isp_afm_iir_gamma_px_15	19:15	rw	'h06	gamma px_15
isp_afm_iir_gamma_px_14	14:10	rw	'h06	gamma px_14
isp_afm_iir_gamma_px_13	9:5	rw	'h06	gamma px_13
isp_afm_iir_gamma_px_12	4:0	rw	'h06	gamma px_12

ISP_AFMV3_GAMMA_PX_3

0x00007b58 = Byte Address

Description: distance of x in gamma curve px18~23

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_23	29:25	rw	'h06	gamma px_23
isp_afm_iir_gamma_px_22	24:20	rw	'h06	gamma px_22
isp_afm_iir_gamma_px_21	19:15	rw	'h06	gamma px_21
isp_afm_iir_gamma_px_20	14:10	rw	'h06	gamma px_20
isp_afm_iir_gamma_px_19	9:5	rw	'h06	gamma px_19
isp_afm_iir_gamma_px_18	4:0	rw	'h06	gamma px_18

ISP_AFMV3_GAMMA_PX_4

0x00007b5c = Byte Address

Description: distance of x in gamma curve px24~29

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_29	29:25	rw	'h06	gamma px_29
isp_afm_iir_gamma_px_28	24:20	rw	'h06	gamma px_28
isp_afm_iir_gamma_px_27	19:15	rw	'h06	gamma px_27
isp_afm_iir_gamma_px_26	14:10	rw	'h06	gamma px_26
isp_afm_iir_gamma_px_25	9:5	rw	'h06	gamma px_25
isp_afm_iir_gamma_px_24	4:0	rw	'h06	gamma px_24

ISP_AFMV3_GAMMA_PX_5

0x00007b60 = Byte Address

Description: distance of x in gamma curve px30~35

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_35	29:25	rw	'h06	gamma px_35
isp_afm_iir_gamma_px_34	24:20	rw	'h06	gamma px_34
isp_afm_iir_gamma_px_33	19:15	rw	'h06	gamma px_33

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isp_afm_iir_gamma_px_32	14:10	rw	'h06	gamma px_32
isp_afm_iir_gamma_px_31	9:5	rw	'h06	gamma px_31
isp_afm_iir_gamma_px_30	4:0	rw	'h06	gamma px_30

ISP_AFMV3_GAMMA_PX_6

0x00007b64 = Byte Address

Description: distance of x in gamma curve px36~41

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_gamma_px_41	29:25	rw	'h06	gamma px_41
isp_afm_iir_gamma_px_40	24:20	rw	'h06	gamma px_40
isp_afm_iir_gamma_px_39	19:15	rw	'h06	gamma px_39
isp_afm_iir_gamma_px_38	14:10	rw	'h06	gamma px_38
isp_afm_iir_gamma_px_37	9:5	rw	'h06	gamma px_37
isp_afm_iir_gamma_px_36	4:0	rw	'h06	gamma px_36

ISP_AFMV3_GAMMA_PX_7

0x00007b68 = Byte Address

Description: distance of x in gamma curve px42~47

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_gamma_px_47	29:25	rw	'h06	gamma px_47
isp_afm_iir_gamma_px_46	24:20	rw	'h06	gamma px_46
isp_afm_iir_gamma_px_45	19:15	rw	'h06	gamma px_45
isp_afm_iir_gamma_px_44	14:10	rw	'h06	gamma px_44
isp_afm_iir_gamma_px_43	9:5	rw	'h06	gamma px_43
isp_afm_iir_gamma_px_42	4:0	rw	'h06	gamma px_42

ISP_AFMV3_GAMMA_PX_8

0x00007b6c = Byte Address

Description: distance of x in gamma curve px48~53

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_gamma_px_53	29:25	rw	'h06	gamma px_53
isp_afm_iir_gamma_px_52	24:20	rw	'h06	gamma px_52
isp_afm_iir_gamma_px_51	19:15	rw	'h06	gamma px_51
isp_afm_iir_gamma_px_50	14:10	rw	'h06	gamma px_50
isp_afm_iir_gamma_px_49	9:5	rw	'h06	gamma px_49
isp_afm_iir_gamma_px_48	4:0	rw	'h06	gamma px_48

ISP_AFMV3_GAMMA_PX_9

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0x00007b70 = Byte Address

Description: distance of x in gamma curve px54~59

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_59	29:25	rw	'h06	gamma px_59
isp_afm_iir_gamma_px_58	24:20	rw	'h06	gamma px_58
isp_afm_iir_gamma_px_57	19:15	rw	'h06	gamma px_57
isp_afm_iir_gamma_px_56	14:10	rw	'h06	gamma px_56
isp_afm_iir_gamma_px_55	9:5	rw	'h06	gamma px_55
isp_afm_iir_gamma_px_54	4:0	rw	'h06	gamma px_54

ISP_AFMV3_GAMMA_PX_10

0x00007b74 = Byte Address

Description: distance of x in gamma curve px60~63

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_63	19:15	rw	'h06	gamma px_63
isp_afm_iir_gamma_px_62	14:10	rw	'h06	gamma px_62
isp_afm_iir_gamma_px_61	9:5	rw	'h06	gamma px_61
isp_afm_iir_gamma_px_60	4:0	rw	'h06	gamma px_60

ISP_AFMV3_GAMMA_Y_ADDR

0x00007b78 = Byte Address

Description: gamma curve y address

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_y_addr	5:0	rw	'h0	gamma curve y address

ISP_AFMV3_GAMMA_Y_DATA

0x00007b7c = Byte Address

Description: gamma curve y data

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_y_data	7:0	rw	'h0	gamma curve y data

ISP_AFMV3_GAMMA_X_ADDR

0x00007b80 = Byte Address

Description: gamma curve x address

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_x_addr	5:0	rw	'h0	gamma curve x address

ISP_AFMV3_GAMMA_X_DATA

0x00007b84 = Byte Address

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Description: gamma curve x data

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_x_data	11:0	rw	'h0	gamma curve x data

ISP_AFMV3_SHARPNESS_ADDR

0x00007b88 = Byte Address

Description: sharpness addreSS

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_sharpness_addr	8:0	rw	'h0	sharpness address

ISP_AFMV3_SHARPNESS_DATA

0x00007b8c = Byte Address

Description: sharpness data

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_sharpness_data	31:0	r	'h0	sharpness data

ISP_AFMV3_HISTOGRAM_ADDR

0x00007b90 = Byte Address

Description: histogram address

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_histogram_addr	8:0	rw	'h0	histogram address

ISP_AFMV3_HISTOGRAM_DATA

0x00007b94 = Byte Address

Description: histogram data

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_histogram_data	31:0	r	'h0	histogram data

ISP_AFMV3_LDG_IIR_HIGHPASS_Y

0x00007b98 = Byte Address

Description: IIR high pass ldg curve y0~y2

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_hldg_r3	23:16	rw	'h10	IIR high pass ldg y2
isp_afm_iir_hldg_r2	15:8	rw	'hff	IIR high pass ldg y1
isp_afm_iir_hldg_r1	7:0	rw	'h40	IIR high pass ldg y0

ISP_AFMV3_LDG_IIR_HIGHPASS_X

0x00007b9c = Byte Address

Description: IIR high pass ldg curve x0~x3

Sub-Field Name	Bits	rw	Reset	Description

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isp_afm_iir_hldg_t4	31:24	rw	'hd0	IIR high pass ldg x3
isp_afm_iir_hldg_t3	23:16	rw	'ha0	IIR high pass ldg x2
isp_afm_iir_hldg_t2	15:8	rw	'h60	IIR high pass ldg x1
isp_afm_iir_hldg_t1	7:0	rw	'h10	IIR high pass ldg x0

ISP_AFMV3_LDG_IIR_LOWPASS_Y

0x00007ba0 = Byte Address

Description: IIR low pass ldg curve y0~y2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lldg_r3	23:16	rw	'h10	IIR low pass ldg y2
isp_afm_iir_lldg_r2	15:8	rw	'hff	IIR low pass ldg y1
isp_afm_iir_lldg_r1	7:0	rw	'h40	IIR low pass ldg y0

ISP_AFMV3_LDG_IIR_LOWPASS_X

0x00007ba4 = Byte Address

Description: IIR low pass ldg curve x0~x3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lldg_t4	31:24	rw	'hd0	IIR low pass ldg x3
isp_afm_iir_lldg_t3	23:16	rw	'ha0	IIR low pass ldg x2
isp_afm_iir_lldg_t2	15:8	rw	'h60	IIR low pass ldg x1
isp_afm_iir_lldg_t1	7:0	rw	'h10	IIR low pass ldg x0

ISP_AFMV3_LDG_FIR_HIGHPASS_Y

0x00007ba8 = Byte Address

Description: FIR high pass ldg curve y0~y2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_hldg_r3	23:16	rw	'h01	FIR high pass ldg y2
isp_afm_fir_hldg_r2	15:8	rw	'h01	FIR high pass ldg y1
isp_afm_fir_hldg_r1	7:0	rw	'h01	FIR high pass ldg y0

ISP_AFMV3_LDG_FIR_HIGHPASS_X

0x00007bac = Byte Address

Description: FIR high pass ldg curve x0~x3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_hldg_t4	31:24	rw	'hff	FIR high pass ldg x3
isp_afm_fir_hldg_t3	23:16	rw	'hb0	FIR high pass ldg x2
isp_afm_fir_hldg_t2	15:8	rw	'h60	FIR high pass ldg x1

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isp_afm_fir_hldg_t1	7:0	rw	'h30	FIR high pass ldg x0

ISP_AFMV3_LDG_FIR_LOWPASS_Y

0x00007bb0 = Byte Address

Description: FIR low pass ldg curve y0~y2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_lldg_r3	23:16	rw	'h01	FIR low pass ldg y2
isp_afm_fir_lldg_r2	15:8	rw	'h01	FIR low pass ldg y1
isp_afm_fir_lldg_r1	7:0	rw	'h01	FIR low pass ldg y0

ISP_AFMV3_LDG_FIR_LOWPASS_X

0x00007bb4 = Byte Address

Description: FIR low pass ldg curve x0~x3

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_lldg_t4	31:24	rw	'hff	FIR low pass ldg x3
isp_afm_fir_lldg_t3	23:16	rw	'hb0	FIR low pass ldg x2
isp_afm_fir_lldg_t2	15:8	rw	'h60	FIR low pass ldg x1
isp_afm_fir_lldg_t1	7:0	rw	'h30	FIR low pass ldg x0

ISP_AFMV3_CORING_IIR_HIGHPASS

0x00007bb8 = Byte Address

Description: IIR high pass coring coeff

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_hcor_slope	23:16	rw	'h10	IIR high pass coring slope
isp_afm_iir_hcor_r1	15:8	rw	'he6	IIR high pass coring r1
isp_afm_iir_hcor_t1	7:0	rw	'h0	IIR high pass coring t1

ISP_AFMV3_CORING_IIR_LOWPASS

0x00007bbc = Byte Address

Description: IIR low pass coring coeff

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lcor_slope	23:16	rw	'h10	IIR low pass coring slope
isp_afm_iir_lcor_r1	15:8	rw	'he6	IIR low pass coring r1
isp_afm_iir_lcor_t1	7:0	rw	'h0	IIR low pass coring t1

ISP_AFMV3_CORING_FIR_HIGHPASS

0x00007bc0 = Byte Address

Description: FIR high pass coring coeff

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

isp_afm_fir_hcor_slope	23:16	rw	'h10	FIR high pass coring slope
isp_afm_fir_hcor_r1	15:8	rw	'he6	FIR high pass coring r1
isp_afm_fir_hcor_t1	7:0	rw	'h0	FIR high pass coring t1

ISP_AFMV3_CORING_FIR_LOWPASS

0x00007bc4 = Byte Address

Description: FIR low pass coring coeff

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_lcor_slope	23:16	rw	'h10	FIR low pass coring slope
isp_afm_fir_lcor_r1	15:8	rw	'he6	FIR low pass coring r1
isp_afm_fir_lcor_t1	7:0	rw	'h0	FIR low pass coring t1

ISP_AFMV3_SHIFT

0x00007bc8 = Byte Address

Description: Shift of fir and block sum

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_data_shift_v	23:20	rw	'h0	iir sum data shift in vertical
isp_afm_fir_data_shift_v	19:16	rw	'h0	fir sum data shift in vertical
isp_afm_fir_lowpass_shift_y	15:12	rw	'h3	fir lowpass shift in y direction
isp_afm_fir_lowpass_shift_x	11:8	rw	'h3	fir lowpass shift in x direction
isp_afm_fir_highpass_shift_y	7:4	rw	'h0	fir highpass shift in y direction
isp_afm_fir_highpass_shift_x	3:0	rw	'h0	fir highpass shift in x direction

ISP_AFMV3_CTRL_SHD

0x00007c00 = Byte Address

Description: afm iir module control shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_hist_thr_shd	31:24	r	'he6	histogram threshold for afm iir shadow
isp_afm_iir_fdata_shift_shd	23:20	r	'h8	FIR filter data shift shadow
isp_afm_iir_idata_shift_shd	19:16	r	'h8	IIR filter data shift shadow
isp_afm_iir_blklevel_raw_shd	15:4	r	'h0	black level on 12 bit raw shadow
isp_afm_iir_gamma_en_shd	1:1	r	'h0	gamma enable signal shadow
isp_afm_iir_en_shd	0:0	r	'h0	afm iir enable signal shadow

ISP_AFMV3_BLK_START_SHD

0x00007c04 = Byte Address

Description: block position of total roi shadow

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Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_blk_ypos_shd	31:16	r	'h0	y position of total roi shadow
isp_afm_iir_blk_xpos_shd	15:0	r	'h0	x position of total roi shadow

ISP_AFMV3_BLK_SIZE_SHD

0x00007c08 = Byte Address

Description: block size of total roi shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_blk_height_shd	31:16	r	'h3c	block height total shadow
isp_afm_iir_blk_width_shd	15:0	r	'h3c	block width total shadow

ISP_AFMV3_FILTER_WEIGHT_SHD

0x00007c0c = Byte Address

Description: filter weight for different filters shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_fir_lshift_weight_shd	31:24	r	'h10	FIR low pass filter weight shadow
isp_afm_fir_hshift_weight_shd	23:16	r	'h10	FIR high pass filter weight shadow
isp_afm_iir_lshift_weight_shd	15:8	r	'h10	IIR low pass filter weight shadow
isp_afm_iir_hshift_weight_shd	7:0	r	'h10	IIR high pass filter weight shadow

ISP_AFMV3_IIR_H_FILTER1_COEFF01_SHD

0x00007c10 = Byte Address

Description: IIR high pass first firstiir coeff0/1 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_hfilt1_coeff1_shd	25:13	r	'h0	IIR high pass filter first firstiir coeff1 shadow
isp_afm_iir_hfilt1_coeff0_shd	12:0	r	'h0	IIR high pass filter first firstiir coeff0 shadow

ISP_AFMV3_IIR_H_FILTER1_COEFF23_SHD

0x00007c14 = Byte Address

Description: IIR high pass first firstiir coeff2/3 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_hfilt1_coeff3_shd	25:13	r	'h0	IIR high pass filter first firstiir coeff3 shadow
isp_afm_iir_hfilt1_coeff2_shd	12:0	r	'h0	IIR high pass filter first firstiir coeff2 shadow

ISP_AFMV3_IIR_H_FILTER2_COEFF01_SHD

0x00007c18 = Byte Address

Description: IIR high pass second firstiir coeff0/1 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_hfilt2_coeff1_shd	25:13	r	'h0	IIR high pass filter second firstiir coeff1 shadow
isp_afm_iir_hfilt2_coeff0_shd	12:0	r	'h0	IIR high pass filter second firstiir coeff0 shadow

ISP_AFMV3_IIR_H_FILTER2_COEFF23_SHD

0x00007c1c = Byte Address

Description: IIR high pass second firstiir coeff2/3 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_hfilt2_coeff3_shd	25:13	r	'h0	IIR high pass filter second firstiir coeff3 shadow
isp_afm_iir_hfilt2_coeff2_shd	12:0	r	'h0	IIR high pass filter second firstiir coeff2 shadow

ISP_AFMV3_IIR_H_FILTER3_COEFF01_SHD

0x00007c20 = Byte Address

Description: IIR high pass lastiir coeff0/1 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_hfilt3_coeff1_shd	25:13	r	'h0	IIR high pass filter lasttiir coeff1 shadow
isp_afm_iir_hfilt3_coeff0_shd	12:0	r	'h0	IIR high pass filter lasttiir coeff0 shadow

ISP_AFMV3_IIR_H_FILTER3_COEFF23_SHD

0x00007c24 = Byte Address

Description: IIR high pass lastiir coeff2/3 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_hfilt3_coeff3_shd	25:13	r	'h0	IIR high pass filter lasttiir coeff3 shadow
isp_afm_iir_hfilt3_coeff2_shd	12:0	r	'h0	IIR high pass filter lasttiir coeff2 shadow

ISP_AFMV3_IIR_L_FILTER1_COEFF01_SHD

0x00007c28 = Byte Address

Description: IIR low pass first firstiir coeff0/1 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_lfilt1_coeff1_shd	25:13	r	'h0	IIR low pass filter first firstiir coeff1 shadow
isp_afm_iir_lfilt1_coeff0_shd	12:0	r	'h0	IIR low pass filter first firstiir coeff0 shadow

ISP_AFMV3_IIR_L_FILTER1_COEFF23_SHD

0x00007c2c = Byte Address

Description: IIR low pass first firstiir coeff2/3 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_lfilt1_coeff3_shd	25:13	r	'h0	IIR low pass filter first firstiir coeff3 shadow

isp_afm_iir_lfilt1_coeff2_shd	12:0	r	'h0	IIR low pass filter first firstiir coeff2 shadow

ISP_AFMV3_IIR_L_FILTER2_COEFF01_SHD

0x00007c30 = Byte Address

Description: IIR low pass second firstiir coeff0/1 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lfilt2_coeff1_shd	25:13	r	'h0	IIR low pass filter second firstiir coeff1 shadow
isp_afm_iir_lfilt2_coeff0_shd	12:0	r	'h0	IIR low pass filter second firstiir coeff0 shadow

ISP_AFMV3_IIR_L_FILTER2_COEFF23_SHD

0x00007c34 = Byte Address

Description: IIR low pass second firstiir coeff2/3 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lfilt2_coeff3_shd	25:13	r	'h0	IIR low pass filter second firstiir coeff3 shadow
isp_afm_iir_lfilt2_coeff2_shd	12:0	r	'h0	IIR low pass filter second firstiir coeff2 shadow

ISP_AFMV3_IIR_L_FILTER3_COEFF01_SHD

0x00007c38 = Byte Address

Description: IIR low pass lastiir coeff0/1 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lfilt3_coeff1_shd	25:13	r	'h0	IIR low pass filter lastiir coeff1 shadow
isp_afm_iir_lfilt3_coeff0_shd	12:0	r	'h0	IIR low pass filter lastiir coeff0 shadow

ISP_AFMV3_IIR_L_FILTER3_COEFF23_SHD

0x00007c3c = Byte Address

Description: IIR low pass lastiir coeff2/3 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lfilt3_coeff3_shd	25:13	r	'h0	IIR low pass filter lastiir coeff3 shadow
isp_afm_iir_lfilt3_coeff2_shd	12:0	r	'h0	IIR low pass filter lastiir coeff2 shadow

ISP_AFMV3_FIR_H_FILTER_COEFF_SHD

0x00007c40 = Byte Address

Description: FIR high pass filter coeff shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_hfilt_vb1_shd	31:24	r	'h10	FIR high pass filter coeff vb1 shadow
isp_afm_fir_hfilt_va1_shd	23:16	r	'h20	FIR high pass filter coeff va1 shadow
isp_afm_fir_hfilt_hb1_shd	15:8	r	'h10	FIR high pass filter coeff hb1 shadow
isp_afm_fir_hfilt_ha1_shd	7:0	r	'h20	FIR high pass filter coeff ha1 shadow

ISP_AFMV3_FIR_L_FILTER_COEFF_HOR_SHD

0x00007c44 = Byte Address

Description: FIR low pass filter horizontal coeff shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_lfilt_hd1_shd	31:24	r	'h07	FIR low pass filter coeff hd1 shadow
isp_afm_fir_lfilt_hc1_shd	23:16	r	'h07	FIR low pass filter coeff hc1 shadow
isp_afm_fir_lfilt_hb1_shd	15:8	r	'h07	FIR low pass filter coeff hb1 shadow
isp_afm_fir_lfilt_ha1_shd	7:0	r	'h09	FIR low pass filter coeff ha1 shadow

ISP_AFMV3_FIR_L_FILTER_COEFF_VER_SHD

0x00007c48 = Byte Address

Description: FIR low pass filter vertical coeff shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_lfilt_vd1_shd	31:24	r	'h09	FIR low pass filter coeff vd1 shadow
isp_afm_fir_lfilt_vc1_shd	23:16	r	'h09	FIR low pass filter coeff vc1 shadow
isp_afm_fir_lfilt_vb1_shd	15:8	r	'h09	FIR low pass filter coeff vb1 shadow
isp_afm_fir_lfilt_va1_shd	7:0	r	'h0a	FIR low pass filter coeff va1 shadow

ISP_AFMV3_GAMMA_PX_0_SHD

0x00007c4c = Byte Address

Description: distance of x in gamma curve px0~5 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_gamma_px_5_shd	29:25	r	'h06	gamma px_5 shadow
isp_afm_iir_gamma_px_4_shd	24:20	r	'h06	gamma px_4 shadow
isp_afm_iir_gamma_px_3_shd	19:15	r	'h06	gamma px_3 shadow
isp_afm_iir_gamma_px_2_shd	14:10	r	'h06	gamma px_2 shadow
isp_afm_iir_gamma_px_1_shd	9:5	r	'h06	gamma px_1 shadow
isp_afm_iir_gamma_px_0_shd	4:0	r	'h06	gamma px_0 shadow

ISP_AFMV3_GAMMA_PX_1_SHD

0x00007c50 = Byte Address

Description: distance of x in gamma curve px6~11 shadow

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Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_11_shd	29:25	r	'h06	gamma px_11 shadow
isp_afm_iir_gamma_px_10_shd	24:20	r	'h06	gamma px_10 shadow
isp_afm_iir_gamma_px_9_shd	19:15	r	'h06	gamma px_9 shadow
isp_afm_iir_gamma_px_8_shd	14:10	r	'h06	gamma px_8 shadow
isp_afm_iir_gamma_px_7_shd	9:5	r	'h06	gamma px_7 shadow
isp_afm_iir_gamma_px_6_shd	4:0	r	'h06	gamma px_6 shadow

ISP_AFMV3_GAMMA_PX_2_SHD

0x00007c54 = Byte Address

Description: distance of x in gamma curve px12~17 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_17_shd	29:25	r	'h06	gamma px_17 shadow
isp_afm_iir_gamma_px_16_shd	24:20	r	'h06	gamma px_16 shadow
isp_afm_iir_gamma_px_15_shd	19:15	r	'h06	gamma px_15 shadow
isp_afm_iir_gamma_px_14_shd	14:10	r	'h06	gamma px_14 shadow
isp_afm_iir_gamma_px_13_shd	9:5	r	'h06	gamma px_13 shadow
isp_afm_iir_gamma_px_12_shd	4:0	r	'h06	gamma px_12 shadow

ISP_AFMV3_GAMMA_PX_3_SHD

0x00007c58 = Byte Address

Description: distance of x in gamma curve px18~23 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_23_shd	29:25	r	'h06	gamma px_23 shadow
isp_afm_iir_gamma_px_22_shd	24:20	r	'h06	gamma px_22 shadow

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isp_afm_iir_gamma_px_21_shd	19:15	r	'h06	gamma px_21 shadow
isp_afm_iir_gamma_px_20_shd	14:10	r	'h06	gamma px_20 shadow
isp_afm_iir_gamma_px_19_shd	9:5	r	'h06	gamma px_19 shadow
isp_afm_iir_gamma_px_18_shd	4:0	r	'h06	gamma px_18 shadow

ISP_AFMV3_GAMMA_PX_4_SHD

0x00007c5c = Byte Address

Description: distance of x in gamma curve px24~29 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_gamma_px_29_shd	29:25	r	'h06	gamma px_29 shadow
isp_afm_iir_gamma_px_28_shd	24:20	r	'h06	gamma px_28 shadow
isp_afm_iir_gamma_px_27_shd	19:15	r	'h06	gamma px_27 shadow
isp_afm_iir_gamma_px_26_shd	14:10	r	'h06	gamma px_26 shadow
isp_afm_iir_gamma_px_25_shd	9:5	r	'h06	gamma px_25 shadow
isp_afm_iir_gamma_px_24_shd	4:0	r	'h06	gamma px_24 shadow

ISP_AFMV3_GAMMA_PX_5_SHD

0x00007c60 = Byte Address

Description: distance of x in gamma curve px30~35 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_gamma_px_35_shd	29:25	r	'h06	gamma px_35 shadow
isp_afm_iir_gamma_px_34_shd	24:20	r	'h06	gamma px_34 shadow
isp_afm_iir_gamma_px_33_shd	19:15	r	'h06	gamma px_33 shadow
isp_afm_iir_gamma_px_32_shd	14:10	r	'h06	gamma px_32 shadow

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isp_afm_iir_gamma_px_31_shd	9:5	r	'h06	gamma px_31 shadow
isp_afm_iir_gamma_px_30_shd	4:0	r	'h06	gamma px_30 shadow

ISP_AFMV3_GAMMA_PX_6_SHD

0x00007c64 = Byte Address

Description: distance of x in gamma curve px36~41 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_gamma_px_41_shd	29:25	r	'h06	gamma px_41 shadow
isp_afm_iir_gamma_px_40_shd	24:20	r	'h06	gamma px_40 shadow
isp_afm_iir_gamma_px_39_shd	19:15	r	'h06	gamma px_39 shadow
isp_afm_iir_gamma_px_38_shd	14:10	r	'h06	gamma px_38 shadow
isp_afm_iir_gamma_px_37_shd	9:5	r	'h06	gamma px_37 shadow
isp_afm_iir_gamma_px_36_shd	4:0	r	'h06	gamma px_36 shadow

ISP_AFMV3_GAMMA_PX_7_SHD

0x00007c68 = Byte Address

Description: distance of x in gamma curve px42~47 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_gamma_px_47_shd	29:25	r	'h06	gamma px_47 shadow
isp_afm_iir_gamma_px_46_shd	24:20	r	'h06	gamma px_46 shadow
isp_afm_iir_gamma_px_45_shd	19:15	r	'h06	gamma px_45 shadow
isp_afm_iir_gamma_px_44_shd	14:10	r	'h06	gamma px_44 shadow
isp_afm_iir_gamma_px_43_shd	9:5	r	'h06	gamma px_43 shadow
isp_afm_iir_gamma_px_42_shd	4:0	r	'h06	gamma px_42 shadow

ISP_AFMV3_GAMMA_PX_8_SHD

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0x00007c6c = Byte Address

Description: distance of x in gamma curve px48~53 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_53_shd	29:25	r	'h06	gamma px_53 shadow
isp_afm_iir_gamma_px_52_shd	24:20	r	'h06	gamma px_52 shadow
isp_afm_iir_gamma_px_51_shd	19:15	r	'h06	gamma px_51 shadow
isp_afm_iir_gamma_px_50_shd	14:10	r	'h06	gamma px_50 shadow
isp_afm_iir_gamma_px_49_shd	9:5	r	'h06	gamma px_49 shadow
isp_afm_iir_gamma_px_48_shd	4:0	r	'h06	gamma px_48 shadow

ISP_AFMV3_GAMMA_PX_9_SHD

0x00007c70 = Byte Address

Description: distance of x in gamma curve px54~59 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_59_shd	29:25	r	'h06	gamma px_59 shadow
isp_afm_iir_gamma_px_58_shd	24:20	r	'h06	gamma px_58 shadow
isp_afm_iir_gamma_px_57_shd	19:15	r	'h06	gamma px_57 shadow
isp_afm_iir_gamma_px_56_shd	14:10	r	'h06	gamma px_56 shadow
isp_afm_iir_gamma_px_55_shd	9:5	r	'h06	gamma px_55 shadow
isp_afm_iir_gamma_px_54_shd	4:0	r	'h06	gamma px_54 shadow

ISP_AFMV3_GAMMA_PX_10_SHD

0x00007c74 = Byte Address

Description: distance of x in gamma curve px60~63 shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_gamma_px_63_shd	19:15	r	'h06	gamma px_63 shadow

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isp_afm_iir_gamma_px_62_shd	14:10	r	'h06	gamma px_62 shadow
isp_afm_iir_gamma_px_61_shd	9:5	r	'h06	gamma px_61 shadow
isp_afm_iir_gamma_px_60_shd	4:0	r	'h06	gamma px_60 shadow

ISP_AFMV3_LDG_IIR_HIGHPASS_Y_SHD

0x00007c88 = Byte Address

Description: IIR high pass ldg curve y0~y2 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_hldg_r3_shd	23:16	r	'h10	IIR high pass ldg y2 shadow
isp_afm_iir_hldg_r2_shd	15:8	r	'hff	IIR high pass ldg y1 shadow
isp_afm_iir_hldg_r1_shd	7:0	r	'h40	IIR high pass ldg y0 shadow

ISP_AFMV3_LDG_IIR_HIGHPASS_X_SHD

0x00007c8c = Byte Address

Description: IIR high pass ldg curve x0~x3 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_hldg_t4_shd	31:24	r	'hd0	IIR high pass ldg x3 shadow
isp_afm_iir_hldg_t3_shd	23:16	r	'ha0	IIR high pass ldg x2 shadow
isp_afm_iir_hldg_t2_shd	15:8	r	'h60	IIR high pass ldg x1 shadow
isp_afm_iir_hldg_t1_shd	7:0	r	'h10	IIR high pass ldg x0 shadow

ISP_AFMV3_LDG_IIR_LOWPASS_Y_SHD

0x00007c90 = Byte Address

Description: IIR low pass ldg curve y0~y2 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lldg_r3_shd	23:16	r	'h10	IIR low pass ldg y2 shadow
isp_afm_iir_lldg_r2_shd	15:8	r	'hff	IIR low pass ldg y1 shadow
isp_afm_iir_lldg_r1_shd	7:0	r	'h40	IIR low pass ldg y0 shadow

ISP_AFMV3_LDG_IIR_LOWPASS_X_SHD

0x00007c94 = Byte Address

Description: IIR low pass ldg curve x0~x3 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_iir_lldg_t4_shd	31:24	r	'hd0	IIR low pass ldg x3 shadow
isp_afm_iir_lldg_t3_shd	23:16	r	'ha0	IIR low pass ldg x2 shadow
isp_afm_iir_lldg_t2_shd	15:8	r	'h60	IIR low pass ldg x1 shadow

isp_afm_iir_lldg_t1_shd	7:0	r	'h10	IIR low pass ldg x0 shadow

ISP_AFMV3_LDG_FIR_HIGHPASS_Y_SHD

0x00007c98 = Byte Address

Description: FIR high pass ldg curve y0~y2 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_hldg_r3_shd	23:16	r	'h01	FIR high pass ldg y2 shadow
isp_afm_fir_hldg_r2_shd	15:8	r	'h01	FIR high pass ldg y1 shadow
isp_afm_fir_hldg_r1_shd	7:0	r	'h01	FIR high pass ldg y0 shadow

ISP_AFMV3_LDG_FIR_HIGHPASS_X_SHD

0x00007c9c = Byte Address

Description: FIR high pass ldg curve x0~x3 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_hldg_t4_shd	31:24	r	'hff	FIR high pass ldg x3 shadow
isp_afm_fir_hldg_t3_shd	23:16	r	'hb0	FIR high pass ldg x2 shadow
isp_afm_fir_hldg_t2_shd	15:8	r	'h60	FIR high pass ldg x1 shadow
isp_afm_fir_hldg_t1_shd	7:0	r	'h30	FIR high pass ldg x0 shadow

ISP_AFMV3_LDG_FIR_LOWPASS_Y_SHD

0x00007ca0 = Byte Address

Description: FIR low pass ldg curve y0~y2 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_lldg_r3_shd	23:16	r	'h01	FIR low pass ldg y2 shadow
isp_afm_fir_lldg_r2_shd	15:8	r	'h01	FIR low pass ldg y1 shadow
isp_afm_fir_lldg_r1_shd	7:0	r	'h01	FIR low pass ldg y0 shadow

ISP_AFMV3_LDG_FIR_LOWPASS_X_SHD

0x00007ca4 = Byte Address

Description: FIR low pass ldg curve x0~x3 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
isp_afm_fir_lldg_t4_shd	31:24	r	'hff	FIR low pass ldg x3 shadow
isp_afm_fir_lldg_t3_shd	23:16	r	'hb0	FIR low pass ldg x2 shadow
isp_afm_fir_lldg_t2_shd	15:8	r	'h60	FIR low pass ldg x1 shadow
isp_afm_fir_lldg_t1_shd	7:0	r	'h30	FIR low pass ldg x0 shadow

ISP_AFMV3_CORING_IIR_HIGHPASS_SHD

0x00007ca8 = Byte Address

Description: IIR high pass coring coeff shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_hcor_slope_shd	23:16	r	'h10	IIR high pass coring slope shadow
isp_afm_iir_hcor_r1_shd	15:8	r	'he6	IIR high pass coring r1 shadow
isp_afm_iir_hcor_t1_shd	7:0	r	'h0	IIR high pass coring t1 shadow

ISP_AFMV3_CORING_IIR_LOWPASS_SHD

0x00007cac = Byte Address

Description: IIR low pass coring coeff shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_lcor_slope_shd	23:16	r	'h10	IIR low pass coring slope shadow
isp_afm_iir_lcor_r1_shd	15:8	r	'he6	IIR low pass coring r1 shadow
isp_afm_iir_lcor_t1_shd	7:0	r	'h0	IIR low pass coring t1 shadow

ISP_AFMV3_CORING_FIR_HIGHPASS_SHD

0x00007cb0 = Byte Address

Description: FIR high pass coring coeff shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_fir_hcor_slope_shd	23:16	r	'h10	FIR high pass coring slope shadow
isp_afm_fir_hcor_r1_shd	15:8	r	'he6	FIR high pass coring r1 shadow
isp_afm_fir_hcor_t1_shd	7:0	r	'h0	FIR high pass coring t1 shadow

ISP_AFMV3_CORING_FIR_LOWPASS_SHD

0x00007cb4 = Byte Address

Description: FIR low pass coring coeff shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_fir_lcor_slope_shd	23:16	r	'h10	FIR low pass coring slope shadow
isp_afm_fir_lcor_r1_shd	15:8	r	'he6	FIR low pass coring r1 shadow
isp_afm_fir_lcor_t1_shd	7:0	r	'h0	FIR low pass coring t1 shadow

ISP_AFMV3_SHIFT_SHD

0x00007cb8 = Byte Address

Description: Shift of fir and block sum shadow

Sub-Field Name	Bits	rw	Reset	Description
isp_afm_iir_data_shift_v_shd	23:20	r	'h0	iir sum data shift in vertical shadow
isp_afm_fir_data_shift_v_shd	19:16	r	'h0	fir sum data shift in vertical shadow
isp_afm_fir_lowpass_shift_y_shd	15:12	r	'h3	fir lowpass shift in y direction shadow

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isp_afm_fir_lowpass_shift_x_shd	11:8	r	'h3	fir lowpass shift in x direction shadow
isp_afm_fir_highpass_shift_y_shd	7:4	r	'h0	fir highpass shift in y direction shadow
isp_afm_fir_highpass_shift_x_shd	3:0	r	'h0	fir highpass shift in x direction shadow

ISP_EE2_CTRL

0x00007d00 = Byte Address

Description: ee2 CTRL

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_yuv_en	16:16	rw	'h0	yuv en
ee2_skin_proc_en	14:12	rw	'h0	skin proc en
ee2_depf_detect_range	11:9	rw	'h7	depf detect range
ee2_depurple_en	8:8	rw	'h0	depurple en
ee2_en	0:0	rw	'h1	en

ISP_EE2_STRENGTH

0x00007d04 = Byte Address

Description: ee2 STRENGTH

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_uv_gain	23:16	rw	'h0	uv gain

ISP_EE2_NR_CTRL

0x00007d08 = Byte Address

Description: ee2 NR CTRL

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_denoise_level	26:24	rw	'h3	denoise level
ee2_use_auxi_dir	16:16	rw	'h1	use auxi dir
ee2_detail_lvl	14:12	rw	'h5	detail lvl
ee2_detail_enhance_str	9:0	rw	'h16	detail enhance str

ISP_EE2_SKIN_MODULE_STRENGTH

0x00007d0c = Byte Address

Description: ee2 SKIN MODULE STRENGTH

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_skin_detect_str	8:0	rw	'hd3	skin detect str

ISP_EE2_SKIN_Y_LIMIT

0x00007d10 = Byte Address

Description: ee2 SKIN Y LIMIT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_skin_hth_y_min	26:16	rw	'h118	skin threshold y min
ee2_skin_hth_y_max	10:0	rw	'h384	skin threshold y max

ISP_EE2_SKIN_CR_LIMIT

0x00007d14 = Byte Address

Description: ee2 SKIN CR LIMIT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_skin_hth_cr_min	27:16	rw	'h3d8	skin threshold cr min
ee2_skin_hth_cr_max	11:0	rw	'h568	skin threshold cr max

ISP_EE2_SKIN_CB_LIMIT

0x00007d18 = Byte Address

Description: ee2 SKIN CB LIMIT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_skin_hth_cb_min	27:16	rw	'h2fc	skin threshold cb min
ee2_skin_hth_cb_max	11:0	rw	'h43c	skin threshold cb max

ISP_EE2_SCALER_CTRL

0x00007d1c = Byte Address

Description: ee2 SCALER CTRL

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_scaler_narrow	13:8	rw	'h4	scaler narrow
ee2_scaler_detail	5:0	rw	'h8	scaler detail

ISP_EE2_MERGE_CURVE_01

0x00007d20 = Byte Address

Description: ee2 MERGE CURVE 01

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_edge_str_t0	26:16	rw	'h8	edge str t0
ee2_edge_str_t1_shift	3:0	rw	'h5	edge str t1 shift

ISP_EE2_MERGE_CURVE_23

0x00007d24 = Byte Address

Description: ee2 MERGE CURVE 23

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_edge_str_t2	26:16	rw	'h80	edge str t2

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ee2_edge_str_t3_shift	3:0	rw	'h8	edge str t3 shift

ISP_EE2_MERGE_CURVE_45

0x00007d28 = Byte Address

Description: ee2 MERGE CURVE 45

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_edge_skin_str_t0	26:16	rw	'h20	edge skin str t0
ee2_edge_skin_str_t1_shift	3:0	rw	'h5	edge skin str t1 shift

ISP_EE2_MERGE_CURVE_67

0x00007d2c = Byte Address

Description: ee2 MERGE CURVE 67

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_edge_skin_str_t2	26:16	rw	'h80	edge skin str t2
ee2_edge_skin_str_t3_shift	3:0	rw	'h8	edge skin str t3 shift

ISP_EE2_NOISE_CUTOFF

0x00007d30 = Byte Address

Description: ee2 NOISE CUTOFF

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_grad_th_edge	26:16	rw	'h8	grad th edge
ee2_grad_th_detail	10:0	rw	'h10	grad th detail

ISP_EE2_SKIN_NOISE_CUTOFF

0x00007d34 = Byte Address

Description: ee2 SKIN NOISE CUTOFF

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_grad_th_edge_skin	26:16	rw	'h18	grad th edge skin
ee2_grad_th_detail_skin	10:0	rw	'h20	grad th detail skin

ISP_EE2_CLARITY_STRENGTH

0x00007d38 = Byte Address

Description: ee2 CLARITY STRENGTH

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_source_strength	24:16	rw	'h100	source strength
ee2_source_strength_skin	8:0	rw	'hc8	source strength skin

ISP_EE2_SHARPEN_CURVE

0x00007d3c = Byte Address

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Description: ee2 SHARPEN CURVE

Sub-Field Name	Bits	rw	Reset	Description
ee2_sharp_curve_lvl	10:8	rw	'h3	sharp curve lvl
ee2_skin_sharp_lvl	2:0	rw	'h2	skin sharp lvl

ISP_EE2_SKIN_Y_GAIN

0x00007d40 = Byte Address

Description: ee2 SKIN Y GAIN

Sub-Field Name	Bits	rw	Reset	Description
ee2_up_gain_skin	26:16	rw	'h50	up gain skin
ee2_down_gain_skin	10:0	rw	'h58	down gain skin

ISP_EE2_Y_GAIN

0x00007d44 = Byte Address

Description: ee2 Y GAIN

Sub-Field Name	Bits	rw	Reset	Description
ee2_up_gain_other	26:16	rw	'h64	up gain other
ee2_down_gain_other	10:0	rw	'h6a	down gain other

ISP_EE2_LIMIT_CTRL

0x00007d48 = Byte Address

Description: ee2 LIMIT CTRL

Sub-Field Name	Bits	rw	Reset	Description
ee2_max_change_type	2:1	rw	'h2	max change type
ee2_max_change_en	0:0	rw	'h1	max change en

ISP_EE2_Y_MAX_CHANGE

0x00007d4c = Byte Address

Description: ee2 Y MAX CHANGE

Sub-Field Name	Bits	rw	Reset	Description
ee2_max_change_up	26:16	rw	'h34	max change up
ee2_max_change_down	10:0	rw	'h3c	max change down

ISP_EE2_SKIN_Y_MAX_CHANGE

0x00007d50 = Byte Address

Description: ee2 SKIN Y MAX CHANGE

Sub-Field Name	Bits	rw	Reset	Description
ee2_max_change_up_skin	26:16	rw	'h24	max change up skin

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ee2_max_change_down_skin	10:0	rw	'h27	max change down skin

ISP_EE2_DEPURPLE_U_LIMIT

0x00007d54 = Byte Address

Description: ee2 DEPURPLE U LIMIT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_depf_limit_down_u	26:16	rw	'h230	depf limit down u
ee2_depf_limit_up_u	10:0	rw	'h386	depf limit up u

ISP_EE2_DEPURPLE_V_LIMIT

0x00007d58 = Byte Address

Description: ee2 DEPURPLE V LIMIT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_depf_limit_down_v	26:16	rw	'h1f4	depf limit down v
ee2_depf_limit_up_v	10:0	rw	'h384	depf limit up v

ISP_EE2_DEPURPLE_Y_CTRL

0x00007d5c = Byte Address

Description: ee2 DEPURPLE Y CTRL

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_depf_detect_luma_th	26:16	rw	'h352	depf detect luma th
ee2_depf_detect_luma_diff	10:0	rw	'h2d0	depf detect luma diff

ISP_EE2_DEPURPLE_COLOR_CTRL

0x00007d60 = Byte Address

Description: ee2 DEPURPLE COLOR CTRL

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_depf_comp_luma_diff	30:20	rw	'h7a	depf comp luma diff
ee2_depf_fix_str	18:10	rw	'ha0	depf fix str
ee2_defp_sat_str	8:0	rw	'h100	defp sat str

ISP_EE2_CTRL_SHD

0x00007d80 = Byte Address

Description: ee2 CTRL shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_yuv_en_shd	16:16	r	'h0	yuv en shadow
ee2_skin_proc_en_shd	14:12	r	'h0	skin proc en shadow
ee2_depf_detect_range_shd	11:9	r	'h7	depf detect range shadow

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ee2_depurple_en_shd	8:8	r	'h0	depurple en shadow
ee2_en_shd	0:0	r	'h1	en shadow

ISP_EE2_STRENGTH_SHD

0x00007d84 = Byte Address

Description: ee2 STRENGTH shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_uv_gain_shd	23:16	r	'h0	uv gain shadow

ISP_EE2_NR_CTRL_SHD

0x00007d88 = Byte Address

Description: ee2 NR CTRL shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_denoise_level_shd	26:24	r	'h3	denoise level shadow
ee2_use_auxi_dir_shd	16:16	r	'h1	use auxi dir shadow
ee2_detail_lvl_shd	14:12	r	'h5	detail lvl shadow
ee2_detail_enhance_str_shd	9:0	r	'h16	detail enhance str shadow

ISP_EE2_SKIN_MODULE_STRENGTH_SHD

0x00007d8c = Byte Address

Description: ee2 SKIN MODULE STRENGTH shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_skin_detect_str_shd	8:0	r	'hd3	skin detect str shadow

ISP_EE2_SKIN_Y_LIMIT_SHD

0x00007d90 = Byte Address

Description: ee2 SKIN Y LIMIT shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_skin_hth_y_min_shd	26:16	r	'h118	skin threshold y min shadow
ee2_skin_hth_y_max_shd	10:0	r	'h384	skin threshold y max shadow

ISP_EE2_SKIN_CR_LIMIT_SHD

0x00007d94 = Byte Address

Description: ee2 SKIN CR LIMIT shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_skin_hth_cr_min_shd	27:16	r	'h3d8	skin threshold cr min shadow
ee2_skin_hth_cr_max_shd	11:0	r	'h568	skin threshold cr max shadow

ISP_EE2_SKIN_CB_LIMIT_SHD

0x00007d98 = Byte Address

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Description: ee2 SKIN CB LIMIT shadow

Sub-Field Name	Bits	rw	Reset	Description
ee2_skin_hth_cb_min_shd	27:16	r	'h2fc	skin threshold cb min shadow
ee2_skin_hth_cb_max_shd	11:0	r	'h43c	skin threshold cb max shadow

ISP_EE2_SCALER_CTRL_SHD

0x00007d9c = Byte Address

Description: ee2 SCALER CTRL shadow

Sub-Field Name	Bits	rw	Reset	Description
ee2_scaler_narrow_shd	13:8	r	'h4	scaler narrow shadow
ee2_scaler_detail_shd	5:0	r	'h8	scaler detail shadow

ISP_EE2_MERGE_CURVE_01_SHD

0x00007da0 = Byte Address

Description: ee2 MERGE CURVE 01 shadow

Sub-Field Name	Bits	rw	Reset	Description
ee2_edge_str_t0_shd	26:16	r	'h8	edge str t0 shadow
ee2_edge_str_t1_shift_shd	3:0	r	'h5	edge str t1 shift shadow

ISP_EE2_MERGE_CURVE_23_SHD

0x00007da4 = Byte Address

Description: ee2 MERGE CURVE 23 shadow

Sub-Field Name	Bits	rw	Reset	Description
ee2_edge_str_t2_shd	26:16	r	'h80	edge str t2 shadow
ee2_edge_str_t3_shift_shd	3:0	r	'h8	edge str t3 shift shadow

ISP_EE2_MERGE_CURVE_45_SHD

0x00007da8 = Byte Address

Description: ee2 MERGE CURVE 45 shadow

Sub-Field Name	Bits	rw	Reset	Description
ee2_edge_skin_str_t0_shd	26:16	r	'h20	edge skin str t0 shadow
ee2_edge_skin_str_t1_shift_shd	3:0	r	'h5	edge skin str t1 shift shadow

ISP_EE2_MERGE_CURVE_67_SHD

0x00007dac = Byte Address

Description: ee2 MERGE CURVE 67 shadow

Sub-Field Name	Bits	rw	Reset	Description

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ee2_edge_skin_str_t2_shd	26:16	r	'h80	edge skin str t2 shadow
ee2_edge_skin_str_t3_shift_shd	3:0	r	'h8	edge skin str t3 shift shadow

ISP_EE2_NOISE_CUTOFF_SHD

0x00007db0 = Byte Address

Description: ee2 NOISE CUTOFF shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_grad_th_edge_shd	26:16	r	'h8	grad th edge shadow
ee2_grad_th_detail_shd	10:0	r	'h10	grad th detail shadow

ISP_EE2_SKIN_NOISE_CUTOFF_SHD

0x00007db4 = Byte Address

Description: ee2 SKIN NOISE CUTOFF shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_grad_th_edge_skin_shd	26:16	r	'h18	grad th edge skin shadow
ee2_grad_th_detail_skin_shd	10:0	r	'h20	grad th detail skin shadow

ISP_EE2_CLARITY_STRENGTH_SHD

0x00007db8 = Byte Address

Description: ee2 CLARITY STRENGTH shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_source_strength_shd	24:16	r	'h100	source strength shadow
ee2_source_strength_skin_shd	8:0	r	'hc8	source strength skin shadow

ISP_EE2_SHARPEN_CURVE_SHD

0x00007dbc = Byte Address

Description: ee2 SHARPEN CURVE shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_sharp_curve_lvl_shd	10:8	r	'h3	sharp curve lvl shadow
ee2_skin_sharp_lvl_shd	2:0	r	'h2	skin sharp lvl shadow

ISP_EE2_SKIN_Y_GAIN_SHD

0x00007dc0 = Byte Address

Description: ee2 SKIN Y GAIN shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_up_gain_skin_shd	26:16	r	'h50	up gain skin shadow

ee2_down_gain_skin_shd	10:0	r	'h58	down gain skin shadow

ISP_EE2_Y_GAIN_SHD

0x00007dc4 = Byte Address

Description: ee2 Y GAIN shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_up_gain_other_shd	26:16	r	'h64	up gain other shadow
ee2_down_gain_other_shd	10:0	r	'h6a	down gain other shadow

ISP_EE2_LIMIT_CTRL_SHD

0x00007dc8 = Byte Address

Description: ee2 LIMIT CTRL shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_max_change_type_shd	2:1	r	'h2	max change type shadow
ee2_max_change_en_shd	0:0	r	'h1	max change en shadow

ISP_EE2_Y_MAX_CHANGE_SHD

0x00007dcc = Byte Address

Description: ee2 Y MAX CHANGE shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_max_change_up_shd	26:16	r	'h34	max change up shadow
ee2_max_change_down_shd	10:0	r	'h3c	max change down shadow

ISP_EE2_SKIN_Y_MAX_CHANGE_SHD

0x00007dd0 = Byte Address

Description: ee2 SKIN Y MAX CHANGE shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_max_change_up_skin_shd	26:16	r	'h24	max change up skin shadow
ee2_max_change_down_skin_shd	10:0	r	'h27	max change down skin shadow

ISP_EE2_DEPURPLE_U_LIMIT_SHD

0x00007dd4 = Byte Address

Description: ee2 DEPURPLE U LIMIT shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
ee2_depf_limit_down_u_shd	26:16	r	'h230	depf limit down u shadow
ee2_depf_limit_up_u_shd	10:0	r	'h386	depf limit up u shadow

ISP_EE2_DEPURPLE_V_LIMIT_SHD

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0x00007dd8 = Byte Address

Description: ee2 DEPURPLE V LIMIT shadow

Sub-Field Name	Bits	rw	Reset	Description
ee2_depf_limit_down_v_shd	26:16	r	'h1f4	depf limit down v shadow
ee2_depf_limit_up_v_shd	10:0	r	'h384	depf limit up v shadow

ISP_EE2_DEPURPLE_Y_CTRL_SHD

0x00007ddc = Byte Address

Description: ee2 DEPURPLE Y CTRL shadow

Sub-Field Name	Bits	rw	Reset	Description
ee2_depf_detect_luma_th_shd	26:16	r	'h352	depf detect luma th shadow
ee2_depf_detect_luma_diff_shd	10:0	r	'h2d0	depf detect luma diff shadow

ISP_EE2_DEPURPLE_COLOR_CTRL_SHD

0x00007de0 = Byte Address

Description: ee2 DEPURPLE COLOR CTRL shadow

Sub-Field Name	Bits	rw	Reset	Description
ee2_depf_comp_luma_diff_shd	30:20	r	'h7a	depf comp luma diff shadow
ee2_depf_fix_str_shd	18:10	r	'ha0	depf fix str shadow
ee2_defp_sat_str_shd	8:0	r	'h100	defp sat str shadow

ISP_EE2_DUMMY_HBLANK

0x00007de4 = Byte Address

Description: ee2 dummy hblank cycle register

Sub-Field Name	Bits	rw	Reset	Description
ee2_dummy_hblank	15:0	rw	'h10	dummy line hblank cycle

ISP_DMSC3_CAC_CTRL

0x00007e00 = Byte Address

Description: demosaic3 CAC CTRL

Sub-Field Name	Bits	rw	Reset	Description
cac_enable	0:0	rw	'h0	cac enable

ISP_DMSC3_CAC_COUNT_START

0x00007e04 = Byte Address

Description: demosaic3 CAC COUNT START

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Sub-Field Name	Bits	rw	Reset	Description
v_count_start	31:16	rw	'h1000	v count start
h_count_start	15:0	rw	'h1000	h count start

ISP_DMSC3_CAC_A

0x00007e08 = Byte Address

Description: demosaic3 CAC A

Sub-Field Name	Bits	rw	Reset	Description
a_blue	31:16	rw	'h0	a blue
a_red	15:0	rw	'h0	a red

ISP_DMSC3_CAC_B

0x00007e0c = Byte Address

Description: demosaic3 CAC B

Sub-Field Name	Bits	rw	Reset	Description
b_blue	31:16	rw	'h0	b blue
b_red	15:0	rw	'h0	b red

ISP_DMSC3_CAC_C

0x00007e10 = Byte Address

Description: demosaic3 CAC C

Sub-Field Name	Bits	rw	Reset	Description
c_blue	31:16	rw	'h0	c blue
c_red	15:0	rw	'h0	c red

ISP_DMSC3_CAC_X_NORM

0x00007e14 = Byte Address

Description: demosaic3 CAC X NORM

Sub-Field Name	Bits	rw	Reset	Description
x_ns	19:16	rw	'h8	x ns
x_nf	4:0	rw	'h10	x nf

ISP_DMSC3_CAC_Y_NORM

0x00007e18 = Byte Address

Description: demosaic3 CAC Y NORM

Sub-Field Name	Bits	rw	Reset	Description
y_ns	19:16	rw	'h8	y ns
y_nf	4:0	rw	'h10	y nf

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ISP_DMSC3_CTRL

0x00007e20 = Byte Address

Description: demosaic3 CTRL

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_denoise_strength	9:4	rw	'h0	denoise strength
dmsc3_sharpen_line_enable	3:3	rw	'h0	sharpen line enable
dmsc3_sharpen_enable	2:2	rw	'h1	sharpen enable
dmsc3_depurple_en	1:1	rw	'h0	depurple en
dmsc3_enable	0:0	rw	'h1	enable

ISP_DMSC3_INTERP_CURVE

0x00007e24 = Byte Address

Description: demosaic3 INTERP CURVE

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_grad_th_max	27:16	rw	'h14	grad th max
dmsc3_grad_th_min	11:0	rw	'h4	grad th min

ISP_DMSC3_CHANNEL_COEF

0x00007e28 = Byte Address

Description: demosaic3 CHANNEL COEF

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_adpt_coef_max	23:16	rw	'h80	adpt coef max
dmsc3_adpt_coef_min	15:8	rw	'h0	adpt coef min
dmsc3_diffcolor_coef	7:0	rw	'h80	diffColor coef

ISP_DMSC3_ADPT_COEF_CTRL

0x00007e2c = Byte Address

Description: demosaic3 ADPT COEF CTRL

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_adpt_coef_th_shift	19:16	rw	'h6	adpt coef th shift
dmsc3_adpt_coef_th_low	15:8	rw	'h18	adpt coef th low
dmsc3_adpt_coef_mode	1:0	rw	'h3	adpt coef mode

ISP_DMSC3_INTERP_G_CTRL0

0x00007e30 = Byte Address

Description: demosaic3 INTERP G CTRL0

Sub-Field Name	Bits	rw	Reset	Description

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dmsc3_interp_dir_strength_level3	31:24	rw	'h10	interp dir strength level3
dmsc3_interp_dir_strength_level2	23:16	rw	'h20	interp dir strength level2
dmsc3_interp_dir_strength_level1	15:8	rw	'h40	interp dir strength level1
dmsc3_interp_dir_strength_level0	7:0	rw	'h20	interp dir strength level0

ISP_DMSC3_INTERP_G_CTRL1

0x00007e34 = Byte Address

Description: demosaic3 INTERP G CTRL1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_interp_corner_en	24:24	rw	'h1	interp corner en
dmsc3_dir0_interp_type	17:16	rw	'h0	dir0 interp type
dmsc3_interp_large_str_3	12:8	rw	'h4	interp large str 3
dmsc3_interp_large_str_2	4:0	rw	'hc	interp large str 2

ISP_DMSC3_INTERP_G_CTRL2

0x00007e38 = Byte Address

Description: demosaic3 INTERP G CTRL2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_corner_shift	19:16	rw	'h6	corner shift
dmsc3_corner_low_th	11:0	rw	'h400	corner low th

ISP_DMSC3_INTERP_RB_CTRL0

0x00007e3c = Byte Address

Description: demosaic3 INTERP RB CTRL0

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_highlight_control_th	27:16	rw	'h400	highlight control th
dmsc3_deburst_mode	14:12	rw	'h3	deburst mode
dmsc3_highlight_control_str	11:8	rw	'h6	highlight control shift
dmsc3_decline_en	4:4	rw	'h1	decline en

ISP_DMSC3_INTERP_WEIGHT

0x00007e44 = Byte Address

Description: demosaic3 INTERP WEIGHT

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_pure_rb_weight_iso	28:24	rw	'h0	pure RB weight ISO

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dmsc3_pure_g_weight_iso	20:16	rw	'h0	pure G weight ISO
dmsc3_pure_rb_weight	12:8	rw	'h0	pure RB weight
dmsc3_pure_g_weight	4:0	rw	'h0	pure G weight

ISP_DMSC3_REFINEG_G

0x00007e48 = Byte Address

Description: demosaic3 REFINEG G

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_refine_g_th	27:16	rw	'h64	refine g th
dmsc3_demoire_refineg_en	0:0	rw	'h1	demoire refineG en

ISP_DMSC3_DE_FALSE_COLOR

0x00007e4c = Byte Address

Description: demosaic3 DE FALSE COLOR

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_gray_protect_str	23:16	rw	'h0	gray protect str
dmsc3_defalsecolor_str	15:8	rw	'h80	DeFalseColor str
dmsc3_cbcr_class_th	6:4	rw	'h5	cbcr class th
dmsc3_defalsecolor_en	0:0	rw	'h1	DeFalseColor en

ISP_DMSC3_DEPURPLE

0x00007e50 = Byte Address

Description: demosaic3 DEPURPLE

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_depurple_color_range_mode	30:28	rw	'h3	depurple color range mode
dmsc3_depurple_sat_shrink	27:24	rw	'h8	depurple sat shrink
dmsc3_depurple_protect_shift	23:20	rw	'h5	depurple protect shift
dmsc3_depurple_protect_th_low	19:12	rw	'h10	depurple protect th low
dmsc3_depurple_thr	11:4	rw	'h28	depurple thr
dmsc3_depurple_cbcr_mode	2:1	rw	'h3	depurple cbcr mode
dmsc3_depurple_grad_control	0:0	rw	'h1	depurple grad control

ISP_DMSC3_SHARP_FAC_LAYER0

0x00007e54 = Byte Address

Description: demosaic3 SHARP FAC LAYER0

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Sub-Field Name	Bits	rw	Reset	Description
dmsc3_sharpen_layer0_factor_black	20:12	rw	'h64	sharpen layer0 factor black
dmsc3_sharpen_layer0_factor_white	8:0	rw	'h64	sharpen layer0 factor white

ISP_DMSC3_SHARP_CLIP_LAYER0

0x00007e58 = Byte Address

Description: demosaic3 SHARP CLIP LAYER0

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_sharpen_layer0_clip_black	22:12	rw	'h1e	sharpen layer0 clip black
dmsc3_sharpen_layer0_clip_white	10:0	rw	'h14	sharpen layer0 clip white

ISP_DMSC3_SHARP_CURVE0_X

0x00007e5c = Byte Address

Description: demosaic3 SHARP CURVE0 X

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_sharpen_layer0_t4_shift	31:28	rw	'h6	sharpen layer0 t4 shift
dmsc3_sharpen_layer0_t3	26:16	rw	'h7a	sharpen layer0 t3
dmsc3_sharpen_layer0_t2_shift	15:12	rw	'h6	sharpen layer0 t2 shift
dmsc3_sharpen_layer0_t1	10:0	rw	'h20	sharpen layer0 t1

ISP_DMSC3_SHARP_CURVE0_Y

0x00007e60 = Byte Address

Description: demosaic3 SHARP CURVE0 Y

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_sharpen_layer0_r3	26:18	rw	'h7e	sharpen layer0 r3
dmsc3_sharpen_layer0_r2	17:9	rw	'h64	sharpen layer0 r2
dmsc3_sharpen_layer0_r1	8:0	rw	'h0	sharpen layer0 r1

ISP_DMSC3_SHARP_FAC_LAYER1

0x00007e64 = Byte Address

Description: demosaic3 SHARP FAC LAYER1

Sub-Field Name	Bits	rw	Reset	Description

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dmsc3_sharpen_layer1_factor_black	20:12	rw	'hc8	sharpen layer1 factor black
dmsc3_sharpen_layer1_factor_white	8:0	rw	'hc8	sharpen layer1 factor white

ISP_DMSC3_SHARP_CLIP_LAYER1

0x00007e68 = Byte Address

Description: demosaic3 SHARP CLIP LAYER1

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer1_clip_black	22:12	rw	'h1e	sharpen layer1 clip black
dmsc3_sharpen_layer1_clip_white	10:0	rw	'h1e	sharpen layer1 clip white

ISP_DMSC3_SHARP_CURVE1_X

0x00007e6c = Byte Address

Description: demosaic3 SHARP CURVE1 X

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer1_t4_shift	31:28	rw	'h5	sharpen layer1 t4 shift
dmsc3_sharpen_layer1_t3	26:16	rw	'h40	sharpen layer1 t3
dmsc3_sharpen_layer1_t2_shift	15:12	rw	'h5	sharpen layer1 t2 shift
dmsc3_sharpen_layer1_t1	10:0	rw	'h8	sharpen layer1 t1

ISP_DMSC3_SHARP_CURVE1_Y

0x00007e70 = Byte Address

Description: demosaic3 SHARP CURVE1 Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer1_r3	26:18	rw	'ha8	sharpen layer1 r3
dmsc3_sharpen_layer1_r2	17:9	rw	'h7e	sharpen layer1 r2
dmsc3_sharpen_layer1_r1	8:0	rw	'h0	sharpen layer1 r1

ISP_DMSC3_SHARP_FAC_LAYER2

0x00007e74 = Byte Address

Description: demosaic3 SHARP FAC LAYER2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer2_factor_black	20:12	rw	'h12c	sharpen layer2 factor black

dmsc3_sharpen_layer2_factor_white	8:0	rw	'h12c	sharpen layer2 factor white

ISP_DMSC3_SHARP_CLIP_LAYER2

0x00007e78 = Byte Address

Description: demosaic3 SHARP CLIP LAYER2

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer2_clip_black	22:12	rw	'h28	sharpen layer2 clip black
dmsc3_sharpen_layer2_clip_white	10:0	rw	'h1e	sharpen layer2 clip white

ISP_DMSC3_SHARP_CURVE2_X

0x00007e7c = Byte Address

Description: demosaic3 SHARP CURVE2 X

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer2_t4_shift	31:28	rw	'h6	sharpen layer2 t4 shift
dmsc3_sharpen_layer2_t3	26:16	rw	'h48	sharpen layer2 t3
dmsc3_sharpen_layer2_t2_shift	15:12	rw	'h6	sharpen layer2 t2 shift
dmsc3_sharpen_layer2_t1	10:0	rw	'h0	sharpen layer2 t1

ISP_DMSC3_SHARP_CURVE2_Y

0x00007e80 = Byte Address

Description: demosaic3 SHARP CURVE2 Y

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer2_r3	26:18	rw	'hc8	sharpen layer2 r3
dmsc3_sharpen_layer2_r2	17:9	rw	'h98	sharpen layer2 r2
dmsc3_sharpen_layer2_r1	8:0	rw	'h0	sharpen layer2 r1

ISP_DMSC3_SHARP_LINE_X

0x00007e84 = Byte Address

Description: demosaic3 SHARP LINE X

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_line_shift1	31:28	rw	'h3	sharpen line shift1
dmsc3_sharpen_line_t1	26:16	rw	'h18	sharpen line t1
dmsc3_sharpen_line_strengt h	11:0	rw	'h8	sharpen line strength

ISP_DMSC3_SHARP_LINE_Y

0x00007e88 = Byte Address

Description: demosaic3 SHARP LINE Y

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_sharpen_line_r2	24:16	rw	'h100	sharpen line r2
dmsc3_sharpen_line_r1	8:0	rw	'h0	sharpen line r1

ISP_DMSC3_DEPURPLE_CTRL1

0x00007e8c = Byte Address

Description: demosaic3 depurple color range control 1

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_depurple_cr_highth_1	31:24	rw	'hc8	depurple cr high threshold 1
dmsc3_depurple_cr_lowth_1	23:16	rw	'h70	depurple cr low threshold 1
dmsc3_depurple_cb_highth_1	15:8	rw	'hdc	depurple cb high threshold 1
dmsc3_depurple_cb_lowth_1	7:0	rw	'h84	depurple cb low threshold 1

ISP_DMSC3_DEPURPLE_CTRL2

0x00007e90 = Byte Address

Description: demosaic3 depurple color range control 2

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_depurple_cr_highth_2	31:24	rw	'h76	depurple cr high threshold 2
dmsc3_depurple_cr_lowth_2	23:16	rw	'h52	depurple cr low threshold 2
dmsc3_depurple_cb_highth_2	15:8	rw	'h7a	depurple cb high threshold 2
dmsc3_depurple_cb_lowth_2	7:0	rw	'h60	depurple cb low threshold 2

ISP_DMSC3_CAC_CTRL_SHD

0x00007f00 = Byte Address

Description: demosaic3 CAC CTRL shadow

Sub-Field Name	Bits	rw	Reset	Description
cac_enable	0:0	r	'h0	cac enable

ISP_DMSC3_CAC_COUNT_START_SHD

0x00007f04 = Byte Address

Description: demosaic3 CAC COUNT START shadow

Sub-Field Name	Bits	rw	Reset	Description
v_count_start	31:16	r	'h1000	v count start
h_count_start	15:0	r	'h1000	h count start

ISP_DMSC3_CAC_A_SHD

0x00007f08 = Byte Address

Description: demosaic3 CAC A shadow

Sub-Field Name	Bits	rw	Reset	Description
a_blue	31:16	r	'h0	a blue
a_red	15:0	r	'h0	a red

ISP_DMSC3_CAC_B_SHD

0x00007f0c = Byte Address

Description: demosaic3 CAC B shadow

Sub-Field Name	Bits	rw	Reset	Description
b_blue	31:16	r	'h0	b blue
b_red	15:0	r	'h0	b red

ISP_DMSC3_CAC_C_SHD

0x00007f10 = Byte Address

Description: demosaic3 CAC C shadow

Sub-Field Name	Bits	rw	Reset	Description
c_blue	31:16	r	'h0	c blue
c_red	15:0	r	'h0	c red

ISP_DMSC3_CAC_X_NORM_SHD

0x00007f14 = Byte Address

Description: demosaic3 CAC X NORM shadow

Sub-Field Name	Bits	rw	Reset	Description
x_ns	19:16	r	'h8	x ns
x_nf	4:0	r	'h10	x nf

ISP_DMSC3_CAC_Y_NORM_SHD

0x00007f18 = Byte Address

Description: demosaic3 CAC Y NORM shadow

Sub-Field Name	Bits	rw	Reset	Description
y_ns	19:16	r	'h8	y ns
y_nf	4:0	r	'h10	y nf

ISP_DMSC3_CTRL_SHD

0x00007f20 = Byte Address

Description: demosaic3 CTRL shadow

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_denoise_strength	9:4	r	'h0	denoise strength
dmsc3_sharpen_line_enable	3:3	r	'h0	sharpen line enable
dmsc3_sharpen_enable	2:2	r	'h1	sharpen enable
dmsc3_depurple_en	1:1	r	'h0	depurple en
dmsc3_enable	0:0	r	'h1	enable

ISP_DMSC3_INTERP_CURVE_SHD

0x00007f24 = Byte Address

Description: demosaic3 INTERP CURVE shadow

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_grad_th_max	27:16	r	'h14	grad th max
dmsc3_grad_th_min	11:0	r	'h4	grad th min

ISP_DMSC3_CHANNEL_COEF_SHD

0x00007f28 = Byte Address

Description: demosaic3 CHANNEL COEF shadow

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_adpt_coef_max	23:16	r	'h80	adpt coef max
dmsc3_adpt_coef_min	15:8	r	'h0	adpt coef min
dmsc3_diffcolor_coef	7:0	r	'h80	diffColor coef

ISP_DMSC3_ADPT_COEF_CTRL_SHD

0x00007f2c = Byte Address

Description: demosaic3 ADPT COEF CTRL shadow

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_adpt_coef_th_shift	19:16	r	'h6	adpt coef th shift
dmsc3_adpt_coef_th_low	15:8	r	'h18	adpt coef th low
dmsc3_adpt_coef_mode	1:0	r	'h3	adpt coef mode

ISP_DMSC3_INTERP_G_CTRL0_SHD

0x00007f30 = Byte Address

Description: demosaic3 INTERP G CTRL0 shadow

Sub-Field Name	Bits	rw	Reset	Description

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dmsc3_interp_dir_strength_level3	31:24	r	'h10	interp dir strength level3
dmsc3_interp_dir_strength_level2	23:16	r	'h20	interp dir strength level2
dmsc3_interp_dir_strength_level1	15:8	r	'h40	interp dir strength level1
dmsc3_interp_dir_strength_level0	7:0	r	'h20	interp dir strength level0

ISP_DMSC3_INTERP_G_CTRL1_SHD

0x00007f34 = Byte Address

Description: demosaic3 INTERP G CTRL1 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_interp_corner_en	24:24	r	'h1	interp corner en
dmsc3_dir0_interp_type	17:16	r	'h0	dir0 interp type
dmsc3_interp_large_str_3	12:8	r	'h4	interp large str 3
dmsc3_interp_large_str_2	4:0	r	'hc	interp large str 2

ISP_DMSC3_INTERP_G_CTRL2_SHD

0x00007f38 = Byte Address

Description: demosaic3 INTERP G CTRL2 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_corner_shift	19:16	r	'h6	corner shift
dmsc3_corner_low_th	11:0	r	'h400	corner low th

ISP_DMSC3_INTERP_RB_CTRL0_SHD

0x00007f3c = Byte Address

Description: demosaic3 INTERP RB CTRL0 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_highlight_control_th	27:16	r	'h400	highlight control th
dmsc3_deburst_mode	14:12	r	'h3	deburst mode
dmsc3_highlight_control_shift	11:8	r	'h6	highlight control str
dmsc3_decline_en	4:4	r	'h1	decline en

ISP_DMSC3_INTERP_WEIGHT_SHD

0x00007f44 = Byte Address

Description: demosaic3 INTERP WEIGHT shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*

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dmsc3_pure_rb_weight_iso	28:24	r	'h0	pure RB weight ISO
dmsc3_pure_g_weight_iso	20:16	r	'h0	pure G weight ISO
dmsc3_pure_rb_weight	12:8	r	'h0	pure RB weight
dmsc3_pure_g_weight	4:0	r	'h0	pure G weight

ISP_DMSC3_REFINEG_G_SHD

0x00007f48 = Byte Address

Description: demosaic3 REFINEG G shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_refine_g_th	27:16	r	'h64	refine g th
dmsc3_demoire_refineg_en	0:0	r	'h1	demoire refineG en

ISP_DMSC3_DE_FALSE_COLOR_SHD

0x00007f4c = Byte Address

Description: demosaic3 DE FALSE COLOR shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_gray_protect_str	23:16	r	'h0	gray protect str
dmsc3_defalsecolor_str	15:8	r	'h80	DeFalseColor str
dmsc3_cbcr_class_th	6:4	r	'h5	cbcr class th
dmsc3_defalsecolor_en	0:0	r	'h1	DeFalseColor en

ISP_DMSC3_DEPURPLE_SHD

0x00007f50 = Byte Address

Description: demosaic3 DEPURPLE shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_depurple_color_range_mode	30:28	r	'h3	depurple color range mode
dmsc3_depurple_sat_shrink	27:24	r	'h8	depurple sat shrink
dmsc3_depurple_protect_shift	23:20	r	'h5	depurple protect shift
dmsc3_depurple_protect_th_low	19:12	r	'h10	depurple protect th low
dmsc3_depurple_thr	11:4	r	'h28	depurple thr
dmsc3_depurple_cbcr_mode	2:1	r	'h3	depurple cbcr mode
dmsc3_depurple_grad_control	0:0	r	'h1	depurple grad control

ISP_DMSC3_SHARP_FAC_LAYER0_SHD

0x00007f54 = Byte Address

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Description: demosaic3 SHARP FAC LAYER0 shadow

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_sharpen_layer0_factor_black	20:12	r	'h64	sharpen layer0 factor black
dmsc3_sharpen_layer0_factor_white	8:0	r	'h64	sharpen layer0 factor white

ISP_DMSC3_SHARP_CLIP_LAYER0_SHD

0x00007f58 = Byte Address

Description: demosaic3 SHARP CLIP LAYER0 shadow

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_sharpen_layer0_clip_black	22:12	r	'h1e	sharpen layer0 clip black
dmsc3_sharpen_layer0_clip_white	10:0	r	'h14	sharpen layer0 clip white

ISP_DMSC3_SHARP_CURVE0_X_SHD

0x00007f5c = Byte Address

Description: demosaic3 SHARP CURVE0 X shadow

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_sharpen_layer0_t4_shift	31:28	r	'h6	sharpen layer0 t4 shift
dmsc3_sharpen_layer0_t3	26:16	r	'h7a	sharpen layer0 t3
dmsc3_sharpen_layer0_t2_shift	15:12	r	'h6	sharpen layer0 t2 shift
dmsc3_sharpen_layer0_t1	10:0	r	'h20	sharpen layer0 t1

ISP_DMSC3_SHARP_CURVE0_Y_SHD

0x00007f60 = Byte Address

Description: demosaic3 SHARP CURVE0 Y shadow

Sub-Field Name	Bits	rw	Reset	Description
dmsc3_sharpen_layer0_r3	26:18	r	'h7e	sharpen layer0 r3
dmsc3_sharpen_layer0_r2	17:9	r	'h64	sharpen layer0 r2
dmsc3_sharpen_layer0_r1	8:0	r	'h0	sharpen layer0 r1

ISP_DMSC3_SHARP_FAC_LAYER1_SHD

0x00007f64 = Byte Address

Description: demosaic3 SHARP FAC LAYER1 shadow

Sub-Field Name	Bits	rw	Reset	Description

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dmsc3_sharpen_layer1_factor_black	20:12	r	'hc8	sharpen layer1 factor black
dmsc3_sharpen_layer1_factor_white	8:0	r	'hc8	sharpen layer1 factor white

ISP_DMSC3_SHARP_CLIP_LAYER1_SHD

0x00007f68 = Byte Address

Description: demosaic3 SHARP CLIP LAYER1 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer1_clip_black	22:12	r	'h1e	sharpen layer1 clip black
dmsc3_sharpen_layer1_clip_white	10:0	r	'h1e	sharpen layer1 clip white

ISP_DMSC3_SHARP_CURVE1_X_SHD

0x00007f6c = Byte Address

Description: demosaic3 SHARP CURVE1 X shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer1_t4_shift	31:28	r	'h5	sharpen layer1 t4 shift
dmsc3_sharpen_layer1_t3	26:16	r	'h40	sharpen layer1 t3
dmsc3_sharpen_layer1_t2_shift	15:12	r	'h5	sharpen layer1 t2 shift
dmsc3_sharpen_layer1_t1	10:0	r	'h8	sharpen layer1 t1

ISP_DMSC3_SHARP_CURVE1_Y_SHD

0x00007f70 = Byte Address

Description: demosaic3 SHARP CURVE1 Y shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer1_r3	26:18	r	'ha8	sharpen layer1 r3
dmsc3_sharpen_layer1_r2	17:9	r	'h7e	sharpen layer1 r2
dmsc3_sharpen_layer1_r1	8:0	r	'h0	sharpen layer1 r1

ISP_DMSC3_SHARP_FAC_LAYER2_SHD

0x00007f74 = Byte Address

Description: demosaic3 SHARP FAC LAYER2 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer2_factor_black	20:12	r	'h12c	sharpen layer2 factor black

dmsc3_sharpen_layer2_factor_white	8:0	r	'h12c	sharpen layer2 factor white

ISP_DMSC3_SHARP_CLIP_LAYER2_SHD

0x00007f78 = Byte Address

Description: demosaic3 SHARP CLIP LAYER2 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer2_clip_black	22:12	r	'h28	sharpen layer2 clip black
dmsc3_sharpen_layer2_clip_white	10:0	r	'h1e	sharpen layer2 clip white

ISP_DMSC3_SHARP_CURVE2_X_SHD

0x00007f7c = Byte Address

Description: demosaic3 SHARP CURVE2 X shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer2_t4_shift	31:28	r	'h6	sharpen layer2 t4 shift
dmsc3_sharpen_layer2_t3	26:16	r	'h48	sharpen layer2 t3
dmsc3_sharpen_layer2_t2_shift	15:12	r	'h6	sharpen layer2 t2 shift
dmsc3_sharpen_layer2_t1	10:0	r	'h0	sharpen layer2 t1

ISP_DMSC3_SHARP_CURVE2_Y_SHD

0x00007f80 = Byte Address

Description: demosaic3 SHARP CURVE2 Y shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_layer2_r3	26:18	r	'hc8	sharpen layer2 r3
dmsc3_sharpen_layer2_r2	17:9	r	'h98	sharpen layer2 r2
dmsc3_sharpen_layer2_r1	8:0	r	'h0	sharpen layer2 r1

ISP_DMSC3_SHARP_LINE_X_SHD

0x00007f84 = Byte Address

Description: demosaic3 SHARP LINE X shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_line_shift1	31:28	r	'h3	sharpen line shift1
dmsc3_sharpen_line_t1	26:16	r	'h18	sharpen line t1
dmsc3_sharpen_line_strengt h	11:0	r	'h8	sharpen line strength

ISP_DMSC3_SHARP_LINE_Y_SHD

0x00007f88 = Byte Address

Description: demosaic3 SHARP LINE Y shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_sharpen_line_r2	24:16	r	'h100	sharpen line r2
dmsc3_sharpen_line_r1	8:0	r	'h0	sharpen line r1

ISP_DMSC3_DEPURPLE_CTRL1_SHD

0x00007f8c = Byte Address

Description: demosaic3 depurple color range control 1 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_depurple_cr_highth_1	31:24	r	'hc8	depurple cr high threshold 1
dmsc3_depurple_cr_lowth_1	23:16	r	'h70	depurple cr low threshold 1
dmsc3_depurple_cb_highth_1	15:8	r	'hdc	depurple cb high threshold 1
dmsc3_depurple_cb_lowth_1	7:0	r	'h84	depurple cb low threshold 1

ISP_DMSC3_DEPURPLE_CTRL2_SHD

0x00007f90 = Byte Address

Description: demosaic3 depurple color range control 2 shadow

Sub-Field Name	*Bits*	*rw*	*Reset*	*Description*
dmsc3_depurple_cr_highth_2	31:24	r	'h76	depurple cr high threshold 2
dmsc3_depurple_cr_lowth_2	23:16	r	'h52	depurple cr low threshold 2
dmsc3_depurple_cb_highth_2	15:8	r	'h7a	depurple cb high threshold 2
dmsc3_depurple_cb_lowth_2	7:0	r	'h60	depurple cb low threshold 2

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Revision History

This section describes differences between document revisions.

Note: This document is not necessarily updated for each patch or minor revision. The information in this document tends to be stable across a revision (nnn) series.

Revision	Date	Compatible Version	Description
0.80	2023-07-18	ISP8000 Series isp8000_6_2_0_rc0s	Initial release

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